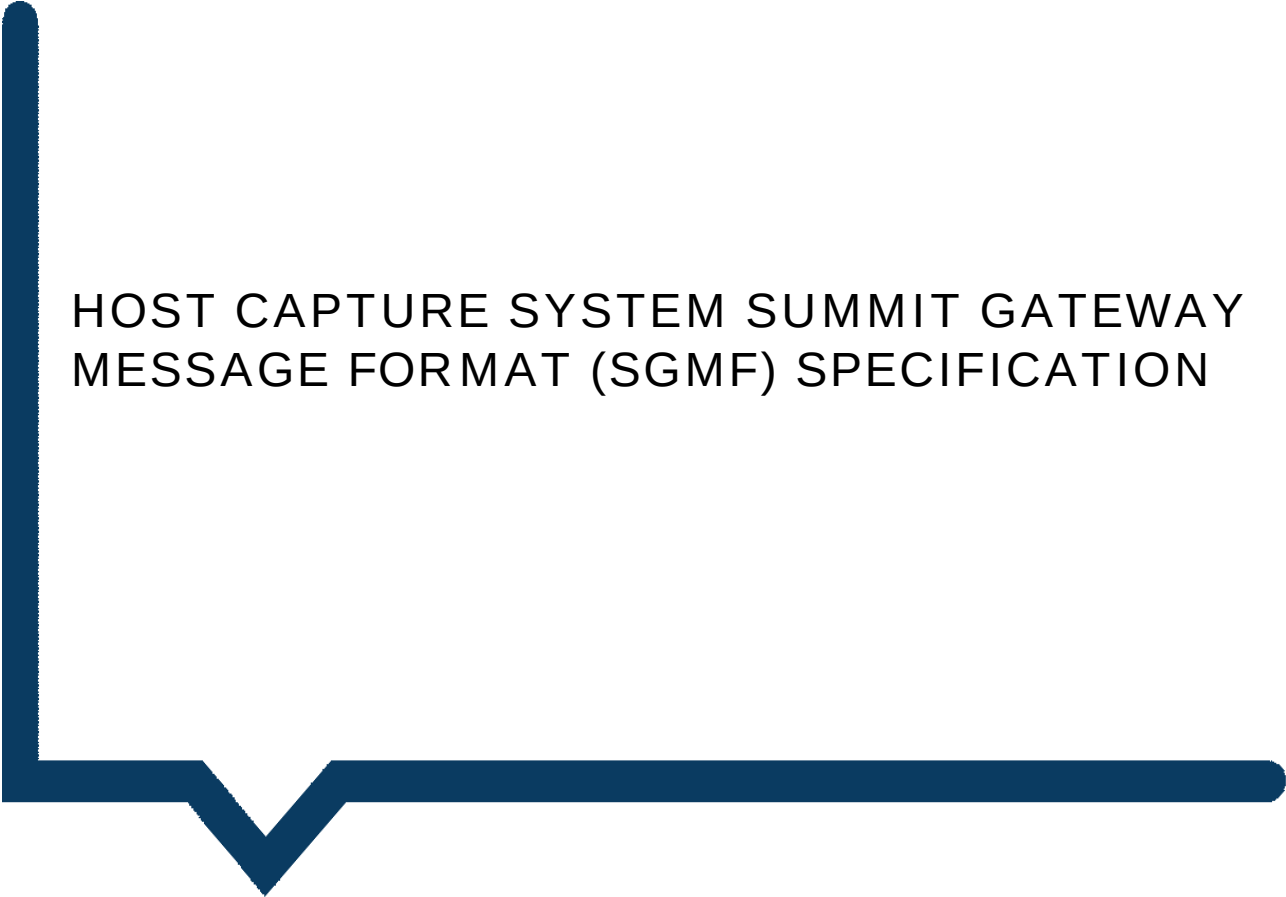




HOST CAPTURE SYSTEM SUMMIT GATEWAY MESSAGE FORMAT (SGMF) SPECIFICATION

VERSION 13.0
FEBRUARY 15, 2021

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Chapter 1

About this document

1.1 Overview

This document describes the proprietary TSYS Acquiring Solutions messaging format for transactions generated by host capture enabled Point-of-Sale (POS) devices processing on the TSYS Acquiring Solutions network. This message format is known as the Summit Gateway Message Format (SGMF). This document is organized into the following chapters:

Chapter	Description
About this document	Provides an overview of the SGMF specifications.
Host Capture Use Cases	Describes the external view of the TSYS Acquiring Solutions host capture use case diagram, showing use cases relevant to the POS device.
Transaction and Message Description	Describes the transaction class structure and its data elements.
Messaging Scenarios	Describes transaction scenarios and includes sequence diagrams to help illustrate the flows.
Field Specifications	Provides definitions for each SGMF field.
Transaction Message Examples	Provides examples of SGMF transaction messages fields specific to the industry, card type or circumstances of use.
Appendix A	Provides transaction codes with other message field values including Program Code, Cardholder Present Data and Payment Code.
Appendix B	Contains information related to EMV.
Appendix C	Contains codes created by Visa during the authorization process.
Appendix D	Lists suggested values for each Mastercard POS Data Subfield when authorizing a Mastercard transaction.
Appendix E	Contains subsections to define the authorization request record character set as well as character sets used for tracks one and two data encoded on the magnetic stripes.
Appendix F	Contains information used with <A139><B2> Document Type and <A139><B1> Ticket Transaction Type.

1.2 Document revisions

All revisions supporting release 13.0 are marked with change bars.

Table 1.1 Version 13.0 document revisions

Date Published	Chapter	Section	Description
February 15, 2021	Transaction and Message Description	3.10.7	Added new Section 3.10.7 Visa System Integrity Fees
	Field Specifications	5.12	Updated the Length and Format in Section 5.12 <A17> Transaction Original Number (TON)
		5.16	Updated description in Section 5.16.14 <A21><B14> Transaction status
			Updated value in Section 5.16.15 <A21><B15> Acceptance device type
		5.49	Removed possible value codes C01, C02 and C05 in Section 5.49 <A54> Transaction Type Identifier
		5.49	Added possible values in Section 5.49 <A54> Transaction Type Identifier
		5.77	Added and Updated response codes in Section 5.77 <A83> Response Code
		5.90	Updated Section 5.90 <A96> Network ID
		5.114	Added code H in Section 5.114 <A120> Authorization Source Code
		5.128	Updated description in Section 5.128.1 <A134><B1> Payment Account Reference
	Appendix C		Added code G5 in Table C.1 Visa card product codes

All revisions supporting release 12.7 are marked with change bars.

Table 1.2 Version 12.7 document revisions

Date Published	Chapter	Section	Description
December 18, 2020	Transaction and Message Description	3.10.6	Added new Section 3.10.6 Account Funding Transactions
	Field Specifications	5.33	Added new possible value of 8 in Section 5.33 <A38> Special Condition Indicator
		5.130	Removed section <A59> Reserved TSYS Acquiring Solutions Gift Card Operator/Clerk ID Added a new Funds Transfer section in Section 5.130 <A136> Business Application Identifier Added code FT - Funds Transfer in Table 5.18

All revisions supporting release 12.6 are marked with change bars.

Table 1.3 Version 12.6 document revisions

Date Published	Chapter	Section	Description
October 5, 2020	About this document	1.1	Added chapter links to the table in Section 1.1 Overview
	Field Specifications	5.16	Updated definition by removing Chip Code Condition as a requirement in Section 5.16.13 <A21><B13> Chip condition code
		5.63	Updated description in Section 5.63 <A69> Transaction Fee Amount
		5.106	Updated the length and description in Section 5.106.3 <A112><B3> Seller Street Address
		5.114	Removed MC, D and AX from the Space description in Section 5.114 <A120> Authorization Source Code
		5.118	Updated the length in Section 5.118.3 <A124><B2> Card Brand Token Assurance Level
		5.121	Added new table in Section 5.121 <A127> Merchant Name/Location Detail
			Updated Usage 3 - Dynamic Merchant Descriptor in Section 5.121 <A127> Merchant Name/Location Detail
			Updated the length in Section 5.121.8 <A127><B8> Foreign Retailer Indicator

All revisions supporting release 12.5 are marked with change bars.

Table 1.4 Version 12.5 document revisions

Date Published	Chapter	Section	Description
August 24, 2020	Field Specifications	5.16	Updated note values from A, I, P and R to A, I, and P in <A21><B1> Cardholder Present Data Removed the note in Code R in <A21><B1> Cardholder Present Data

All revisions supporting release 12.4 are marked with change bars.

Table 1.5 Version 12.4 document revisions

Date Published	Chapter	Section	Description
August 17, 2020	Transaction and Message Description	3.10.5	Added new Section 3.10.5 Consumer Bill Payment Service

Date Published	Chapter	Section	Description
	Field Specifications	5.16	Added information to the description in Section 5.16.1 <A21><B1> Cardholder Present Data Updated descriptions for values 4, A, I, P and R in <A21><B1> Cardholder Present Data Added information to the description in Section 5.16.3 <A21><B3> Card Data Input Mode Added information about Discover MIT in Section 5.16.14 <A21><B14> Transaction status Added new values D, E, N, S and U in Section 5.16.14 <A21><B14> Transaction status
		5.38	Update the note in Section 5.38.5 <A43><B5> Additional Amount
		5.72	Updated description in Section 5.72 <A78> Transaction ID
		5.107	Updated definition for code 2104 in Section 5.107 <A113> Message Reason Code
		5.114	Added codes and updated code descriptions in Section 5.114 <A120> Authorization Source Code
		5.121	Updated Section 5.121 <A127> Merchant Name/ Location Detail and added Table 5.13
		5.130	Added <A127><B8> Foreign Retailer Indicator Updated description in Section 5.130 <A136> Business Application Identifier Added Code CB in Table 5.15
	Appendix B		Updated description for Tag 9C in Table B.2 Added Table B.3

All revisions supporting release 12.3 are marked with change bars.

Table 1.6 Version 12.3 document revisions

Date Published	Chapter	Section	Description
June 16, 2020	Field Specifications	5.62	Added information to description in Section 5.62 <A68> Settlement Amount
		5.63	Updated acquirer to merchant in the description in Section 5.63 <A69> Transaction Fee Amount
		5.121	Updated definition for Usage 1 - Payment Facilitators and Usage 2 - Visa Marketplace in Section 5.121 <A127> Merchant Name/Location Detail

All revisions supporting release 12.2 are marked with change bars.

Table 1.7 Version 12.2 document revisions

Date Published	Chapter	Section	Description
May 18, 2020	Transaction and Message Description	3.10.4	Added new Section 3.10.4 Purchase Return Authorization
	Field Specifications	5.28	Removed information about effective February 2020 Release in <A33><B5> Digital Payment Cryptogram
		5.38	Added information to description in <A43><B4> Sign
		5.119	Updated Value Description for 101 and removed 102 in <A125><B1> Mastercard Wallet Identifier
		5.121	Updated description with effective date of April 2021 for Visa in <A127><B2>Location Detail Name

All revisions supporting release 12.1 are marked with change bars.

Table 1.8 Version 12.1 document revisions

Date Published	Chapter	Section	Description
March 06, 2020	Host Capture Use Cases	2.3	Updated description for Adjust Transaction in Table 2.1
	Transaction and Message Description	3.10.3	Added new Section 3.10.3 Mastercard High Risk Purchase Identifiers
	Field Specifications	5.16	Added information to description in <A21><B14> Transaction status
		5.49	Updated heading from <54> Payment Transaction Type Identifier to <A54> Transaction Type Identifier
		5.73	Updated transaction code 13 in Section 5.73 <A79> Use of TSYS Token Request
		5.106	Updated description for <A112><B5> Seller City
		5.121	Updated description for Usage 3 - Dynamic Merchant Descriptor in Section 5.121 <A127> Merchant Name/ Location Detail
			Added description for Section 5.121.4 <A127><B4> Location Detail City in Section 5.121 <A127> Merchant Name/Location Detail
	Transaction Message Examples	6.12	Updated link for <A54> Transaction Type Identifier

All revisions supporting release 12.0 are marked with change bars.

Table 1.9 Version 12.0 document revisions

Date Published	Chapter	Section	Description
February 17, 2020	Transaction and Message Description	3.10.2	Added new Section 3.10.2 Enhanced Authentication for Online Gambling Merchants
	Messaging Scenarios	4.3	Updated definition for Section 4.3 Refund transaction
	Field Specifications	5.16	Added information to description in <A21><B14> Transaction status
		5.26	Added to the description in Section 5.26 <A32> MOTO/e-Commerce Indicator
		5.28	Added to the description for 3-D Secure CAVV and 3-D Secure CAVV, Revised in Section 5.28 <A33> Authentication Data and removed position descriptions
		5.34	Added to the description for position 3 in Table 5.12
		5.49	Updated description and added P70- Cryptocurrency and P71 - High Risk Securities in Section 5.49 <A54> Transaction Type Identifier
		5.73	Added Mastercard and Discover to transaction code 13 in Section 5.73 <A79> Use of TSYS Token Request
		5.121	Updated the title from Payment Facilitator to Merchant Name/Location Detail and updated information in Section 5.121 <A127> Merchant Name/Location Detail

Date Published	Chapter	Section	Description
			Updated Section 5.121.1 <A127><B1> Payment Facilitator Identifier
			Updated Section 5.121.2 <A127><B2> Location Detail Name
			Updated Section 5.121.3 <A127><B3> Sub-Merchant Identifier
			Updated Section 5.121.4 <A127><B4> Location Detail City
			Updated Section 5.121.5 <A127><B5> Location Detail State/Province Code
			Updated Section 5.121.6 <A127><B6> Location Detail Country Code
		5.124	Updated Section 5.121.7 <A127><B7> Location Detail Postal Code
			Added information in description for Section 5.124 <A130> Authorization Indicator

All revisions supporting release 11.11 are marked with change bars.

Table 1.10 Version 11.11 document revisions

Date Published	Chapter	Section	Description
December 19, 2019	Field Specifications	5.41	Added description to Tag A46 Partial Authorization Indicator for Visa only (estimated amount)
		5.107	Updated information Section 5.107 <A113> Message Reason Code
		5.109	Renamed the header to Mastercard Payment Indicators for Section 5.109 <A115> Mastercard Payment Indicators
			Updated description for Section 5.109.1 <A115><B1> Domain Server
			Updated description and renamed header for Section 5.109.2 <A115><B2> Device Type
	Transaction Message Examples	6.29	Added Tag name <A93> Card Type in Section 6.29 ACH Transaction

All revisions supporting release 11.10 are marked with change bars.

Table 1.11 Version 11.10 document revisions

Date Published	Chapter	Section	Description
November 09, 2019	Field Specifications	5.6	Removed Code A - ACH from Section 5.6 <A11> Program Code
		5.7	Added A - Automated Clearing House for a possible value in Section 5.7 <A12> Payment Code
		5.16	Removed V, D and AX from Possible value 6 in Section 5.16.3 <A21><B3> Card Data Input Mode
		5.18	Updated the note description in Section 5.18 <A23> Track 1
		5.19	Updated the note description in Section 5.19 <A24> Track 2
		5.90	Updated description in Section 5.90 <A96> Network ID
		5.105	Updated Tag DF79 description in Section 5.105 <A111> Chip Card Data
	Transaction Message Examples		Removed PIN-less debit bill pay Removed PIN-less debit bill pay with token

Table 1.12 Version 11.9 document revisions

Date Published	Chapter	Section	Description
September 25, 2019	Field Specifications	5.16	Updated possible card brands for value 6 in the description in <A21><B3> Card Data Input Mode
		5.121	Updated format from NUM to A/N and updated 15-digit to 15- byte in the description in <A127><B3> Sub-Merchant Identifier

All revisions supporting release 11.8 are marked with change bars.

Table 1.13 Version 11.8 document revisions

Date Published	Chapter	Section	Description
September 10, 2019	Messaging Scenarios	4.2	Updated headers and definitions in Section 4.2 Request authorization
		4.4	Updated the header and definition in Section 4.4 Adjust transaction Updated step 6 of flow and image for Summit Split Shipment Flow and added section No Response to Split Shipment in Section 4.4 Adjust transaction
	Field Specifications	5.6	Updated Note for length Y in <A11> Program Code
		5.28	Added new Table 5.1 SLI valid scenarios in Section 5.28 <A33> Authentication Data <A33><B3> Security Level Indicator (SLI)
		5.122	Updated description and Table 5.14 and Table 5.15 in Section 5.122 <A128> ACH Transaction Data

All revisions supporting release 11.7 are marked with change bars.

Table 1.14 Version 11.7 document revisions

Date Published	Chapter	Section	Description
August 20, 2019	Field Specifications	5.107	Updated code for Deferred Authorizations from 2506 to 5206 in Section 5.107 <A113> Message Reason Code

Date Published	Chapter	Section	Description
August 19, 2019	Field Specifications	5.27	Added description in Section 5.27 <A33> Transaction ID (XID)
		5.28	Added new Section 5.28 <A33> Authentication Data
		5.29	Added description in Section 5.29 <A34> Cardholder Authentication for Secure e-Commerce Transactions
		5.34	Updated description in Section 5.34 <A39> Electronic Commerce Security Level Indicator
		5.35	Added description in Section 5.35 <A40> Authentication Data
		5.107	Updated Section 5.107 <A113> Message Reason Code for Deferred Authorizations
		5.118	Updated DES Enhancement to Support E-Commerce Transactions description in Section 5.118 <A124> Card Brand Tokenization
		5.122	
		5.136	Added Table 5.14
			Updated description in Section 5.136 <A142> Authentication Data

All revisions supporting release 11.6 are marked with change bars.

Table 1.15 Version 11.6 document revisions

Date Published	Chapter	Section	Description
June 21, 2019	Field Specifications	5.34	Updated valid value 21 description in Section 5.34 <A39> Electronic Commerce Security Level Indicator
		5.63	Updated description for length 9 in Section 5.63 <A69> Transaction Fee Amount
		5.83	Removed column labeled Condition (each condition must be present) in Section 5.83 <A89> Transaction Data Condition Code
			Updated code 04, 40, and 90 in Section 5.83 <A89> Transaction Data Condition Code
		5.119	Updated value 216 description from Android Pay to Google Pay in <A125><B1> Mastercard Wallet Identifier
		5.121	Updated description in <A127><B4> Location Detail City
	Transaction Message Examples	6.33	Added new Section 6.33 Recurring / Installment

All revisions supporting release 11.5 are marked with change bars.

Table 1.16 Version 11.5 document revisions

Date Published	Chapter	Section	Description
May 24, 2019	Messaging Scenarios	4.9	Renamed Verified by Visa to Visa Secure
	Field Specifications	5.26	Renamed Verified by Visa to Visa Secure in possible value 5
		5.29	Renamed Verified by Visa to Visa Secure in Position 1, 3-D Secure Authentication Results Code

Date Published	Chapter	Section	Description
	Transaction Message Examples	6.33	Added new Section 6.33 Recurring / Installment

All revisions supporting release 11.4 are marked with change bars.

Table 1.17 Version 11.4 document revisions

Date Published	Chapter	Section	Description
May 03, 2019	Messaging Scenarios	4.9	Added new table in Section 4.9 3D Secure Authentication
	Field Specifications	5.16	Updated description for possible values 5 and E in Section 5.16.6 <A21><B6> Card data input capability
		5.24	Removed MC from possible value 3 in Section 5.16.7 <A21><B7> Cardholder authentication capability
		5.34	Updated definition in Section 5.24 <A30> Address Verification Data
		5.38	Added a new possible value of 3 in Table 5.12
		5.65	Updated the description for possible values 80 and 81 in Section 5.38.2 <A43><B2> Amount Type
		5.109	Updated definition in Section 5.65 <A71> Amex Airline Data
		5.128	Added a note in Section 5.109.2 <A115><B2> Device Type
			Updated description for length 35 in Section 5.128.1 <A134><B1> Payment Account Reference

Date Published	Chapter	Section	Description
		5.121	Updated description in Usage 1 - Payment Facilitators in Section 5.121<A127> Merchant Name/Location Detail<A77><B1> Updated description for: • Section 5.121.1 <A127><B1> Payment Facilitator Identifier • Section 5.121.2 <A127><B2>Location Detail Name • Section 5.121.3 <A127><B3> Sub-Merchant Identifier • Section 5.121.4 <A127><B4> Location Detail City • Section 5.121.5 <A127><B5> Location Detail State/Province Code • Section 5.121.6 <A127><B6> Location Detail Country Code • Section 5.121.7 <A127><B7> Location Detail Postal Code

All revisions supporting release 11.3 are marked with change bars.

Table 1.18 Version 11.3 document revisions

Date Published	Chapter	Section	Description
April 12, 2019	Field Specifications	5.16	Added note to the description for Possible value 4 in Section 5.16.1 <A21><B1> Cardholder Present Data and added note to the description in Section 5.16.3 <A21><B3> Card Data Input Mode
		5.34	Updated position 1 and 2 valid value 21 in Section 5.34 <A39> Electronic Commerce Security Level Indicator
		5.62	Updated description for Section 5.62 <A68> Settlement Amount
		5.88	Updated the description and add two new possible values O and I in Section 5.88 <A94> Requested ACI

Table 1.20 Version 11.1 document revisions

Date Published	Chapter	Section	Description
February 25, 2019	Transaction and Message Description	3.10	Updated Section 3.10 General Industry Requirements
	Field Specifications	5.16	Added (MC) to the possible value 3 in <A21><B7> Cardholder authentication capability Updated description in <A21><B14> Transaction status
		5.35	Updated Section 5.35 <A40> Authentication Data
		5.136	Updated description for Section 5.136.1 <A142><B1> Program Protocol and Section 5.136.2 <A142><B2> Directory Server Transaction ID

All revisions supporting release 11.0 are marked with change bars.

Table 1.21 Version 11.0 document revisions

Date Published	Chapter	Section	Description
January 10, 2019	Messaging Scenarios	4.4	Updated graphics for Summit Split Shipment Flow and Incremental Flows
		4.9	Updated Section 4.9 3D Secure Authentication
	Field Specifications	5.16	Updated Section 5.16 <A21> POS Entry Data remove only from possible value <A21><B3> Card Data Input Mode , <A21><B6> Card data input capability , and <A21><B9> Operating environment
		5.17	Updated Section 5.17 <A22> Market Specific Data Identifier
		5.16.15.16.15.16.15.16.15.335.16.1	Updated Section 5.33 <A38> Special Condition Indicator
		5.58	Updated Section 5.58 <A64> e-Commerce Goods Indicator
		5.88	Updated Section 5.88 <A94> Requested ACI
		5.91	Updated Section 5.91 <A97> Address Verification Result Code
		5.99	Updated <A105><B19> Discount Rate Exponent

All revisions supporting release 10.9 are marked with change bars.

Table 1.22 Version 10.9 document revisions

Date Published	Chapter	Section	Description
December 18, 2018	Field Specifications	5.13	Updated description in Section 5.13 <A18> Merchant Category Code (MCC)
		5.16.15.16.15.16.15.16.15.775.16.1	Updated response code R1 and added response code R3 in Section 5.77 <A83> Response Code

All revisions supporting release 10.8 are marked with change bars.

Table 1.23 Version 10.8 document revisions

Date Published	Chapter	Section	Description
October 16, 2018	Messaging Scenarios	4.8	Updated description in Section 4.8 Using encryption and tokens
	Field Specifications	5.16.15.16.15.16.15.16.15.735.16.1 5.123	Added and removed some information in Section 5.73 <A79> Use of TSYS Token Request Updated description in Section 5.123.5 <A129><B5> Tax Type Applied

All revisions supporting release 10.7 are marked with change bars.

Table 1.24 Version 10.7 document revisions

Date Published	Chapter	Section	Description
September 28, 2018	Field Specifications	5.16.15.16.15.16.15.16.5.16.1	Added a note in Section 5.16.3 <A21><B3> Card Data Input Mode
		5.345.16.1	Updated description in Section 5.34 <A39> Electronic Commerce Security Level Indicator
		5.121	Removed some information in each description in the following: Section 5.121.1 <A127><B1> Payment Facilitator Identifier Section 5.121.2 <A127><B2> Location Detail Name Section 5.121.3 <A127><B3> Sub-Merchant Identifier Section 5.121.4 <A127><B4> Location Detail City Section 5.121.5 <A127><B5> Location Detail State/Province Code Section 5.121.6 <A127><B6> Location Detail Country Code Section 5.121.7 <A127><B7> Location Detail Postal Code
		5.124	Updated description for Code P and F in Section 5.124 <A130> Authorization Indicator

All revisions supporting release 10.6 are marked with change bars.

Table 1.25 Version 10.6 document revisions

Date Published	Chapter	Section	Description
August 24, 2018	Field Specifications	5.16	Added Discover only to I - Incremental Authorization in Section 5.16.1 <A21><B1> Cardholder Present Data Added P in the description in Section 5.16.13 <A21><B13> Chip condition code
August 13, 2018	Field Specifications	5.16.15.16.15.16.15.16.1 5.215.16.1 5.25 5.71 5.91 5.118	Added possible value of I - Incremental Authorization in Section 5.16.1 <A21><B1> Cardholder Present Data Added information in the description in Section 5.21 <A26> Manually Entered Expiry Date Added information in the description in Section 5.25 <A31> Verification Code (CID, CVC2, CVV2) Updated description for Section 5.71 <A77> Mastercard Transit Program Added information in the description in Section 5.91 <A97> Address Verification Result Code Added information in the description in Section 5.118 <A124> Card Brand Tokenization

All revisions supporting release 10.5 are marked with change bars.

Table 1.26 Version 10.5 document revisions

Date Published	Chapter	Section	Description
July 11, 2018	Field Specifications	5.16	Updated definition for possible value 7 Section 5.16.3 <A21><B3> Card Data Input Mode

All revisions supporting release 10.4 are marked with change bars.

Table 1.27 Version 10.4 document revisions

Date Published	Chapter	Section	Description
June 19, 2018	Messaging Scenarios	4.4	Added two new sections Split Shipment Flow and No Response to Split Shipment
	Field Specifications	5.5	Added two new transaction codes 110 and 111 in Section 5.5 <A10> Transaction Code
	Transaction Message Examples	6.31 6.32	Added new Section 6.31 Hold Capture Original Request Message Added new Section 6.32 Split Shipment Request Message

All revisions supporting release 10.3 are marked with change bars.

Table 1.28 Version 10.3 document revisions

Date Published	Chapter	Section	Description
May 04 2018	Field Specifications	5.72	Added information in the description in Section 5.72 <A78> Transaction ID

All revisions supporting release 10.2 are marked with change bars.

Table 1.29 Version 10.2 document revisions

Date Published	Chapter	Section	Description
April 24, 2018	Field Specifications	5.124	Updated description in Section 5.124 <A130> Authorization Indicator
		5.134	Updated description in Section 5.134 <A140> Transaction Integrity Class

All revisions supporting release 10.1 are marked with change bars.

Table 1.30 Version 10.1 document revisions

Date Published	Chapter	Section	Description
March 16, 2018	Field Specifications	5.16	Updated information in Section 5.109.2 <A115><B2> Device Type

All revisions supporting release 10.0 are marked with change bars.

Table 1.31 Version 10.0 document revisions

Date Published	Chapter	Section	Description
February 12, 2018	Field Specifications	5.16	Added new possible value of 7 in Section 5.16.3 <A21><B3> Card Data Input Mode
		5.26	Updated possible values, 4, 5, 6 7 R and T in Section 5.26 <A32> MOTO/e-Commerce Indicator
		5.29	Updated Table 5.8 in Section 5.29 <A34> Cardholder Authentication for Secure e-Commerce Transactions
		5.65	Updated Section 5.65 <A71> Amex Airline Data Added a new Section 5.65.13 <A71><B13>Customer IP and a new Section 5.65.14 <A71><B14>Customer Email Address
		5.83	Updated Section 5.83 <A89> Transaction Data Condition Code
		5.84	Updated possible values in Section 5.84 <A90> CAVV Result Code
		5.88	Updated the definition for the possible value R in Section 5.88 <A94> Requested ACI
		5.90	Removed P as a possible value in Section 5.90 <A96> Network ID
		5.114	Removed Code 3 and added Space in Section 5.114 <A120> Authorization Source Code
		5.129	Added information in Section 5.129 <A135> POS Environment Indicator

All revisions supporting release 9.9 are marked with change bars.

Table 1.32 Version 9.9 document revisions

Date Published	Chapter	Section	Description
December 12, 2017	Field Specifications	5.122	Remove ARC from Section 5.122.1 <A128><B1> ACH SEC Code
		5.134	Updated Section 5.134 <A140> Transaction Integrity Class

All revisions supporting release 9.8 are marked with change bars.

Table 1.33 Version 9.8 document revision

Date Published	Chapter	Section	Description
October 18, 2017	Messaging Scenarios	4.9	Added Section 4.9 3D Secure Authentication
	Field Specifications	5.7	Updated Section 5.16.6 <A21><B6> Card data input capability
		5.131	Added Section 5.131 <A137> Alternate Hierarchy
	Transaction Message Examples	6.30	Added Section 6.30 3D Secure Transaction

All revisions supporting release 9.7 are marked with change bars.

Table 1.34 Version 9.7 document revision

Date Published	Chapter	Section	Description
September 12, 2017	Field Specifications	5.16	Updated Section 5.16.6 <A21><B6> Card data input capability
			Removed Visa and Mastercard from value M, P, Q and R in Section 5.16.9 <A21><B9> Operating environment
			Updated Section 5.16.10 <A21><B10> Card data output capability
		5.63	Updated Section 5.63 <A69> Transaction Fee Amount
		5.75	Updated Section 5.75 <A81> Local Transaction Date
		5.77	Added new response code M0 in Section 5.77 <A83> Response Code
		5.76	Update Section 5.76 <A82> Local Transaction Time
		5.129	Update Code C in Section 5.129 <A135> POS Environment Indicator
		5.134	Updated Section 5.134 <A140> Transaction Integrity Class
		5.135	Updated Section 5.135 <A141> Mastercard Service Parameters

All revisions supporting release 9.6 are marked with change bars.

Table 1.35 Version 9.6 document revision

Date Published	Chapter	Section	Description
August 31, 2017	Field Specifications	5.34	Updated Section 5.34 <A39> Electronic Commerce Security Level Indicator
		5.124	Section 5.124 <A130> Authorization Indicator
	Appendix B		Updated and added information in Appendix B

All revisions supporting release 9.5 are marked with change bars.

Table 1.36 Version 9.5 document revision

Date Published	Chapter	Section	Description
August 31, 2017	Field Specifications	5.34	Updated Section 5.34 <A39> Electronic Commerce Security Level Indicator
		5.124	Section 5.124 <A130> Authorization Indicator
	Appendix B		Updated and added information in Appendix B

Visa Check Acceptance is no longer supported effective October 13, 2017. All references to Check Acceptance have been removed from the specification.

Date Published	Chapter	Section	Description
August 14, 2017	Field Specifications	5.25	Added Visa Checkout Usage in the description for position 3-6 in Section 5.25 <A31> Verification Code (CID, CVC2, CVV2)
		5.34	Updated Section 5.34 <A39> Electronic Commerce Security Level Indicator
		5.114	Added code B in Section 5.114 <A120> Authorization Source Code
		5.118	Updated Section 5.118 <A124> Card Brand Tokenization Removed Card Brand Token Request Removed Card Brand Token Status inquiry and Maintenance Removed <A124><B5> through <A124><A124><B9>
		5.134	Added Section 5.134 <A140> Transaction Integrity Class
		5.135	Added Section 5.135 <A141> Mastercard Service Parameters
	Appendix B		Added Tag9F24

All revisions supporting release 9.4 are marked with change bars.

Table 1.37 Version 9.4 document revision

Chapter	Section	Description
Field Specifications	5.66	Added and updated Section 5.66 <A72> Passenger Transportation Data Group
	5.131	Added Section 5.131 <A137> Alternate Hierarchy
	5.132	Added Section 5.132 <A138> American Express Travel Cruise Addendum
	5.133	Added Section 5.133 <A139> American Express Airline Addendum this was previously <A137>. <A137> is now for Internal Use Only
Yes Appendix F		Added Yes Appendix F

All revisions supporting release 9.3 are marked with change bars.

Table 1.38 Version 9.3 document revision

Chapter	Section	Description
Messaging Scenarios	4.3	Added information in Section 4.3 Refund transaction to include Purchase Return Authorization for Visa
Field Specifications	5.129	Updated Section 5.129 <A135> POS Environment Indicator

All revisions supporting release 9.2 are marked with change bars.

Table 1.39 Version 9.2 document revision

Chapter	Section	Description
Field Specifications	5.107	Added information in Section <A113> Message Reason Code at the end of the Message Reason Code table

All revisions supporting release 9.1 are marked with change bars.

Table 1.40 Version 9.1 document revision

Chapter	Section	Description
Field Specifications	5.16	Updated descriptions for Section 5.16 <A21> POS Entry Data
	5.18	Updated description for Section 5.18 <A23> Track 1
	5.19	Updated description for Section 5.19 <A24> Track 2
	5.26	Updated possible values in Section 5.26 <A32> MOTO/e-Commerce Indicator
	5.73	Updated description for in Section 5.73 <A79> Use of TSYS Token Request
	5.85	Added 04 as a possible value in Section 5.85 <A91> Merchant Advice Code
	5.127	Updated description for Section 5.127 <A133> Lane ID
	5.128	Updated description for <A134><B1> Payment Account Reference
Transaction Message Examples	6.3 - 6.15	Updated Examples that use <A21><B7>, <A21><B8>, <A21><B10>, <A21><B11> and <A21><B12> for the following sections: 6.3 - 6.15 , 6.18 - 6.19 and 6.28 - 6.29
	6.18 - 6.19	
	6.28 - 6.29	
Appendix E		Added new Appendix E

All revisions supporting release 9.0 are marked with change bars.

Table 1.41 Version 9.0 document revision

Chapter	Section	Description
Field Specifications	5.16	Added new possible value of 'T' - Discover PayButton in <A21><B1> Cardholder Present Data
	5.27	Updated the description in <A33> Transaction ID (XID)
	5.29	Updated and added information in Section 5.29 <A34> Cardholder Authentication for Secure e-Commerce Transactions
	5.128	Added Mastercard in the description in <A134><B1> Payment Account Reference
	5.130	Added new section Section 5.130 <A136> Business Application Identifier
Appendix C		Added new description for X^ in Table C.1

All revisions supporting release 8.8 are marked with change bars.

Table 1.42 Version 8.8 document revision

Chapter	Section	Description
Field Specifications	5.119	Updated the description in Section 5.119.1 <A125><B1> Mastercard Wallet Identifier

All revisions supporting release 8.7 are marked with change bars.

Table 1.43 Version 8.7 document revision

Chapter	Section	Description
Field Specifications	5.38	Added Visa only and removed Discover only from Value 43 in Section 5.38.2 <A43><B2> Amount Type

All revisions supporting release 8.6 are marked with change bars.

Table 1.44 Version 8.6 document revision

Chapter	Section	Description
Field Specifications	5.26	Add Amex as a card brand applicable to value A in Section 5.26 <A32> MOTO/e-Commerce Indicator

All revisions supporting release 8.5 are marked with change bars.

Table 1.45 Version 8.5 document revision

Chapter	Section	Description
Field Specifications	5.34	Updated Section 5.34 <A39> Electronic Commerce Security Level Indicator
	5.35	Updated description in Section 5.35 <A40> Authentication Data
	5.49	Add definition to Section 5.49 <A54> Transaction Type Identifier and removed the description from the table
	5.50	Updated the description in Section 5.50<A55> Real-time Substantiation Indicator<A77><B1>
	5.68	Added information in definition in Section 5.68 <A74> Cardholder Verification Results
	5.70	Updated Section 5.70 <A76> Mastercard EMS Fraud Service Group , Section 5.70.2 <A76><B2> EMS Service Response Indicator , Section 5.70.3 <A76><B3> EMS Result Code , Section 5.70.4 <A76><B4> EMS Additional Information , Section 5.70.5 <A76><B5> EMS Risk Score and Section 5.70.6 <A76><B6> EMS Risk Score Reason Code
	5.71	Updated definition in Section 5.71 <A77> Mastercard Transit Program
	5.109	Updated information in Section 5.109.1 <A115><B1> Domain Server and Section 5.109.2 <A115><B2> Device Type
	5.124	Updated description in Section 5.124 <A130> Authorization Indicator
	5.126	Updated description in Section 5.126 <A132> Fraud Enhanced Data

All revisions supporting release 8.4 are marked with change bars.

Table 1.46 Version 8.4 document revision

Chapter	Section	Description
Field Specifications	5.10	Added information to the description in Section 5.10 <A15> Transaction Amount
	5.16	Added the value of A as a possible value in Section 5.16.1 <A21><B1> Cardholder Present Data Updated possible value 3 in Section 5.16.5 <A21><B5> Authentication Entity
	5.26	Updated possible value 5 in Section 5.26 <A32> MOTO/e-Commerce Indicator
	5.27	Updated description in Section 5.27 <A33> Transaction ID (XID)
	5.29	Updated description in Section 5.29 <A34> Cardholder Authentication for Secure e-Commerce Transactions
	5.34	Updated Section 5.34 <A39> Electronic Commerce Security Level Indicator from UCAF Collection Indicator to Mastercard Electronic Commerce Security Indicator and added information to the section
	5.38	Added note in Section 5.38 <A43> Additional Amounts Group Added 43 as a possible value in Section 5.38.2 <A43><B2> Amount Type
	5.41	Updated Section 5.41 <A46> Partial Authorization Indicator
	5.47	Updated the description for 17 in Section 5.47<A52> Reversal Request Code<A77><B1>
	5.72	Added information in description in Section 5.72 <A78> Transaction ID
	5.77	Added response code D3 in Section 5.77 <A83> Response Code
	5.84	Added possible values in Section 5.84 <A90> CAVV Result Code
	5.90	Removed V-VisaNet from the possible values
	5.107	Added and updated information in Section 5.107 <A113> Message Reason Code
	5.128	Added Section 5.128 <A134> Payment Specific Data
	5.129	Added Section 5.129 <A135> POS Environment Indicator

All revisions supporting release 8.3 are marked with change bars.

Table 1.47 Version 8.3 document revision

Chapter	Section	Description
Field Specifications	5.16	Updated definition in Section 5.16.13 <A21><B13> Chip condition code
	5.34	Updated Section 5.34 <A39> Electronic Commerce Security Level Indicator
	5.90	Added possible values in Section 5.90 <A96> Network ID

All revisions supporting release 8.2 are marked with change bars.

Table 1.48 Version 8.2 document revision

Chapter	Section	Description
Field Specifications	5.26	Clarified the definitions for the values of 5 and 6 in Section 5.26 <A32> MOTO/e-Commerce Indicator
	5.53	Clarified the information under Section 5.53 <A58> Visa Contactless Data Group

All revisions supporting release 8.1 are marked with change bars.

Table 1.49 Version 8.1 document revision

Chapter	Section	Description
Field Specifications	5.16	Updated <A21><B3> and <A21><B6> values X and Y and added value Z in <A21><B3>

All revisions supporting release 8.0 are marked with change bars.

Table 1.50 Version 8.0 document revision

Chapter	Section	Description
Field Specifications	5.127	Added Section 5.127 <A133> Lane ID NOTE: The Lane ID will not be effective until April 2016.

All revisions supporting release 7.9 are marked with change bars

Table 1.51 Version 7.9 document revision

Chapter	Section	Description
Field Specifications	5.127	Added Section 5.127 <A133> Lane ID

All revisions supporting release 7.8 are marked with change bars.

Table 1.52 Version 7.8 document revision

Chapter	Section	Description
Field Specifications	5.46	Updated length and format in Section 5.46.2 <A51><B2> Reserved

All revisions supporting release 7.7 are marked with change bars.

Table 1.53 Version 7.7 document revision

Chapter	Section	Description
Field Specifications	5.123	Updated description in Section 5.123 <A129> Tax Data

All revisions supporting release 7.6 are marked with change bars.

Table 1.54 Version 7.6 document revision

Chapter	Section	Description
Field Specifications	5.118	Updated length in Section 5.118.2 <A124><B1> Card Brand Token Requestor ID
	5.124	Updated description in Section 5.124 <A130> Authorization Indicator

All revisions supporting release 7.5 are marked with change bars.

Table 1.55 Version 7.5 document revision

Chapter	Section	Description
Field Specifications	5.26	Updated Section 5.26 <A32> MOTO/e-Commerce Indicator
	5.106	Updated Section 5.106.5 <A112><B5> Seller City

All revisions supporting release 7.4 are marked with change bars.

Table 1.56 Version 7.4 document revision

Chapter	Section	Description
Field Specifications	5.26	Updated Section 5.26 <A32> MOTO/e-Commerce Indicator
	5.27	Updated Section 5.27 <A33> Transaction ID (XID)
	5.29	Updated Section 5.29 <A34> Cardholder Authentication for Secure e-Commerce Transactions
	5.34	Updated Section 5.34 <A39> Electronic Commerce Security Level Indicator
	5.43	Updated Section 5.43 <A48> Amex Merchant DBA Name
	5.44	Updated Section 5.44 <A49> Amex Station Location Code
	5.45	Updated Section 5.45 <A50> Amex Seller/Vendor Code
	5.46	Updated Section 5.46 <A51> Digital Entity Identifier
	5.83	Updated Section 5.83 <A89> Transaction Data Condition Code
	5.106	Updated Section 5.106 <A112> Card Acceptor Data
	5.109	Updated Section 5.109.2 <A115><B2> Device Type
	5.123	Added new Section 5.123 <A129> Tax Data
	5.124	Updated Section 5.124 <A130> Authorization Indicator
	5.125	Updated Section 5.125 <A131> Multi-Clearing Indicator
	5.126	Updated Section 5.126 <A132> Fraud Enhanced Data
Appendix D		Added new Appendix D

All revisions supporting release 7.3 are marked with change bars.

Table 1.57 Version 7.3 document revision

Chapter	Section	Description
Messaging Scenarios	4.2	Added new scenarios - Offline Authorization and ACH Transaction
Field Specifications	5.6	Added code “A” in Section 5.6 <A11> Program Code
	5.122	Added new Section 5.122 <A128> ACH Transaction Data
Transaction Message Examples	6.29	Added new Section 6.29 ACH Transaction

All revisions supporting release 7.2 are marked with change bars.

Table 1.58 Version 7.2 document revision

Chapter	Section	Description
Field Specifications	5.90	Removed Alaska Option in Section 5.90 <A96> Network ID
	5.106	Updated description in Section 5.106 <A112> Card Acceptor Data
	5.109	Updated Section <A115><B2> Mobile Device Type
	5.120	Added an “A” to the header
	5.121	Added an “A” to the header
Appendix C		Updated information for description for value S1 in Table C.1

All revisions supporting release 7.1 are marked with change bars.

Table 1.59 Version 7.1 document revision

Chapter	Section	Description
Field Specifications	5.106	Removed <A112><B5> Seller City in Section 5.106 <A112> Card Acceptor Data

All revisions supporting release 7.0 are marked with change bars.

Table 1.60 Version 7.0 document revision

Chapter	Section	Description
Field Specifications	5.16	Removed possible value of 9 from Section 5.16.1 <A21><B1> Cardholder Present Data and Section 5.16.2 <A21><B2> Card Present Data
	5.68	Added new possible values in Section 5.68.3 <A74><B3>Billing Name Result Code
	5.106	Added Section 5.107 <A113> Message Reason Code
	5.118	Updated Section 5.118 <A124> Card Brand Tokenization
	5.120	Added Section 5.120 <A126> ISO Identifier
	5.121	Added Section 5.121 <A127> Merchant Name/Location Detail

All revisions supporting release 6.5 are marked with change bars.

Table 1.61 Version 6.5 document revision

Chapter	Section	Description
Field Specifications	5.16	Added new possible values in Section 5.16.1 <A21><B1> Cardholder Present Data
	5.34	Added new possible values in Section 5.34 <A39> Electronic Commerce Security Level Indicator
	5.118	Updated Section Response Value : Updated Section 5.118 <A124> Card Brand Tokenization

All revisions supporting release 6.4 are marked with change bars.

Table 1.62 Version 6.4 document revision

Chapter	Section	Description
Field Specifications	5.43	Updated Section 5.43 <A48> Amex Merchant DBA Name
	5.44	Updated Section 5.44 <A49> Amex Station Location Code
	5.106	Updated Section 5.106 <A112> Card Acceptor Data and removed <A112><B3> Seller Business Name

All revisions supporting release 6.3 are marked with change bars.

Table 1.63 Version 6.3 document revision

Chapter	Section	Description
Section Messaging Scenarios	4.8	Updated Section 4.8 Using encryption and tokens
Field Specifications	5.5	Added value of 21 in Section 5.5 <A10> Transaction Code
	5.16	Updated Section 5.16 <A21> POS Entry Data
	5.27	Updated Section 5.27 <A33> Transaction ID (XID)
	5.29	Updated Section 5.29 <A34> Cardholder Authentication for Secure e-Commerce Transactions
	5.44	Updated Section 5.44 <A49> Amex Station Location Code
	5.45	Updated Section 5.45 <A50> Amex Seller/Vendor Code
	5.73	Updated Section 5.73 <A79> Use of TSYS Token Request
	5.89	Updated Section 5.89 <A95> Return ACI
	5.100	Updated Section 5.100 <A106> Encrypted Transaction
	5.106	Updated Section 5.106 <A112> Card Acceptor Data
	5.108	Updated Section 5.108 <A114> Alternate Account ID 1
	5.110	Updated Section 5.110 <A116> Mastercard Mapping Service
	5.115	Updated Section 5.115 <A121> Spend Qualified Indicator
	5.118	Added Section 5.118 <A124> Card Brand Tokenization

All revisions supporting release 6.2 are marked with change bars.

Table 1.64 Version 6.2 document revision

Chapter	Section	Description
Field Specifications	5.117	Added Section 5.117 <A123> Gateway Administrative Message

All revisions supporting release 6.1 are marked with change bars.

Table 1.65 Version 6.1 document revision

Chapter	Section	Description
Field Specifications	5.103	Removed Section <A104><B19> Line Item Level Discount Treatment
	5.104	Added Section 5.99.20 <A105><B20> Line Item Level Discount Treatment

All revisions supporting release 6.0 are marked with change bars.

Table 1.66 Version 6.0 document revision

Chapter	Section	Description
Field Specifications	5.24	Updated <A30><B6> Cardholder Billing First Name and <A30><B7> Cardholder Billing Last Name
	5.34	Updated <A39> Electronic Commerce Security Level Indicator Updated Response Value :
	5.49	Updated <A54> Transaction Type Identifier
	5.68	Updated <A74> Cardholder Verification Results and added values to <A74><B3>Billing Name Result Code
	5.98	Added <A104><B17>Invoice Level Discount Treatment , <A104><B18>Tax Treatments , and <A105> Level III Line Item Details Group
	5.115	Added <A121> Spend Qualified Indicator
	5.116	Added <A122> Goods Sold
Appendix C		Updated Table C.1

All revisions supporting release 5.3 are marked with change bars.

Table 1.67 Version 5.3 document revision

Chapter	Section	Description
Transaction and Message Description	3.2	Updated Section 3.2Message structure<A77><B1>
	3.3	Added Section 3.3Transaction Data Elements<A77><B1>
	3.7	Renamed Section 3.7Serial Line Protocol<A77><B1>
	3.8	Renames Section 3.8Serial Line Access Options<A77><B1>
	3.9	Added Section 3.9SGMF Direct Connect<A77><B1>
Field Specifications	5.54	Renamed Section 5.54<A60> Tip Amount<A77><B1>
	5.78	Updated Section 5.78<A84> Approval Code<A77><B1>
	5.111	Added Section 5.111<A117> TSYS Acquiring Solutions Gateway ID<A77><B1>
	5.112	Added Section 5.112<A118> Authorization Only Indicator<A77><B1>
	5.113	Added Section 5.113<A119> Visa Validation Code<A77><B1>
	5.114	Added Section 5.114<A120> Authorization Source Code<A77><B1>
	6.15	Updated Section 6.15Quasi cash credit card<A77><B1>
	6.15	Updated Section 6.15Quasi cash credit card<A77><B1>
	6.18	Updated Section 6.18Amex Airline Passenger Transport (APD) Transaction<A77><B1>
	6.19	Updated Section 6.19Passenger Transport Data Transaction<A77><B1>
	6.20	Updated Section 6.20Transaction reversal<A77><B1>
	6.28.1	Updated Section 6.28.1Credit Request with Token Request - Card Not Present - Manually Entered<A77><B1>
	6.28.2	Updated Section 6.28.2Encrypted retail credit<A77><B1>
	6.28.2	Updated Section 6.28.2Encrypted retail credit<A77><B1>
	6.28.3	Updated Section 6.28.3Encrypted card not present credit card<A77><B1>
Appendix B		Updated

All revisions supporting release 5.2 are marked with change bars.

Table 1.68 Version 5.2 document revision

Chapter	Section	Description
Messaging Scenarios	4.4.4	Updated Section 4.4.4 Split Shipment Flow
Field Specifications	5.16	Updated Section 5.16.3 <A21><B3> Card Data Input Mode , Section 5.16.6 <A21><B6> Card data input capability and Section 5.16.9 <A21><B9> Operating environment
	5.25	Updated Section 5.25 <A31> Verification Code (CID, CVC2, CVV2)
	5.41	Updated Section 5.41 <A46> Partial Authorization Indicator
	5.46	Updated Section 5.46 <A51> Digital Entity Identifier
	5.48	Updated Section 5.48 <A53> Discover Promotional Code
	5.62	Updated Section 5.62 <A68> Settlement Amount
	5.86	Updated Section 5.86 <A92> Card Product Codes
	5.108	Updated Section 5.108 <A114> Alternate Account ID 1
	5.109	Added Section 5.109 <A115> Mastercard Payment Indicators
	5.110	Added Section 5.110 <A116> Mastercard Mapping Service
Transaction Message Examples	5.144	Updated Section 5.144 <A325> Batch Summary Group
	6.3	Updated Section 6.3 Retail credit card
	6.4	Updated Section 6.4 Credit Return
	6.8	Updated Section 6.8 Credit authorization only or card authentication
	6.18	Updated Section 6.18 Amex Airline Passenger Transport (APD) Transaction
	6.19	Updated Section 6.19 Passenger Transport Data Transaction
	6.21	Updated Section 6.21 Adjust transaction
	6.28	Updated Section 6.28 Encryption and tokenization
Appendix B	6.28.2	Updated Section 6.28.2 Encrypted retail credit
		Updated
Appendix C		Added Appendix C

All revisions supporting release 5.1 are marked with change bars.

Table 1.69 Version 5.1 document revision

Chapter	Section	Description
Field Specifications	5.105	Updated Section 5.105 <A111> Chip Card Data

All revisions supporting release 5.0 are marked with change bars.

Table 1.70 Version 5.0 document revision

Chapter	Section	Description
Field Specifications	5.5	Updated code 20 in Section 5.5 <A10> Transaction Code
	5.63	Updated Section 5.63 <A69> Transaction Fee Amount
	5.71	Updated Section 5.71<A77> Mastercard Transit Program<A77><B1>
	5.77	Updated Section 5.77 <A83> Response Code
	5.78	Updated Section 5.78 <A84> Approval Code
	5.86	Added value Section 5.86 <A92> Card Product Codes
	5.105	Updated Section 5.105 <A111> Chip Card Data
Transaction Message Examples	5.108	Added Section 5.108 <A114> Alternate Account ID 1
	6.3	Updated Section 6.3 Retail credit card
	6.4	Updated Section 6.4 Credit Return
	6.7	Updated Section 6.7 Card not present credit card
	6.8	Updated Section 6.8 Credit authorization only or card authentication

All revisions supporting release 4.4 are marked with change bars.

Table 1.71 Version 4.4 document revision

Chapter	Section	Description
Field Specifications	5.16	Updated Section 5.16.3 <A21><B3> Card Data Input Mode Removed value from Section 5.16.2 <A21><B2> Card Present Data
	5.105	Updated Section 5.105 <A111> Chip Card Data
	5.107	Added Section 5.107 <A113> Message Reason Code
Appendix B		Added

Table 1.72 Version 4.3 document revision

Chapter	Section	Description
Field Specifications	5.39	Updated Section 5.39 <A44> Healthcare Service Group , Section 5.39.3 <A44><B3> Payer ID/Carrier ID and added Section 5.39.4 <A44><B4> Approval or Reject Reason Code
	5.86	Updated the description of N1 in Section 5.86 <A92> Card Product Codes
	5.100	Updated Section 5.100.1 <A106><B1>Encryption Type
	5.101	Updated Section 5.101 <A107> Encrypted Track 1
	5.102	Updated Section 5.102 <A108> Encrypted Track 2
	5.103	Updated Section 5.103.3 < A109><B3> Encrypted Manually Entered Verification Code (CID, CVC2, CVV2)

All revisions supporting release 4.2 are marked with change bars.

Table 1.73 Version 4.2 document revision

Chapter	Section	Description
Field Specifications	5.16	Updated Section 5.16 <A21> POS Entry Data
	5.17	Updated Section 5.17 <A22> Market Specific Data Identifier
	5.78	Updated Section 5.78 <A84> Approval Code
	5.86	Updated Section 5.86 <A92> Card Product Codes
	5.105	Added new Section 5.105 <A111> Chip Card Data
	5.106	Added new Section 5.106 <A112> Card Acceptor Data
Transaction Message Examples	6.3	Updated Section 6.3 Retail credit card
	6.4	Updated Section 6.4 Credit Return
	6.5	Updated Section 6.5 Retail/Restaurant debit
	6.15	Updated Section 6.15 Quasi cash credit card
Appendix B		Added new Appendix B

All revisions supporting release 4.1 are marked with change bars.

Table 1.74 Version 4.1 document revision

Chapter	Section	Description
Messaging Scenarios	4.8	Added new Section 4.8 Using encryption and tokens
Field Specifications	5.5	Added Transaction code in Section 5.5 <A10> Transaction Code
	5.41	Add new table in Section 5.41 <A46> Partial Authorization Indicator
	5.73	Added new information in Section 5.73 <A79> Use of TSYS Token Request
	5.77	Added new information in Section 5.77 <A83> Response Code
	5.81	Added row to Section 5.81 <A87> CCI Response (Commercial Card Indicator)
	5.100-5.104	Added new Section 5.100 <A106> Encrypted Transaction through Section 5.104 <A110> Token Authentication
Transaction Message Examples	6.28	Added new Section 6.28 Encryption and tokenization

All revisions supporting release 4.0 are marked with change bars.

Table 1.75 Version 4.0 document revisions

Chapter	Section	Description
Field Specifications	5.10	Added information in the description in Section 5.10 <A15> Transaction Amount
	5.36	Updated Note in Section 5.36 <A41> TSYS Acquiring Solutions Developer ID
	5.37	Updated Note in Section 5.37 <A42> TSYS Acquiring Solutions Version ID
	5.41	Added new table Section 5.41 <A46> Partial Authorization Indicator
	5.71	Added new Section 5.71 <A77> Mastercard Transit Program
	5.72	Added new Section 5.72 <A78> Transaction ID
	5.90	Added information in the description in Section 5.90 <A96> Network ID

Chapter	Section	Description
Transaction Message Examples	6.1	Added information to Section 6.1 Authenticate POS device
	6.2	Added information to Section 6.2 Deactivate POS device
	6.3	Added information to Section 6.3 Retail credit card
	6.4	Added information to Section 6.4 Credit Return
	6.5	Added information to Section 6.5 Retail/Restaurant debit
	6.6	Added information to Section 6.6 Retail EBT
	6.7	Added information to Section 6.7 Card not present credit card
	6.9	Added information to Section 6.9 Credit prepaid activation
	6.10	Added information to Section 6.10 Credit prepaid load
	6.11	Added information to Section 6.11 Offline authorization
	6.12	Added information to Section 6.12 Account funding transaction
	6.13	Added information to Section 6.13 Cardholder funds transfer
	6.14	Added information to Section 6.14 Bill pay credit/debit card
	6.15	Added information to Section 6.15 Quasi cash credit card
	6.15	Added information to Section 6.15 Quasi cash credit card
	6.15	Added information to Section 6.16 Level III line item detail transaction
	6.16	Added information to Section 6.17 Level III enhanced data transaction
	6.17	Added information to Section 6.18 Amex Airline Passenger Transport (APD) Transaction
	6.18	Added information to Section 6.19 Passenger Transport Data Transaction
	6.19	Added information to Section 6.20 Transaction reversal
	6.20	Added information to Section 6.21 Adjust transaction
	6.21	Added information to Section 6.25 Batch detail inquiry
	6.25	Added information to Section 6.26 Transaction detail inquiry
	6.26	

Chapter	Section	Description
	6.28	Added information to Section 6.28 Encryption and tokenization
	6.28	Added information to Section 6.28 Encryption and tokenization

All revisions supporting release 3.2 are marked with change bars.

Table 1.76 Version 3.2 document revisions

Chapter	Section	Description
Field Specifications	5.47	Added new value in Section 5.47 <A52> Reversal Request Code
	5.70	Added new Section 5.70 <A76> Mastercard EMS Fraud Service Group
	5.86	Updated Section 5.86 <A92> Card Product Codes

All revisions supporting release 3.1 are marked with change bars.

Table 1.77 Version 3.1 document revisions

Chapter	Section	Description
Field Specifications	5.10	Updated paragraph in description

All revisions supporting release 3.0 are marked with change bars.

Table 1.78 Version 3.0 document revisions

Chapter	Section	Description
Field Specifications	5.10	Added paragraph in description
	5.24	Updated Section 5.24 <A30> Address Verification Data
Transaction Message Examples	6	Updated the following sections: 6.3 , 6.7 , 6.8 , 6.12 , 6.13 , 6.14 , 6.15 , 6.18 , 6.19 , 6.28 .

All revisions supporting release 2.3 are marked with change bars.

Table 1.79 Version 2.3 document revisions

Chapter	Section	Description
Host Capture Use Cases	2.3	Replaced description in the Delay Capture row in Table 2.1
Messaging Scenarios	4.4	Added Delay Capture Adjustment in Section 4.4 Adjust transaction
Field Specifications	5.67	Updated Note in the description
	5.90	Added values in the description in Section 5.90 <A96> Network ID
	5.92	Added Note in the description
	5.139	Updated <A303> Response Text field for <A302> Code 00

All revisions supporting release 2.2 are marked with change bars.

Table 1.80 Version 2.2 document revisions

Chapter	Section	Description
Messaging Scenarios	4.4	Replaced verbiage in Section 4.4 Adjust transaction Replaced drawing
Field Specifications	5.12	Replaced description in Section 5.12 <A17> Transaction Original Number (TON)
	5.38	Added new possible value in Section 5.38.2 <A43><B2> Amount Type
	5.54	Updated description in Section 5.54 <A60> Tip Amount
	5.78	Added two new values in Section 5.78 <A84> Approval Code
Transaction Message Examples	6.5	Updated Section 6.5 Retail/Restaurant debit
	6.6	Updated Section 6.6 Retail EBT
	6.21	Updated Section 6.21 Adjust transaction
	6.22	Updated Section 6.22 Tip adjustment
	6.28	Updated Section 6.28 Encryption and tokenization

Table 1.81 Version 2.1 document revisions

Chapter	Section	Description
Field Specifications	5.69	Updated Section <A75> Custom Identifier and Section <A75><B2> Custom Identifier Detail

All revisions supporting release 2.0 are marked with change bars.

Table 1.82 Version 2.0 document revisions

Chapter	Section	Description
Host Capture Use Cases	2.3	Updated Adjust Transaction description
Messaging Scenarios	4.4	Updated text for Adjust Transaction and replaced with revised drawing Added new sub section Hold Capture Adjustment
Field Specifications	5.67 5.68 5.69 5.78	Added new section <A73> Hold Capture Indicator Added new section <A74> Cardholder Verification Results Added new section <A75> Custom Identifier Added new table for Discover
Transaction Message Examples	6.7 6.21	Added new rows in Section 6.7 Added new rows in Section 6.21

All revisions supporting release 1.5.4 are marked with change bars.

Table 1.83 Version 1.5. 4 document revisions

Chapter	Section	Description
All references to Diners have been removed from the spec.		
Field Specifications	5.26 5.50	Updated value 5 description Added a new value

All revisions supporting release 1.5.3 are marked with change bars.

Table 1.84 Version 1.5. 3 document revisions

Chapter	Section	Description
Field Specifications	5.90	Added ATM possible values and updated existing possible values. Removed note in Section <A96> Network ID

Chapter	Section	Description
Transaction Message Examples	6.8 - 6.16	Added new values to Section 6.8 , Section 6.9 , Section 6.10 , Section 6.11 , Section 6.12 , Section 6.13 , Section 6.14 , Section 6.15 and Section 6.16 .

All revisions supporting release 1.5.2 are marked with change bars.

Table 1.85 Version 1.5.2 document revisions

Chapter	Section	Description
Host Capture Use Cases	2.2	Updated diagram
	2.3	Replaced table in Section 2.3
Messaging Scenarios	4.2	Updated section Request authorization
Field Specifications	5.5	Added new transaction codes in Section 5.5
	5.11	Added note to <A16> Secondary Amount
	5.16	Updated the values in <A21><B12> PIN capture capability
	5.53	Added new section <A58><B8> Card Sequence Number
	5.81	Added note and replaced last paragraph in Section 5.81
Transaction Message Examples	6.3	Added new section in 6.3
	6.7	Added three new rows in Section 6.7
	6.8	Added new section in 6.8
	6.12	Added new section in 6.12
	6.13	Added new section in 6.13
	6.14	Added new section in 6.14
	6.15	Added new section in 6.15
	6.18	Removed A87 from Section 6.18
	6.19	Added three new rows in Section 6.19

All revisions supporting release 1.5.1 are marked with change bars.

Table 1.86 Version 1.5.1 document revisions

Chapter	Section	Description
Transaction and Message Description	3.2	Added section 3.2 Message Structure

Chapter	Section	Description
Field Specifications	5.5	Added code to ≤A10> Transaction Code
	5.78	Added values to A, B & D in ≤A84> Approval Code Added codes to ≤A84> Approval Code
	5.86	Added values to A, B & D in ≤A92> Card Product Codes

All revisions supporting release 1.5 are marked with change bars.

Table 1.87 Version 1.5 document revisions

Chapter	Section	Description
Host Capture Use Cases	2.3	Added information to description.
Transaction and Message Description		Removed section 3.2 Message Structure
Messaging Scenarios		Added new section Authorization with Multiple Level III Line Items
Field Specifications	5.5	Added rows to ≤A10> Transaction Code
	5.7	Removed {Reserved}note from value G
	5.54	Removed {Reserved}note from Header
	5.64	Added definition
	5.65	Added section ≤A71> Amex Airline Data
	5.77	Added row to ≤A83> Response Code
	5.66	Added section ≤A72> Passenger Transportation Data Group
	5.98	Added section ≤A104> Level III Enhanced Data Group
	5.99	Added section ≤A105> Level III Line Item Details Group

Chapter	Section	Description
Transaction Message Examples	6.2	Added Deactivate POS device
	6.3	Updated field A98 in section Retail credit card
	6.17	Added Level III enhanced data transaction
	6.16	Added Level III line item detail transaction
	6.18	Added Amex Airline Passenger Transport (APD) Transaction
	6.19	Added Passenger Transport Data Transaction
	6.21	Added rows to Adjust transaction
	6.28	Added Encryption and tokenization
	6.28	Added Encryption and tokenization
Appendix A		Removed {Reserved} note from all field descriptions on rows where A12 value is “G”.

All revisions supporting release 1.4 are marked with change bars.

Table 1.88 Version 1.4 document revisions

Chapter	Section	Description
Field Specifications	5.9	Added additional instruction for sending a repeat transaction
	5.5	Added EBT usage to “Code 11-Cash Advance”
	5.17	Removed the <space> as a possible value
	5.22	Clarified the content to be clear about valid combinations of fields
	5.41	Clarified default value and Discover card usage descriptions
	5.53.2, 5.53.3, 5.53.7	Updated the description.
	5.53.4, 5.53.5, 5.53.6	Updated the length, format and description.
	5.65	Clarified for the concept of the 'expected' Batch Number
	5.77	New responses code “RT” added
	5.78	Removed retired Mastercard values. Updated description with a note.
	5.144.1	

Chapter	Section	Description
Transaction Message Examples	6.4	Added new section Credit Return
	6.3,6.7	Added <A101> Amex CPC group tags
	6.5,6.6	Removed <A30> Address Verification Data tags
	6.6	<A26> Manually Entered Expiry Date is "O" optional is the request
	6.21,6.22,6.23	Removed<A18> and <A48> from three examples
	6.23	Updated <A80> Batch Number is "--" not required in the request
	6.24	Updated the response values for <A304> and <A305>

All revisions supporting release 1.3 are marked with change bars.

Table 1.89 Version 1.3 document revisions

Chapter	Section	Description
Messaging Scenarios		Updated the description and diagram example.
Field Specifications	5.2	Updated the length of <A2> Authentication Code
	5.11	Updated the description of <A16> Secondary Amount
	5.16.9	Updated the description of <A21><B9> Operating environment
	5.42.6	Updated the description of <A47><B6> Shipping Method
	5.50	Updated the description of <A55> Real-time Substantiation Indicator
	5.52	Updated the format of <A57> NTIA UPC/SKU Data
	5.54	Updated the description of <A60> Tip Amount
	5.89	Updated the description of <A95> Return ACI
	5.92	Renamed to <A98> Delayed Capture Indicator and updated the description.
	5.141	Updated the field length of <A304> Local Batch Close Date .
	5.144.2	Added a value to <A325><B1> Summary Type table

Chapter	Section	Description
Transaction Message Examples	6.6	Added <A29> to Retail EBT section and updated the comment for <A2>
	6.16	Updated Level III line item detail transaction section
	6.21	Updated Adjust transaction section
	6.22	Updated Tip adjustment section
	6.23	Updated Void transaction section

All revisions supporting release 1.2 are marked with change bars.

Table 1.90 Version 1.2 document revisions

Chapter	Section	Description
Field Specifications	5.17	Added value to <A22> Market Specific Data Identifier
	5.24	Updated value to <A29> Security Control Info
	5.49	Added value to <A54> Transaction Type Identifier
	5.77	Updated <A83> Response Code
	5.139	Updated <A302> Batch Response Code
Transaction Message Examples	6.5	Added <A29> to Retail/Restaurant debit section
	6.22	Updated Adjust transaction section
	6.23	Updated Tip adjustment section
	6.27	Updated Void transaction section
		Removed rows <A325>, <A325><B1>, <A325><B2>, <A325><B3>, and <A325><B4>

All revisions supporting release 1.1 are marked with change bars.

Table 1.91 Version 1.1 document revisions

Chapter	Section	Description
Host Capture Use Cases	2.2	Use Case diagram revised to add Gateway Entry use case, and update relationships appropriately.
	2.3	The Batch Inquiry use case description clarifies that those transactions pending adjustment will be counted as a 'pre-auth' card type in the batch summary response.
	2.3	The Close Batch use case description clarifies that the close batch event will auto-initiate a when the batch is 48 hours old.
Transaction and Message Description	3.2	Message Structure diagram updated with various field changes outlined in the history.
	3.5 , 3.6	TDVD and XML formatted message displays the correct routing header fields.
Messaging Scenarios	All	- All diagrams correctly reference the Batch Number field. - The Batch Detail Inquiry sequence shows the max. number of transactions=5. - Adjustment transactions scenarios no longer reference the A17 TON as an alternate request value to using the RNN. - The correct batch number used to make the request noted on the diagram. - The correct Batch Summary response is noted on the diagram
Field Specifications	5.10	Corrected field length
	5.54	A59 TSYS Gift Card Operator/Clerk ID is listed as a reserved field.
	5.63	Added <A69> Transaction Fee Amount field.
	5.77 , 5.79 , 5.86 , 5.140 , 5.146.2	A83, A85, A93, A303, and A325.B1 response text is specified as ALL CAPS.
	5.84	A90 CAVV result code displays the correct field description.
	5.95	A101.B2 Shipped to Zip and A101.B3-B6 Charge Descriptor 1-4 added to Amex CPC group.
	5.108	A301 Batch Status Description displays the correct field description.
	5.143	A307 Requested Transaction Count correctly shows field length = '1' and the maximum number of transactions = '5'.
	5.146.6 5.146.7	A330.B6 and.B7 added clarifying language for partial authorizations.

Chapter	Section	Description
Transaction Message Examples	previous 6.5 6.3,6.7,6.8,6.12,6.13 6.3-6.7,6.8-6.16 6.5,6.6,6.14,6.16,6.26 6.11 6.21 6.24 6.26 6.27	- Retail TSYS Gift Card example removed as a result of reserving A59 for future use. - Removed A53.B1, A53.B2, A53.B3. - Correctly shows A21.B6 Card Data Input Capability and A21.B9 Operating Environment as 'mandatory' - Updated examples with new row for <A69> Transaction Fee Amount. - Updated field A92 - Added A101 group to Adjust Transaction example; removed incorrect A101 references from all Authorization examples. - A326 Pending Transaction Count is correctly shown in Batch Summary Inquiry. - Transaction Detail Inquiry removed A43, A44, and A67 fields. - Manual Close Batch example updated to show all the correct fields and conditions.
Appendix A		- Added 'credit return' row. - Updated all 'G' payment code rows to indicate that gift card is reserved for future use.
Appendix B		Added Appendix B—Alphabetical Index of Field Names

Chapter 2

Host Capture Use Cases

NOTE This chapter describes use cases of the TSYS Acquiring Solutions' Summit Gateway host capture system.

2.1 Context

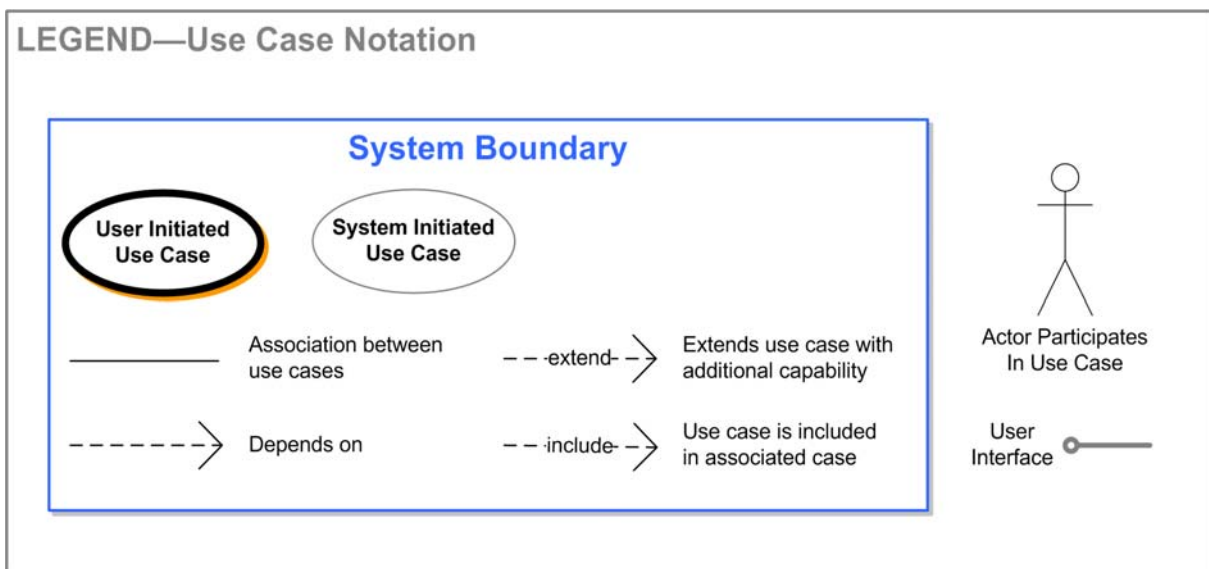
Host capture capabilities have a dramatic affect on how the POS device interacts with TSYS Acquiring Solutions. The POS user, however, has a nearly unchanged experience operating the POS device, benefiting from fewer downloads to the POS device and no transaction uploads when closing a current batch.

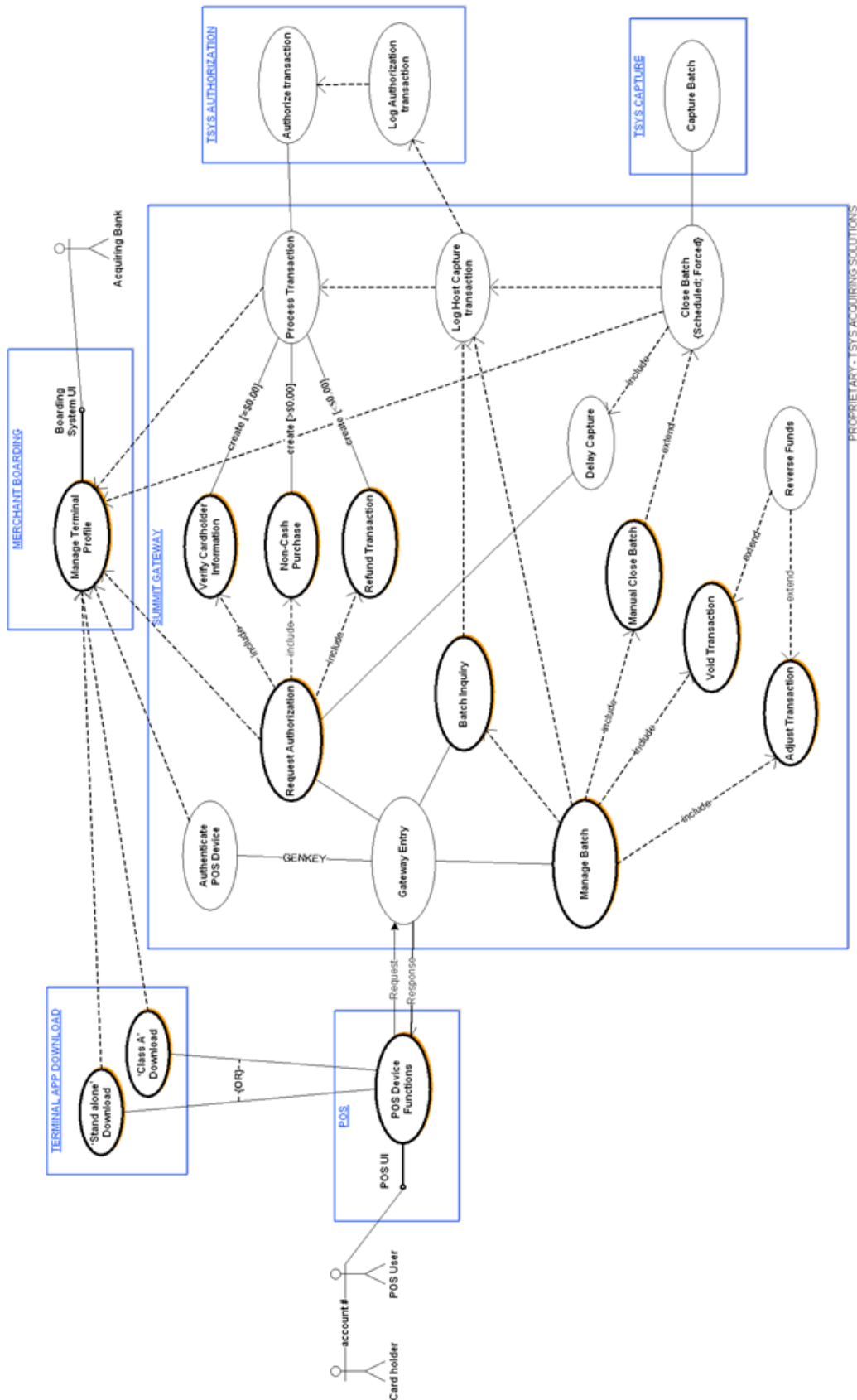
The POS device runs a much thinner client where the ability to call the function from the host is embedded in the POS device application, but not the function itself. This hub-and-spoke services model gives increased assurance of software compliance to each participating merchant.

Most host capture system use cases are dependant on the TSYS Acquiring Solutions Merchant System to authenticate POS devices and augment merchant information.

2.2 Diagram

The use case diagram below depicts the relationships between use cases contained within system boundaries. Descriptions of the diagram follow in section [2.3 Description](#).





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2.3 Description

In general, the narrative of the host capture use case diagram reads from left-to-right, begins with a POS user initiating a transaction and concludes with a closed batch of transactions being prepared for settlement processing.

Table 2.1 Use case description for Summit Gateway Host Capture system

Object	Type	Description
POS	System boundry	The POS System comprises a POS User and a POS Device. An authenticated POS Device and up-to-date application enables access to the TSYS Acquiring Host Capture System.
Cardholder	Actor	The Cardholder provides an account number in a variety of ways, with a variety of cards, to a POS User.
POS User	Actor	The POS User uses the Host Capture enabled POS Device to perform several POS Device functions. A POS User is usually a person, but in some cases it could be a machine, e.g. banking automatic teller, or pay-by-phone services.
POS UI	User interface	The POS User Interface enables the POS User to interact with the terminal application to perform the device functions.
POS Device Functions	Use case	The POS device can perform host capture use cases: • Authenticate POS Device • Request Authorization • Batch Inquiry • Manage Batch • Download Application
Terminal Application Download	System boundry	The POS User downloads a new application version to the POS device. Compliance updates are the most common reason why a POS User needs to download an application.
Class A Download	Use case	The software application resident in TSYS provided POS Devices periodically needs to be updated. POS Devices are supported by the TSYS Download System, a.k.a. Software Download Interface (SDI). POS Devices that were provisioned by TSYS are also known as Class A POS devices.
Stand Alone Download	Use case	The software application resident in 3rd Party-provided POS Devices periodically needs to be updated. 3rd Party POS Devices are supported by the 3rd Party Download System which performs functions like the TSYS Download System, but it not owned or operated by TSYS. POS Devices that were provisioned by Value Added Resellers (VAR) and other POS Device-related services organizations are also known as Class B POS Devices.

Object	Type	Description
Merchant Boarding	System boundary	The TSYS Acquiring Merchant Boarding System enables the boarding and maintenance of Merchant and Terminal Profiles that are a dependency for Host Capture use cases. This system boundary includes both the Merchant Management System (MMS) and Express.
Acquiring Bank	Actor	The Acquiring Bank is a business entity that owns the contractual relationship between the Merchant, POS Devices, and TSYS Acquiring. The Acquiring Bank is responsible for managing Merchant, Store and POS Device data in the TSYS Acquiring Merchant Boarding System.
Boarding System UI	User interface	There are several interfaces to the TSYS Acquiring Merchant Boarding system, including terminal emulation MMS, web-based Express, remote updates via XML requests.
Manage Terminal Profile	Use case	The Terminal Profile is a critical data facet. Many of the Host Capture System use cases depend on the ability to access the Terminal profile (and its parent Merchant profile) in order to identify a POS Device and thereby perform necessary functions.
Summit Gateway	System boundary	The TSYS Acquiring Host Capture System accepts requests from and responds to the POS system. The Host Capture System depends on the TSYS Acquiring Merchant Boarding System for Merchant data to authenticate POS devices and augment transactions, controls the message translation to and from the TSYS Acquiring Authorization System, and enables the transfer of Closed Batches to the TSYS Acquiring Capture System.
Gateway Entry	Use case	A transaction request message is received and routed to the appropriate end-point or system process, and the response message is routed to the requesting POS Device.
Authenticate POS Device	Use case	The TSYS Acquiring Host Capture System requires that a POS Device is authenticated before it can perform any other use case. The POS User is required to provide an Authentication Code and an Authentication Factor when verifying the POS Device identity with the Host Capture System; and once verified then a unique 'Gen key' is sent to the POS Device. The Gen key is sent by the POS Device in all subsequent messages to Host. The POS Device can be deactivated. Sending in a deactivation transaction code will invalidate the GenKey held by the POS Device. The POS Device must perform the Authenticate POS Device use case to enable processing on the host.
Request Authorization	Use case	With a card account number and other Cardholder data, the POS User performs one of the following use cases: • Verify Cardholder Information • Non-Cash Purchase • Refund Transaction
Verify Cardholder Information	Use case	The POS User wants to validate the card and Cardholder, and lookup other related information. The use case has sub-functions that enable checking of several information elements.

Object	Type	Description
Non-Cash Purchase	Use case	A POS User is transacting a purchase using the Cardholder's account number, not cash. This is the most commonly performed use case.
Refund Transaction	Use Case	A POS User is refunding some or all funds for a transaction that is no longer in the Current Batch. If the transaction was still in the Current Batch, then the POS User can perform the Void Transaction or Adjust Transaction use cases.
Process Transaction	Use case	Authorization transactions are augmented by adding or deriving from data elements already existing in the TSYS Merchant Boarding system. The message is converted to and from TSYS Authorization message specifications. The authorization request is resolved with an approval response.
Log Host Capture Transaction	Use case	The Host Capture System ensures a complete and accurate listing of all transactions occurring in the host for each requesting POS device. Logging transactions enables several business and system processes, including merchant and client billing for Host Capture services rendered by TSYS.
Batch Inquiry	Use case	The POS User is looking for a single or group of transactions for that POS Device. The Batch Inquiry use case has sub-functions that enable three data views: • Batch summary: allows viewing of batch status # purchases, # refunds, \$ net amount by card type for the Current Batch or one of the five prior Closed Batches. Delayed Capture transactions are listed like a card type. • Batch detail: allows viewing of up-to five transaction summaries for the Current or last Closed Batch, including the Transaction Original Number. • Transaction detail: allows viewing detailed data about a specific transaction in the Current or last Closed Batch.
Manage Batch	Use case	The POS User is performing one of several functions to manage the Current Batch on the TSYS Acquiring Host Capture System. The Manage Batch use case depends on the Batch Inquiry use case to enable the correct selection of a specific transaction in the Current Batch. Manage Batch includes these use cases: • Void Transaction • Adjust Transaction • Manual Close Batch
Void Transaction	Use case	The POS User wants to change the Settlement Amount of an original Authorization transaction to the amount of \$0.00. A void transaction can only be performed on an authorization transaction in the current open batch.

Object	Type	Description
Adjust Transaction	Use case	The POS User adjusts financial or non-financial data of an authorization transaction that has not been captured and settled. The amount must be specified, even if it is zero. Financial adjustments of the settlement amount must be greater than a non-zero amount. Any ensuing adjustment for the same tag takes precedence over previous adjustments.
Reverse Funds	Use case	Summit Gateway generates a reversal of funds to restore the Available Funds (a.k.a. the Open-to-Buy) of the Cardholder's account. The Reverse Funds use case extends the Adjust Transaction and Void Transaction use cases. - When the user performs the Adjust Transaction use case on a credit, debit, cash benefit, food stamp, gift card, or private label transaction then the Summit Gateway will send a partial reversal to the card brand. - When the user performs the Void Transaction use case on a credit, debit, cash benefit, food stamp, gift card, or private label transaction then the Summit Gateway will send a full reversal to the card brand.
Manual Close Batch	Use case	The POS User initiates the Close Batch use case with a 'batch close' request message from the POS device. A manual close batch request has no effect on Close Batch events scheduled in the Terminal Profile. The Manual Close Batch use case extends the Close Batch use case.
Close Batch	Use case	Transactions in a Current Batch are bundled into a Closed Batch and passed to the TSYS Acquiring Capture System. The Close Batch use case can be initiated by the user or by the system. There are three initiation methods: <i>Table 2.2 Manual: The POS User performs the Manual Close Batch use case (see above).</i> <i>Table 2.3 Scheduled: A Host Capture Terminal Profile allows up to six scheduled Close Batch times to be specified for each 24-hour period. Each scheduled time is set to the local time by the system. The host will close the current open batch at the scheduled times each day.</i> <i>Table 2.4 Forced: The system will force close a batch when approximately two days have elapsed since the last batch close event to minimize late transaction interchange impacts to the merchant.</i>

Object	Type	Description
Delay Capture	Use case	The host uses a Capture Delay option set in the boarding system Host Capture terminal profile to determine how to handle a transaction during a batch close event. There are three Capture Delay values that can be selected for host capture terminals. The values are "None", "Delay 1 Batch", and "Hold till Release". • <u>Capture Delay = "None"</u> - (default value for new profiles) No optional capture delay feature selected. The host will ignore tag A98 and will reject transactions that include tag A73. • <u>Capture Delay = "Delay 1 Batch"</u> - Selecting the capture delay value "Delay 1 Batch" in the boarding system will flag a merchant's terminal as a delay capture participant. The host will enable delay capture processing for transactions that include tag A98, and will reject transactions that include tag A73. • <u>Capture Delay = "Hold till Release"</u> - Selecting the capture option "Hold till Release" in the boarding system will flag a merchant's terminal as a hold capture participant. The host will ignore tag A98, and will not capture transactions sent with tag A73 HOLD until the original is adjusted with tag A73 CAPTURE. Transactions with the Commercial Card Indicator: Will delay capture one batch close unless the Level II data is present in the original or the Level II data is sent in an adjustment, regardless of the capture delay option selected. The Delay 1 Batch indicator is typically used: • When a restaurant will adjust the original authorization to add a tip amount. • When Level III data will be included with the authorization. The Hold till Release indicator is typically used: • When a direct marketing firm cannot capture the transaction until the product ships.
TSYS Authorization	System boundary	Existing Auth Systems, including embedded processes that interact with Associations, Issuers, etc.
TSYS Capture	System boundary	Existing Capture platform.

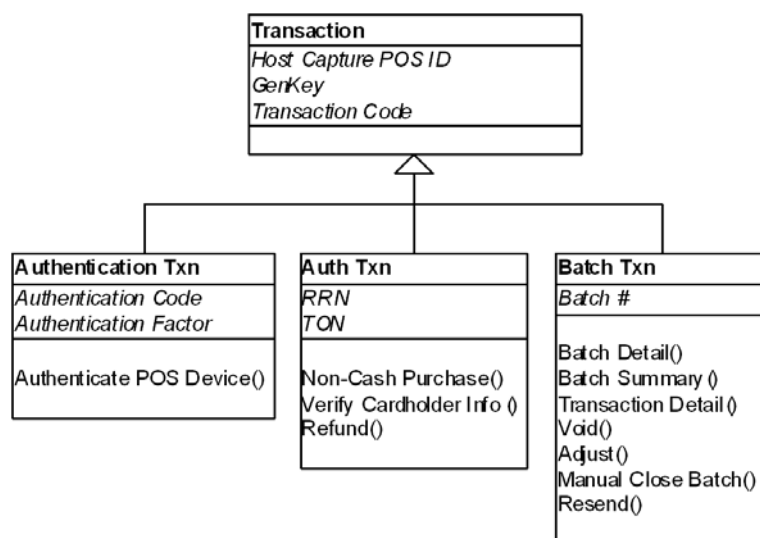
Chapter 3

Transaction and Message Description

NOTE This chapter explains the construction principles for SGMF formatted transaction messages generated by the POS device performing host capture use cases.

3.1 Transaction specialization and major attributes

The TSYS Acquiring Solutions host capture system is based on the request/response model. The SGMF transaction message is a tagged, variable length tag-delimiter-value-delimiter (TDVD) construction.



A transaction is a generalized class and each instance of the three specialized SGMF transactions inherit the host capture POS ID, GenKey, and transaction code attributes. An authentication transaction results from the authenticate POS device use case. An authorization transaction (a.k.a. auth transaction) results from the non-cash purchase, refund, and verify cardholder information use cases. A batch transaction results from batch inquiry, void, adjust, resend, and close batch use cases. The major attributes of the class are described in the table below.

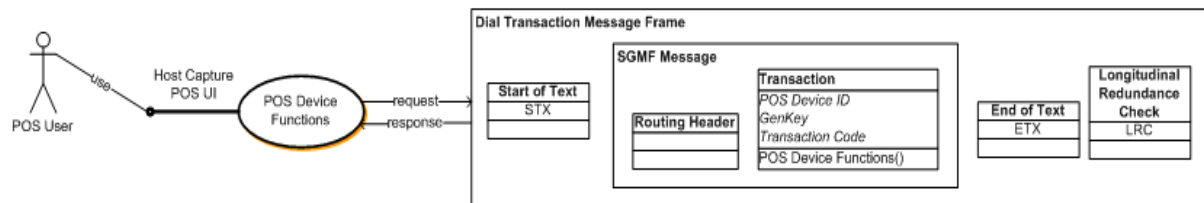
Transaction attributes	Description
Host capture POS ID	Unique ID that will associate the POS device to a unique merchant hierarchy. NOTE: This must be sent with every request to the TSYS Acquiring Solutions host.
GenKey	The GenKey is produced by successfully performing the authenticate POS device use case. This unique value identifies the POS device to the host capture system. The GenKey is sent in every transaction request message to the host capture system.

Transaction attributes	Description
Transaction code	The transaction code enables certain processing logic depending on value.
RRN	A new Retrieval Reference Number (RRN) is generated by the host system for every auth transaction. The RRN is the primary number used by the POS user to locate transactions during a batch inquiry, and to identify the original auth transaction during void, adjust, and resend.
TON	The Transaction Original Number (TON) is an attribute of the original auth transaction that enables an alternate lookup number in circumstances when communications to the host are disrupted. The TON is generated by the POS device for every original auth transaction request and is echoed in the auth transaction response.
Authentication code	The POS user must enter the authentication code as a precondition to receiving a GenKey from the host capture system.
Authentication factor	The authentication factor is a secondary merchant-specific number to further ensure the POS device is a secure source of transaction data.
Batch #	The batch number is an attribute used to identify a current or previous batch in request and response messages.

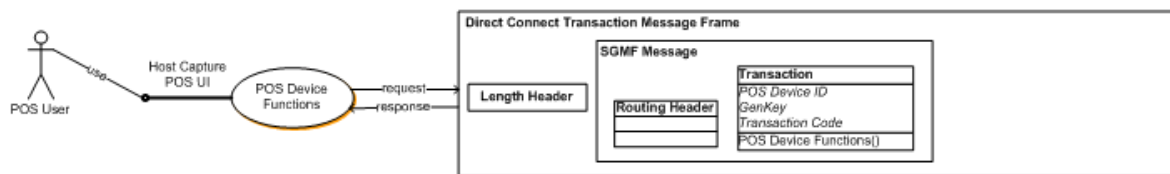
3.2 Message structure

The class diagrams below describe the SGMF message structures that are supported by the Summit Gateway host. The transaction message frame comprises all network protocol data elements. The SGMF message frame comprises the routing header and transaction data elements.

3.2.1 Dial

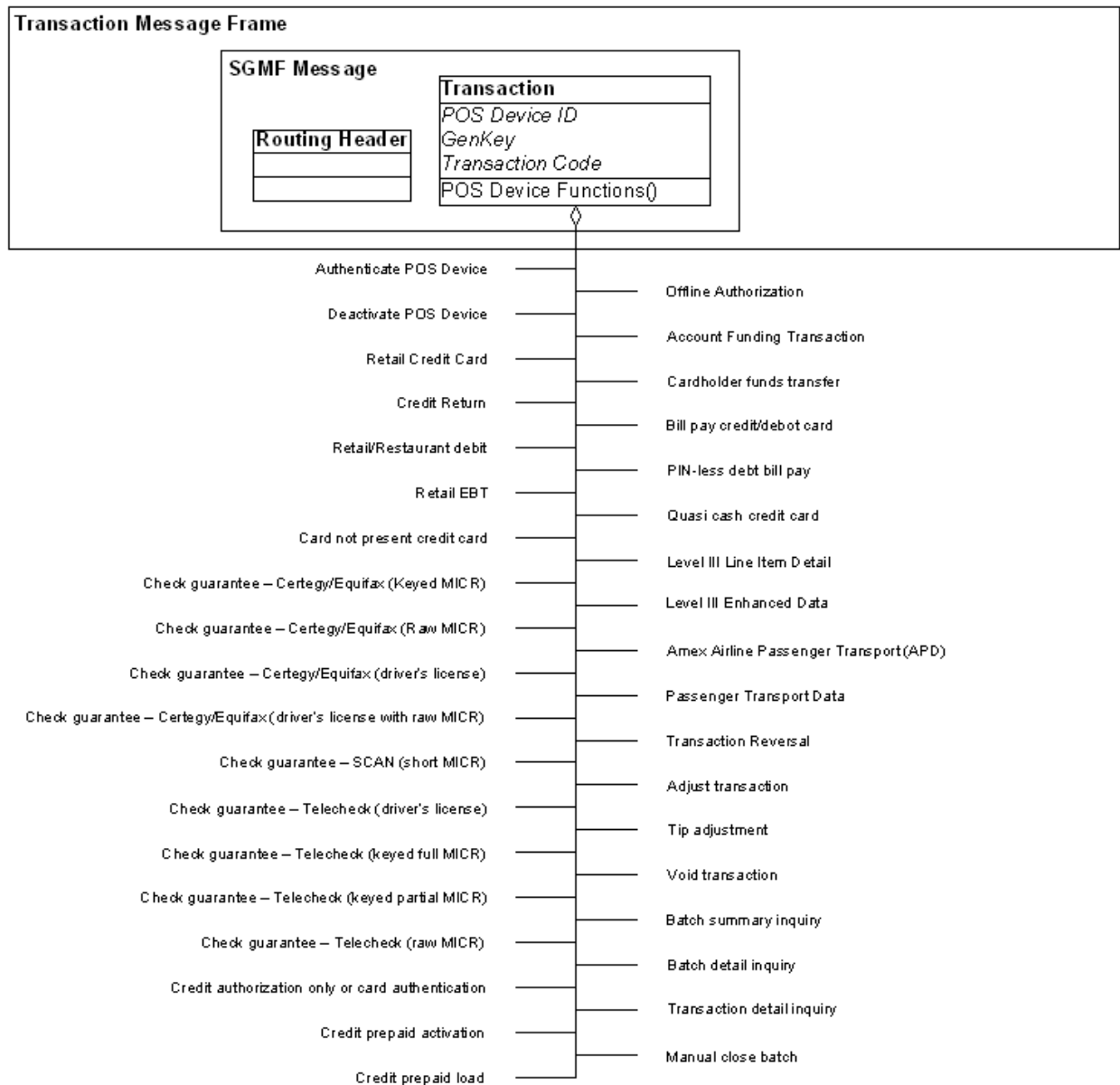


3.2.2 Direct Connect



3.3 Transaction Data Elements

The class diagram below describes the various transaction types that are supported by the host for both the dial and direct connect message types.



3.4 Routing header field definitions

The routing header comprises the following fields in the request and response messages.

NOTE The routing header fields are not tagged. The request header will have the format “H0.ROUTID” and the response header will have the format “I0”.

3.4.1 Request

The routing header is nine characters in fixed length. The following table identifies the fields in the request.

Name	Value	Length	Form	Description
Record Format	H	1	Alpha	Always ‘H’ for host capture request.
Application Type	0, 2 or 4	1	Num	0 = single tran 2 = multi-tran 4 = interleaved
Delimiter		1	Special	Always a period
Routing ID	nnnnnn	6	A/N	Routing ID assigned during approval testing

3.4.2 Response

The response header is three characters in fixed length. The following table identifies the fields in the response.

Name	Value	Length	Form	Description
Record Format	I	1	Alpha	Always ‘I’ for host capture request.
Application Type	0, 2 or 4	1	Num	0 = single tran 2 = multi-tran 4 = interleaved
Delimiter		1	Special	Always a period

3.5 TDVD format

Field tagged data can be TDVD format. The field IDs are defined in the [Field Specifications](#).

- Tags are case-sensitive.
- SGMF is a tagged message format and there is no required ordering of tagged fields within a message, except <A1> Host Capture POS ID must be the first tag sent in all request messages.
- Developers should ignore unknown tags by default so that future SGMF tag additions do not hinder POS device operation.
- First level field tags begin with the upper-case letter ‘A’ followed by a numeric field ID value. First level delimiters are the ASCII file separator character (x1C, control-\\).

- Second level subfield tags begin with the upper-case letter 'B' followed by a numeric field ID value. Second level delimiters are the ASCII group separator character (x1D, control-J).

3.5.1 Manually entered card present purchase request with AVS data example

```
H2.RRRRRRA1^\11111222233333^\A3^\1A2B3C4D5E6F7A8B9C0A1B2C^\A10^\10^\A12^\C^\A15^\1250^\A17^\99^\A21^\B1^]0^]B2^]1^]B3^]6^]B4^]5^]B5^]4^]^\A25^\7777666655554444^\A26^\0909^\A30^\B1^]85284^]B2^]8320 SOUTH HARDY DR^]^\
```

3.5.2 Partially authorized response example

```
I2.A80^\234^\A17^\99^\A81^\041508^\A82^\112233^\A83^\10^\A84^\123456^\A85^\PARTIAL APPROVAL^\A86^\333444555666^\A93^\Private Label^\A94^\K^\A96^\Y^\A43^\B1^]00^]B2^]10^]B3^]840^]B4^]C^]B5^]650^]^\
```

3.6 XML format

Field tagged data can be well-formed XML. The field IDs are defined in the [Field Specifications](#) chapter.

- Tags are case sensitive.
- SGMF is a tagged message format and there is no required ordering of tagged fields within a message, except <A1> Host Capture POS ID must be the first tag sent in all request messages.
- Developers should ignore unknown tags by default so that future SGMF tag additions don't hinder POS device operation. The root element for requests is SGREQ. The root element for responses is SGRSP.
- First level field tags begin with upper-case letter 'A' followed by a numeric field ID value.
- Second level subfield tags begin with the upper-case letter 'B' followed by a numeric field ID value.

3.6.1 Manually entered card present purchase request with AVS data example

```
H2.RRRRRR<SGREQ><A1>11111222233333</A1><A3>1A2B3C4D5E6F7A8B9C0A1B2C</A3><A10>10</A10><A12>C</A12><A15>1250</A15><A17>99</A17><A21><B1>0</B1><B2>1</B2><B3>6</B3><B4>5</B4><B5>4</B5><A25>7777666655554444</A25><A26>0909</A26><A30><B1>85284</B1><B2>8320 SOUTH HARDY DR</B2></A30></SGREQ>
```

3.6.2 Partially authorized response example

```
I2.<SGRSP><A80>234</A80><A17>99</A17><A81>041508</A81><A82>112233</A82><A83>10</A83><A84>123456</A84><A85>PARTIAL APPROVAL</A85><A86>333444555666</A86><A93>Private Label</A93><A94>K</A94><A96>Y</A96><A43><B1>00</B1><B2>10</B2><B3>840</B3><B4>C</B4><B5>650</B5></A43></SGRSP>
```

3.7 Serial Line Protocol

The published TSYS Acquiring Solutions EIS 1051 Protocol specification will support the TSYS Acquiring Solutions host capture system with the following bit-length exceptions:

All framing, control and data will use 8-bit ASCII formatting to eliminate the need to convert data to odd or even parity prior to transmission.

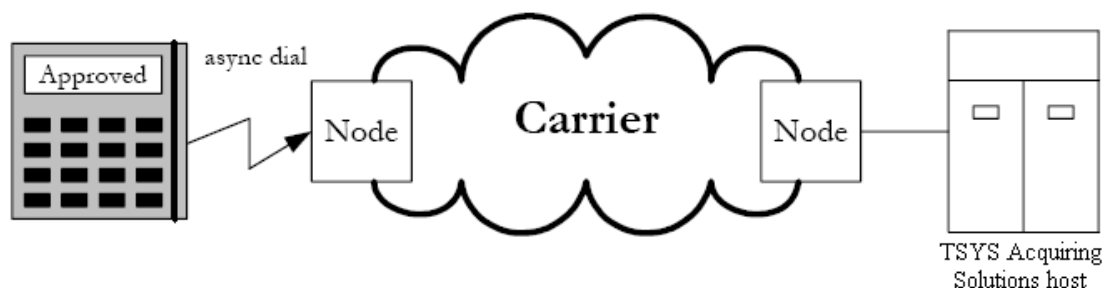
The table below describes the Control-character summary: 8 data bits, no parity (N), one stop bit (1).

Character	Binary	HEX	DEC	Description
STX	00000010	02	02	Start-of-Text, the first framing character signaling the start of a message.
ETX	00000011	03	03	End-of-Text, the ending frame character that precedes the LRC.
EOT	00000100	04	04	End-of-Transmission, used by the host to signal disconnect to the POS device.
DLE EOT	00010000 00000100	10 04	16 04	This pair of characters is sent by the POS device to request a disconnect.
ENQ	00000101	05	05	- When transmitted by the host, an enquiry is transmitted as an invitation to the POS device to transmit a new transaction request. - When transmitted by the POS device, an enquiry is a request for the host to retransmit the last message or protocol character it sent.
ACK	00000110	06	06	Affirmative acknowledgement signifies correct reception of a message or command.
NAK	00010101	15	21	Negative acknowledgement used to indicate that the command or message was not understood or was received with errors.

3.8 Serial Line Access Options

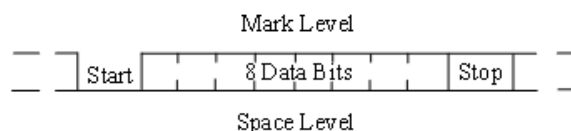
3.8.1 Dial access (asynch)

Dial communications utilize Bell modem standards in North America and Consultative Committee for International Telephony and Telegraphy (CCITT) modem standards for other global regions. TSYS Acquiring Solutions dial connections currently support communication speeds of 300, 1200 and 2400 bps. In North America 9600 bps access is available via toll free dial. The following table provides information on relevant modem standards and their line speed capabilities.



Speed	U.S. and Canada	Other countries
300 bps	Bell 103	V.21
1200 bps	Bell 212A	V.22
2400 bps	V.22bis	V.22bis
9600 bps	V.32	N/A

Information transmitted over TSYS Acquiring Solutions dial access for host capture must be represented as 1 start-bit, 8 data-bits and 1 stop-bit. This applies to both dial and non-dial network access methods.



3.8.2 Dedicated access (Point-to-Point private line)

There are additional documents for use with the dedicated access options. Please contact the TSYS Acquiring Solutions POS Development Lab to obtain copies of these documents.

3.8.3 Internet SSL/TLS (HTTPS and sockets)

There are additional documents for use with internet SSL access options. Please contact the TSYS Acquiring Solutions POS Development Lab to obtain copies of these documents.

3.8.4 Internet VPN (IPSec)

There are additional documents for use with the Internet VPN access options. Please contact the TSYS Acquiring Solutions POS Development Lab to obtain copies of these documents.

3.8.5 Wireless SSL/TLS (GPRS, CDMA)

There are additional documents for use with Wireless SSL access options. Please contact the TSYS Acquiring Solutions POS Development Lab to obtain copies of these documents.

3.9 SGMF Direct Connect

SGMF Direct Connect supports both request and response messages in ASCII format.

3.9.1 Protocol

SGMF Direct Connect uses TCP/IP.

3.9.2 TCP Port

When using TCP/IP to send and receive authorization request and responses, the merchant's software is responsible for establishing and maintaining an active socket connection for the duration of the anticipated usage.

The connection should stay connected between messages to minimize the overhead caused by repeatedly establishing and breaking connection.

In addition, the merchant's software should include logic to periodically check for a closed connection and automatically re-establish connection.

NOTE IP address and TCP Ports will be assigned by TSYS Acquiring Solutions Authorizations for connection to the SGMF platform.

3.9.3 Message Length Indicator

TSYS Acquiring Solutions authorization applications are record-oriented; therefore, every message received must include a length field prior to the standard message header. This requirement facilitates extracting multiple messages within a buffer.

The length field consists of a right justified, 2 byte binary length representation of the length value. Every message received from the socket must be preceded by this 2 byte header.

3.9.4 Network Management Messages

SGMF Direct Connect provides additional non-financial network management messages.

3.9.5 Keep Alive

The host capture system supports a keep alive message.

Keep alive messages are initiated by the client to either keep the connection open, or ping the host to determine availability. The host will not monitor any connection for keep alive activity, or disconnect any connection due to inactivity.

3.9.6 Example

Request: "x00x16TAS_KEEP_ALIVE_REQUEST"

Response: "x00x17TAS_KEEP_ALIVE_RESPONSE"

3.9.7 Soft Shut

The host capture system supports a soft shut message.

Soft shut request messages are initiated by the host to alert the client that transaction traffic should be routed to the other coast. Soft shut messages precede a network disconnect, and do not require a soft shut response message from the client. In the soft shut state, all in-flight responses will be returned until the client is disconnected, and new transaction requests will be ignored.

3.9.8 Example

Request: "x00x15TAS_SOFT_SHUT_REQUEST"

3.10 General Industry Requirements

Please refer to the following for more information on various industry rules.

3.10.1 Negative Option Billing Merchant Business Model

A negative option billing merchant refers to a type of retailer that provides a sampling of goods or services at a free or reduced price to consumers. The merchant requires the consumer to provide payment card information in order to receive the sample and then, calling it a subscription, the merchant bills the consumer at a future date unless the consumer proactively cancels the subscription.

Mastercard has several rules regarding negative option billing merchants. Among them are these:

- MCC must be 5968 – Direct Marketing—Continuity/Subscription Merchants
- For e-commerce credit transactions, negative option billing merchants must include their website URL in the DBA Name field in the boarding platforms.
- The Customer Service Phone field in the boarding platform must contain the merchant's contact telephone number that is valid and accessible by all consumers worldwide.

3.10.2 Enhanced Authentication for Online Gambling Merchants

Mastercard has rules for promoting the use of enhanced authentication tools for reducing fraud for online gambling transactions. An online gambling transaction is defined as a card-not-present transaction associated with any of the following Mastercard MCCs:

- MCC 7995 (Gambling Transactions)
- MCC 7800 (Government Owned Lottery [U.S. Region Only])
- MCC 7801 (Internet Gambling [U.S. Region Only])
- MCC 7802 (Government Licensed Horse/Dog Racing [U.S. Region Only])
- MCC 9406 (Government Owned Lottery [Specific Countries])

An online gambling transaction must ensure that at least one of the following occurs in the authorization request message:

- The CVC2 value is populated
- EMV 3-D Secure (3DS) authentication has occurred
- With respect to a recurring payment transaction, Identity Check Insights are shared. Use Security Level Indicator Value of 214.

3.10.3 Mastercard High Risk Purchase Identifiers

There is a surge within the transaction processing and payments industry to support and identify purchases considered as high risk. Issuers must use this information to assess levels of risk, approve or decline transactions accordingly, and then report and monitor activity as necessary. Currently issuers base transaction approval/decline decisions on MCC in the transaction. Because MCCs can be used for a variety of merchants, this is not necessarily a reliable indicator that the transaction contains the purchase of cryptocurrency or high-risk securities.

Merchants that sell cryptocurrencies must use MCC 6051 (Quasi Cash – Merchant) and those that sell high-risk securities must use MCC 6211 (Securities – Brokers/Dealers).

Cryptocurrencies

Cryptocurrency is a digital currency, virtual currency, or alternative currency system with decentralized control. It is not controlled or managed by a centralized bank or economic system such as the Federal Reserve System, corporate boards, or governments which control the supply of currency.

There are currently over 4,000 variants of cryptocurrencies in existence, most notably Bitcoin. The legal status of cryptocurrencies varies by country. Some countries allow trade of cryptocurrencies and its use for the purchase of goods and services, while other countries have banned or restricted it.

High-Risk Securities

A high-risk securities transaction is when a consumer uses a Mastercard account to purchase, sell, or broker a financial instrument, including but not limited to derivatives such as forwards, futures, options, and swaps.

Mastercard defines a high-risk securities merchant as any securities merchant that facilitates one or more of the following activities:

- Binary options trading
- Contracts for difference (CFD)
- Foreign exchange (Forex) currency options trading
- Cryptocurrency options trading
- Initial coin offerings (ICOs)

3.10.4 Purchase Return Authorization

Visa, Mastercard and Discover use online authorizations for purchase return transactions to improve the cardholder experience and reduce merchant service inquiries about returns.

An authorized purchase return that is pending settlement will be included in an online statement, just like an authorized purchase. This enables a cardholder to view a return (credit) in their banking application in real time and an issuer can include the transaction in services such as text alerts.

When a purchase return authorization (PRA) is submitted for authorization, the issuer can validate the cardholder account. This enables issuers to decline potentially fraudulent cards and minimize chargebacks for non-existent or closed accounts.

Purchase returns are not applicable to ATM cash withdrawals, manual cash disbursements, fund transfers, money orders, sales of gambling values, or other sales of cash-like instruments since these types of transactions are not sales for goods or services.

Refer to the following with [<A12> Payment Code](#) value of 13:

- [<A130> Authorization Indicator](#)
- [<A21><B14> Transaction status](#)

Mastercard PRA: Note that cryptography data is not mandatory for PRA transactions. However, if a refund transaction has been initiated through the dip or tap of a card or contactless-enabled access device at a POS terminal, cryptography should be included.

3.10.5 Consumer Bill Payment Service

Consumers often have to log on to multiple biller sites to pay their bills, many of which do not accept card payments. Visa Consumer Bill Payment Service (CBPS) providers simplify the process by allowing consumers to pay all bills via card and through a single channel. Often CBPS providers offer a mobile application or e-commerce site for cardholders to manage and make bill payments. The CBPS provider, on behalf of the cardholder, then completes the transaction with the biller separately by passing funds via ACH or another payment method.

3.10.6 Account Funding Transactions

For Visa, the Account Funding Transaction (AFT) is a transaction where funds are pulled from a Visa account and are subsequently used to fund another Visa or non-Visa account; for example, loading or topping up prepaid card accounts, moving funds into another financial account such as a savings deposit account, or loading third party digital wallets.

Samsung Pay Cash (Stored Value Digital Wallet) is a feature that allows users to create a digital debit card that is stored in Samsung Pay. Users are able to easily add funds to this card either with an existing card within the app, or by transferring funds directly from a bank account.

Instructions for Use:

- The proper [<A10> Transaction Code](#) and [<A11> Program Code](#) must be set for AFT: 10 – Purchase and W – Account Funding, respectively.
- [<A136> Business Application Identifier](#) value must be **FT** – Funds Transfer.
- The proper combination of fields [5.16.1 <A21><B1> Cardholder Present Data](#), [5.16.2 <A21><B2> Card Present Data](#), and/or [<A32> MOTO/e-Commerce Indicator](#) value must be used to indicate that the customer is or is not present and the card is not present, or the transaction is an e-Commerce request.

- If service providers or merchants use [<A69> Transaction Fee Amount](#), they must include those amounts in [<A15> Transaction Amount](#).
- The [<A18> Merchant Category Code \(MCC\)](#) must be set to one of these values:
 - 4829 - Wire Transfer Money Order
 - 6540 - Non-Financial Institution - Stored Value Card Purchase/Load
 - 6012 - Financial Institution/Merchandise
- For POS Data Code [<A21><B3> Card Data Input Mode](#) Valid values include:
 - 1 - Manual input
 - 2 - Magnetic stripe read
 - C - Online chip
 - M - PAN auto-entry via contactless chip
 - 7 - Payment Credential stored on file

3.10.7 Visa System Integrity Fees

Visa is introducing new rules and fees to ensure businesses properly use and action declined transaction response codes that will serve to improve authorization and mitigate invalid transactions. These changes are designed to drive improvements across the payments ecosystem by encouraging good business practices that will reduce fraud, improve authorization approval rates and reduce operational costs for the ecosystem. The fees are effective with the April 2022 release.

Visa will reposition and cluster decline response codes into four categories:

- Category 1 – Issuer will never approve. This is a sub-set of decline codes that indicates the card is blocked for use or never existed and means there is no circumstance in which the issuer will grant an approval—for example, in the case of a lost or stolen card.
- Category 2 – Issuer cannot approve at this time. This is a sub-set of decline codes that indicates the issuer may approve, but cannot do so now, perhaps due to a system issue, detection action or a lack of funds. This cluster covers temporary decline decisions made by issuers that may change over time. It occurs when the issuer is prepared to approve a transaction, but is unable to do so at the time and would welcome a further authorization attempt in the future—for example, as in the case of a non-sufficient funds decline code.
- Category 3 – Issuers cannot approve with these details. This is a sub-set of decline codes that indicates the issuer cannot approve based on the details provided, such as an invalid account number or incorrect Card Verification Value (CVV), CVV2 or expiration date.
- Category 4 – Generic Response Codes. While the great majority of declines fall into the above categories, issuers may use some specialized codes for certain circumstances. However, their usage will remain minimal.

Table 3.1 Visa Response Code Categories

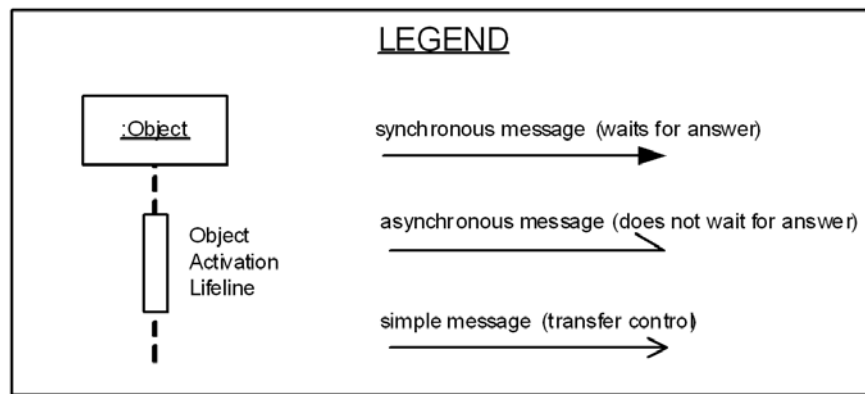
Decline Response Category	Response Codes	Merchant Reattempt Limit
Category 1 - Issuer will never approve	Possible values: 4 - Pick Up Card 7 - Pick Up Card, Special Condition 12 - Invalid Transaction 15 - No Such Issuer 41 - Lost Card 43 - Stolen Card 46 - Closed Account 57 - Trans. not Permitted to Cardholder R0 - Stop Payment Order R1 - Revocation of Auth Order R3 - Revocation of all Authorization	Reattempt not permitted
Category 2 - Issuer cannot approve at this time	Possible values: 03 - Invalid merchant 19 - Re-enter transaction 51 - Not sufficient funds 59 - Suspected Fraud 61 - Exceeds approval amount limit 62 - Restricted card (card invalid in region or country) 65 - Exceeds withdrawal frequency limit 75 - Allowable number of PIN-entry tries exceeded 78 - Blocked, first used 86 - Cannot Verify PIN 91 - Issuer or switch inoperative 93 - Transaction cannot be completed – violation of law 96 - System malfunction N3 - Cash service not available N4 - Cash request exceeds issuer or approved limit	Reattempt permitted up to 15 attempts in 30 days

Decline Response Category	Response Codes	Merchant Reattempt Limit
Category 3 - Issuer cannot approve with these details	Possible values: 14 - Invalid Account 54 - Expired card or expiration date missing 55 - PIN incorrect or missing 70 - PIN data required 82 - Negative Online CAM, dCVV, iCVV or CVV results IA - Additional customer authentication required N7 - Decline for CVV2 Failure	Reattempt permitted up to 15 attempts in 30 days
Category 4 - Generic Response	All other Decline Response codes	

Chapter 4

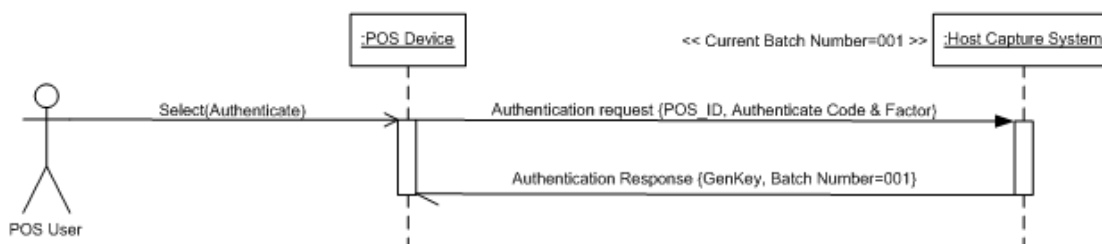
Messaging Scenarios

This chapter outlines the scenarios for POS system and TSYS Acquiring Solutions host capture system exchanges of SGMF formatted transaction messages generated by the POS device performing host capture use cases. The following messaging scenarios and sequence diagrams are organized by use case names, as described in [Host Capture Use Cases](#).



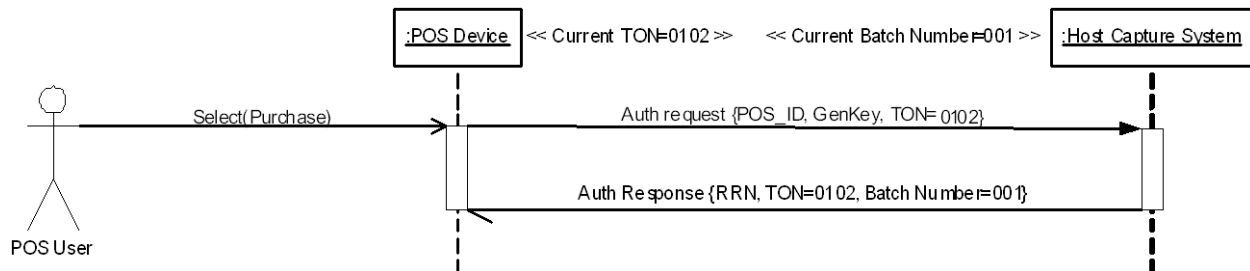
4.1 Authenticate POS device

The POS device achieves a ready state when it has successfully received a GenKey from the host capture system. The authentication request includes the <A1> Host Capture POS ID, a clear text <A2> Authentication Code and an <A4> Authentication Factor. The user-entered Authentication Code and Factor enhance the security of the POS device. The authentication response includes the <A3> GenKey and the <A80> Batch Number of the current batch.



4.2 Request authorization

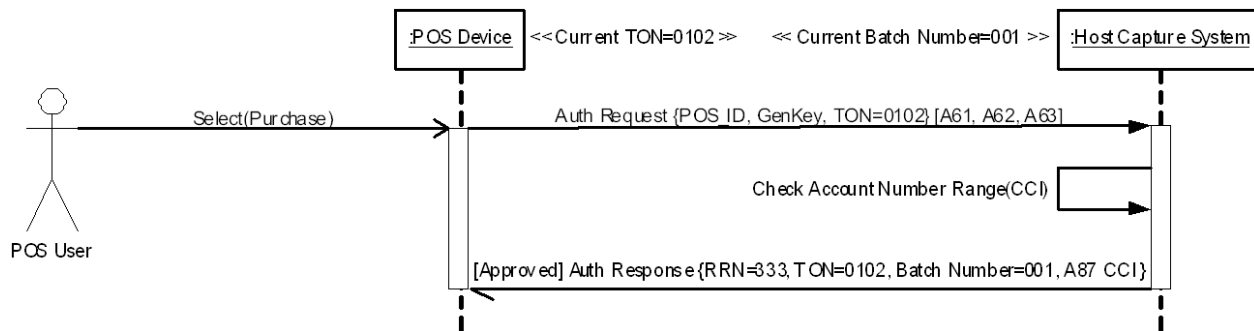
Cardholder data elements are transmitted from the POS device as an auth transaction request with a new <A17> TON. The host capture system sends an authorization response that includes a new <A86> RRN, the <A80> Current Batch Number and echoes the TON.



4.2.1 Commercial card authorization

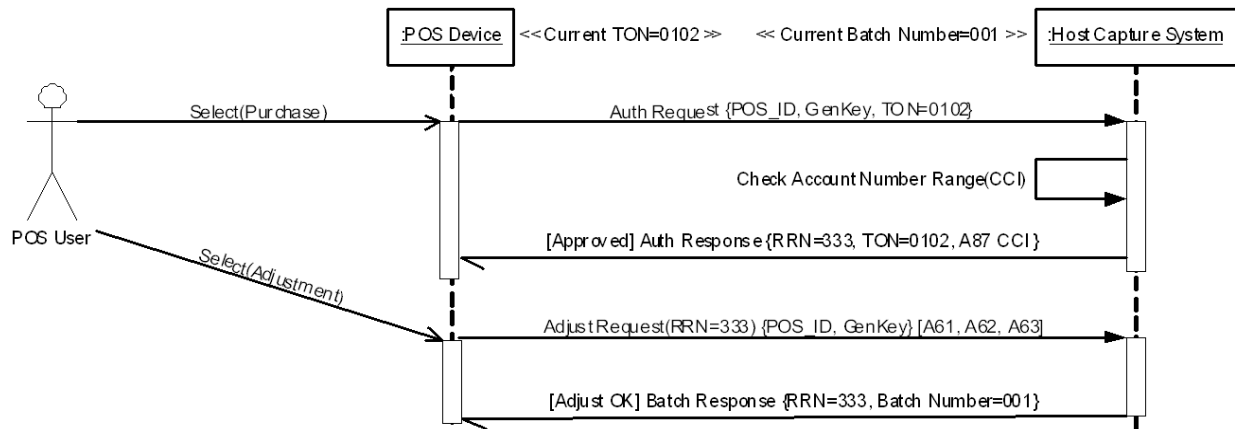
There are two messaging options for a commercial card authorization with Level II data elements.

1. The POS device sends the Level II data elements in the original authorization request message. The Cardholder data elements are transmitted from the POS device as an authorization transaction request with a new <A17> TON, and the message includes the A61 Tax Amount, A62 Tax Amount Identifier and A63 Purchase Order Number. The host capture system sends a response with the <A86> RRN, <A87> CCI Response (Commercial Card Response Indicator) (e.g. {CCI=R}), the <A80> Batch Number, and echoes the TON.



2. The POS device does not send the Level II data elements in the original authorization request message. Cardholder data elements are transmitted from the POS device as an auth transaction request with a new <A17> TON. The host capture system sends a response with the <A86> RRN, <A87> CCI Response (Commercial Card Response Indicator) (e.g. {CCI=R}) and echoes the TON. The authorization response does not include <A80> Batch Number.

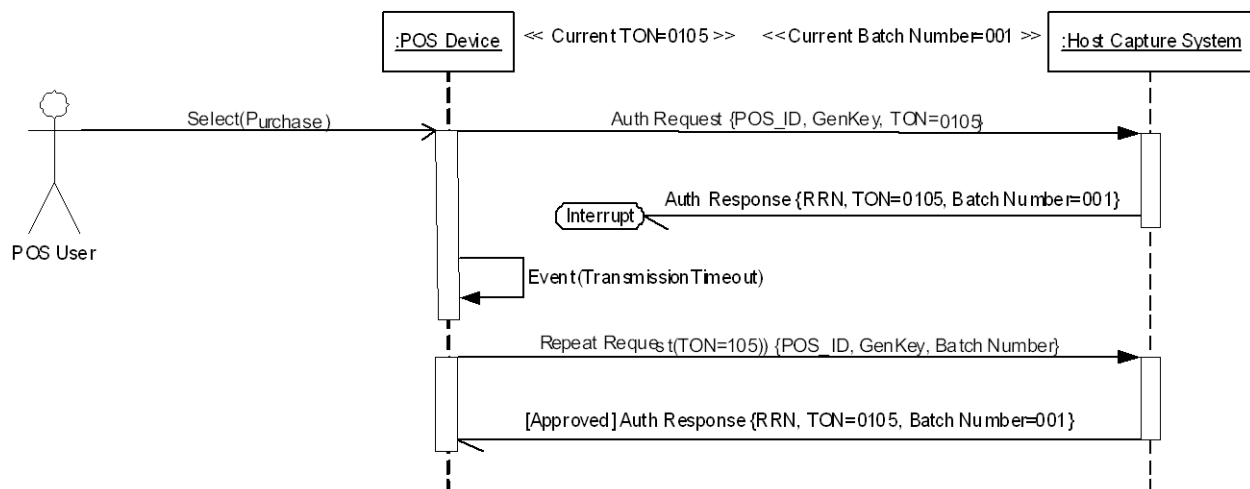
The POS user adjusts the original auth transaction to append the Level II data by using the RRN to locate the transaction. The host capture system sends the adjustment response with the current batch number and echoes the RRN. If the authorization is not adjusted before the current batch closes, it will be captured in the next batch with or without adjustment.



4.2.2 No response to authorization

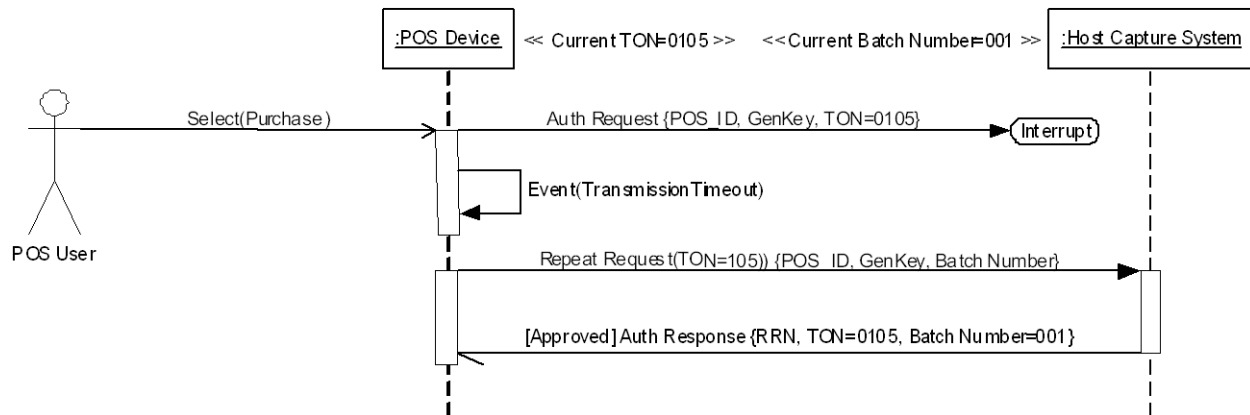
There are two recovery paths for no response to authorization with repeat authorization depending on whether the host capture system failed to respond to the POS device, or the POS device failed to request from the host capture system.

1. If the POS device does not receive an auth transaction response from the host capture system and the transaction timeout occurs then the authorization request can be resent by using the <A17> TON of the original auth transaction and the <A14> Repeat Indicator. The host capture system has already logged this auth transaction and will respond by resending the existing <A84> Approval Code and the associated <A85> Authorization Response Text.



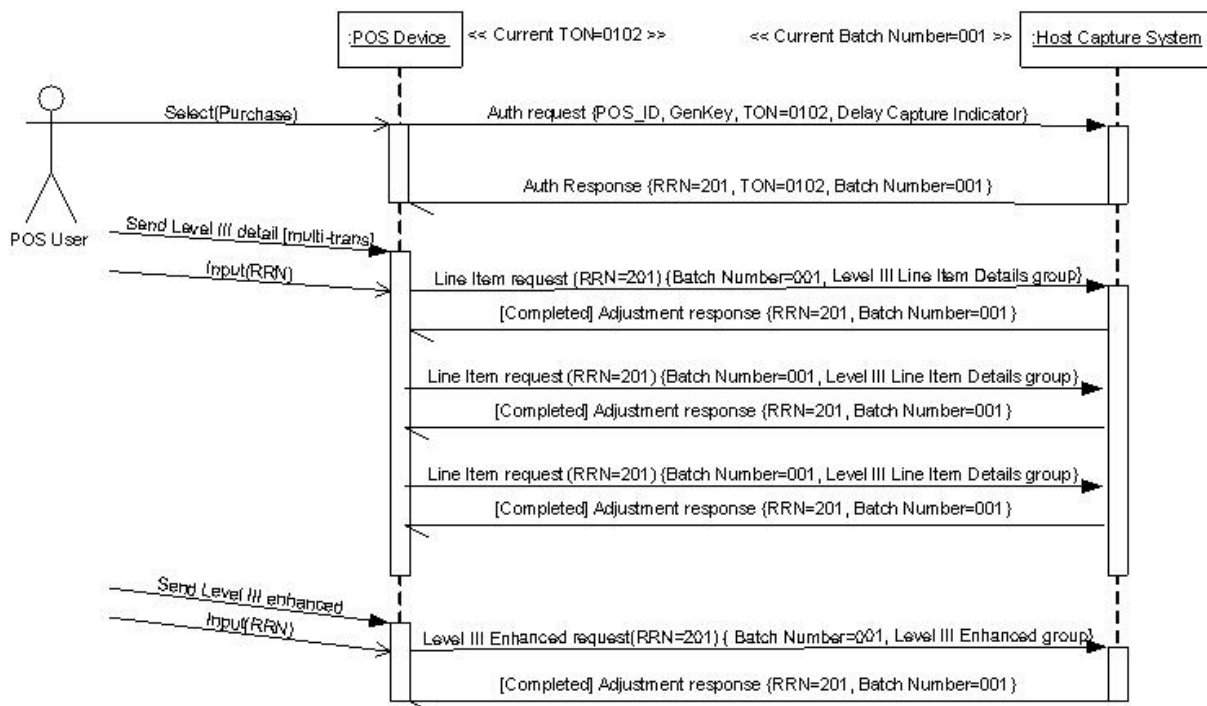
2. If an auth transaction request is not received by the host capture system and the POS device transmission timeout occurs then the request can be resent by using the <A17> TON of the original auth transaction and the <A14> Repeat Indicator. The host capture system will not find an auth transaction with the input

TON because it was never received by the host. The host capture system will attempt to process this auth transaction using the normal authorization path.



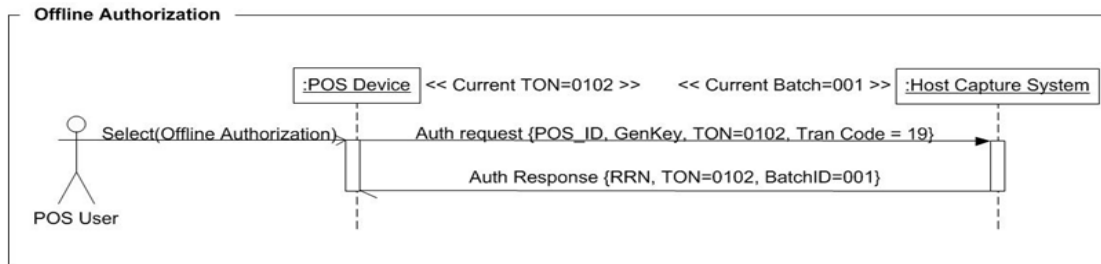
4.2.3 Auth with Multiple Line Items

When an authorization will be followed by level III data, then cardholder data elements are transmitted from the POS device as an authorization request with a new <A17> TON and the <A98> Delay Capture Indicator 'DLY'. The host sends a response with the <A86> RRN, the <A80> Current Batch Number, and echoes the TON. The Level III Line Items are sent next using a 'multi-transaction' application type. Each line item is sent as its own transaction with <A10> Transaction Code=105. When all line items have been sent, then the POS sends the Level III Enhanced Data request message with the <A10> Transaction Code=104 to conclude the Level III transaction. When the Level III Enhanced Data is received for an authorization the transaction will be captured by the next batch close event.



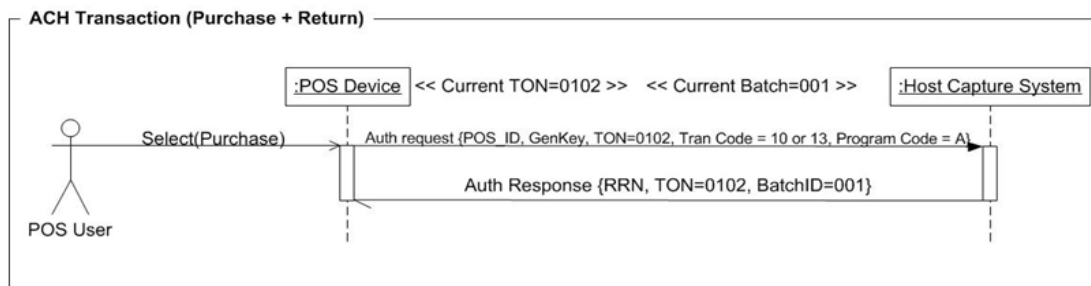
4.2.4 Offline authorization

An offline authorization is a capture only transaction that is authorized elsewhere and captured on the TSYS Acquiring Solutions host. The cardholder data elements are transmitted from the POS device as an auth transaction request with a new <A17> TON, and <A10> Transaction Code of 19. The host capture system sends an authorization response that includes a new <A86> RRN, the <A80> Current Batch Number and echoes the TON.



4.2.5 ACH transaction

An ACH transaction is a capture only transaction that is captured on the TSYS Acquiring Solutions host. The cardholder data elements are transmitted from the POS device as an auth transaction request with a new <A17> TON, and <A10> Transaction code of either 10 (purchase) or 13 (return), and <A11> Program Code of A. The host capture system sends an authorization response that includes a new <A86> RRN, the <A80> Current Batch Number and echoes the TON.



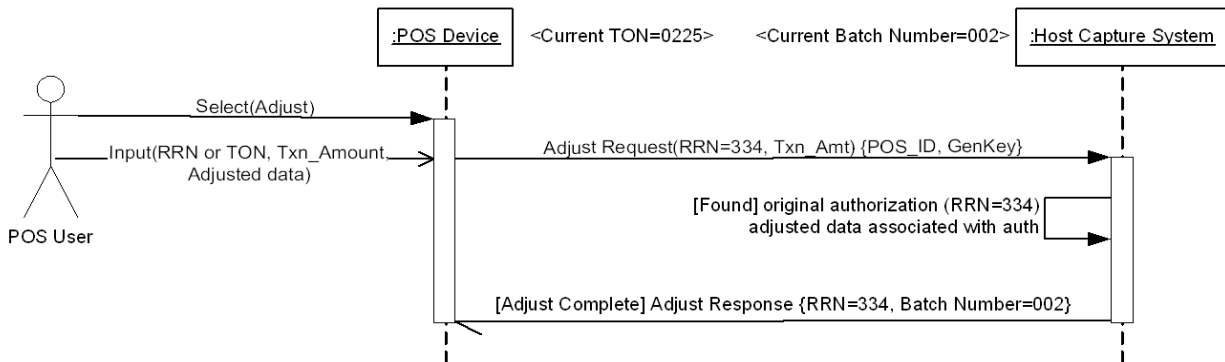
4.3 Refund transaction

This auth transaction credits a cardholder's account. Cardholder data elements are transmitted from the POI as an auth transaction request with a new <A17> TON. The host capture system sends a response with a new <A86> RRN and <A80> Current Batch Number and echoes the TON.

A purchase return authorization may be declined by the issuer, thereby causing it not to post in settlement.

4.4 Adjust transaction

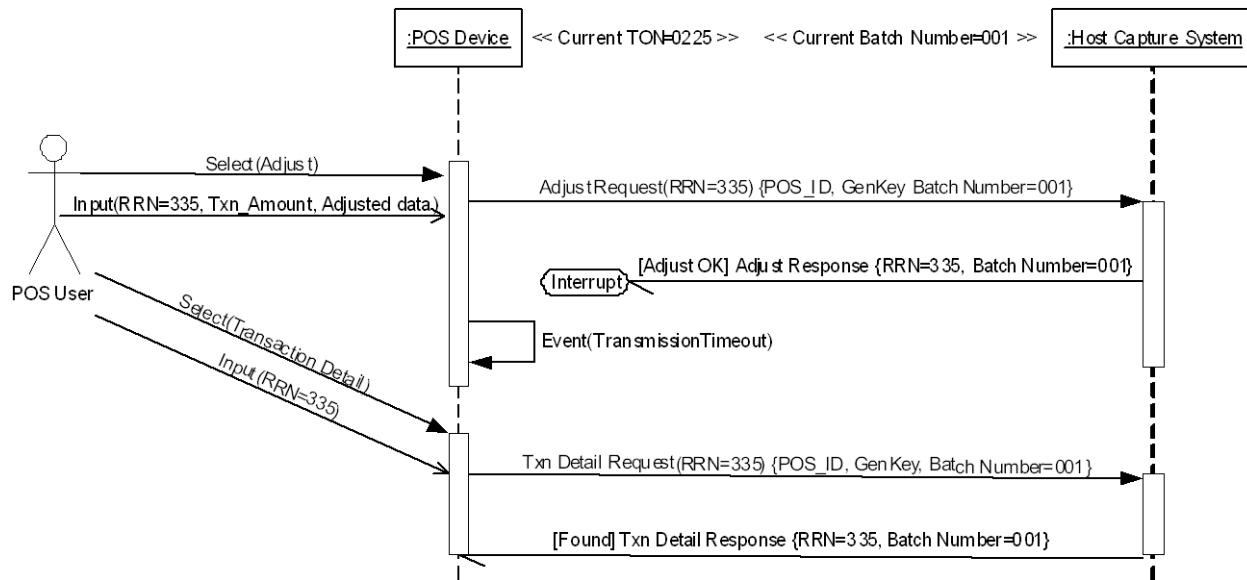
The adjustment request associates with an original authorization transaction in the current or prior batch by matching either the <A86> RRN (or alternately the <A17> TON) values. The host capture system uses the input RRN or TON value to find the authorization transaction and apply the adjusted data to this authorization. The POS user inputs the RRN or the TON, the original <A15> Transaction Amount, and the adjusted data. The adjustment response includes the <A80> Current Batch Number and echoes the RRN and the TON of the original authorization.



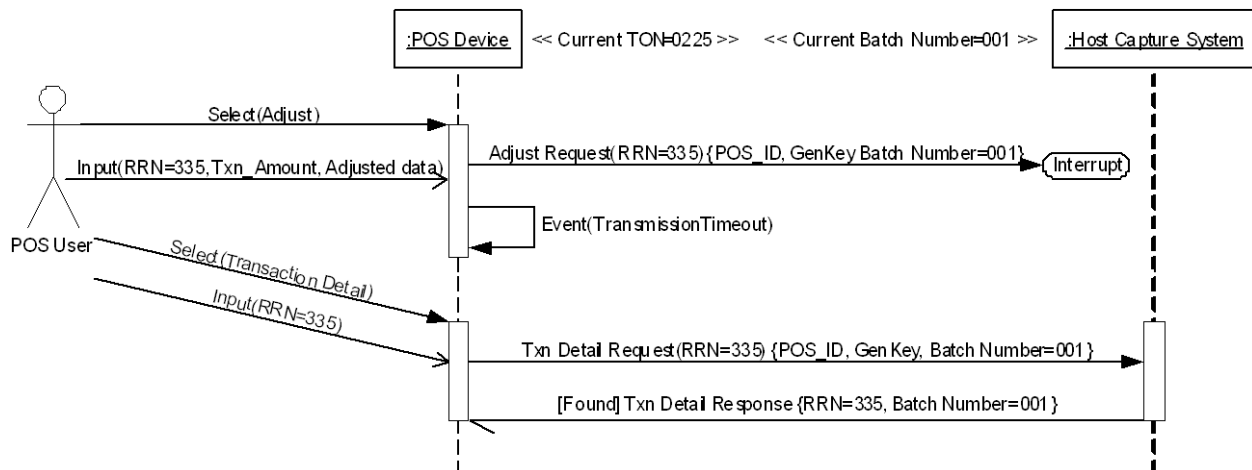
4.4.1 No response to adjustment

There are two recovery paths for no adjustment with transaction detail depending on whether the host capture system failed to respond to the POS device or the POS device failed to request from the host capture system.

1. If the POS device does not receive an adjust transaction response from the host capture system and the transmission timeout occurs then a transaction detail request is sent by using the <A86> RRN and the <A80> Batch Number of the original auth transaction. The host capture system has already logged this adjust transaction and responds with the Batch Number, the RRN, the <A17> TON and sends the transaction details showing the prior adjusted values.



2. If an adjust request is not received by the host capture system and the POS device transmission timeout occurs then a transaction detail request is sent using the <A17> TON and the <A80> Batch Number of the original auth transaction. The host capture system has not logged the original adjust request and responds with the <A80> Batch Number, the <A86> RRN, echoes the TON and sends the transaction details which show the adjusted values were not updated.



4.4.2 Delay Capture Adjustment

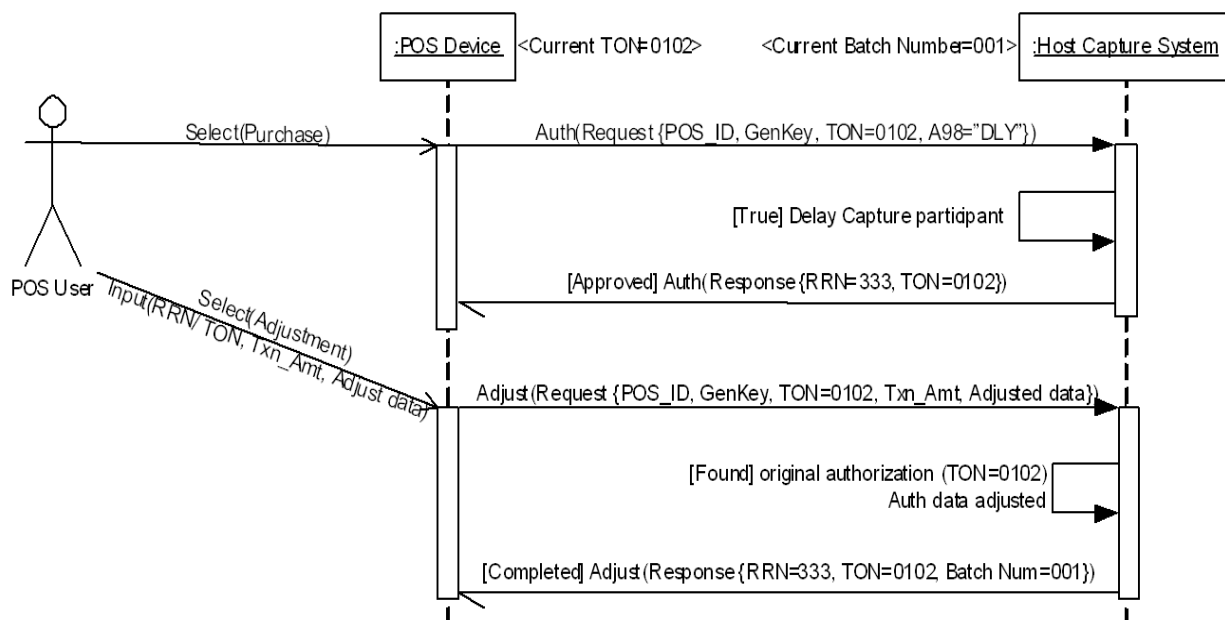
A merchant terminal that is a 'delay capture participant' and uses tag <A98>= "DLY" in an authorization request will prevent the authorization from being captured in the next batch close event, unless the merchant adjusts the original authorization. An authorization can be delayed one batch close event using tag <A98>.

If the original authorization included tag <A98> and the merchant adjusts the original authorization before the next batch close, then the authorization will be captured when the next batch close occurs.

If the original authorization included tag <A98> and the merchant does not adjust the original authorization before the next batch close, then the authorization will not be captured when the next batch close occurs. This authorization will be captured in the subsequent batch close with or without adjustment.

The adjust transaction response includes the <A80> Current Batch Number and echoes the RRN and the TON.

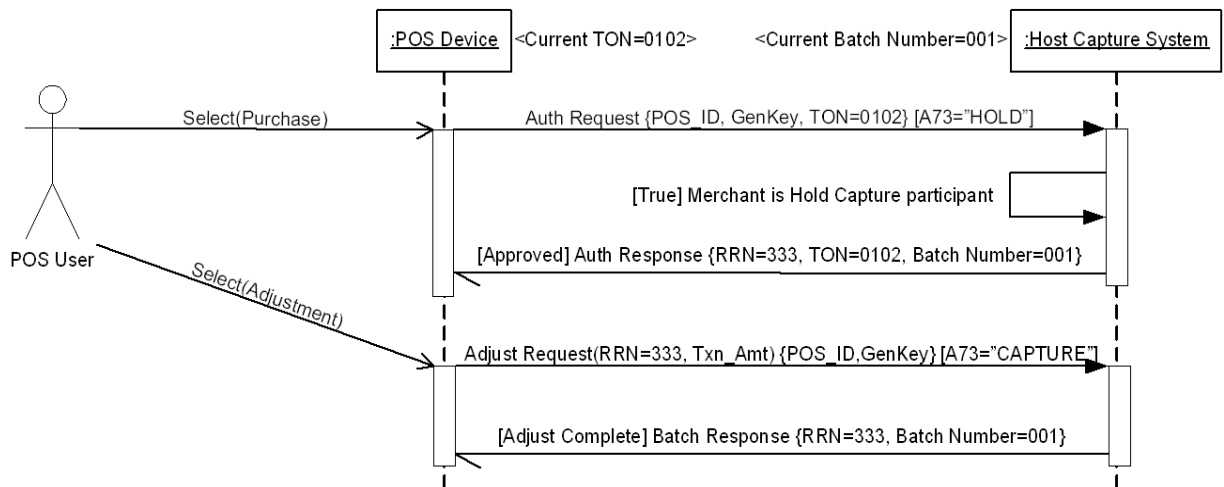
NOTE Tag <A98> will be ignored by the system if sent by a merchant who is not a 'delay capture participant'. An Authorization request from a nonparticipating merchant that includes <A98> will be captured in the next batch close event.



4.4.3 Hold Capture Adjustment

If a merchant is a 'hold capture participant' then the POS device can include tag <A73>= "HOLD" in the authorization request. Including this authorization tag will prevent the transaction from being captured and settled. When the POS device subsequently sends an adjustment transaction with the RRN, the transaction amount of the original authorization and the tag <A73> = "CAPTURE", then the held authorization will be released and will be captured during the next batch close. The adjust transaction response includes the <A80> Current Batch Number and echoes the RRN.

NOTE If a merchant is not a 'hold capture participant' and the POS device sends tag <A73> then the host will reject the transaction and respond with "Not Allowed".



4.4.4 Split Shipment Flow

Summit Split Shipment operates paired with the concept of HOLD Capture.

1. The Point of Interaction (POI) sends an original authorization with A73 = HOLD.
2. A request to the card brand is created by Summit.
3. Summit receives a response from the card brand.
4. Summit creates a Capture data record from request and response elements as well as field values from the merchant profile that is held in the system indefinitely.
5. Summit re-interprets the card brand response back into SGMF and returns to the POI with a system generated Reference Retrieval Number (RRN).
6. A Split Shipment transaction (A10 = 110) is sent with the RRN from the Original transaction, but a TON unique to the Split Shipment transaction. The A15 value should reflect the Authorization Amount from the original, and the A68 tag should be included to reflect the settlement amount of the Split Shipment. The A99 (Multiple Clearing Sequence Number) and A100 (Multiple Clearing Sequence Count) tags should be included. Summit will respond with the RRN of the original transaction. The Split Shipment will create a separate Capture record that will be marked for settlement in the next settlement batch close.

One or more additional Split Shipment transactions could be sent with the above criteria.

It should be noted that at no point after the first Split Shipment should the Original transaction be captured using the A73=CAPTURE tag. Validation in the Summit system will prevent this from occurring.

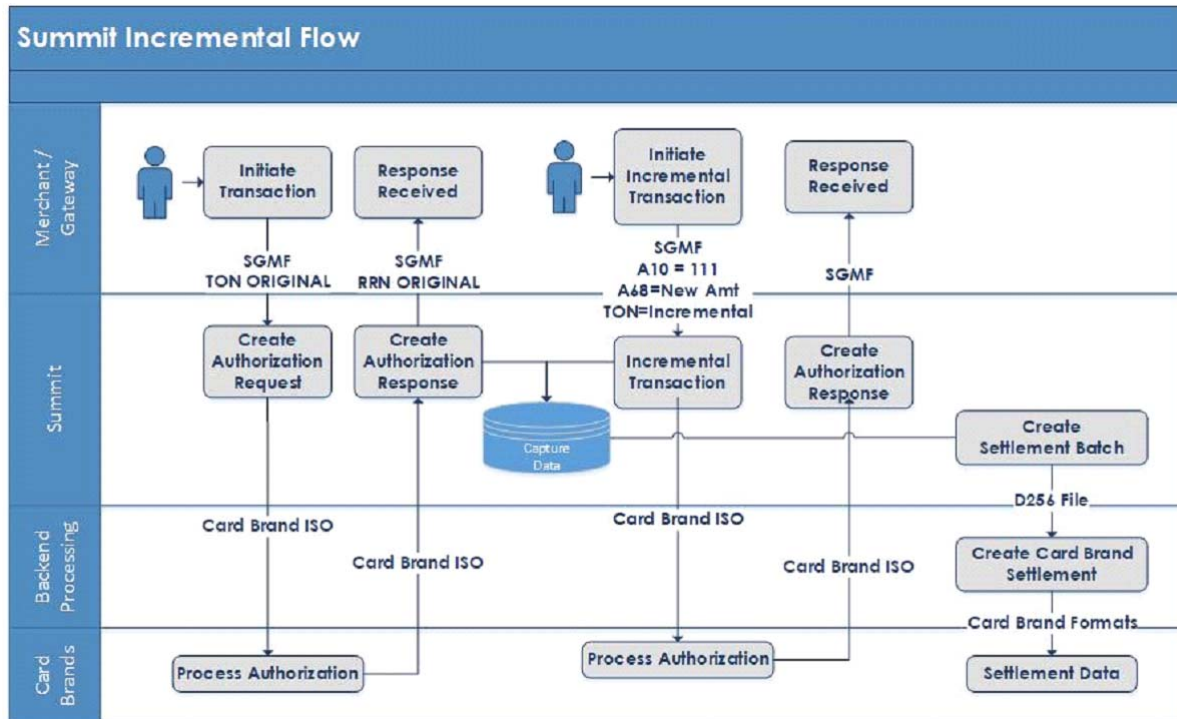
or the POI. If the original request is not found, the split shipment will be processed as noted previously, as a new Split Shipment transaction.

4.4.6 Incremental Flow

Incremental transactions are a means of incrementing the amount of an existing Authorization. Given that Summit does not allow an adjustment to increase the Settlement Amount, an incremental is performed by sending an Incremental adjustment to the original authorization.

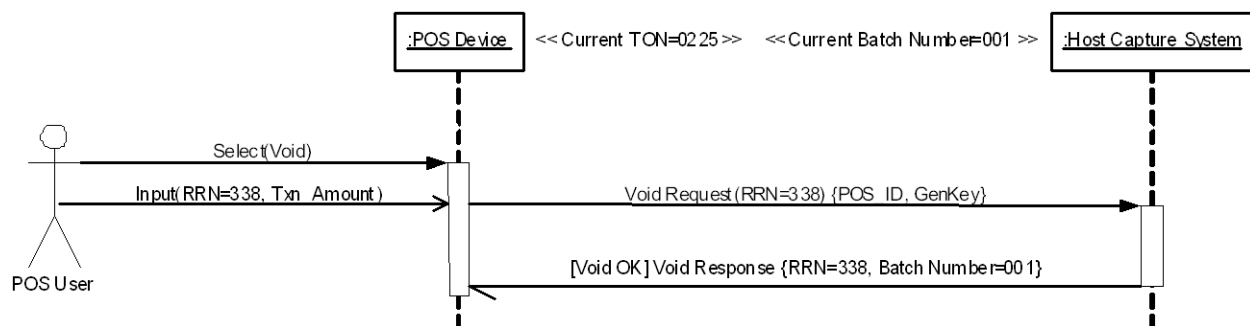
1. The Point of Interaction (POI) sends an original authorization.
2. A request to the card brand is created by Summit.
3. Summit creates a Capture data record from request and response elements as well as field values from the merchant profile that is held in the system indefinitely.
4. Summit re-interprets the card brand response back into SGMF and returns to the POI with a system generated Reference Retrieval Number (RRN).
5. An Incremental transaction (A10 = 111) is sent with the TON from the Original transaction. The A15 value should reflect the Authorization Amount from the original, and the A68 tag should be included to reflect the incremented settlement amount.
6. Summit will create an incremental authorization to the Card Brands using the original System Trace Number, Retrieval Reference Number, and Transaction Identifier.
7. Summit creates an adjustment of the settlement amount to the original transaction.

The transaction will be captured and settled regularly with the rest of the open batch.



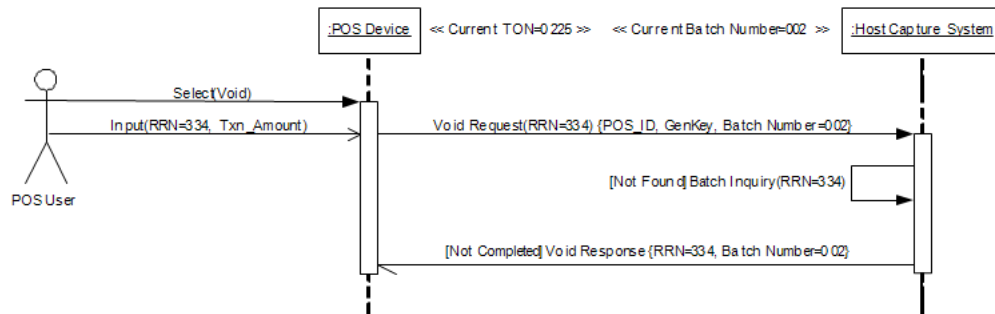
4.5 Void transaction

This void request associates with an original auth transaction in the current batch by its <A86> RRN (or alternately the <A17> TON). The host capture system finds the associated auth transaction in the current batch then sends a Void response with the <A80> Current Batch Number, echoes the RRN and TON and indicates that the void was successful.



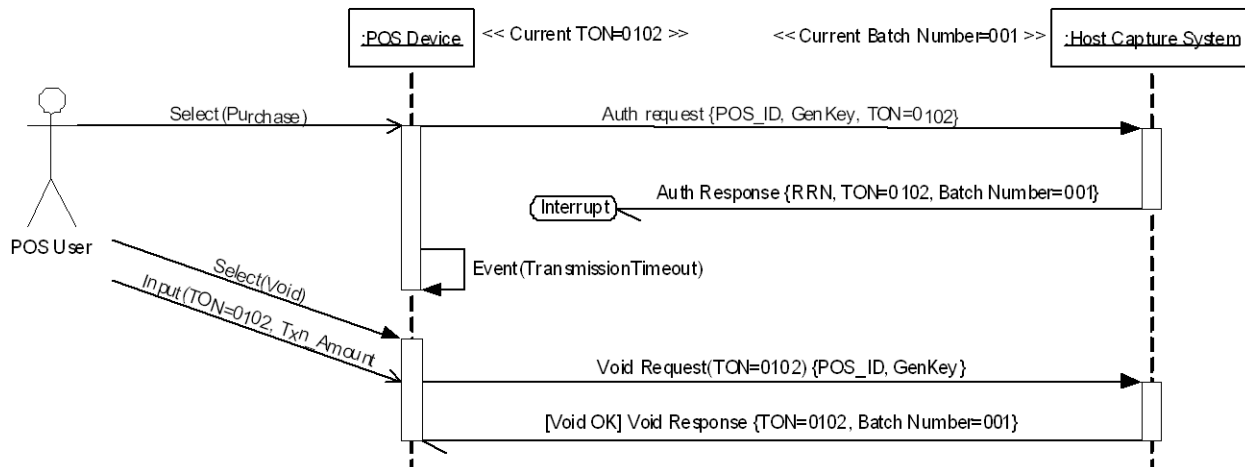
4.5.1 Void a transaction not in the current batch

This adjust transaction is identical to the Adjust transaction above, except the requested adjustment references an RRN or TON that is not in the current batch and responds with “Not Completed”.



4.5.2 No response to authorization with void recovery

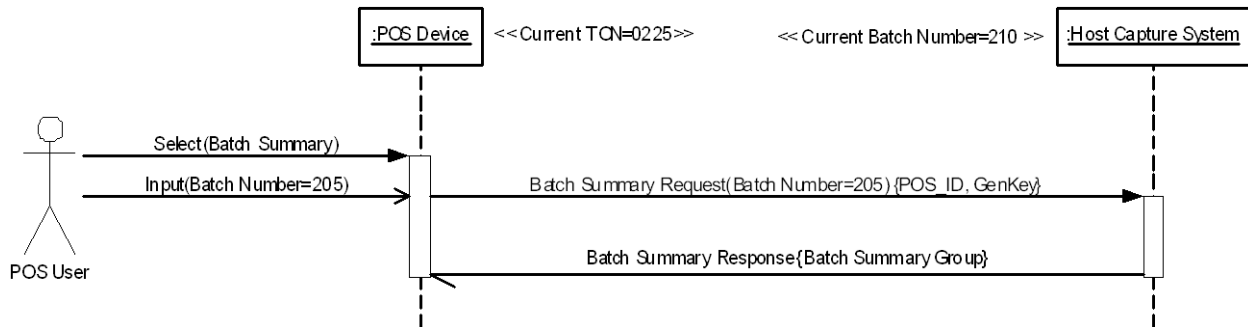
If an auth transaction response is sent by the host capture system and is not received by the POS device and the transmission timeout occurs then the void transaction use case can be performed.



4.6 Batch inquiry

4.6.1 Batch summary

This batch summary request is sent to the host capture system with a <A80> Batch Number value equal to the current batch or one of the five prior Batches. The host capture system responds with the requested data elements included with the <A325> Batch Summary Group and its sub-data elements, and are grouped by card type values in the <A325><B1> Summary Type.

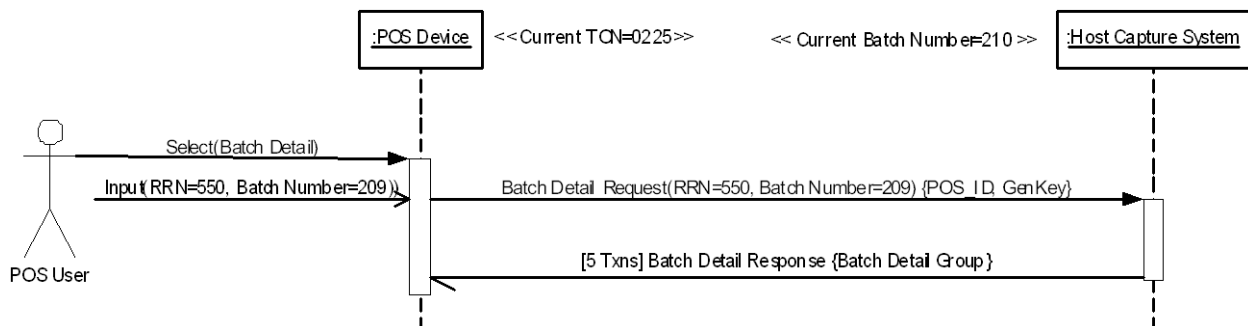


4.6.2 Batch detail

The batch detail request is sent to the host capture system with a <A80> Batch Number value equal to the current or last batch number, <A307> Requested Transaction Count and the <A86> RRN (or alternately the <A17> TON). The RRN is used to locate the correct range of transactions within the batch and the list will display in descending order starting at the transaction prior to the requested RRN. The host capture system responds with the requested data elements included with the <A330> Batch Detail Group and its sub-data elements.

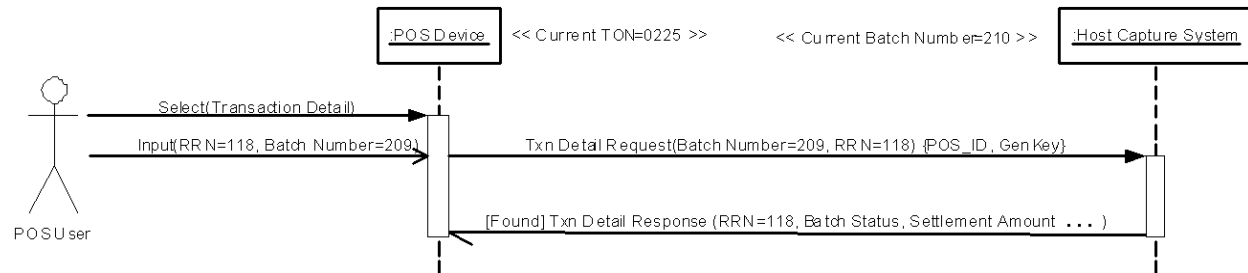
If the RRN is not sent in the request then the system will default to the most recent RRN. If the batch number is not sent in the request the system will default to the current batch. If the requested transaction Count is not included in the request then the response will send five transactions. The maximum number of transactions that can be requested is five.

The typical sequence of use: a batch detail inquiry request is sent with no RRN, transaction count, or batch number data; the host returns the most recent 5 transactions in the current batch; another batch detail inquiry request is sent including the RRN of the last transaction in the first detail request; the five transactions prior to the requested RRN are returned.



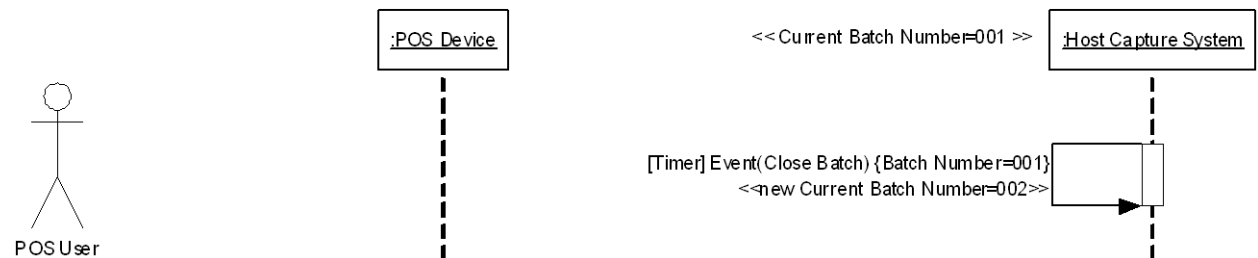
4.6.3 Transaction detail

The transaction detail request is sent to the host capture system with a <A80> Batch Number value equal to the current or last batch number and the <A86> RRN (or alternately the <A17> TON). The RRN is used to locate the correct transaction within the batch. The host capture system responds with the requested data elements.



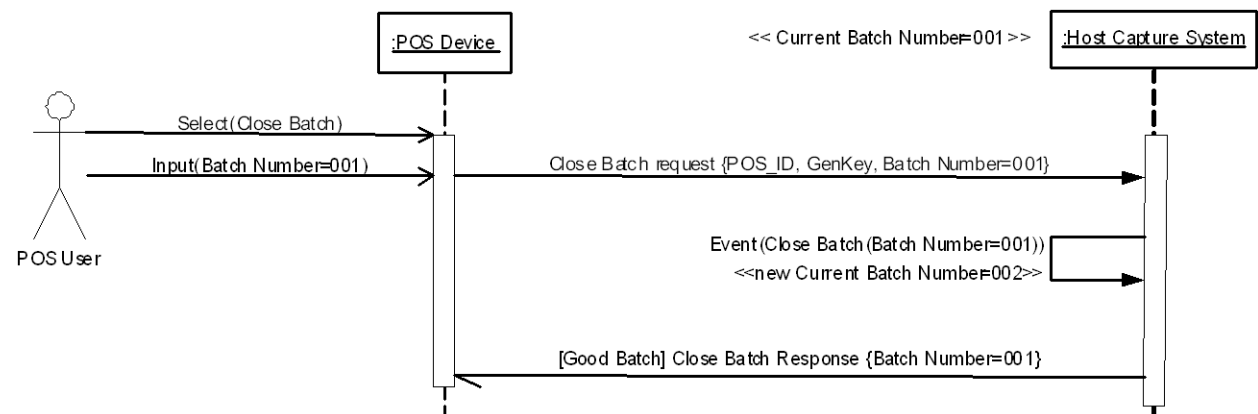
4.7 Batch close

The batch close use case is associated with an event timer parameter read from the TSYS Acquiring merchant boarding system. When the timer event occurs the host capture system initiates the batch close use case. Any auth or batch transactions (e.g. pre-authorization, commercial card) that cannot be settled are withheld from the closed batch and placed in the new current batch.



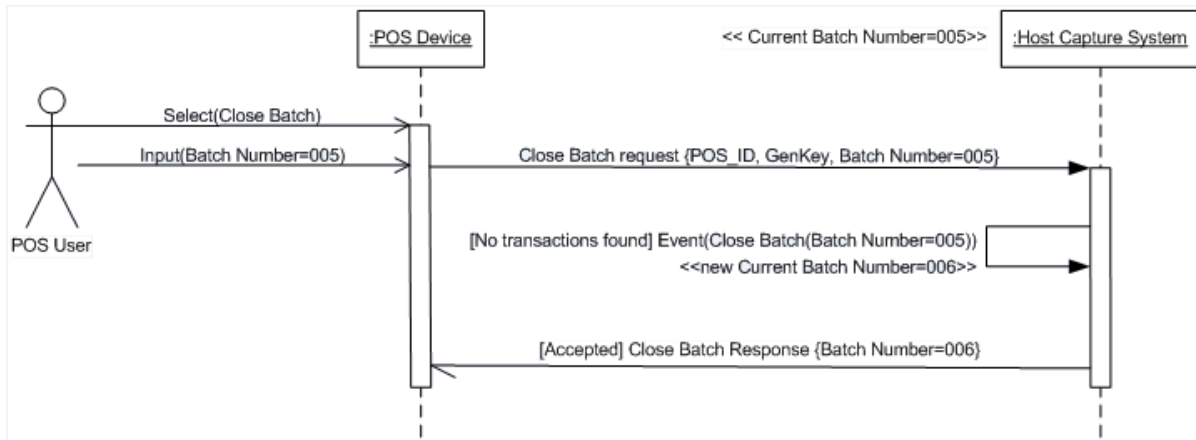
4.7.1 Manual batch close

The manual batch close request is sent to the host capture system with the <A80> Current Batch Number to initiate the batch close use case. The response includes the batch number and indicates that the batch was successfully closed.



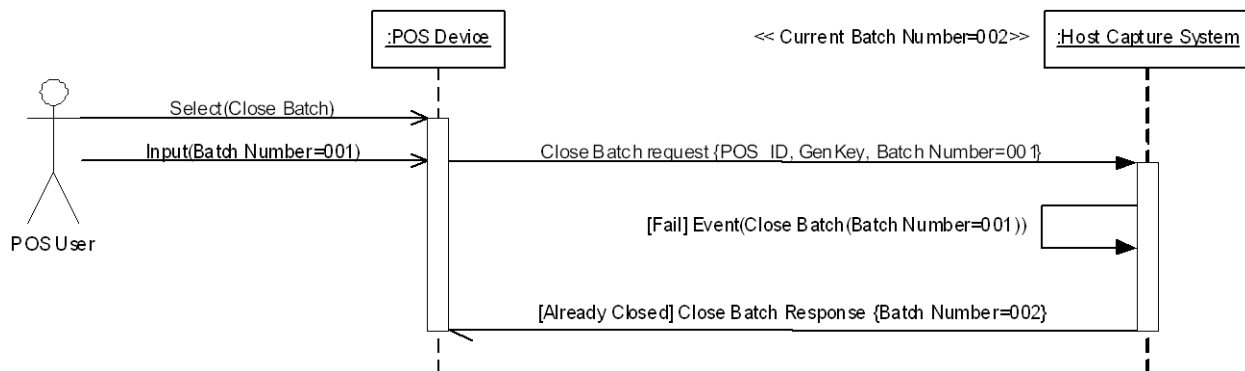
4.7.2 Manual batch close request for empty current batch

If the POS user performs the manual batch close use case and there are no transactions associated with the current open batch, then the host increments the batch number and responds with the new <A80> Current Batch number, the <A302> Batch Response Code and <A303> Batch Response Text “ACCEPTED”.



4.7.3 Manual batch close for already closed batch

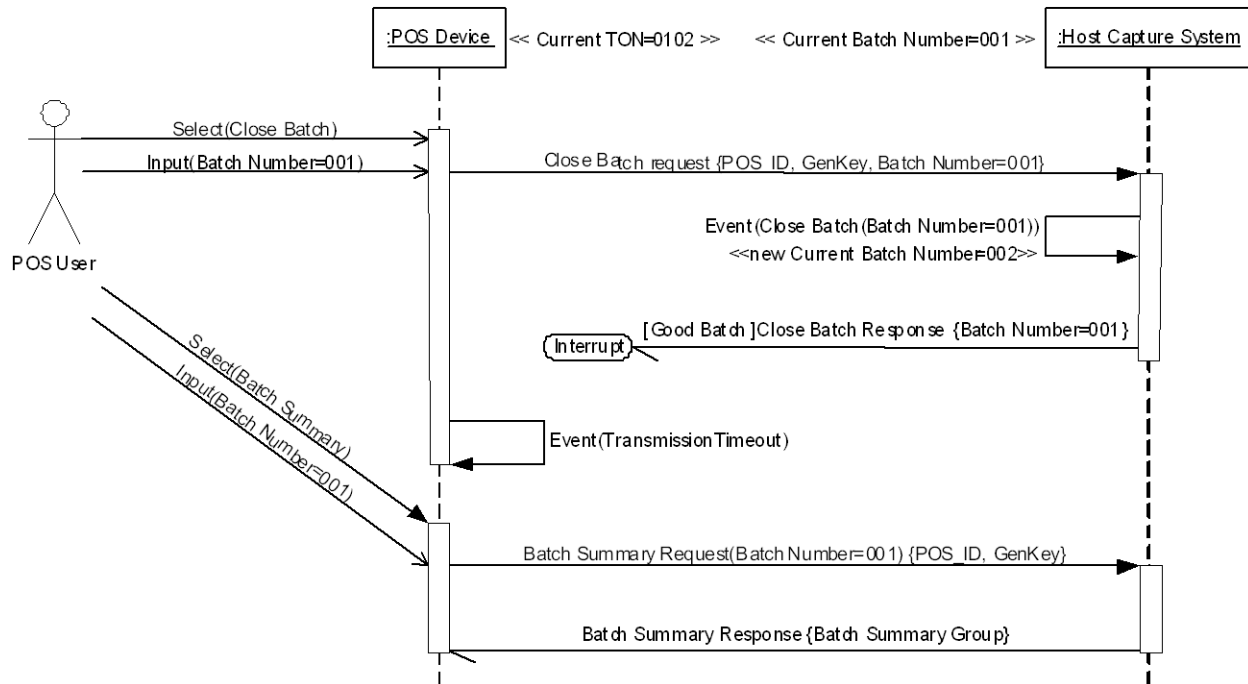
If the POS user attempts to perform the manual batch close use case on a batch number that has already been closed, then the system responds with the new <A80> Current Batch number, the <A302> Batch Response Code and <A303> Batch Response Text “Already Closed”.



4.7.4 No response to manual batch close with batch summary recovery

If a manual batch close response is not received by the POS device then the POS user requests a batch summary using the <A80> Batch Number of the batch requested to close.

- If the batch status indicates that it is the current batch, the POS user can send the manual batch close request again.
- If the batch status indicates that it is the last batch, then the batch close was successful and the POS user does not need to perform the manual close batch use case again



4.8 Using encryption and tokens

Security features include the use of encryption and tokens. Encryption allows the transaction to hold encrypted data, which ensures transmission security. TSYS Tokens can be requested in a transaction and in some cases, returned with a response. Simply requesting tokens does not enhance transmission security but it allows a receiver to store unique card-type data that then reduces PCI Audit Scope for the merchant. Please note that the TSYS token may include alphanumeric characters.

Tokenization comes in two types, TSYS Tokenization and Card Brand Tokenization, and the merchant should decide which solution is best for the line of business.

Please refer to the following for more information on Tokens and Encryption

- TSYS Tokenization - Tag A79
- Card Brand Tokenization - Tag A124
- Encryption:
 - Encryption Type and Encryption Transmission Block (ETB) - Tag A106
 - Track 1 - Tag A107
 - Track 2 - Tag A108

Type of Transaction	Mastercard	Visa	Discover	Amex
Digital Secure Remote Payments (DSRP) with UCAF (eCommerce)	Tag A32 Tag A40 Tag A21 (SF3=S) Tag A39	-	-	-
Visa Secure	-	Tag A32 Tag A34	-	-
Amex SafeKey / Tokenization	-	-	-	Tag A32 Tag A34
Discover ProtectBuy	-	-	Tag A32 Tag A34	-

Chapter 5

Field Specifications

This chapter describes each UTML tag in detail. The chapter is organized by tags in ascending order {<A1>, <A2> ... <An>}. Routing Header field definitions are found in [Routing header field definitions](#).

5.1 <A1> Host Capture POS ID

Length	Format	Description
15 fixed	A/N	A Unique ID that associates the POS device with a unique merchant hierarchy. NOTE: This field must be the first tag sent in all request messages to the TSYS Acquiring Solutions host.

5.2 <A2> Authentication Code

Length	Format	Description
6-10	A/N	An alphanumeric code provided to the POS user for input when authenticating a POS device during the POS device authentication process.

5.3 <A3> GenKey

Length	Format	Description
24 fixed	ASCII representation of HEX	Randomly generated string of alphanumeric characters that will be returned in the POS device authentication response and associated with the host capture POS ID. The GenKey must be stored in the POS device, must be sent with every request to the TSYS Acquiring Solutions host after authentication, and will be checked against the host capture POS ID.

5.4 <A4> Authentication Factor

Additional piece of merchant specific information used to identify a merchant during the POS device authentication process. Either the merchant zip/postal code or phone number may be transmitted.

5.4.1 <A4><B1> Merchant Postal Code

Length	Format	Description
5 or 6	A/N	The merchant's 5 digit US zip code or 6 character postal code.

5.4.2 <A4><B2>Merchant Phone Number

Length	Format	Description
7 to 10	Numeric	Left zero fill to 7 digits if the phone number is shorter than 7 digits. If phone number is longer than 10 digits, send the right-most 10 digits. North american phone numbers should include the area code.

5.5 <A10> Transaction Code

Length	Format	Description
1 to 3	A/N	This field contains a code identifying the type of function being requested of the TSYS Acquiring Solutions host. The Summary of transaction types contains a table of transaction code mappings with <A11> Program Code and <A12> Payment Code.

Code	Transaction	Description
1	Authentication	Authenticate a POS device on the TSYS Acquiring host capture system.
2	Deactivation	Deactivate an authenticated POS device. The POS must Authenticate again before processing any transaction on the host.
10	Purchase	Standard authorization request.
11	Cash advance	Use in banks, financial institutions, and EBT only NOTE Visa, Mastercard, Discover, and EBT only.
12	Quasi cash	Purchase of semi-cash items such as casino chips, travelers checks and stamps. NOTE Visa and Mastercard
13	Return	Return of goods purchased.
14	Balance inquiry	Request an available balance or limit.
15	Prepaid Activation	Activate a prepaid card.
16	Prepaid load	Load or increase value on a prepaid product.
17	Prepaid unload	Unload or decrease value on a prepaid product.
18	Prepaid close	Close or deactivate a prepaid product.
19	Offline authorization	Transaction authorized elsewhere and stored for capture on TSYS Acquiring Solutions host.
20	Card and AVS Authentication	Transaction authorized but will not be captured.

Code	Transaction	Description
Y	Bill pay	Transaction is a bill payment NOTE U.S. and Visa only. Other card brands and non-U.S. bill pay should use A10 Transaction Code 10 and no A11 Program Code.
Z	Auto-substantiation	Transaction substantiates participation in a benefits program.

5.7 <A12> Payment Code

Length	Format	Description
1 fixed	A/N	This one character code indicates the card payment application for the transaction. Possible value: A - Automated Clearing House C - Credit D - Debit E - Cash Benefit F - Food Stamp G - Gift Card P - Private Label S - SMS Cobrand transactions (this value is reserved for contracted SMS Visa Cobrand transactions)

5.8 <A13> Reversal Indicator

Length	Format	Description
3 fixed	Alpha	Indicates the transaction is a reversal. This must contain the value 'RVR'.

5.9 <A14> Repeat Indicator

Length	Format	Description
3 fixed	Alpha	Indicates the transaction is a repeat of a previous transaction attempt. Must contain the value 'RPT'. To generate a repeat transaction, add the <A14> tag to the original transaction and resend. NOTE Repeat transactions are not allowed for TSYS Acquiring Solutions gift card and debit transactions.

5.10 <A15> Transaction Amount

Length	Format	Description
1 to 12	Numeric	This field contains the transaction amount to be authorized. The transaction amount is to be presented with implied decimal point. The allowable number of significant digits and the positioning of any implied decimal point are dictated by the designated currency code. EXAMPLE USD \$.01 could be represented as 1, 01, or 001. NOTE For authorization reversal transactions, this field contains the total authorization amount (the total of all previous authorizations performed for a particular transaction or associated with a single transaction ID). Effective June 25, 2011, for account verification, including Account Status Inquiry Service requests or recurring payment account status check authorization requests, the transaction amount must be zero (0). Effective March 1, 2012, for Product Eligibility requests, the transaction amount must be zero (0). Mastercard authorizations originally coded as a preauthorization may require a longer chargeback protection period. To increase the effective duration of the chargeback protection period, the merchant may submit incremental preauthorization requests for the same transaction on later dates. Incremental preauthorizations for an additional amount may be used to increase the authorized amount held against the card account and to extend the chargeback protection period associated with the original preauthorization. Incremental preauthorizations for a zero amount may be used to extend only the chargeback protection period associated with the original preauthorization. When the chargeback protection period of a preauthorization is extended because of an incremental preauthorization, it is extended for 30 days from the date of the latest approved incremental preauthorization. Please refer to the Mastercard Customer Interface Specification for complete information.

5.11 <A16> Secondary Amount

Length	Format	Description
1 to 12	Numeric	This field contains a secondary transaction amount related to the transaction type indicated. The amount is to be presented with an implied decimal point. The allowable number of significant digits, as well as the positioning of any implied decimal point, are dictated by the currency code designated. EXAMPLE USD \$.01 could be represented as 1, 01, or 001. NOTE In industries where cash back is permitted on purchase transactions, this field could contain a cash back amount. NOTE For all Authorization Reversal transactions, this field must contain the final transaction amount which will or has been settled.

5.12 <A17> Transaction Original Number (TON)

Length	Format	Description
1 to 32	A/N	This is a POS device generated number that can uniquely identify an original authorization transaction within a batch. The TON can be used by the POS Device to perform Void, Adjustment, Tip Adjustment transactions, or to request transaction details. <u>TON Uniqueness:</u> The TON is intended as a backup in cases when the <A86> RRN is not available for message matching. However, if the terminal application is designed to consistently use the TON instead of the RRN then the uniqueness rules for the TON must be more rigorous. TSYS suggests the following application rules: • Format the TON as "DDSSS"; • DD is the current day of the month; SSS is a sequence of numbers from 001-999. • Configure the sequence number for 2- 5 positions so the sequence number can increment through a single day without rolling over. • Allow the sequence to increment and roll-over. Do not program the TON to reset when a batch close occurs. • The application could prompt for the TON before prompting for the RRN.

5.13 <A18> Merchant Category Code (MCC)

Length	Format	Description
4 fixed	Numeric	This four-character numeric field contains a number that identifies the merchant's industry classification. If a merchant participates in the Card Brand MCC program, the MCC used during authorization is pulled from profile data and may not be the same as the value sent in by the Terminal. If the MCC used to populate the authorization request was taken from the profile, the new value is returned to the Point of Interaction (POI).

5.14 <A19> Language Indicator

Length	Format	Description
2 fixed	Numeric	This field contains a two-character numeric language indicator. This value designates the language to be used in formatting the authorization response text message. The language can be selected at the POS. Possible value: 00 - English 01 - Spanish 02 - Portuguese 03 - Reserved 04 - Reserved 05 - Reserved 06 - Reserved 07 - Reserved

5.15 <A20> Forwarding Institution ID (FIID)

Length	Format	Description
6 fixed	Numeric	This field contains a code identifying the institution forwarding the request to the TSYS Acquiring Solutions host. The code can be a BIN or a prearranged institution ID. NOTE In PIN-based debit transactions for cash disbursements or balance inquiries, this field contains the aggregator institution ID. The FIID is required only when the merchant's BIN does not point to the Acquirer Working Key (AWK) as arranged with the TSYS Acquiring Solutions host.

5.16 <A21> POS Entry Data

This group contains subfields that indicate the circumstances in which the cardholder's card number was entered into the POS Device as well as the state of the POS Device and Application at the time of the transaction.

The fields come in two types, static or dynamic.

Static Fields:

Static fields have the same value for every transaction. They do not typically change once the software and hardware are considered together in the environment in which they are deployed.

Example:

In Subfield B6 (Card Data Input Capability), say a terminal/device, its software, and the environment in which it is deployed have the value set at 2 - Magnetic stripe reader capability. For most cardholders, the magnetic stripe will work just fine. However, for a scenario where a cardholder's magnetic stripe is damaged and it will not swipe properly, the merchant has to enter it manually. This does not change the capability of the terminal, software and environment.

Dynamic Values:

Dynamic fields can change based on the transaction scenario.

Example:

In Subfield 1 (Cardholder Present Data), for one transaction, the cardholder may be present (value 0), and for the next, the transaction is initiated by phone (value 3).

The definitions of subfields 1 through 12 and the allowed values are provided in the following subparagraphs.

NOTE Several of the examples in Chapter 6 - Transaction Message Examples indicate that Tags B7,B8, B10, B11 and B12 are optional and can be augmented by the Host System unless received by the POS Software. It should be noted that this data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.

Please see [Appendix D](#) for suggested values for Mastercard transactions.

5.16.1 <A21><B1> Cardholder Present Data

This subfield indicates if the Cardholder is present at the point of service and if not, the reason why. The value is dynamic and is configured based on the Cardholder presence at the time of the transaction.

NOTE Values A, I, and P are deprecated with the October 2020 Release and must not be used for new development after that date. The values will be removed in a future release; continued use may result in unexpected behavior and/or potential compliance fines. The values are applicable in Transaction Status (<A21><B14>).

Length	Format	Description
1 fixed	A/N	Possible value: 0 - Cardholder present 1 - Cardholder not present, unspecified reason 2 - Cardholder not present, mail transaction 3 - Cardholder not present, phone transaction 4 - Cardholder not present, standing (recurring) transaction such as a subscription or Merchant Initiated Transaction (MIT) Mastercard: Card Data Input Mode <A21><B3> value 7 (payment credential stored on file) is required for all recurring transactions. 5 - Cardholder not present, electronic commerce 8 - Cardholder not present, recurrent billing such as a loan or installment A - Reauthorization for full amount (original transaction was token-based with a valid cryptogram; Discover only) Note: Deprecated October 2020 I - Incremental authorization (Discover only) Note: Deprecated October 2020 P - Partial shipment purchase transaction (original transaction was token-based with valid cryptogram) Note: Deprecated October 2020 R - Recurring purchase transaction (original transaction was token-based with valid cryptogram) T - PayButton (Discover only)

5.16.2 <A21><B2> Card Present Data

This subfield indicates if the card is present at the point of service. The value is dynamic and configured based on the card presence at the time of the transaction.

Length	Format	Description
1 fixed	A/N	Possible value: 0 - Card not present 1 - Card present W - Transponder [AX] X - Contactless chip transactions, including Amex Expresspay Z - Digital Wallet [AX]

5.16.3 <A21><B3> Card Data Input Mode

This subfield indicates the method used to capture information from the card. The value is dynamic and configured based on the input mode at the time of the transaction.

Discover Merchant Initiated Transactions (MIT):

- PAN Entry Mode 7 is required for card sales that are processed using stored card account information.
- PAN Entry Mode 7 is optional for MIT, which include recurring payments, installment payments, unscheduled payments.
- PAN Entry Mode 7 must not be assigned to QR Code/Bar Code/Payment Code Transactions, Contactless Card Transactions (including those using a Mobile Payment Device), or Card Sales with Card Account information stored by an Issuer or provisioned in a digital wallet.

Length	Format	Description
1 fixed	A/N	Possible value: 0 - Unspecified, data not available (INVALID - Internal Use Only) 1 - Manual input, no terminal; Voice auth/ARU 2 - Magnetic stripe reader input 3 - Bar code/Payment code 6 - Key entered input 7 - Payment Credential stored on file Note: This value is required for all Mastercard recurring transactions. A - PAN auto-entry via contactless magnetic stripe B - Magnetic stripe reader input, track data captured and passed unaltered C - Online chip F - Offline chip M - PAN auto-entry via contactless Chip (EMV Mode) N - Track data read and sent unaltered, chip capable terminal, chip data could not be read P - Empty candidate list fallback R - PAN entry via electronic commerce, including remote chip S - Electronic commerce, no security, channel encrypted, or SET without cardholder certificate V - Manually entered with keyed CID (AX, JCB Canada) W - Swiped transaction with keyed CID (AX, JCB Canada) X - Reserved for private use Y - Reserved for private use Z - Contactless Interface Change. Identifies when a chip card transaction with a dual-interface card switches from a contactless to a contact chip card transaction. (D)

NOTE AMEX defines ApplePay with value C - Online Chip

5.16.4 <A21><B4> Cardholder Authentication Method

This subfield indicates the method used for verifying the Cardholder's identity. The value is dynamic and configured based on the Cardholder Authentication Method at the time of the transaction.

Length	Format	Description
1 fixed	A/N	Possible value: 0 - Not authenticated 1 - PIN 2 - Electronic signature analysis 5 - Manual signature verification 6 - Other manual verification (such as driver's license number) 9 - Unspecified, data not available (INVALID - Internal Use Only) S - Other systematic verification T - Electronic ticket environment (Amex)

5.16.5 <A21><B5> Authentication Entity

This subfield indicates the component or person who verified Cardholder Identity reported in Cardholder Authentication <A21><B4>. The value is dynamic and configured based on the Entity that performed the Authentication at the time of the transaction.

Length	Format	Description
1 fixed	A/N	Possible value: 0 - Not authenticated 1 - ICC - Offline PIN 2 - Card acceptance device (CAD) 3 - Authorizing agent - Online PIN 4 - Merchant/card acceptor - signature 5 - Other 9 - Unspecified, data not available (INVALID - Internal Use Only)

5.16.6 <A21><B6> Card data input capability

This subfield indicates the highest-level capability of the device and its software used to originate this transaction. The value should be static and configured based on the card data input capabilities in the environment in which it is deployed.

Length	Format	Description
1 fixed	A/N	Possible value: 0 - Unspecified; data not available (INVALID - Internal Use Only) 1 - Manual; no terminal; Voice auth/ARU 2 - Magnetic stripe reader capability only 3 - Bar Code / Payment Code 4 - Optical Character Reader (OCR) capability (MC, AX) 5 - Integrated Circuit Card (ICC) capability and MSR 6 - Key entry only capability; Mastercard recurring and installment transactions A - PAN auto-entry via contactless magnetic stripe B - Magnetic stripe reader and key entry capability C - Magnetic stripe reader, ICC and key entry capability D - Magnetic stripe reader and ICC capability E - ICC entry capability H - ICC reader and contactless capability, Magnetic stripe and manual entry implied M - PAN auto-entry via contactless chip V - Other capability (MC) X - Reserved for private use

5.16.7 <A21><B7> Cardholder authentication capability

This subfield indicates the highest-level capability of the device and its software used to verify the Cardholder's identity at this terminal. The value is static and configured based on the terminal cardholder-authentication capabilities in the environment in which it is deployed.

NOTE Several of the examples in Chapter 6 indicate that this Tag is optional and can be augmented by the Host System. It should be noted that this data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.

Length	Format	Description
1 fixed	A/N	Possible value: 0 - No electronic authentication capability 1 - PIN entry capability 2 - Electronic signature analysis capability 3 - mPOS software-based PIN entry capability 5 - Electronic authentication capability is inoperative 6 - Other 9 - Unspecified, data not available (INVALID - Internal Use Only)

5.16.8 <A21><B8> Card capture capability

This subfield indicates if the terminal is capable of pulling the card into the device while the transaction is taking place. The value is static and configured based on the terminal card-capture capabilities in the environment in which it is deployed.

NOTE Several of the examples in Chapter 6 indicate that this Tag is optional and can be augmented by the Host System. It should be noted that this data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.

Length	Format	Description
1 fixed	A/N	Possible value: 0 - No capture capability 1 - Card capture capability 9 - Unknown; data not available

5.16.9 <A21><B9> Operating environment

This subfield indicates the terminal's location and indicates whether it is attended by the card acceptor. The value is static and configured based on the Payment Application's operating capabilities in the environment in which it is deployed.

Length	Format	Description
1 fixed	A/N	Possible value: 0 - No terminal used; Voice auth/ARU; Mastercard recurring/installment transactions 1 - On card acceptor premises; attended 2 - On card acceptor premises; unattended 3 - Off card acceptor premises; attended 4 - Off card acceptor premises; unattended 5 - On cardholder premises; unattended 6 - Off cardholder premises; unattended 9 - Unspecified, data not available (INVALID - Internal Use Only) M - Off card acceptor premises; merchant mobile POS environment, including mPOS P - On card acceptor premises; merchant mobile POS environment, including mPOS Q - Off card acceptor premises; cardholder mobile environment, including home PC, mobile phone, PDA R - On card acceptor premises; cardholder mobile environment, including home PC, mobile phone, PDA S - Electronic delivery of product (AX) T - Physical delivery of product (AX)

5.16.10 <A21><B10> Card data output capability

This subfield indicates the ability of the terminal and software to update or change the card. The value should be static and configured based on the terminal and software capabilities in the environment in which it is deployed.

NOTE Several of the examples in Chapter 6 indicate that this Tag is optional and can be augmented by the Host System. It should be noted that this data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.

Length	Format	Description
1 fixed	A/N	Possible value: 0 - Unspecified; data not available (INVALID - Internal Use Only) 1 - None 2 - Magnetic strip write 3 - ICC S - Other

5.16.11 <A21><B11> Terminal data output capability

This subfield indicates the ability of the terminal and/or software to print and/or display messages. This value is static and configured based on the terminal and/or software capabilities in the environment in which it is deployed.

NOTE Several of the examples in Chapter 6 indicate that this Tag is optional and can be augmented by the Host System. It should be noted that this data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.

Length	Format	Description
1 fixed	A/N	Possible value: 0 - Unspecified, data not available (INVALID - Internal Use Only) 1 - None 2 - Printing capability only 3 - Display capability only 4 - Printing and display capability

5.16.12 <A21><B12> PIN capture capability

This subfield indicates the length of the PIN that the terminal is capable of capturing. The value is static and configured based on the capability of the terminal in the environment in which it is deployed.

NOTE Several of the examples in Chapter 6 indicate that this Tag is optional and can be augmented by the Host System. It should be noted that this data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.

Length	Format	Description
1 fixed	A/N	Possible value: 0 - No PIN capture capability 1 - Unspecified, data not available (INVALID - Internal Use Only) 2 - Reserved 3 - Reserved 4 - PIN capture capability four characters maximum 5 - PIN capture capability five characters maximum 6 - PIN capture capability six characters maximum 7 - PIN capture capability seven characters maximum 8 - PIN capture capability eight characters maximum 9 - PIN capture capability nine characters maximum A - PIN capture capability ten characters maximum B - PIN capture capability eleven characters maximum C - PIN capture capability twelve characters maximum

5.16.13 <A21><B13> Chip condition code

This one-character field provides information about fallback transactions, which are magnetic stripe read transactions using chip capable cards at chip capable POS devices. It is used when the chip card's magnetic stripe is read instead of the chip. When Service Code value (from Track 1 or Track2) is 2 or 6, this indicates the card is chip capable.

Length	Format	Description
1 fixed	A/N	Possible value: 0 - Use this value when the service code does not begin with 2 or 6 (indicates this is not a chip capable card). 1 - Use this value when the transaction was initiated from a magnetic swipe with a service code that begins with 2 or 6 (indicating chip capable card) and the last read from the chip capable terminal was a successful chip read or was not a chip transaction. 2 - Use this value when the transaction was initiated at a chip capable terminal from a magnetic swipe that contains a service code beginning with 2 or 6 (indicating chip capable card) and the previous transaction initiated by that terminal was an unsuccessful chip read.

5.16.14 <A21><B14> Transaction status

Length	Format	Description
1 fixed	A/N	This is a 1 - character code indicating the purpose or status of the request. The field will be accepted for any card brand, but will be used in Mastercard or Discover transactions only, as indicated below. This field is not required for a \$0.00 transaction. This field is required for Transit merchants who wish to identify Transit Aggregated Transactions. Mastercard and Discover Purchase Return Authorization Requests: Value must be 0 Discover MIT: Usage of values A, D, E, I, N, P, R, S, and U is mandatory for Merchant-Initiated Transactions (MIT) using a Card Brand Token and for MIT transactions using a PAN. It should be noted that if this value is not sent in by the POI, it will be derived internally by TSYS software using other elements of the transaction. Note that this can result in inaccuracies based on the transaction values and the nature of such derivation.

Value	Description	MC	D
0	Normal request (original presentment)	X	X
2	SecureCode Phone Order	X	
3	ATM Installment Inquiry	X	
4	Preauthorized request	X	X
5	Time Based Payment Authorization Request-Brazil domestic transactions	X	
6	ATC Update	X	
8	Account Status Inquiry Service (ASI)	X	
9	Tokenization Request/Notification	X	
A	Re-authorize for Full Amount		X
D	Delayed Card Sale		X
E	Resubmission of Card Sale		X
G	Transit Aggregated Transaction		X
I	Incremental Authorization		X
N	No-Show Charge		X
P	Partial Shipment		X
R	Recurring Payment		X

5.17 <A22> Market Specific Data Identifier

Length	Format	Description
1 fixed	A/N and space	The Market Specific Data Identifier is used to indicate which industry data is being submitted. When this data is supplied in an incremental transaction, it should use the same value that was requested in the original authorization transaction. This field is specific to Visa. Possible value: A - Auto rental (for future use) B - Bill payment transaction E - eCommerce Transaction Aggregation H - Hotel/Lodging (for future use) J - B2B Invoice payments N - Failed market specific data (response only) M - Healthcare T - Transit (for future use)

5.18 <A23> Track 1

Length	Format	Description
1 to 76	A/N and special characters	The Track 1 data read from the cardholder's card is checked for parity and LRC errors and then converted from the six-bit characters encoded on the card to seven bit characters as defined in ANSI X3.4. The character set definitions are provided in Appendix E for reference. As part of the conversion, the POS device must remove the framing characters (start sentinel, end sentinel, and LRC characters). The separators must be converted to either an ASCII “^” (HEX 5E) or ASCII <US> (HEX 1F) characters. NOTE The entire unaltered track (excluding framing characters) must be provided in the original authorization or pre-authorization request message or else an error may occur. The track must not be included in responses, completions, or reversal transactions.

5.19 <A24> Track 2

Length	Format	Description
1 to 37	A/N and special characters	The Track 2 data read from the cardholder's card is checked for parity and LRC errors and then converted from the four-bit characters encoded on the card to seven bit characters as defined in ANSI X3.4. The character set definitions are provided in Appendix E for reference. As part of the conversion, the POS device must remove the start sentinel, end sentinel, and LRC characters. The separators must be converted to either an ASCII "=" (HEX 3D) or ASCII "D" (HEX 44) characters. NOTE The entire unaltered track (excluding framing characters) must be provided in the original authorization or pre-authorization request message or else an error may occur. The track must not be included in responses, completions, or reversal transactions.

5.20 <A25> Manually Entered Card Number

Length	Format	Description
1 to 19	A/N	The customer credit card data can be key entered when the card magnetic stripe cannot be read, if the card is not physically present, or when a card reader is not available. NOTE PIN-based debit transactions cannot be manually entered. A manual entry is allowed for Electronic Benefits Transfer.

5.21 <A26> Manually Entered Expiry Date

Length	Format	Description
4 fixed	Numeric	This four-character numeric field contains the card expiration date in the form MMYY (month-month-year-year). EBT/Electronic Voucher transactions can utilize this field in MMYY format. Mastercard Dynamic Expiration Date: Include the Dynamic Expiration Date in the Customer Data Field for a tokenized e-commerce purchase transaction. Please refer to Tag A124 Card Brand Tokenization for more information.

5.22 <A28> PIN Block

Length	Format	Description
16 fixed	0-9, A-F (ASCII representation of HEX)	This field contains the encrypted PIN block as transmitted from the PIN Pad.

5.23 <A29> Security Control Info

Length	Format	Description
16	0-9, A-F (ASCII representation of HEX)	This field contains either DUKPT security information or ANSI TDES security information.

DUKPT Security Information

When a PIN is entered on a PIN Pad supporting DUKPT (Derived Unique Key Per Transaction) methodology, the POS device forwards the following characters as provided by the PIN pad. TSYS Acquiring Solutions certified PIN Pads use the first 16 characters. Other PIN Pads may use the additional characters.

kkkkkkk	ssssssssss[ssssssssss]
Key Set Identifier (KSID) A unique, six-character numeric Bank Identification Number. For TSYS Acquiring Solutions-certified PIN Pads the KSID characters can have the following values: Character one can either be 4 or 5. Character two: A - TSYS Acquiring Solutions B - Third party injection C - Travel Money D - TSYS Acquiring Solutions E - TSYS Acquiring Solutions F - TSYS Acquiring Solutions Character three can be 0 - 9 or A - F. Character four can be 0 - 9 or A - F. Character five can be 0 - 9 or A - F. Character six can be 0 - 9 or A - F.	Expanded TRSM ID (PIN Pad Serial Number) & Expanded Transaction Counter TSYS Acquiring Solutions PIN Pads use a 10 character TRSM ID. The PIN Pad serial number is stored as five hex digits minus one bit for a total of 19 bits of data. The transaction counter is stored as five hex digits plus one bit for a total of 21 bits of data. These two fields concatenated together contain 40 bits. ANSI TDES Security Information is fixed-length with five subfields. • Positions 1 - 2, Security Format Code: This code defines the security technique used. • Positions 3 - 4, PIN Encryption Algorithm Identifier: This code defines the encryption technique used. • Positions 5 - 6, PIN Block Format Code: This code defines the PIN block format. • Positions 7 - 8, Zone Key Index: This value indicates which key was used to encrypt the PIN. In Dynamic Key Exchange (DKE) messages, this subfield is used to indicate which key is being changed. • Positions 9 - 10, PIN Data Type: This subfield is zero-filled. • Positions 11 - 16, Reserved: This subfield is zero-filled. ANSI TDES Security Information must contain the following subfield values. • Positions 1 - 2 must be 20 • Positions 3 - 4 are not edited • Positions 5 - 6 must be 01, 02, or 03 • Positions 7 - 8 must be 01 or 02 • Positions 9 - 16 must be zeros in outgoing requests

5.24 <A30> Address Verification Data

This group contains subfields for the cardholder's billing and shipping address information.

Address Verification Data must be included for Amex account verification/balance inquiry transactions.

5.24.1 <A30><B1> Cardholder Billing Postal Code

Length	Format	Description
5 to 9	A/N	For addresses in the USA, this is a numeric 5+4 ZIP, or a five-digit ZIP. For international addresses, the postal code may vary in length and contain alpha characters. NOTE Alpha characters must be upper case.

5.24.2 <A30><B2> Cardholder Billing Address

Length	Format	Description
1 to 20	A/N and spaces	This is up to the first 20 characters of the cardholder billing address (including the unit, apartment, flat, or suite number).

5.24.3 <A30><B3> Cardholder Billing City

Length	Format	Description
1 to 13	Alpha	This up to the first 13 characters of the cardholder billing city. NOTE Amex usage only.

5.24.4 <A30><B4> Cardholder Billing State/Province

Length	Format	Description
2 fixed	Alpha	This is the cardholder billing 2 character state/province code. NOTE Amex usage only.

5.24.5 <A30><B5> Cardholder Billing Country

Length	Format	Description
3 fixed	A/N	This is the cardholder billing 3 character ISO alpha country code. NOTE Amex usage only.

5.24.6 <A30><B6> Cardholder Billing First Name

For Discover, this field represents the 'Cardholder First Name'. For Amex, this field represents the 'Cardholder Billing First Name.'

Length	Format	Description
1 to 35	Alpha	This field contains up to the first 35 characters of the cardholder's billing first name. For Amex, the field should be truncated to the first 15 characters.

5.24.7 <A30><B7> Cardholder Billing Last Name

For Discover, this field represents the 'Cardholder Last Name'. For Amex, this field represents the 'Cardholder Billing Last Name.'

Length	Format	Description
1 to 35	Alpha	This field contains up to the first 35 characters of the cardholder billing last name. For Amex, the field should be truncated to the first 30 characters.

5.24.8 <A30><B8> Cardholder Billing Phone Number

Length	Format	Description
7 to 10	Numeric	This is the cardholder billing phone number. If phone number is shorter than 7 digits, left zero fill to 7 digits. If phone number is longer than 10 digits, send rightmost 10 digits. NOTE Amex usage only.

5.24.9 <A30><B9> Ship to Postal Code

Length	Format	Description
5 to 9	A/N	For addresses in the USA, this is a numeric 5+4 ZIP; or a five-digit ZIP. For international addresses, the postal code may vary in length and contain alpha characters. Alpha characters must be upper case. NOTE Amex usage only.

5.24.10 <A30><B10> Ship to Address

Length	Format	Description
1 to 20	A/N and spaces	This is up to the first 20 characters of the cardholder ship-to address (including the unit, apartment, flat, or suite number). NOTE Amex usage only.

5.24.11 <A30><B11> Ship to City

Length	Format	Description
1 to 13	Alpha	This up to the first 13 characters of the cardholder ship-to city. NOTE Amex usage only.

5.24.12 <A30><B12> Ship to State/Province

Length	Format	Description
2 fixed	Alpha	This is the cardholder ship-to 2 character state/province code. NOTE Amex usage only.

5.24.13 <A30><B13> Ship to Country

Length	Format	Description
3 fixed	A/N	This is the cardholder ship-to 3 character ISO alpha country code. NOTE Amex usage only.

Position	Code	Description
1	0	Verification code is intentionally not provided
	1	Verification code is present
	2	Verification code is present but illegible
	9	Cardholder states that no verification code is present on the card
2	0	Only the normal Response Code should be returned
	1	Response code and verification code result code should be returned
3-6		Verification code as printed on the card, right justify/space fill entry NOTE If position 1 is 0, 2 or 9, positions 3-6 should be space filled. May contain a 3-digit Dynamic Token Verification Value [DTVV] for tokenized e-commerce transactions that originate from Mastercard or Visa Checkout. Please refer to Tag <A124> Card Brand Tokenization for more information.

NOTES

- If position 1 = 1, then position 2 should be set to 0 or 1.
- If position 1 = 0, 2 or 9, then position 2 should be set to 0.

5.26 <A32> MOTO/e-Commerce Indicator

Length	Format	Description
1 fixed	A/N	This field contains a one-character transaction indicator identifying the type of direct marketing transaction being authorized. Note that if this field contains a value other than 5, 6, or 7, the transaction will not qualify for CAVV processing for electronic commerce. Possible value: SPACE - not a mail/telephone order transaction 1 - One time occurrence of a mail/telephone order transaction 2 - Recurring mail/telephone order transaction 3 - Installment payment of a mail/telephone order transaction 4 - Unknown classification/ Other mail order 5 - Secure electronic commerce transaction This is used for fully authenticated CAVV Verification transactions using the 3-D Secure Protocol for Visa Secure, Mastercard SecureCode, Amex Safekey, and Discover ProtectBuy. 6 - This is used for non-authenticated security transactions at a 3-D Secure-capable merchant. The merchant attempted to authenticate the cardholder using 3-D Secure. 7 - Non-authenticated Security Transaction 8 - Non-secure transaction A - In-App Authentication (Discover, Mastercard and AMEX) R - First recurring SecureCode phone order transaction (Mastercard Only) T - Single non-recurring SecureCode Phone Order transaction (Mastercard Only)

5.27 <A33> Transaction ID (XID)

Tag A33 has two usages. This is usage 1, which is deprecated effective with the October 2019 release and must not be used for new development after that date. It will be retired October 2020; continued use may result in unexpected behavior and/or potential compliance fines.

Length	Format	Description
40 fixed	0-9, A-f (ASCII representation of HEX)	This tag contains a unique Secure Electronic Commerce Number - the Transaction ID (XID) - generated by the merchant server to identify the transaction. The XID is used in conjunction with Tag A34. Tag A33 is present in a CAVV Verification service credit authorization or debit financial request. The tag is not required in 0400 reversals. Visa and Discover CAVV Verification Service: This tag is present in full authentication requests according to Tag A34, usage 3-D Secure CAVV; the XID is sent in Tag A33 and the CAVV is sent in Tag A34. This field is not required if Tag A34, 3-D Secure CAVV, Revised is being used. CPS/Electronic Commerce: This field is required in Preferred fee program requests that require 3-D Secure encryption. The acquirer transfers the value to the request as generated by the issuer's ACS. This field is not used in the Basic fee program. American Express: SafeKey: Tag A33 is required in authorization requests for SafeKey processing with e-Commerce transactions only if the Amex SafeKey Transaction ID (XID) is present. The XID is determined by the Merchant during the Amex SafeKey payment authentication. Tokens: For Tokens, this field contains the Token Block B Data and is required only if the cryptographic value is greater than 20 bytes. Tag A34 must be present when Tag A33 is present in the request.

5.28 <A33> Authentication Data

Tag A33 has two usages. This is usage 2, which is effective with the October 2019 release. It is used for 3D Secure, eCommerce, Masterpass, and Digital Secure Remote Payment transactions. Please refer to the individual tags for further information.

5.28.1 <A33><B1> CAVV

Length	Format	Description
40 fixed	0-9, A-f (ASCII representation of HEX)	This is a multi-use field for Secure Electronic Commerce transactions. It contains encrypted data for verification purposes depending on the service involved. The following usages apply: • 3-D Secure CAVV • 3-D Secure CAVV, Revised • American Express SafeKey • American Express Token See below for details

5.28.2 3-D Secure CAVV

This field usage is applicable for Visa 3-D Secure and contains the Cardholder Authentication Verification Value (CAVV) for 3-D Secure transactions. It applies to a CAVV Verification Service for a credit authorization request that is for full authentication; the [<A33><B2> Transaction ID \(XID\)](#) must also be included. It is not returned in credit responses. It is not present in reversal requests. The CAVV is a cryptographic value calculated by the issuer's Access Control Server (ACS) using the issuer's encryption key and related elements according to Protocol 1.0.1. The CAVV value is unique to the cardholder and to the transaction that was authenticated. The endpoint transfers the ACS data to this field when preparing the authorization request. Visa or the issuer verifies the CAVV to ensure that the issuer's ACS authenticated the cardholder for the transaction and that its contents have not been altered.

See [3-D Secure CAVV, Revised](#) for information on sending the CAVV in compressed format with ATN, which replaces the need for the XID.

Refer to documentation from your 3-D Secure Provider for details on the below subfields.

Positions:

1	2	3	4	5
3-D Secure Authentication Results Code	Second Factor Authentication Code	CAVV Key Indicator	CAVV Value	CAVV Unpredictable Number
Byte 1	Byte 2	Byte 3	Bytes 4-5	Bytes 6-7

6.1	6.2	6.3
Card Sequence Number	Card Verification Results	Reserved
Byte 8-9	Byte 10-13	Byte 14-20

5.28.3 3-D Secure CAVV, Revised

This field usage is applicable for Visa 3-D Secure and Discover ProtectBuy and applies to a Cardholder Authentication Verification Value (CAVV) Verification Service credit authorization request being submitted as an attempt or as a full authentication where the Authentication Tracking Number (ATN) is being used in place of the Transaction Identifier (XID). The field is not present in subsequent reversals. It is not returned in response messages.

CAVV Revised, can contain the Token Authentication Verification Value (TAVV).

DTVV: For eligible 0100 authorization and 0200 full-financial requests, this field contains a static value generated from the DTVV cryptogram and is sent to the issuer.

The field contains an ATN and the CAVV in compressed format for CAVV Verification Service transactions. The CAVV is a cryptographic value calculated by the issuer's Access Control Server (ACS) using the issuer's encryption key and related elements. The CAVV value is unique to the cardholder and to the transaction that was authenticated. The ATN replaces the need for the [≤A33><B2> Transaction ID \(XID\)](#).

See [3-D Secure CAVV](#) for sending only the CAVV in uncompressed format.

This field can be used for different versions of the 3-D Secure CAVV in revised format. Field layout and values allowed in any position of this usage can change according to these revised versions. Refer to documentation from your 3-D Secure Provider for details on the below subfields.

Positions:

1	2	3	4	5
3-D Secure Authentication Results Code	Second Factor Authentication Code or Authentication Method	CAVV Key Indicator	CAVV Value	Unpredictable Number
Byte 1	Byte 2	Byte 3	Bytes 4-5	Bytes 6-7
6	7	8		
Authentication Tracking Number	(CAVV) Version and Authentication Action or 3DS Protocol Version Number	IP Address in Hex Format or Reserved		
Byte 8-15	Byte 16	Byte 17-20		

5.28.4 American Express SafeKey

American Express SafeKey is an industry-standard Authentication method that provides greater security by authenticating the Cardmember during an online purchase and protecting payment card information as it is transmitted via the Internet.

American Express does not offer a breakdown of the cryptogram detail.

The SafeKey Cardholder Verification Value (CAVV) is a cryptographic value derived by the Issuer during the Amex SafeKey payment authentication that can provide evidence of the results of payment authentication during an online purchase. In this case, a valid value should be sent in [<A32> MOTO/e-Commerce Indicator](#). If the XID is present, it must be included in the XID Field in [<A33><B2> Transaction ID \(XID\)](#).

5.28.5 American Express Token

The **American Express Payment Token** transaction processing solution is based on an industry aligned and interoperable tokenization system that offers increased protection against fraud by using a Payment Token. A Payment Token will be used in place of sensitive Cardmember data such as Primary Account Number (PAN) to originate payment transactions.

American Express does not offer a breakdown of the cryptogram detail.

Token Data Block A, which contains bytes 1-20 of the cryptographic value is required. Token Data Block B <A33><B2> Transaction ID (XID) is optionally present when Token Data Block A is used.

5.28.6 <A33><B2> Transaction ID (XID)

Length	Format	Description
40 fixed	0-9, A-f (ASCII representation of HEX)	This tag contains a unique Secure Electronic Commerce Number - the Transaction ID (XID) - generated by the merchant server to identify the transaction. The XID is used in conjunction with Tag A33.B1. XID is present in a CAVV Verification service credit authorization or debit financial request. The tag is not required in reversals. Visa and Discover CAVV Verification Service: This tag is present in full authentication requests according to Tag B1, usage 3-D Secure CAVV. This tag is not required if Tag B1, 3-D Secure CAVV, Revised is being used. CPS/Electronic Commerce: This field is required in Preferred fee program requests that require 3-D Secure encryption. The acquirer transfers the value to the request as generated by the issuer's ACS. This field is not used in the Basic fee program. American Express: SafeKey: XID is required in authorization requests for SafeKey processing with e-Commerce transactions only if the Amex SafeKey Transaction ID (XID) is present. The XID is determined by the Merchant during the Amex SafeKey payment authentication. Tokens: For Tokens, this field contains the Token Block B Data and is required only if the cryptographic value is greater than 20 bytes. Tag B1 must be present when XID is present in the request.

5.28.7 <A33><B3> Security Level Indicator (SLI)

This Mastercard field contains the electronic commerce indicators representing the security level and cardholder authentication associated with the transaction. Valid values are listed below.

In general, the values in this tag apply to the type of authentication data in the Mastercard UCAF Data field <A33><B4>. If Mastercard UCAF Data is not present, but the <A33><B5> Digital Payment Cryptogram is, the SLI must be set appropriately for a DSRP transaction.

Table 5.1 SLI valid scenarios

When the merchant submits:		MC will validate/modify SLI to:
<A33><B4> UCAF Data	<A33><B5> Digital Payment Cryptogram	<A33><B3> Security Level Indicator (SLI)
DSRP Cryptogram	N/A	DSRP Cryptogram
N/A	DSRP Cryptogram	DSRP Cryptogram
AAV (fully authenticated or attempt)	DSRP Cryptogram	AAV

When the merchant submits:		MC will validate/modify SLI to:
AAV (fully authenticated or attempt)	N/A	AAV

Mastercard's processing rules allow merchants to gain a liability shift and interchange benefit if they submit their electronic commerce transactions for cardholder authentication. If merchants provide a value in this tag that indicates the transaction is SecureCode but do not pass the appropriate authentication detail in Tag A33.B4 - UCAF Authentication Data, demonstrating they actually authenticated or attempted to authenticate the cardholder, Mastercard will downgrade the Authorization Request transaction and forward to the issuer. The correct values will be returned in the Tag [≤A33><B3>](#) response message, whether downgraded or not.

The SecureCode functionality is not applicable for Mastercard Full Financial (Debit) Transactions.

Request Value:

This field must be present in all Auth Request messages for electronic commerce transactions. The field is conditional in the following messages:

- All other types of Authorization Requests
- Authorization Advice
- Reversal Request

Response Value:

The Electronic Commerce Security Level Indicators contained in the response for all Mastercard e-Commerce Credit transactions, contains the electronic commerce indicators representing the security level and cardholder authentication associated with the transaction.

Mastercard will also reject clearing/settlement records if the indicators do not match the authorization indicators. The response field contains the correct values needed for settlement, whether the transaction was downgraded or not.

This field is not applicable for Mastercard Full Financial Debit transactions.

Table 5.2 Security Level Indicators - Request and Response

Position	Length	Format	Description
1 and 2	2 fixed	Numeric	Security Protocol/Cardholder Authentication Valid values: 21 - Channel encryption; cardholder certificate not used (preferred value for Mastercard SecureCode, Identity Check, and all eCommerce) 22 - Masterpass-generated transaction 24 - Digital Secure Remote Payment (DSRP) with UCAF Data 91 - No security protocol; cardholder certificate not used

Position	Length	Format	Description
			7 - Partial shipment or recurring payment (<A40> Authentication Data only required) for Identity Check); Liability will depend on the original UCAF values provided and matching with the initial transaction 8-9 - Reserved for future use

5.28.8 <A33><B4> UCAF Data

Length	Format	Description
up to 32	A/N/S	Usage 1: UCAF Authentication Data This Mastercard only variable length issuer generated security field is up to 32 bytes in length. When sending data in this field, the Security Level Indicator <A33><B3> must be set to the proper value. It contains the encoded Mastercard SecureCode issuer or cardholder-generated authentication data (collected by the merchant) resulting from all SecureCode fully authenticated or attempts transactions, and any data associated with the 3-D Secure Electronic Commerce Verification Service. Usage 2: Accountholder Authentication Value (AAV) AAV is part of Mastercard's SecureCode-specific implementation of UCAF related to issuer authentication. This is a security method designed to authenticate cardholders when they pay online. The values shown below are the leading one or two characters of the value. <u>3D Secure 1.0EMV 3D Secure</u> j kG, kH, kJ, kK h kL, kM

5.28.9 <A33><B5> Digital Payment Cryptogram

Length	Format	Description
28 fixed	A/N/S	Digital Payment Cryptogram This Mastercard only field is used to send a Digital Secure Remote Payment (DSRP) cryptogram for DSRP transactions submitted as electronic commerce. When sending data in this field, the associated electronic commerce <A33><B3> Security Level Indicator must be sent.

5.28.10 <A33><B6> Program Protocol

Length	Format	Description
1 fixed	Numeric	Program Protocol Program Protocol is a one-byte alphanumeric field and must be provided by the merchant if they are participating in Mastercard Identity Check and have previously authenticated using the Mastercard Identity Check or SecureCode program. It identifies the type of 3D-Secure program. Valid Values: 1 - 3D Secure Version 1.0 (3DS 1.0) 2 - EMV 3D Secure (3DS 2.0)

5.28.11 <A33><B7> Directory Server Transaction ID

Length	Format	Description
36 fixed	A/N/S	Directory Server Transaction ID The Directory Server Transaction ID is a 36-byte alphanumeric value generated by the EMV 3DS Mastercard Directory Server during the authentication transaction and passed back to the merchant with the authentication results. This field allows the merchant to pass the Directory Server Transaction ID during authorization to link authentication and authorization data.

5.28.12

5.29 <A34> Cardholder Authentication for Secure e-Commerce Transactions

Tag A34 is deprecated effective with the October 2019 release and must not be used for new development after that date. It will be retired October 2020; continued use may result in unexpected behavior and/or potential compliance fines.

Length	Format	Description
40 fixed	0-9, A-f (ASCII representation of HEX)	This is a multi-use field for Secure Electronic Commerce transactions. It contains encrypted data for verification purposes depending on the service involved. The following usages apply: • 3-D Secure CAVV • 3-D Secure CAVV, Revised • American Express SafeKey • American Express Token See below for details.

5.29.1 3-D Secure CAVV

This field usage is applicable for Visa 3-D Secure and contains the Cardholder Authentication Verification Value (CAVV) for 3-D Secure transactions. It applies to a CAVV Verification Service for a credit authorization request that is for full authentication; the [<A33> Transaction ID \(XID\)](#) must also be included. It is not returned in credit responses. It is not present in reversal requests. The CAVV is a cryptographic value calculated by the issuer's Access Control Server (ACS) using the issuer's encryption key and related elements according to Protocol 1.0.1. The CAVV value is unique to the cardholder and to the transaction that was authenticated. The endpoint transfers the ACS data to this field when preparing the request. Visa or the issuer verifies the CAVV to ensure that the issuer's ACS authenticated the cardholder for the transaction and that its contents have not been altered.

See [3-D Secure CAVV, Revised](#) for information on sending the CAVV in compressed format with ATN, which replaces the need for the XID.

Positions:

1	2	3	4	5
3-D Secure Authentication Results Code	Second Factor Authentication Code	CAVV Key Indicator	CAVV Value	CAVV Unpredictable Number
Byte 1	Byte 2	Byte 3	Bytes 4-5	Bytes 6-7

See description below.

6.1	6.2	6.3
Card Sequence Number	Card Verification Results	Reserved
Byte 8-9	Byte 10-13	Byte 14-20

Position 1, 3-D Secure Authentication Results Code: This 1-byte/2-BCD value is a 1-digit code indicating the result of the issuer's ACS authentication decision. A leading zero is required to pad the first unused half-byte of the 3D Authentication Results Code. The valid values are in [Table 5.3](#).

Table 5.3 CAVV, Position 1 values

Code	Description
0	Authentication successful (status Y)

Position 2, Second Factor Authentication Code: This 1-byte/2-BCD value is a 2-digit code, determined by the issuer's ACS based on the type of additional authentication performed. This value may indicate when a VSDC card is used. This value is determined by the second factor authentication. The valid values are in [Table 5.4](#).

Table 5.4 CAVV, Position 2 values

Code	Description
00	Not present
11	VSDC card used, cryptogram failed
12	VSCD card used, cryptogram passed

Position 3, CAVV Key Indicator: This 1-byte/1-BCD value is a 1-digit code indicating the CAVV key set used to calculate the CAVV value. A leading zero is required to pad the first unused half-byte of the CAVV Key Indicator. This value is determined by the V.I.P. key ID. The valid values are in [Table 5.5](#).

Table 5.5 CAVV, Position 3 values

Code	Description
01	CAVV key set 1
02	CAVV key set 2

Position 4, CAVV: This 2-byte/3-BCD value is a 3-digit code generated by the issuer's ACS that may be used by the issuer to validate the authentication response during authorization. A leading zero is required in byte 4 to pad the first unused half-byte of the CAVV, for example, **0456**. This value is determined by the ACS and the keys loaded in V.I.P.

Position 5, CAVV Unpredictable Number: This 2-byte/4-BCD value is a 4-digit code used by the issuer's ACS to generate the CAVV.

Position 6.1 and Position 6.2: When a 3D-Secure transaction involves another method of authentication, such as a VSDC card, the positions 6.1 and 6.2 are formatted as shown below. Otherwise, the rest of the field (byte 8-20) is filled with binary zeroes. If the first digit of the Second Factor Authentication Code is **1**, it indicates that a VSDC card was used and position 6 will contain the following VSDC authentication data.

Position 6.1, Card Sequence Number: This 2-byte/3-BCD value is a 3-digit code identifying the VSDC card's sequence number that distinguishes it from other cards having the same primary account number. A leading zero in byte 8 is required to pad the first unused half-byte of the Card Sequence Number, for example, **0123**.

When the number of digits is less than 3 digits, zero-fill byte 8 and pad the first unused half-byte of byte 9 with a zero, for example, **0002**. This value is determined by the second factor authentication.

Positions 6.2, Card Verification Results: This position is 4 bytes (binary). It contains a series of card-recorded offline and online processing indicators. This value is determined by the chip terminal.

Positions 6.3, Reserved: Not used. For VSDC-bytes 14-20 are zero-filled. [Table 5.6](#) is an example of field CAVV with 3-D Secure CAVV data.

Table 5.6 CAVV Example with 3-D Secure CAVV Data

Field	Value	Meaning
3-D Secure Authentication Results Code	00	Authentication successful
Second Factor Authentication Code	00	Non-VSDC card used
CAVV Output	0114	CAVV
CAVV Unpredictable Number	7993	
Card Sequence Number	0000	
Card Verification Results (CVR)	00000000	
Zero-fill	0000000000000000	

5.29.2 3-D Secure CAVV, Revised

This field usage is applicable for Visa 3-Domain (3-D) Secure and Discover ProtectBuy and applies to a Cardholder Authentication Verification Value (CAVV) Verification Service credit authorization request being submitted as an attempt or as a full authentication where the Authentication Tracking Number (ATN) is being used in place of the Transaction Identifier (XID). The field is not present in subsequent reversals. It is not returned in response messages.

Visa Token Service: This field is required with a value of **3** in position 7, byte 16 - version and authentication in e-commerce messages containing token data.

Visa Token Convert Service: This field is required with a value of **3** in position 7, byte 16 - version and authentication, for application-based e-commerce and NFC Visa payWave messages using the Visa Token Convert Service.

CAVV Revised, can contain the Token Authentication Verification Value (TAVV).

The field contains an ATN and the CAVV in compressed format for CAVV Verification Service transactions. The CAVV is a cryptographic value calculated by the issuer's Access Control Server (ACS) using the issuer's encryption key and related elements. The CAVV value is unique to the cardholder and to the transaction that was authenticated. The ATN replaces the need for the [≤A33> Transaction ID \(XID\)](#).

See [3-D Secure CAVV](#) for sending only the CAVV in uncompressed format.

Positions:

1	2	3	4	5
3-D Secure Authentication Results Code	Second Factor Authentication Code	CAVV Key Indicator	CAVV Value	Unpredictable Number
Byte 1	Byte 2	Byte 3	Bytes 4-5	Bytes 6-7

See description below.

6	7	8
Authentication Tracking Number	Version and Authentication Action	IP Address in Hex Format
Byte 8-15	Byte 16	Byte 17-20

Position 1, 3-D Secure Authentication Results Code: This 1-byte/2-BCD value is a 1-digit code indicating the result of the issuer's ACS authentication decision. A leading zero is required to pad the first unused half-byte of the Visa Secure Authentication Results Code. The value is determined from the Payer Authentication Request's transaction status. [Table 5.7](#) contains position 1 valid values.

First BCD digit = **0**.

Table 5.7 CAVV, Revised Position 1, 3-D CAVV Secure Authentication Results Code

Status	CAVV ACS Result	CAVV ACS Result Definition	Associated MOTO/ECI
Y (Success)	0	Authentication successful (status Y)	5
U (Unable)	5	Authentication could not be performed (status U)	7
N (Failed)	9	Authentication failed (status N)	n/a issuer unable to authenticate cardholder; merchants are not permitted to submit these transactions for authorization
A (Attempt)	7	Acquirer attempt (status A); proof of authentication attempt generated for non-participating issuer or cardholder	6
N (Attempt - issuer)	8	Acquirer attempt, issuer ACS not available (status A); proof of authentication attempt generated for participating issuer with server unavailable (Visa Proof of Attempts STIP)	6

NOTE: Note: Visa sets the CAVV Results Code to 0 when position 1 is 5 or 9.

Position 2, Second Factor Authentication Code: This 1-byte/2-BCD value is a 2-digit code, determined by the issuer's ACS based on the type of additional authentication performed. This value may indicate when a Visa Smart Debit/Smart Credit (VSDC) card is used. This value is determined by the second factor authentication. [Table 5.8](#) contains position 2 valid values.

Table 5.8 CAVV, Revised Position 2 values

Code	Description
00	3DS 1.0.2, or prior, All authentication methods
01	3DS 2.0 Challenge flow using static passcode
02	3DS 2.0 Challenge flow using One Time Passcode (OTP) via SMS method
03	3DS 2.0 Challenge flow using OTP via key fob or card reader method
04	3DS 2.0 Challenge flow using OTP via App method
05	3DS 2.0 Challenge flow using OTP via any other method
06	3DS 2.0 Challenge flow using Knowledge Based Authentication (KBA) method
07	3DS 2.0 Challenge flow using Out of Band (OOB) authentication with biometric method

Code	Description
08	3DS 2.0 Challenge flow using OOB authentication with App login method
09	3DS 2.0 Challenge flow using OOB authentication with any other method
10	3DS 2.0 Challenge flow using any other authentication method
11	Deprecated ¹
12	Deprecated ²
97	3DS 2.0 Frictionless flow, risk-based authentication (RBA) review
98	3DS 2.0 Attempts server responding
99	3DS 2.0 Frictionless flow
Unrecognized value	3DS unrecognized authentication method

1. This value is deprecated, formerly known as 11 (VSDC card used; cryptogram failed).

2. This value is now deprecated and was formerly known as 12 (VSDC card used; cryptogram passed).

Position 3, CAVV Key Indicator: This 1-byte/1-BCD value is a 1-digit code indicating the CAVV key set used to calculate the CAVV value. A leading zero is required to pad the first unused half-byte of the CAVV Key Indicator. This value is determined by the V.I.P. key ID. [Table 5.9](#) contains position 3 valid values.

Table 5.9 CAVV, Revised Position 3 values

Code	Description
01	CAVV key set 1
02	CAVV key set 2
03-09	Reserved for Visa keys(s)
10	US Region attempt server Visa key #1
11	US Region attempt server Visa key #2
12-99	Reserved for Visa key (s)

Position 4, CAVV: This 2-byte/3-BCD value is a 3-digit code generated by the issuer's ACS that may be used by the issuer to validate the authentication response during authorization. A leading zero is required in byte 4 to pad the first unused half-byte of the CAVV, for example, **0456**. This value is determined by the ACS and the keys loaded in V.I.P (attempts only).

Position 5, Unpredictable Number: This 2-byte/4-BCD value is a 4-digit code that contains the four least significant digits for the authentication tracking number. The value is derived from the authentication tracking number by the ACS.

Position 6, Authentication Tracking Number (ATN): This 8-byte/16-BCD value is a 16-digit code generated by the issuer's ACS to identify the transaction.

Position 7, Version and Authentication Action: The left nibble of this 1-byte/1-BCD value identifies a version; the right nibble identifies the authentication action. [Table 5.10](#) contains position 7 valid values.

Table 5.10 CAVV, Revised Position 7 values

Code	Description
Version	
0	Authentication action and cardholder IP address not present
1	Authentication action and cardholder IP address present
3	Visa Token Service Payment Requirements
Authentication Value	
0	Standard authentication performed (no ADS or FYP performed).
1	ADS-registration authentication performed.
2	Forgot your password (FYP)-re-registration/re-authorization performed. Note that if an invalid value is submitted for this position, <A90> CAVV Result Code will be populated with a 0 (CAVV authentication results invalid) for U.S. issuers.
4	Token authentication verification value (TAVV)-Token authentication verification performed.

Position 8, IP Address in Hex Format: This 4-byte value identifies the client IP address submitted in the authorization message from ACS. The IP address must be in hexadecimal format to fit in the field.

5.29.3 American Express SafeKey

American Express SafeKey is an industry-standard Authentication method that provides greater security by authenticating the Cardmember during an online purchase and protecting payment card information as it is transmitted via the Internet.

American Express does not offer a breakdown of the cryptogram detail.

The SafeKey Cardholder Verification Value (CAVV) is a cryptographic value derived by the Issuer during the Amex SafeKey payment authentication that can provide evidence of the results of payment authentication during an online purchase. In this case, a valid value should be sent in [<A32> MOTO/e-Commerce Indicator](#). If the XID is present, it must be included in the XID Field in [<A33> Transaction ID \(XID\)](#).

5.29.4 American Express Token

The **American Express Payment Token** transaction processing solution is based on an industry aligned and interoperable tokenization system that offers increased protection against fraud by using a Payment Token. A Payment Token will be used in place of sensitive Cardmember data such as Primary Account Number (PAN) to originate payment transactions.

American Express does not offer a breakdown of the cryptogram detail.

Token Data Block A, which contains bytes 1-20 of the cryptographic value is required. Token Data Block B [<A33>](#) is optionally present when Token Data Block A is used.

5.30 <A35> EBT Electronic Voucher Serial Number

Length	Format	Description
1 to 15	A/N	This field contains the Electronic Voucher Serial Number provided by the Electronic Benefits Transfer (EBT) voice authorization system because of a POS system or EBT network unavailable condition. NOTE Cash benefits transactions do not use the Electronic Voucher Serial Number, Food and Consumer Identifier (FCS ID) and Voucher Approval Code fields. Food stamp transactions must use the FCS ID. Electronic voucher transactions require Electronic Voucher Serial Number, FCS ID, and Voucher Approval Code fields.

5.31 <A36> EBT Voucher Approval Code

Length	Format	Description
6 fixed	A/N	This field contains the voice authorization approval code sent in the request message for EBT/electronic voucher transactions. NOTE Cash benefits transactions do not use the Electronic Voucher Serial Number, FCS ID and Voucher Approval Code fields. Food stamp transactions must use the FCS ID. Electronic voucher transactions require Electronic Voucher Serial Number, FCS ID, and Voucher Approval Code fields.

5.32 <A37> Service Development Indicator

Length	Format	Description
1 fixed	Numeric	The Service Development Indicator indicates merchant participation in a Visa Service Development program. Possible value: 5 - Transponder Indicator - transactions from participating merchants use radio frequency transponder to exchange information with the POS device. 6 - Relationship Participant Indicator - indicates merchant's participation in the Visa Relationship Manager Service. 7 - Deferred Billing Indicator - notifies issuer that transaction being submitted bills the cardholder for merchandise that was received within the past 90 days.

5.33 <A38> Special Condition Indicator

Length	Format	Description
1 fixed	Numeric	This field describes selected special conditions at the point of sale. Possible values: 7 - The transaction is related to purchase of cryptocurrency. 8 - The transaction is for quasi-cash, a product or item that is representative of actual cash such as money orders, traveler's checks, foreign currency, lottery tickets, or casino gaming chips. 9 - The transaction includes payment on an existing debt.

One usage of this tag is part of the Visa Debt Repayment program. It is only one of a few key tags. The other required tags and values are described in the table below.

Table 5.11 Key Tags and Values for Visa Debt Repayment Program

Location	Field Name	Value
<A10>	Transaction Code	10-Purchase
<A11>	Program Code	Y - Bill Pay
<A22>	Market Specific Data Identifier	B- Bill Payment Transaction

5.34 <A39> Electronic Commerce Security Level Indicator

Tag A39 is deprecated effective with the October 2019 release and must not be used for new development after that date. It will be retired October 2020; continued use may result in unexpected behavior and/or potential compliance fines.

This Mastercard field contains the electronic commerce indicators representing the security level and cardholder authentication associated with the transaction. This field must be present in all Auth Request/0100 messages for electronic commerce transactions. The field is conditional in the following messages:

- All other types of Authorization Requests
- Authorization Advice
- Reversal Request

In general, the values in this tag apply to the type of authentication data in the Mastercard UCAF Data field [<A33><B4>](#) or [<A40>](#). If Mastercard UCAF Data is not present, but the [<A33><B5> Digital Payment Cryptogram](#) is, the SLI must be set appropriately for a DSRP transaction.

Mastercard's processing rules allow merchants to gain a liability shift and interchange benefit if they submit their electronic commerce transactions for cardholder authentication. Effective November 8, 2016, if merchants provide a value in Tag A39 that indicates the transaction is SecureCode but do not pass the appropriate authentication detail in Tag A40-UCAF Authentication Data, demonstrating they actually authenticated or attempted to authenticate the cardholder, Mastercard will downgrade the Authorization Request/0100 transaction and forward to the issuer. The correct values will be returned in the Tag A39 response message, whether downgraded or not.

The SecureCode functionality is not applicable for Mastercard Full Financial/0200 (Debit) Transactions.

Request Value:

This request has two usages. Usage 1 is present only for backwards compatibility. Effective October 13, 2017, it is recommended Usage 2 be utilized.

- Usage 1 requires only the UCAF Collection Indicator and is 1 position, fixed. The Security Protocol and Cardholder Authentication values are not included in this usage, and will be derived internally by TSYS software using other elements of the transaction. Note that this can result in inaccuracies based on the transaction values and the nature of such derivation.
- Usage 2 requires all three Positions. This usage is recommended.

The request is required on all Mastercard SecureCode e-Commerce transactions.

Response Value:

The Electronic Commerce Security Level Indicators contained in the response for all Mastercard e-Commerce Credit transactions, contains the electronic commerce indicators representing the security level and cardholder authentication associated with the transaction.

Mastercard will also reject clearing/settlement records if the indicators do not match the authorization indicators. The response field contains the correct values needed for settlement, whether the transaction was downgraded or not.

This field is not applicable for Mastercard Full Financial Debit transactions.

Table 5.12 Mastercard e-Commerce Security Level Indicators - Request and Response

Position	Length	Format	Description
1 and 2	2 fixed	Numeric	Security Protocol/Cardholder Authentication Valid values: 21 - Channel encryption; cardholder certificate not used (preferred value for Mastercard SecureCode, Identity Check, and all eCommerce) 22 - Masterpass-generated transaction 24 - Digital Secure Remote Payment (DSRP) with UCAF Data 91 - No security protocol; cardholder certificate not used

Position	Length	Format	Description
			7 - Partial shipment or recurring payment (<A40> Authentication Data only required for Identity Check); Liability will depend on the original UCAF values provided and matching with the initial transaction. 8-9 - Reserved for future use

5.35 <A40> Authentication Data

Tag A40 is deprecated effective with the October 2019 release and must not be used for new development after that date. It will be retired October 2020; continued use may result in unexpected behavior and/or potential compliance fines.

Length	Format	Description						
Up to 32	A/N and special characters	5.35.1 Usage 1: UCAF Authentication Data This Mastercard only variable length issuer generated security field is up to 32 bytes in length. When sending data in this field the UCAF Collection Indicator <A39> must be set to the proper value. It contains the encoded Mastercard SecureCode issuer or cardholder-generated authentication data (collected by the merchant) resulting from all SecureCode fully authenticated or attempts transactions, and any data associated with the 3-D Secure Electronic Commerce Verification Service. 5.35.2 Usage 2: Accountholder Authentication Value (AAV) AAV is part of Mastercard's SecureCode-specific implementation of UCAF related to issuer authentication. This is a security method designed to authenticate cardholders when they pay online. The values shown below are the leading one or two characters of the value. <table><tr><th><u>3D Secure 1.0</u></th><th><u>EMV 3D Secure</u></th></tr><tr><td>j</td><td>kG, kH, kJ, kK</td></tr><tr><td>h</td><td>kL, kM</td></tr></table>	<u>3D Secure 1.0</u>	<u>EMV 3D Secure</u>	j	kG, kH, kJ, kK	h	kL, kM
<u>3D Secure 1.0</u>	<u>EMV 3D Secure</u>							
j	kG, kH, kJ, kK							
h	kL, kM							

5.36 <A41> TSYS Acquiring Solutions Developer ID

Length	Format	Description
6 fixed	A/N	The Developer ID provides a unique value to identify the developer of POS software. This number is provided by TSYS Acquiring Solutions during approval testing. NOTE This field is mandatory in all transactions.

5.37 <A42> TSYS Acquiring Solutions Version ID

Length	Format	Description
4 fixed	A/N	The Version ID provides a unique value to identify the version of POS software. This number is provided by TSYS Acquiring Solutions during approval testing. NOTE This field is mandatory in all transactions.

5.38 <A43> Additional Amounts Group

Group <A43> contains 5 subfields that allow additional amounts to be transmitted in requests and responses. Group <A43> may be repeated up to 4 times. All 5 subfields should be included when sending group <A43>.

5.38.1 <A43><B1> Account Type

Length	Format	Description
2 fixed	Numeric	Possible value: 00 - Not specified 10 - Savings 20 - Checking 30 - Credit card 40 - Universal 60 - Stored value account 96 - Cash Benefits Account (for use by Electronic Benefits Transfer transactions only) 98 - Food Stamps Account (for use by Electronic Benefits Transfer transactions only)

5.38.2 <A43><B2> Amount Type

Length	Format	Description
2 fixed	A/N	Possible values: 01 - Deposit account: Ledger balance Credit card account: Open to buy 02 - Deposit account: Available balance Credit card account: Credit limit 05 - Amex Prepaid Card available amount remaining (Amex only) 10 - Partial authorized amount 43 - Total cumulative authorized amount for series of incremental authorization transactions (Visa & Discover only) 57 - Original amount 80 - Amount cash over (Discover only) In a request, this value indicates the requested cash over amount. In a response, this indicates the approved amount. 81 - Original Amount Cash Over (Discover Only) In a response, this value indicates the originally requested amount in situations with a partial approval. 3S - Amount co-payment 4S - Amount Healthcare (Visa and Mastercard authorization requests only) 4T - Amount Transit (Visa authorization request only) 4U - Amount Prescription/Rx (Visa and Mastercard authorization requests) 4V - Amount Vision/Optical (Visa authorization request only) 4W - Amount clinic/other qualified medical (Visa authorization request only) 4X - Amount Dental (Visa authorization request only)

5.38.3 <A43><B3> Currency Code

Length	Format	Description
3 fixed	Numeric	This subfield contains the 3 digit numeric currency code for the additional amount.

5.38.4 <A43><B4> Sign

Length	Format	Description
1 fixed	Alpha	Possible value: C - Positive balance D - Negative balance As an example, for healthcare auto-substantiation request values such as prescription or dental amounts, use a value of C.

5.38.5 <A43><B5> Additional Amount

Length	Format	Description
1 to 12	Numeric	This is a variable length field one to 12-character numeric in length. This field contains the additional amount. The amount is presented with implied decimal point. EXAMPLE \$.01 could be represented as 1, 01, or 001. The allowable number of significant digits as well as the positioning of any implied decimal point, are dictated by the designated currency code.

NOTE:

NOTE A Discover Merchant Initiated Transaction (MIT) is an addendum to a card sale in which the Merchant initiates the card sale per a pre-agreed and documented consent from the Cardholder. For Discover MIT partial shipment, recurring, and reauthorization of full amount transactions, the original transaction amount and original transaction ID are required, as well as <A43><B2> Additional Amount – Amount Type value of 57 – Original Amount..

5.39 <A44> Healthcare Service Group

Group <A44> contains subfields used to authorize healthcare services. Group <A44> may be repeated up to 4 times. Subtags <B1> and <B2> are present in group <A44> in the request message and are echoed in the response message. Subtags <B3> and <B4> are present in group <A44> in the response message.

Code	Definition
D	Merchandise must be fully approved or declined Cash over must be fully approved (if merchandise is approved) or declined

Visa Only (estimated amount)

These values must be used on the initial transaction when the amount is estimated. This is applicable for merchant types such as lodging, vehicle rental, cruise lines, bars, restaurants, and transportation for merchants such as buses and taxi cabs.

E	Estimated Amount
F	Estimated Amount (partial approval supported)

5.42 <A47> Amex ITD Data

This group contains subfields for providing Internet/Telephone Data (ITD) for Amex direct marketing purchases.

5.42.1 <A47><B1> Secondary ID

Length	Format	Description
3 fixed	Alpha	This subfield must be “TTD”.

5.42.2 <A47><B2> Customer e-Mail

Length	Format	Description
Up to 60	A/N and special characters	This subfield contains the customer e-mail address.

5.42.3 <A47><B3> Customer Host Name

Length	Format	Description
Up to 60	A/N and special characters	This subfield contains the customer host name.

5.42.4 <A47><B4> HTTP Browser Type

Length	Format	Description
Up to 60	A/N and special characters	This subfield contains the customer browser type and version information.

5.42.5 <A47><B5> Ship to Country

Length	Format	Description
Up to 20	A/N and special characters	This subfield contains the ship-to country. The field can be ISO alpha or numeric codes or literal. EXAMPLES USA, 840 and United States.

5.42.6 <A47><B6> Shipping Method

Length	Format	Description
2 fixed	A/N and special characters	This subfield contains a 2 digit shipment type code. Possible value: 01 - Same day 02 - Overnight, next day 03 - Priority, 2-3 days 04 - Ground, 4+ days 05 - Electronic delivery 06 - Ship to store 07-ZZ - Future use

5.42.7 <A47><B7> Merchant Product SKU

Length	Format	Description
Up to 15	A/N and special characters	This subfield contains the merchant product SKU.

5.42.8 <A47><B8> Customer IP Address

Length	Format	Description
15 fixed	A/N and special characters	This subfield contains the customer IP address in nnn.nnn.nnn.nnn format.

5.43 <A48> Amex Merchant DBA Name

Length	Format	Description
Up to 30	A/N and special characters	This subfield contains the Merchant DBA name. This field is required for Amex transactions that are submitted by an Aggregator. It is recommended that this be sent for all Amex transactions. Spaces should be included and the name should be truncated, if necessary, rather than abbreviated. See tag <A112> for additional information.

5.44 <A49> Amex Station Location Code

Length	Format	Description
Up to 20	A/N and special characters	This field contains the unique merchant-assigned station location code for Oil Company Industry, including Card Acceptor Terminal (CAT) transactions, where a single Service Establishment Number is not used for each physical location. It is required for all Amex Oil Company Industry transactions. If <A50> Amex Seller/Vendor Code is populated, tag <A49> will not be used. See tag <A112> for additional information.

5.45 <A50> Amex Seller/Vendor Code

Length	Format	Description
Up to 20	A/N and special characters	This field contains a unique seller/vendor code assigned by the Payment Service Provider/Aggregator. In American Express transactions, this is known as the Seller ID. This field is required for merchants processing through a Payment Service Provider/Aggregator. For Oil Company Industry, use tag <A49> instead of tag <A50>. See tag <A112> for additional information.

5.46 <A51> Digital Entity Identifier

A unique identifier that is assigned by Visa at the time of authorization that identifies transactions that originate from Visa Checkout.

5.46.1 <A51><B1> Digital Entity Identifier

Length	Format	Description
5 fixed	A/N	A unique identifier that is assigned by Visa at the time of authorization that identifies transactions that originate from Visa Checkout.

5.46.2 <A51><B2> Reserved

Length	Format	Description
0, 12	NUM	The point of sale does not need to send this field. This subfield is reserved for internal use only.

5.47 <A52> Reversal Request Code

Length	Format	Description
2 fixed	Numeric	This code is present in Mastercard reversal requests to signify the reason for the request. 17 - Valid for Mastercard <u>Non-Debit</u> Reversal Requests Only - used in a reversal message to indicate that the cardholder cancelled the transaction after an authorization request was sent. 32 - Partial reversal 34 - Suspect fraud

5.48 <A53> Discover Promotional Code

Length	Format	Description
10 fixed	A/N and special characters	The Discover/PayPal Network may establish promotional relationships with merchants. This field contains the promotion code indicating the promotion to be used by the merchant for rewarding the cardholder at the point of service. The value sent in the authorization request will be echoed back in the authorization response. This field may be repeated up to 5 times to support up to 5 promotion codes. Promotional Codes can be used by PayPal. It should be noted that not all Discover authorization scenarios apply to PayPal. For example, card not present scenarios are not allowed with PayPal.

5.49 <A54> Transaction Type Identifier

This Mastercard field indicates the type of additional transaction purpose. It can be used for the following message types.

- Auth Request/0100
- Auth Response/0110
- Auth Advice/0120
- Reversal Request/0400
- Reversal Response/0410
- Reversal Advice/0420

Length	Format	Description
3 fixed	A/N	Possible value: C03 - rePower Load value C04 - Gaming repay C06 - Payment of Credit Card Balance with Cash or Check C07 - General Person-to-Person Transfer C52 - General Transfer to Own Account C53 - Agent Cash Out C54 - Payment of Own Credit Card Bill C55 - Business Disbursement C56 - Government/Non-profit Disbursement C57 - Rapid Merchant Settlement C65 - General Business-to-Business Transfer (MCC 6538) F07 - General Person-to-Person Transfer F52 - General Transfer to Own Account F53 - Agent Cash Out F54 - Payment of Own Credit Card Bill F61 - Transfer to Own Staged Digital Wallet Account F64 - Transfer to Own Debit or Prepaid Card Account P70 - Cryptocurrency P71 - High Risk Securities

5.50 <A55> Real-time Substantiation Indicator

Length	Format	Description
1 fixed	Numeric	This one character Mastercard only field is used to indicate whether the merchant terminal verified the purchased items against an Inventory Information Approval System (IIAS). This field is used for Authorization Request/0100 and Authorization Advice/0120 transactions only. 0 = Merchant POS device did not verify the purchased items against an IIAS 1 = Merchant POS device verified the purchased items against an IIAS 2= Merchant claims exemption from IIAS based on the 90 percent rule

5.51 <A56> Electromagnetic Signature

Length	Format	Description
128 fixed	0-9, A-f (ASCII representation of HEX)	This field contains the digital value of the magnetic signature from the card if it is captured when the card is swiped. This data will not be returned in the response, and it must not be stored after authorization. This must not be submitted in contactless or chip transactions. This field is only allowed on Visa and Debit transactions.

5.52 <A57> NTIA UPC/SKU Data

Length	Format	Description
1 to 32	A/N and special characters	This variable length field is used for UPC/SKU data used in conjunction with National Telecommunications and Information Administration (NTIA) issued coupons. These coupons are used toward the purchase of a digital-to-analog converter box. This field can only be sent in an Authorization Only Transaction.

5.53 <A58> Visa Contactless Data Group

This group contains eight subfields that support data from Visa Contactless 'payWave' cards. Not all subfields are required to be present.

5.53.1 <A58><B1> Amount Authorized

Length	Format	Description
12 fixed	Numeric	This subfield contains the amount of the transaction (cryptogram amount) used by the card to generate the cryptogram. For U.S. processors, the expected amount value is 0.

5.53.2 <A58><B2> Application Cryptogram

Length	Format	Description
16 fixed	A/N	This subfield contains the cryptogram used for authentication of the transaction. ASCII representation of HEX.

5.53.3 <A58><B3> Application Transaction Counter

Length	Format	Description
4 fixed	A/N	This subfield contains a count of the transactions performed within the card application. The count is incremented by one each time a transaction is initiated. Counter implemented in HEX. Send value as ASCII representation of HEX.

5.53.4 <A58><B4> Customer Exclusive Data

Length	Format	Description
2 to 64	A/N	Description of this subfield will be included in the program documentation. ASCII representation of HEX.

5.53.5 <A58><B5> Form Factor Indicator

Length	Format	Description
2 to 8	A/N	This subfield contains indicators about the attributes of cardholder's device and the technology used for communication between the cardholder's device and the acquiring POS device. ASCII representation of HEX.

5.53.6 <A58><B6> Issuer Application Data

Length	Format	Description
2 to 64	A/N	This subfield contains the issuer application data transmitted from the card to the issuer and is updated by the issuer in the response messages. ASCII representation of HEX.

5.53.7 <A58><B7> Unpredictable Number

Length	Format	Description
8 fixed	A/N	This subfield contains the number used in the generation of the cryptogram for contactless transactions. ASCII representation of HEX.

5.53.8 <A58><B8> Card Sequence Number

Length	Format	Description
1 to 3	N	This subfield contains a sequence number that distinguishes between separate cards having the same primary account number.

5.54 <A60> Tip Amount

Length	Format	Description
1 to 12	Numeric	This field contains the tip (gratuity) amount of the transaction being settled. If there is no gratuity amount associated with the transaction, A60 can be sent with a value of “0”.

5.55 <A61> Tax Amount (Commercial Card)

Length	Format	Description
1 to 12	Numeric	This field contains the amount as defined in Tax Amount Identifier <A62>. If the value in field <A62> is set to ‘0’, <A61> should contain ‘0’. Any non-zero sales tax must have a minimum value of 0.1% and a maximum value of 22% of the sale amount.

5.56 <A62> Tax Amount Identifier (Commercial Card)

Length	Format	Description
1 fixed	Numeric	This field contains a character identifying the characteristics of the amount appearing in Tax Amount <A61>. 0 = Not used 1 = Local sales tax amount 2 = Tax exempt

5.57 <A63> Purchase Order Number (Commercial Card)

Length	Format	Description
Up to 25	A/N	This field contains a Purchase Order Number supplied by the Visa or Mastercard Purchasing Card cardholder.

5.58 <A64> e-Commerce Goods Indicator

Length	Format	Description
1 to 0	Alpha	This field is used to identify the type of goods purchased over the internet. The following table provides the possible indicator values for this field. This field should only be sent if <A32> MOTO/e-Commerce Indicator has a value of 5, 6, 7, or 8. D = Digital goods P = Physical goods This field is supported in capture only and although it is accepted for any card brand, it is sent only to Visa.

5.63 <A69> Transaction Fee Amount

Length	Format	Description
9	A/N	Transaction Fee Amount is used in PIN POS and credit transactions to carry the merchant-assessed surcharge for informational purposes only. This optional tag is nine alphanumeric characters in length. The format of the nine characters is "annnnnnnn", where "a" is either "D" for debit or "C" for credit and "nnnnnnnn" is the numeric fee amount with the decimal implied. This tag may also be present in reversal messages. This field is applicable for Visa, Mastercard, and Discover credit transactions only. Example: "D00000150" is a \$1.50 transaction fee amount debited against the cardholder's account. A prefix of 'C' would put the amount back to the cardholder account. The same value must be used in originals and reversals.

5.64 <A70>RIID

Length	Format	Description
6	Num	Receiving Institution ID. Will be augmented at the host unless sent in the request message.

5.65 <A71> Amex Airline Data

This tag contains data to satisfy American Express Airline data in two different formats. Only one of these formats may be submitted at a time. The airline record will be accepted regardless of MCC but the settlement records will be produced for Amex only if the merchant profile is set up with an Amex MCC Format Code of '14'.

- Internet Airline Customer (IAC) data is specific to airline industry merchants that submit card not present data. This is the recommended format, as it is more comprehensive. For these merchants, IAC subfields may contain travel-specific information, including the departure date, passenger name, travel origin and destination, routing cities, airline carriers, fare basis, number of passengers, and customer IP and email addresses.
- Airline Passenger Data (APD) is specific to airline industry merchants that submit card not present data. This format has been available for a long time and, **effective with the April 2018 release, should not be used for new development.**

5.65.5 <A71><B5> Origin (Origin Airport)

Length	Format	Description
5	A/N	This field contains the first segment of the travel origination airport. The five-byte code sequence allows for anticipated expansion of the current three character airport code. This field applies to IAC and APD formats. EXAMPLE XYZ

5.65.6 <A71><B6> Dest (First Segment Travel Destination Airport)

Length	Format	Description
5	A/N	This field contains the destination airport for the first travel segment of trip; not necessarily the final destination. For example, if the passenger flies from STL to MIA with layover at JFK, the Destination Airport for the first segment is JFK. The five-byte code sequence allows for anticipated expansion of the current three character airport code. This field applies to IAC and APD formats. EXAMPLE XYZ

5.65.7 <A71><B7> Routing Cities

Length	Format	Description
1-5	A/N	This field contains the Routing Airport or City Codes for each leg on the ticket (including ORIGIN and DESTINATION). This tag can be repeated up to 10 times. This field applies to IAC and APD formats.

5.65.8 <A71><B8>Airline Carriers

Length	Format	Description
1-5	A/N	This field contains the Airline Carrier Code for each leg on the ticket (including Origin and Destination Airports). Each leg must have an Airline Carrier Code entry, even if multiple (or all) legs are on the same Airline. This tag can be repeated up to 9 times. This field applies to IAC and APD formats.

5.65.13 <A71><B13>Customer IP

Length	Format	Description
15	A/N	This field contains the customer's Internet IP address, up to 15 bytes. The format is nnn.nnn.nnn.nnn This field applies to IAC format only. EXAMPLE 1: 127.142.151.223 EXAMPLE 2: 127.142.5.56 EXAMPLE 3: 12.142.49.190

5.65.14 <A71><B14>Customer Email Address

Length	Format	Description
15	A/N	This field contains the Customer's email address. This field applies to IAC format only. EXAMPLE CFFROST@EMAILADDRESS.COM

5.66 <A72> Passenger Transportation Data Group

This record is used to report additional descriptive details for the Airline and Travel/Cruise Industries. This information may become part of the descriptive bill on the Cardholder's statement or be used to resolve billing inquiries and disputes.

5.66.1 <A72><B1>Ticket Number

Length	Format	Description
1-13	A/N	Visa: This field contains a ticket number comprised of a three-character Carrier Code (ARC/TWA accounting code) followed by a 10-character Ticket Identifier. AMEX: For Travel/Cruise transactions, this field contains the ticket or document number printed on the source document. This field is required for Airline transactions if <A139> <B1> Ticket Transaction Type is 'TKT' or 'EXC'. It is recommended this field be populated for all Transaction Types. If only the booking reference number is available, this field contains the booking reference number.

5.66.2 <A72><B2>Passenger Name

Length	Format	Description
1-20	A/N	Visa: This field must contain the name of the airline passenger. If not available, include the cardholder's name. AMEX: This field contains the Passenger Name and is required if <A139> <B1> Ticket Transaction Type is 'TKT' or 'EXC'.

5.66.3 <A72><B3>Restricted Ticket Indicator

Length	Format	Description
1	A/N	This field contains an indicator utilized to specify special restrictions associated with an issued airline ticket. Possible value: 0 - no restriction 1 - Restricted Ticket/ Non- Refundable

5.66.4 <A72><B4>Departure Date

Length	Format	Description
6	NUM	This field contains the airline departure data in MMDDYY format. The date supplied may be in the future. If not used, default the field to "000000".

5.66.5 <A72><B5>Origination City/Airport Code

Length	Format	Description
3	A/N	This field contains the three character code of the city/airport where the trip originates. Visa: This field is required and may not contain all spaces or all zeros for originals and reversals. AMEX: This field is required and may not contain all spaces or all zeros if <A139> <B1> Ticket Transaction Type is 'TKT' or 'EXC'.

5.66.6 <A72><B6>Trip Leg Information

Length	Format	Description
8	A/N	Each Trip Leg is comprised of ten separate fields. This subfield can be repeated up to 4 times, with the first leg being mandatory for Passenger Transport. The table below shows the format for the fields: Either the first 8 positions are populated, or all 50 positions are populated. <u>Unused fields must be character space filled.</u>
Positions	Format	Description
1	N	Trip Leg Valid Values: (1, 2, 3, or 4)
2-3	A/N	Carrier Code This field contains the Carrier Code for this trip segment, which is the two-character, IATA Carrier Code that corresponds to the airline carrier that provided the transportation for this segment. Refer to the <i>IATA Airline Coding Directory</i> for valid codes. Visa: The entry may not contain all spaces or zeros for originals and reversals of Passenger Transport transactions requesting Electronic Interchange Reimbursement Fee (EIRF) or any custom payment service. AMEX: For Travel/Cruise, this field contains the IATA Carrier Code (a.k.a., designator code or carrier code) that corresponds to the carrier that issued the ticket, which may or may not be the actual transportation provider. For Airline, this field is required for Trip Leg 1, if <A139><B1> Ticket Transaction Type is 'TKT' or 'EXC'. This field is conditional for Trip Legs 2-4, depending on whether additional Trip Legs are <A139><B1> Ticket Transaction Type 'TKT' or 'EXC'.

Length	Format	Description
4	A/N	Service Class This field contains the Service Class for this trip segment, which is a code that corresponds to the fare class (e.g., A, B, C, etc.) within the overall class of service (e.g., First Class, Business Class, and Economy/Coach Class) applicable to this segment. Refer to the <i>LATA Airline Coding Directory</i> for valid codes. AMEX: The Class of Service Code is required for Trip Leg 1, if Trip Leg 1 <A139> <B1> Ticket Transaction Type is 'TKT' or 'EXC'. This field is conditional for Trip Legs 2-4, depending on whether additional Trip Legs are included in this travel itinerary, and if <A139> <B1> Ticket Transaction Type is 'TKT' or 'EXC'.
5	A/N	Stopover Code This value identifies whether a Stopover is permitted with the purchased ticket. A stopover indicates when a traveler spends more than four (4) hours at a destination before the next leg of travel begins. Possible value: Space - Destination Point O - Stopover permitted (letter 'O', not zero) X - Connection The following example shows the usage of the field for travel between New York, NY and Sydney, Australia. The last segment of the trip should always be indicated with a space. • Stopover code 1: X Connection in NYC to DFW • Stopover code 2: O stopover in DFW on route to LAX • Stopover code 3: O stopover in LAX on route to HNL • Stopover code 4: O stopover in HNL on route to SYD • Stopover code 5: (space) terminate travel in SYD

Length	Format	Description
6-8	A/N	Destination Airport Code This field contains the Arrival/Destination City/Location Code for this trip segment, which is the standard airline routing code (a.k.a., airport code or location identifier) applicable to the arrival portion of this segment. Refer to the <i>LATA Airline Coding Directory</i> for valid codes. AMEX: This field is required for Segment 1, if <A139> <B1> Ticket Transaction Type is 'TKT' or 'EXC'. This field is conditional for Segments 2-4, depending on whether additional segments are included in this travel itinerary, and if <A139> <B1> Ticket Transaction Type is 'TKT' or 'EXC'.
9-11	A/N	Departure Airport Code This field contains the Departure City/Location Code for this Trip Leg, which is the standard airline routing code (a.k.a., airport code or location identifier) applicable to the departure portion of this segment. Refer to the <i>LATA Airline Coding Directory</i> for valid codes. AMEX: This field is required for Trip Leg 1, if <A139> <B1> Ticket Transaction Type is 'TKT' or 'EXC'. This field is conditional for Trip Legs 2-4, depending on whether additional Trip Legs are included in this travel itinerary, and if <A139> <B1> Ticket Transaction Type is 'TKT' or 'EXC'.
12-19	N	Departure Date This field contains the Departure Date for this Trip Leg. The format is: CCYYMMDD where CC = Century YY = Year MM = Month DD = Day For example, February 16, 2011, would be entered as: 20110216 AMEX: This field is required for Trip Leg 1, if <A139> <B1> Ticket Transaction Type is 'TKT' or 'EXC'. This field is conditional for Trip Legs 2-4, depending on whether additional segments are included in this travel itinerary, and if <A139> <B1> Ticket Transaction Type is 'TKT' or 'EXC'.

Length	Format	Description
20-23	A/N	Flight Number This field contains the Flight Number for this Trip Leg, which is the carrier-assigned, advertised Flight Number for this trip leg; or the literal "OPEN", if the Trip Leg is open and there is no pre-assigned flight number. AMEX: This field is required for Trip Leg 1, if <A139> <B1> Ticket Transaction Type is "TKT" or "EXC". This field is conditional for Trip Legs 2-4, depending on whether additional Trip Legs are included in this travel itinerary, and if <A139> <B1> Ticket Transaction Type is "TKT" or "EXC".
24-38	A/N	Fare Basis This field contains the Fare Basis for this Trip Leg, which is the Fare Basis Code that corresponds to the specific fare level for which the passenger is entitled for this segment. Refer to the <i>IATA Airline Coding Directory</i> for valid codes. Field contents may vary from the mandatory, one-byte, Prime Code (left justified and character space filled) up to an entry containing values for all optional subfields, plus additional airline-specific identifiers, up to 15 bytes, maximum. AMEX: This field is required for Trip Leg 1, if <A139> <B1> Ticket Transaction Type is "TKT" or "EXC". This field is conditional for Trip Legs 2-4, depending on whether additional Trip Legs are included in this travel itinerary, and if <A139> <B1> Ticket Transaction Type is "TKT" or "EXC".
39-50	N	Fare This field contains the total Fare for the this trip segment.

5.66.7 <A72><B7>Ticketing Carrier Code

Length	Format	Description
3	A/N	This code indicates the name of the carrier (United Airlines, Jet Blue, etc.) Refer to the <i>LATA Airline Coding Directory</i> for valid codes. AMEX: For Travel/Cruise, this field contains the IATA Carrier Code (a.k.a., designator code or carrier code) that corresponds to the carrier that issued the ticket, which may or may not be the actual transportation provider. For Airline, this field is required and contains the Airline Process Identifier, which is the IATA Carrier Code of the airline that sold and ticketed the travel; or if the travel was sold through a travel agency, the airline on which the travel will predominantly be conducted.

5.66.8 <A72><B8>Travel Agency ID Code

Length	Format	Description
8	A/N	Travel Agency ID Code This field contains the Travel Agency Identifier Code (a.k.a. IATA Numeric Code) for the travel agency where the ticket booking was made. Refer to the <i>LATA Airline Coding Directory</i> for valid codes. This field is required if the booking was made through a travel agency and the travel agency is an IATA member.

5.67 <A73> Hold Capture Indicator

Length	Format	Description
6	A/N	This tag enables merchants to prevent the capture of an authorization transaction until it is released by the merchant. A merchant setup for the use of the Summit Gateway hold capture feature can send a 'HOLD' indicator on an original authorization request message, and then release the transaction to be captured by sending a 'CAPTURE' adjustment to the original authorization transaction. Possible values: (all caps) HOLD - Excludes authorization from batch close processing when sent in original authorization. CAPTURE - Removes batch processing exclusion when sent in adjustment of the original authorization. NOTE Merchants must be established as Hold Capture participants to be able to utilize this tag. Selecting the Capture Delay value "Hold till Release" in the boarding system will flag a merchant's terminal as a hold capture participant.

5.68 <A74> Cardholder Verification Results

This group contains subfields that carry verification response codes for cardholder identification data sent in the authorization request.

For Amex transactions, Amex will respond to an authorization request with verification results for cardholder identification data. The data is arranged in order of 5 bytes, one for each result of a cardholder-identification data element.

For Discover, only the Billing Name Result Code is sent.

If a cardholder-verification data element is not present in the request then the associated tag will not be included in the authorization response.

5.68.1 <A74><B1>Billing ZIP Code

Length	Format	Description
1	A/N	Possible values: Y - Data Matches U - Data Unchecked N - No Match S - Service not allowed R - Retry

5.68.2 <A74><B2>Billing Street Match Code

Length	Format	Description
1	A/N	Possible values: Y - Data Matches U - Data Unchecked N - No Match S - Service not allowed R - Retry

5.68.3 <A74><B3>Billing Name Result Code

This code has different possible values depending on the card type.

Length	Format	Description
1	A/N	Possible values: Amex Y - Data Matches U - Data Unchecked N - No Match S - Service not allowed R - Retry Possible values: Discover B - No response due to blank input K - Unknown P - Not processed M - First Name and Last Name match F - First Name matches, Last name does not match L - First Name does not match, Last Name matches N - Nothing matches W - No data from Issuer/Authorization system U - Retry, system unable to process

5.68.4 <A74><B4>Telephone Number Match Code

Length	Format	Description
1	A/N	Possible values: Y - Data Matches U - Data Unchecked N - No Match S - Service not allowed R - Retry

5.68.5 <A74><B5>E-mail Address Match Code

Length	Format	Description
1	A/N	Possible values: Y - Data Matches U - Data Unchecked N - No Match S - Service not allowed R - Retry

5.69 <A75> Custom Identifier

This field is used to send Mastercard custom identifier information and has a maximum length of 82 bytes in two sub-tags. Both sub-tag must be sent.

5.69.1 <A75><B1> Custom Identifier Type

Length	Format	Description
6	A/N	Possible values: CUSORD - Customer Order CUSTPO - Customer Purchase Order FOLNUM - Folio Number INVACT - Invoice or Account Number for Bill Payments OTHER1 - Other PMTREF - Payment Reference Number PRCHID - Purchase Identification RECLOC - Record Locator RESNUM - Reservation Number RNTAGR - Rental Agreement SUPINV - Supplier Invoice SUPORD - Supplier Order TKTNUM - Ticket Number TRANID - Transaction Identification TRKNUM - Tracking Number

5.69.2 <A75><B2> Custom Identifier Detail

Length	Format	Description
76	A/N	Free form text of information referenced in Custom Identifier Type

5.70 <A76> Mastercard EMS Fraud Service Group

This Mastercard-only field is used by participating Card Not Present merchants to request and receive results on a predictive risk score by the Expert Monitoring System (EMS) that may assist in determining if a CNP transaction is fraudulent. It is valid for Mastercard Authorization Request/0100 transactions only.

5.70.1 <A76><B1> EMS Service Request Indicator

Length	Format	Description
1	A/N	This field is included in the request, and indicates whether the merchant requested the transaction to be scored by the EMS. Valid values: 0 = No action required 1 = Transaction to be scored

5.70.2 <A76><B2> EMS Service Response Indicator

Length	Format	Description
2	N	This field is included in the response, and indicates that EMS scoring was performed. If this tag is not received, no scoring was performed. Valid Value: 90 = EMS Scoring was performed

5.70.3 <A76><B3> EMS Result Code

Length	Format	Description
1	A/N	This field is included in the response, and indicates the EMS result code. Valid Values: C = EMS Scoring is successful I = Invalid, transaction does not qualify for EMS due to: • Transaction is card present, or • Card was issued outside of valid issuing region U = Scoring was not successfully performed

5.70.4 <A76><B4> EMS Additional Information

Length	Format	Description
1	A/N	This field is included in the response, and indicates if the merchant's acquirer is enrolled in the service and the whether the service was requested. The value of 'N' will be sent if the score was not delivered but a request was received for the transaction to be scored. Valid Values: N = Not qualified for EMS scoring service Blank = No value present

5.70.5 <A76><B5> EMS Risk Score

Length	Format	Description
3	A/N	This field is included in the response, and indicates the transaction risk score. Score values range from 001 through 999 where 001 indicates the least likely fraudulent transaction and 999 indicates the most likely fraudulent transaction.

5.70.6 <A76><B6> EMS Risk Score Reason Code

Length	Format	Description
2	A/N	This field is included in the response, and indicates the key factors that influenced the EMS score. (Participating merchants may contact the acquirers for a list of reason codes.)

5.71 <A77> Mastercard Transit Program

This Mastercard field used in Auth Request/0100 and Auth Advice/0120 transactions only, provides additional information regarding transit transactions. Specifically it is used only for modes of transportation that use contactless transactions.

5.71.1 <A77><B1> Transit Transaction Type Indicator

Length	Format	Description
2	Numeric	This field identifies the transit transaction type. Valid Values: 01 - Prefunded 02 - Real-time authorized 03 - Post authorized, aggregated 04 - Authorized aggregated split clearing 05 - Other 07 - Debt recovery 06, 08-99 - Reserved for future use

5.71.2 <A77><B2> Transportation Mode Indicator

Length	Format	Description
2	Numeric	This field identifies the mode of transportation. Valid Values: 00 - Unknown 01 - Urban bus 02 - Interurban bus 03 - Light train mass transit (underground metro, LTR) 04 - Train 05 - Commuter train 06 - Water borne vehicle 07 - Toll 08 - Parking 09 - Taxi 10 - High speed train 11 - Rural bus 12 - Express commuter train 13 - Para transit 14 - Self drive vehicle 15 - Coach 16 - Locomotive 17 - Powered motor vehicle 18 - Trailer 19 - Regional train 20 - Inter city 21 - Funicular train 22 - Cable car 23-99 - Reserved for future use

5.72 <A78> Transaction ID

Length	Format	Description
15	AN	This field is used to support incremental authorization requests. Incremental transactions are associated with a single cardholder, and must reference the original transaction. Response This field contains the Transaction ID provided by the card brand. It is used to identify the original transaction in later requests. Request This field contains the Transaction ID sent in the original response. Merchant Initiated Transactions (MIT) A Discover MIT is an addendum to a card sale in which the Merchant initiates the card sale per a pre-agreed and documented consent from the Cardholder. For Discover MIT partial shipment, recurring, and reauthorization of full amount transactions, the original transaction amount and original transaction ID are required, as well as <A43><B5> Additional Amount – Amount Type value of 57 – Original Amount. Visa Usage 1: Standing Instruction A standing instruction is an agreement between the cardholder and the merchant for the provision of goods or services. This MIT must contain the original transaction identifier and is for the following types of merchant initiated transactions: credential on file, recurring, and installment. Refer to <A135> - POS Environment Indicator for a description of valid values. Usage 2: Industry Practice An Industry Practice MIT is a follow-up to an original cardholder merchant interaction that could not be completed with a single transaction. If an Industry Practice MIT transaction is sent, it must include the transaction identifier of the original transaction. The message reason code is also required for the Industry Practice MIT and can be type: incremental, resubmission, delay charges, reauthorization, and no show. Refer to <A113> - Message Reason Code for a description of valid values.

5.73 <A79> Use of TSYS Token Request

Length	Format	Description
Variable	AN	This tag enables the device/POS Application to request a TSYS token. Upon request, the value noted below should be included. With the response, this tag will contain the TSYS token. Use of transaction code 21 will send a TSYS token without performing any card validation. Use with any other Transaction Code performs the usual transaction request as specified. Possible value: (all caps) TKR - Indicates the response should contain a TSYS token. For more information, please refer to Section 4.8 Using encryption and tokens

Purchase with a TSYS Token

In general, a TSYS token is used for processing recurring payments for existing debt or repeat business. This functionality is not available for debit transactions or card present scenarios.

NOTE A purchase with a TSYS Token can still be voided or adjusted through the normal void/adjust process.

The TSYS token must have been requested previously with a valid PAN. For these types of authorization requests, the following restrictions apply:

- <A10> Transaction Code must be one of the following:
 - 10 - Purchase
 - 13 - Return (Purchase Return Authorization for all card brands except Amex and TSYS Private Label)
 - 14 - Balance Inquiry
 - 20 - Authorization only
- <A21> POS Entry Data subfields:
 - <A21><B1> Cardholder Present Data must **not** be 0 - Cardholder present
 - <A21><B2> Card Present Data must be 0 - Card not present
 - <A21><B3> Card Data Input Mode must be one of the following:
 - 0 - Unspecified, data not available [default]
 - 1 - Manual input, no POS device
 - 6 - Key entered input
 - S - Electronic commerce, no security, channel encrypted, or SET without cardholder certificate
 - V - Manually entered with keyed CID (Amex)
 - <A21><B4> Cardholder Authentication Method must be one of the following
 - 0 - Not authenticated
 - 9 - Unknown, data not available [default]

- <A21><B5> Authentication Entity must be one of the following
 - 0 - Not authenticated
 - 5 - Other (MC)
 - 9 - Unknown, data not available [default]
- <A21><B6> Card data input capability must be one of the following
 - 0 - Unknown; data not available
 - 1 - Manual; no POS device
 - 6 - Key entry only capability
- Either of <A32> MOTO/e-Commerce Indicator OR <A38> Existing Debt indicator must be present
- <A93> Card Type must not be DEBIT or EBT

5.74 <A80> Batch Number

Length	Format	Description
1 to 3	Numeric	Incrementing ID assigned by TSYS Acquiring Solutions host to identify a merchant batch. The batch number that is returned by the host is the current batch number, not necessarily the final batch number when captured.

5.75 <A81> Local Transaction Date

Length	Format	Description
6 fixed	Numeric	This field contains a Local Transaction Date (MMDDYY) calculated by the authorization center using the time zone differential. This field can be used as the transaction date printed on the receipt. For authorization Reversals, use the Transaction Date of the original purchase that is being reversed. For incremental transactions, the date from the original authorization should be used.

5.76 <A82> Local Transaction Time

Length	Format	Description
6 fixed	Numeric	This six-character numeric field contains the Local Transaction Time returned by the authorizing system (HHMMSS). For direct debit, the transaction time should be printed on the receipt. For incremental transactions, the date from the original authorization should be used.

5.77 <A83> Response Code

Length	Format	Description
2 or 3	Numeric	This field contains a response code indicating the status of the authorization request. The POS system must evaluate the response code and not the response text to determine the nature of a response message. NOTE Do not interpret all non-approved response codes as “DECLINED.”

<A83> Response code	<A85> Response text	Definition
00	APPROVAL	Approved
01	CALL	Refer to issuer
02	CALL	Refer to issuer - special condition
03	TERM ID ERROR	Invalid Merchant ID

<A83> Response code	<A85> Response text	Definition
04	HOLD-CALL OR PICK UP CARD	Pick up card (no fraud)
05	DECLINE	Do not honor
06	ERROR XXXX	General error
07	HOLD-CALL OR PICK UP CARD	Pick up card, special condition (fraud account)
08	APPROVAL	Honor Mastercard with ID
10	PARTIAL APPROVAL	Partial approval for the authorized amount returned in <A43>
11	APPROVAL	VIP approval
12	INVALID TRANS	Invalid transaction
13	AMOUNT ERROR	Invalid amount
14	CARD NO ERROR	Invalid card number
15	NO SUCH ISSUER	No such issuer
19	RE ENTER	Re-enter transaction
21	NO ACTION TAKEN	Unable to back out transaction
28	NO REPLY	File is temporarily unavailable
39	NO CREDIT ACCT	No credit account
41	HOLD-CALL OR PICK UP CARD	Lost card, pick up (fraud account)
43	HOLD-CALL OR PICK UP CARD	Stolen card, pick up (fraud account)
46	CLOSED ACCOUNT	Closed account
51	DECLINE	Insufficient funds
52	NO CHECK ACCOUNT	No checking account
53	NO SAVE ACCOUNT	No savings account
54	EXPIRED CARD	Expired card
55	WRONG PIN	Incorrect PIN
57	SERV NOT ALLOWED	Transaction not permitted - Card
58	SERVE NOT ALLOWED	Transaction not permitted - POS device
59	SUSPECTED FRAUD	Suspected Fraud
61	EXCEEDS APPROVAL AMT LIMIT	Exceeds approval amount limit
62	DECLINE	Invalid service code, restricted
63	SEC VIOLATION	Security violation

<A83> Response code	<A85> Response text	Definition
65	EXCEEDS WITHDRAWAL FREQUENCY LIMIT	Exceeds withdrawal frequency limit
6P	VERIFICATION DATA FAILED	Verification data failed
75	PIN EXCEEDED	Allowable number of PIN entry tries exceeded
76	UNSOLIC REVERSAL	Unable to locate, no match
77	NO ACTION TAKEN	Inconsistent data, rev. or repeat
78	NO ACCOUNT	Blocked, first used transaction from new cardholder, and card not properly unblocked
79	ALREADY REVERSED	Already reversed at switch
80	NO IMPACT	No Financial impact (used in reversal responses to declined originals).
81	ENCRYPTION ERROR	Cryptographic error
82	INCORRECT CVV	CVV data is not correct Or Offline PIN authentication interrupted
83	CAN'T VERIFY PIN	Cannot verify PIN
85	CARD OK	No reason to decline
86	CAN'T VERIFY PIN	Cannot verify PIN
91	NO REPLY	Issuer or switch is unavailable
92	INVALID ROUTING	Destination not found
93	DECLINE	Violation, cannot complete
94	DUPLICATE TRANS	Unable to locate, no match
96	SYSTEM ERROR	System malfunction
100	ACTIVATED	POS device authentication successful
101	NOT ACTIVATED	POS device authentication not successful
102	POS ID ERROR	Could not validate HC POS ID<A1>
103	DEACTIVATED	POS device deactivation successful
200	COMPLETED	Adjustment successful
201	NOT COMPLETED	Adjustment not successful
B1	SRCHG NOT ALLOWED	Surcharge amount not permitted on debit cards or EBT food stamps
B2	SRCHG NOT ALLOWED	Surcharge amount not supported by debit network issuer
CV	FAILURE CV CARD	Type verification error

<A83> Response code	<A85> Response text	Definition
D3	3D - SECURE CRYPTOGRAM FAILURE	Transaction failure due to missing or invalid 3D - Secure Cryptogram
E1	ENCR NOT CONFIGD	Encryption is not configured
E3	DECRYPT FAILURE	Data could not be decrypted
EA	ACCT LENGTH ERR	Verification error
EB	CHECK DIGIT ERR	Verification error
EC	CID FORMAT ERROR	Verification error
K0	TOKEN RESPONSE	Token request was processed
K1	TOKEN NOT CONFIGD	Tokenization is not configured
K3	TOKEN FAILURE	Data could not be de-tokenized
HV	FAILURE HV	Hierarchy Verification Error
M0	DOMESTIC DEBIT NOT ALLOWED	Mastercard: Canada region-issued Domestic Debit Transaction not allowed
N3	CASHBACK NOT AVL	Cashback service is not allowed
N4	DECLINE	Exceeds issuer withdrawal limit
N7	CVV2 MISMATCH	CVV2 value supplied invalid
R0	STOP RECURRING	Customer requested stop of specific recurring payment
R1	STOP ALL RECUR	Customer requested stop of all recurring payment transactions from specific merchant
R3	STOP ALL RECUR	All recurring payments have been canceled for the card number in the request
RT	SEND TXN AGAIN	Send the original transaction again. First transaction was not received and could not be reversed.

5.78 <A84> Approval Code

Length	Format	Description
6 fixed	A/N	This six-character field contains an authorization code when a transaction has been approved. If the Response Code <A83> returned indicates that the transaction is not approved, the contents of the field should be ignored. NOTE For offline requests, if less than 6 A/N characters were provided by the issuer, this field should be left justified and spaced filled to length 6.

Effective October 12, 2012, the Card Product Value in the sixth position of the approval code will no longer be supported by Visa.

Mastercard supports card product identification at the card-level for U.S. Consumers Credit products. The last position of the approval code will contain the Card Product Value.

Value	Card product	Description
B	Mastercard Enhanced Value Platform	Enhanced (Consumer) Account qualifies for Mastercard Enhanced Value Platform only
C	Core Value Small Business (MCB, MEB, and MPC)	Account qualifies for Mastercard Core Value Small Business
D	Core Value Small Business (MCB, MEB, and MPC) and Product Graduation	Account qualifies for Mastercard Core Value Small Business and Mastercard Product Graduation
E	Business World (MWB)	Account qualifies for Mastercard Small Business
F	Business World (MWB) and Product Graduation	Account qualifies for Mastercard Small Business and Mastercard Product Graduation
G	Business World Elite (MAB)	Account qualifies for Mastercard Small Business
H	Business World Elite (MAB) and Product Graduation	Account qualifies for Mastercard Small Business and Mastercard Product Graduation
J	N/A	Account qualifies for Mastercard Small Business
K	N/A	Account qualifies for Mastercard Small Business and Mastercard Product Graduation
M	Mastercard Enhanced Value Platform and Mastercard Product Graduation	Account qualifies for Mastercard Enhanced Value Platform and Mastercard Product Graduation
P	Mastercard Product Graduation Only	Account qualifies for Mastercard Product Graduation Only
S	Mastercard High Value	Account qualifies for Mastercard High Value
T	Mastercard Product Graduation/ Mastercard High Value	Account qualifies for Mastercard Product Graduation and Mastercard High Value

Value	Card product	Description
Z	N/A	Mastercard provided default value indicating that the specific cardholder account found in DE 2 (Primary Account Number [PAN]) of the transaction does not participate in Account Level Management processing for Mastercard Enhanced Value Platform or Mastercard Product Graduation

Discover and PayPal support card product identification at the card-level for U.S. Consumers Credit products. The last position of the approval code will contain the Card Product Value.

Value	Card product	Description
C	Consumer Credit	Core Card
R	Consumer Credit	Rewards Card
P	Consumer Credit	Premium Card
Q	Consumer Credit	Premium Plus Card
B	Commercial Credit Card	Commercial Credit Card
E	Commercial Credit Card	Commercial Executive Business Credit Card
Z	Unspecified Card Product	NA

5.79 <A85> Authorization Response Text

Length	Format	Description
1 to 16	A/N and spaces	This field contains a response or display text message. This message can be used by the POS device to display the authorization result. The display text must not be used to determine the nature of a response message. TSYS Acquiring Solutions translates the responses according to the Language Indicator submitted in the authorization request message. The table in the description for <A83> provides the (U.S.) English message summary.

5.80 <A86> Retrieval Reference Number (RRN)

Length	Format	Description
12 fixed	Numeric	The host capture system generates an RRN unique per transaction in the current batch and returns this new value in every auth transaction response. The RRN is the recommended value to use when performing Batch Management and Batch Inquiry use cases to ensure the correct transactions are selected.

5.81 <A87> CCI Response (Commercial Card Indicator)

If the card used is a commercial card, the TSYS Acquiring Solutions host responds with one of the following response indicators identifying the card being processed. These values will only be returned for Mastercard and Visa commercial cards.

If this field is present in the response this transaction will not be captured in the current batch without adjustment. If the transaction is adjusted before the current batch closes, then this transaction will be captured in the current batch. If the transaction is not adjusted before the current batch closes, it will be captured in the next batch with or without adjustment.

NOTE If the request contains A61 Tax Amount, A62 Tax Amount Identifier and A63 Purchase Order Number then the transaction will be captured in the current batch without adjustment.

Value	Description
B	Business card
D	Visa commerce (reserved for future use)
L	Card eligible for B2B settlement match edits
R	Corporate card
S	Purchasing card

When a response indicator has been received indicating that the approved card is a Business, Corporate or Purchasing Card, the POS application should prompt for addendum data elements (A61, A62, A63) as needed for settlement and transmit an adjustment. If the A61, A62, A63 data elements were sent in the original authorization request the POS applications does not need to prompt for or transmit an adjustment.

5.82 <A88> Card Verification Result (CID/CVC2/CVV2)

Length	Format	Description
1 fixed	A/N	This field contains the issuer returned CID/CVC2/ CVV2 Result Code. Possible value: M - CVV2/CVC2/CID - Match N - CVV2/CVC2/CID - No Match P - Not processed S - Merchant has indicated that Verification Code is not present on card U - Issuer is not certified and/or has not provided Visa encryption keys

5.83 <A89> Transaction Data Condition Code

Length	Format	Description
2 fixed	Numeric	This value indicates the Transaction Data Condition Code in the request when specified conditions are present.

Code	Description
02	Card Present-Magnetic Stripe/Discover ZIP
03	Card Present - Magnetic Stripe/Discover Zip
04	Invalid - CVV/DCVV/iCVV contains some or all blanks
05	Card Present - Contactless Chip Card
06	Card Present - Digital 1
20	Card Present - Magnetic Stripe/Discover Zip
30	Card Present –Magnetic Stripe/Discover Zip
40	Invalid - CVV/DCVV/iCVV contains some or all blanks
50	Card Present - Chip Card
60	E-commerce - Discover ProtectBuy
70	Card Not Present - Digital 2
71	Card Not Present - Digital 4
72	Card Not Present - Digital 3
74	Card Not Present - Digital (Other)
80	Card Present - Digital 2
81	Card Present - Digital 4
82	Card Present - Digital 3
84	Card Present - Digital (Other)
99	Invalid - POS Entry Mode/MCC is invalid
00	Keyed Entry/Card Not Present/E-commerce

5.84 <A90> CAVV Result Code

Length	Format	Description
2 fixed	Numeric	The CAVV Results Code indicates whether the authentication value submitted by the merchant can be validated by the Card Brand or the issuer. Possible values: Blank or not present - CAVV not present or CAVV not verified, issuer has not selected CAVV verification option 0 - CAVV could not be verified or CAVV data was not provided when expected 1 - CAVV failed verification - cardholder authentication 2 - CAVV passed verification - cardholder authentication 3 - CAVV passed verification - attempted authentication. A 3-D Secure Authentication Results Code value of 07 from the Issuer Attempts Server indicates that authentication was attempted. Issuer attempts CAVV key was used to generate the CAVV. 4 - CAVV failed verification - attempted authentication A 3-D Secure Authentication Results Code value of 07 from the Issuer Attempts Server indicates that authentication was attempted. Issuer attempts CAVV key was used to generate the CAVV. 5 - Reserved for future use 6 - Reserved for future use 7 - CAVV failed verification-attempted authentication A 3-D Secure Authentication Results Code value of 07 from Visa Attempts Service indicates that an authentication attempt was performed. Visa attempts CAVV key was used to generate the CAVV. NOTE: The Results Code value of 7 is also valid for American Express SafeKey responses, including those for non-U.S. cards.

Length	Format	Description
		8 - CAVV passed verification-attempted authentication A 3-D Secure Authentication Results Code value of 07 from Visa Attempts Service indicates that an authentication attempt was performed. Visa attempts CAVV key was used to generate the CAVV. NOTE: The Results Code value of 8 is also valid for American Express SafeKey responses, including those for non-U.S. cards. 9 - CAVV failed verification-attempted authentication A 3-D Secure Authentication Results Code value of 08 from Visa Attempts Service indicates that an authentication attempt was performed when the Issuer ACS was not available. Visa attempts CAVV key was used to generate the CAVV. NOTE: The Results Code value of 9 is also for American Express SafeKey responses, including those for non-U.S. cards. A - CAVV passed verification-attempted authentication A 3-D Secure Authentication Results Code value of 08 from Visa Attempts Service indicates that an authentication attempt was performed when the Issuer ACS was not available. Visa attempts CAVV key was used to generate the CAVV. NOTE: The Results Code value of A is also for American Express SafeKey responses, including those for non-U.S. cards. B - CAVV passed verification-attempted authentication, no liability shift Only Visa generates this code; issuers do not. C - CAVV was not verified-attempted authentication (for Visa use only) If 3-D Secure Authentication Results Code value is 07 in the CAVV and the issuer did not return a CAVV results code in the authorization response, or Tag <A90> = 0 in the response message and the CAVV encryption keys do not exist in V.I.P., V.I.P. sets the value to C in Tag <A90>.

Length	Format	Description
		D - CAVV was not verified-cardholder authentication (for Visa use only) If 3-D Secure Authentication Results Code value is 00 in the CAVV and the issuer did not return a CAVV results code in the authorization response, or Tag <A90> = 0 in the response message and the CAVV encryption keys do not exist in V.I.P., V.I.P. sets the value to D in Tag <A90>.
		U - CAVV Unchecked [AX]

5.85 <A91> Merchant Advice Code

Length	Format	Description
2 fixed	Numeric	This field contains the Merchant Advice Code returned by Mastercard MIP. Possible value: 01 - New account information 02 - Try again later 03 - Do not try again for recurring payment transaction 04 - Token requirements not fulfilled for this token type 21 - Recurring payment cancellation

5.86 <A92> Card Product Codes

Length	Format	Description
2	A/N	This field contains a code created by Visa during the authorization process. This field is conditional in the response. It will only be returned if the values are non-zero. Refer to Appendix C for values.

5.87 <A93> Card Type

Length	Format	Description
Variable	A/N and spaces	This field indicates the type of card sent in the authorization request. Possible value: • VISA • Mastercard • AMEX • DISCOVER • PAYPAL • JCB • PRIVATE LABEL • GIFT CARD • DEBIT • EBT

5.88 <A94> Requested ACI

Length	Format	Description
1 fixed	A/N	This one-character field contains the Requested Authorization Characteristics Indicator (ACI) used to identify an authorization request as potentially qualifying for CPS (Custom Payment Services) and Mastercard Merit programs. If a merchant chooses not to participate in CPS, the Requested ACI value should default to an "N." Below are the codes currently supported by both Visa and Mastercard. Possible values: N - Device is not CPS capable Y - Device is CPS capable P - CPS Capable - Manually Keyed Hotel/Lodging transaction or Preferred Customer Authorization Request for Hotel, Auto Rental, and Passenger Transport Industries I - CPS Capable - Incremental Authorization Request; Incremental authorization transactions are not supported on AMEX R - No AVS requested Visa: Meets Direct Marketing recurring payment qualification without address verification request, U.S. only. Healthcare and select developing market MCCs may submit the ACI of R to bypass AVS requirements. Mastercard: Recurring Payment; SHIP - (Service Industries Interchange Program) for telecommunications and cable/satellite MCCs

5.89 <A95> Return ACI

Length	Format	Description
1 fixed	A/N	This one-character field contains the Returned Authorization Characteristics Indicator (ACI). This value provides information concerning the transaction's CPS qualification status, and will only be sent if <A94> was included in the request. It is not recommended that the POS system attempt to interpret the meaning of this value. Possible value: A - CPS qualified B - Tokenized e-Commerce with mobile device E - CPS qualified and Card Acceptor Data was supplied in the authorization request F - CPS qualified for Visa Account Funding Transactions I - CPS qualified incremental authorization request C - CPS qualified for a self-service automated fuel dispense J - Card not present, recurring bill payment K - CPS qualified and included an address verification request in the authorization request (Unable to read magnetic stripe) M - Meets national payment service requirements with no address verification: Direct Marketing N - Not CPS qualified P - CPS qualified and accepted for Preferred Customer qualification or 3D Secure validation failure for T&E transaction R - Recurring or Installment Payments S - CPS attempted for Preferred e-Commerce (3D Secure) T - Transaction cannot participate in CPS U - CPS qualified for Preferred e-Commerce (3D Secure) V - CPS qualified and included an address verification request in the authorization request W - CPS qualified for Basic e-Commerce (Non-3D Secure) <SPACE> - If "Y" sent and transaction not qualified (VAS downgrade)

5.90 <A96> Network ID

Length	Format	Description
1 fixed	A/N	This field contains the identification code of the network on which the transaction was authorized. The Network ID must be printed on the receipt. The table below contains a listing of Network ID codes. Possible value: ! - ATH/Evertec 3 - Alaska Option 5 - Visa Check Card II 7 - Shazam 8 - Maestro A - NETS C - Culliance D - NETS E - Accel F - NYCE G - Interlink H - PULSE K - EBT L - Pulse M - STAR N - STAR Q - STAR W S - PULSE U - AFFN V - Visa Programs, Private Label, Non-Visa Auth Gateway W - STAR SE Y - NYCE Z - STAR NE

5.91 <A97> Address Verification Result Code

Length	Format	Description
1 fixed	A/N	This field contains the Address Verification Result Code. An Address Verification Result Code can provide additional information concerning the authentication of a particular transaction for which cardholder Address Verification was requested. The merchant should expect to receive a Card Validation Code Result when submitting a Dynamic Token Verification Value in Tag <A31> for Mastercard or Visa tokenized e-commerce transactions. Please refer to Tag <A124> Card Brand Tokenization for more information.

<A97> Code	Response text	Definition	U.S.	International	Card Brand
A	Address Match	The street addresses match but the postal/ZIP code does not, or the request does not include the postal/ZIP code.	X	X	V, AX, D, PP, JCB
B	Address Match	Street Addresses match; Postal Code not verified because of incompatible formats (both street address and Postal Code were sent)	X	X	V
C	Serv Unavailable	Street address and Postal Code not verified because of incompatible formats (both street address and Postal Code were sent)	X	X	V
D	Exact Match	Street Addresses and postal codes match		X	V
F	Exact Match	Street Address and Postal Code match. Applies to UK- issued cards	X	X	V
G	Ver Unavailable	Issuer is not an AVS participant, or AVS data was present in the request but issuer did not return an AVS result, or Visa performs AVS on behalf of the issuer and there was no address record on file for this account.		X	V
I	Ver Unavailable	Address information not verified		X	V, D, PP, JCB
M	Exact Match	Street Address and postal code match		X	V
N	No Match	No match. Transaction contained Postal/ZIP code only, or street address only, or postal code and street address. Also used when transaction requests AVS but sends no AVS data.	X	X	V, MC, AX, D, PP, JCB

<A97> Code	Response text	Definition	U.S.	International	Card Brand
P	ZIP Match	Postal Codes match; Postal code and street address were sent, but street address not verified due to incompatible formats.	X	X	V
R	Retry	Retry, system unable to process	X		V, MC, AX, D, PP, JCB
S	Serv Unavailable	Service not supported	X		MC, AX, D, PP, JCB
U	Ver Unavailable	No data from Issuer/Authorization platform	X		V, MC, AX, D, PP, JCB
W	ZIP Match	Nine character postal code matches; address does not	X	X (MC)	MC, D, PP, JCB
X	Exact Match	Nine character postal code and address match	X	X (MC)	MC, D, PP, JCB
Y	Exact Match	Street address and postal code match	X		V, MC, AX, D, PP, JCB
Z	ZIP Match	Postal code matches; street address does not match or street address not included in request.	X	X (V)	V, MC, AX, D, PP, JCB
1	APPROVAL	Cardholder name and ZIP match	X	N/A	AX
2	APPROVAL	Cardholder name, address, and ZIP match	X	N/A	AX
3	APPROVAL	Cardholder name and address match	X	N/A	AX
4	APPROVAL	Cardholder name matches	X	N/A	AX
5	APPROVAL	Cardholder name incorrect; ZIP matches	X	N/A	AX
6	APPROVAL	Cardholder name incorrect; address and ZIP match	X	N/A	AX
7	APPROVAL	Cardholder name incorrect; address matches	X	N/A	AX
8	NO MATCH	Cardholder name, address, and ZIP do not match	X	N/A	AX

5.92 <A98> Delayed Capture Indicator

Length	Format	Description
3 fixed	Alpha	The value of this field must be “DLY”. This field indicates that this transaction should not be captured in the current batch without adjustment. • If the transaction is adjusted before the current batch closes, then this transaction will be captured in the current batch. • If the transaction is not adjusted before the current batch closes, it will be captured in the next batch with or without adjustment. NOTE Merchants must be established as Delay Capture participants to be able to utilize this tag. Selecting the Capture Delay value "Delay 1 Batch" in the boarding system will flag a merchant's terminal as a delay capture participant.

5.93 <A99> Multiple Clearing Sequence Number

Length	Format	Description
1 to 2	Numeric	Card Not Present merchants using installment payments should use this field to indicate which payment number this particular payment represents in the total count of installment payments to be made.

EXAMPLE

Record	<A99> Sequence Number	<A100> Sequence Count
First Payment of 3	1	3
Second Payment of 3	2	3
Third Payment of 3	3	3

5.94 <A100> Multiple Clearing Sequence Count

Length	Format	Description
1 to 2	Numeric	This two-character numeric field contains the total number of transactions associated with this sequence of transactions. This is a required field in the last transaction of the sequence. For all other transactions in a sequence, this field is not required. Card Not Present merchants should utilize this field when processing installment payments. In this case, this field identifies the total number of installment payments to be made in the series.

5.95 <A101> Amex Corporate Purchasing Card Group

Group <A101> contains subfields used to transmit Corporate Purchasing Card (Level II) data for applicable Amex transactions.

5.95.1 <A101><B1> Requester Name

Length	Format	Description
1 to 38	A/N	This subfield contains the name of the requester, if the Card member is not the person that ordered or purchased the product or service.

5.95.2 <A101><B2> Shipped to ZIP

Length	Format	Description
6	A/N	This field contains the ZIP code to which the commodity was shipped. If the destination ZIP code is not available, provide the ZIP code of the Merchant where the purchase took place.

5.95.3 <A101><B3> Charge Descriptor 1

Length	Format	Description
1 to 40	A/N	This is a free text field provided for description of charges.

5.95.4 <A101><B4> Charge Descriptor 2

Length	Format	Description
1 to 40	A/N	This is a free text field provided for description of charges.

5.95.5 <A101><B5> Charge Descriptor 3

Length	Format	Description
1 to 40	A/N	This is a free text field provided for description of charges.

5.95.6 <A101><B6> Charge Descriptor 4

Length	Format	Description
1 to 40	A/N	This is a free text field provided for description of charges.

5.96 <A102> Amex CAPN Retail Department Name

Length	Format	Description
1 to 40	A/N	This field is used to identify the name of the department within the location where the transaction took place.

5.97 <A103> Amex CAPN Retail Group

Group <A103> contains subfields used to provide Amex CAPN related information for Amex Retail transactions that apply. This group and its subfields can be repeated up to 6 times.

5.97.1 <A103><B1> Retail Item Amount

Length	Format	Description
1 to 12	Numeric	This contains the unit cost that corresponds to the retail item described in this record.

5.97.2 <A103><B2> Retail Item Description

Length	Format	Description
1 to 19	A/N	This contains a description of the retail item that was purchased.

5.97.3 <A103><B3> Retail Item Quantity

Length	Format	Description
1 to 3	Numeric	This contains the quantity purchased that corresponds to the retail item described in this line item record.

5.98 <A104> Level III Enhanced Data Group

5.98.1 <A104><B1> National Tax Amount

Length	Format	Description
1 to 12	Numeric	National Tax included in the transaction amount.

5.98.2 <A104><B2> Freight Amount

Length	Format	Description
1-12	Numeric	Freight or shipping portion of the total transaction amount.

5.98.3 <A104><B3> Duty Amount

Length	Format	Description
1-12	Numeric	Fee amount associated with the import of the purchased goods.

5.98.4 <A104><B4> Ship from Postal/ZIP Code

Length	Format	Description
5-10	A/N	Postal/ZIP code of the address from where the purchased goods are being shipped.

5.98.5 <A104><B5> Destination Postal/ZIP Code

Length	Format	Description
5-10	A/N	Postal/ZIP code of the address where purchased goods will be delivered.

5.98.6 <A104><B6> Destination Country Code

Length	Format	Description
3	A/N	Code of the country where the goods are being shipped.

5.98.7 <A104><B7> Line Item Count

Length	Format	Description
1-3	Numeric	Total number of line item detail records associated with the transaction to be captured. Maximum value for Visa is 998. Maximum value for Mastercard is 98.

5.98.8 <A104><B8> Merchant VAT Registration Number

Length	Format	Description
1-20	A/N	Government assigned tax identification number of the merchant from whom goods or services were purchased.

5.98.9 <A104><B9> Customer VAT Registration Number

Length	Format	Description
1-13	A/N	Value Added Tax registration number supplied by the Commercial Card cardholder.

5.98.10 <A104><B10>Summary Commodity Code

Length	Format	Description
1-4	A/N	This field contains an international description code of the overall goods or services being supplied. The acquirer bank or processor should provide the merchant an updated listing of currently defined codes.

5.98.11 <A104><B11>Discount Amount

Length	Format	Description
1-12	Numeric	Total amount of any discount applied by the merchant. This field should not be included if the total amount reflects a discount price or if no discount applies.

5.98.12 <A104><B12>Unique VAT Invoice Reference Number

Length	Format	Description
1-15	A/N	Invoice number that is associated with the VAT invoice.

5.98.13 <A104><B13>Order Date

Length	Format	Description
6	Numeric	Purchase order date. YYMMDD format

5.98.14 <A104><B14>VAT/TAX Amount (Frt/Ship)

Length	Format	Description
1-12	Numeric	Amount of the Value Added Tax for Visa. For Visa, two decimal places are implied.

5.98.15 <A104><B15>VAT/TAX Rate (Frt/Ship)

Length	Format	Description
1-4	Numeric	Decimal amount representing the Value Added Tax applied for Visa. For Visa, two decimal places are implied.

5.98.16 <A104><B16>Alternate Tax Amount

Length	Format	Description
1-9	Numeric	Second tax amount in countries where more than one type of tax can be applied to purchases.

5.98.17 <A104><B17>Invoice Level Discount Treatment

Indicates how the merchant is managing discounts.

Length	Format	Description
1	N	Possible values: 0 - No invoice level discount provided 1 - Tax was calculated on the post-discount invoice total 2 - Tax was calculated on the pre-discount invoice total

5.98.18 <A104><B18>Tax Treatments

Indicates how the merchant is handling taxes.

Length	Format	Description
1	N	Possible values: 0 - NLL-Net prices with tax calculated at line item level. 1 - NIL-Net prices with tax calculated at invoice level 2 - GLL-Gross prices given with tax information provided at line item level 3 - GIL-Gross prices given with tax information provided at invoice level 4 - NON-No tax applies (small supplier) on the invoice for the transaction

5.99 <A105> Level III Line Item Details Group

5.99.1 <A105><B1> Item Commodity Code

Length	Format	Description
1-12	A/N	International description code indicating the individual good or service being supplied. NOTE: The acquirer's bank will provide the merchant an updated list of currently defined codes. It is recommended to use a National or International list of standardized codes.

5.99.2 <A105><B2> Item Description

Length	Format	Description
1-35	A/N/Space	Description of the item(s) being supplied. Not all spaces; do not send all spaces.

5.99.3 <A105><B3> Item Product Code

Length	Format	Description
1-12	A/N	Identifier assigned by the merchant.

5.99.4 <A105><B4> Item Quantity

Length	Format	Description
1-12	A/N	Quantity of the non-fuel item(s) being purchased. Implied four decimal places.

5.99.5 <A105><B5> Item Unit of Measure

Length	Format	Description
1-12	A/N	Identifier assigned by the merchant.

5.99.6 <A105><B6> Item Unit Cost

Length	Format	Description
1-12	Numeric	Unit cost of the non-fuel item purchased. Four decimal places are implied.

5.99.7 <A105><B7> Item Discount Amount

Length	Format	Description
1-12	Numeric	Total discount amount applied against the line item total. Field should be all zeros if the unit cost reflects a discount price or if no line item discount applies.

5.99.8 <A105><B8> Item Total Amount

Length	Format	Description
1-12	Numeric	Purchase amount associated with the specific line item detail record.

5.99.9 <A105><B9> Item Alternate Tax Identifier

Length	Format	Description
1-15	A/N	Tax identification number of the merchant that reported the alternate tax amount.

5.99.10 <A105><B10> Item VAT/TAX Rate

Length	Format	Description
1-4	Numeric	Decimal amount representing the Value Added Tax applied. For Visa, two decimal places are implied.

5.99.11 <A105><B11> Item Tax Type

Length	Format	Description
4	Numeric	Type of value-added taxes that are being used.

5.99.12 <A105><B12> Item VAT/Tax Amount

Length	Format	Description
1-12	Numeric	Amount of value-added tax. For Visa, two decimal places are implied.

5.99.13 <A105><B13> Net/Gross Indicator

Length	Format	Description
1	A/N	Value for the method that indicates whether the extended item amount included the tax.

Indicator value	Description
Y	Item amount does not include tax amount.
N	Item amount includes tax amount.

5.99.14 <A105><B14> Extended Item Amount

Length	Format	Description
1-9	Numeric	Total amount of the individual item, typically calculated as price multiplied by quantity. Two decimal digits are implied.

5.99.15 <A105><B15> Debit/Credit Indicator

Length	Format	Description
1	Numeric	Indicates the sign of the "Extended Item Amount."
Indicator value		Description
C		Extended item amount is credit.
D		Extended item amount is debit.

5.99.16 <A105><B16>Item Discount Indicator

Length	Format	Description
1	A/N	Indicates whether a discount was applied to the purchase price of the item purchased.
Indicator value		Description
Y		Amount is discounted.
N		Amount is not discounted.

5.99.17 <A105><B17>Item Discount Rate

Length	Format	Description
1-5	Numeric	Discount rate for the line item.

5.99.18 <A105><B18>Quantity Exponent Indicator

Length	Format	Description
1	A/N	Number of decimal places in the Item Quantity field.

5.99.19 <A105><B19> Discount Rate Exponent

Length	Format	Description
1	A/N	If the Item Discount Indicator is Y, this field must contain the number of decimal places in the Discount Amount field in the range 0 – 5.

5.99.20 <A105><B20> Line Item Level Discount Treatment

Indicates how the merchant is handling the discount at the line item level. Visa will insert 0 (zero) if an invalid value is submitted.

Length	Format	Description
1	N	Possible values: 0 - No line level discount provided 1 - Tax was calculated on the post-discount line item total 2 - Tax was calculated on the pre-discount line item total

5.100 <A106> Encrypted Transaction

When including encrypted data in lieu of Track or manually entered card number data, this needs to be identified in specific ways. Including encrypted data requires addition of <A107> Encrypted Track 1, <A108> Encrypted Track 2, or <A109> Encrypted Manual Entry depending on what is encrypted. <A106> Encrypted Transaction is also required to include the Encryption Type and Encryption Transmission Block so that the data may be decrypted. Refer to details later in this document.

5.100.1 <A106><B1>Encryption Type

This describes the type of encryption used in the transaction.

Length	Format	Description
1	A/N	Encryption Type

This indicates a listing of encryption types accepted.

Value	Description
V	Voltage TEP2 Encryption

5.100.2 <A106><B2>Encryption Transmission Block (ETB)

This is the Encryption Transmission Block (ETB) TSYS uses to decrypt encrypted data. The ETB must be Base-64 encoded before transmission to TSYS Acquiring Solutions.

Length	Format	Description
250-300	A/N, "+", "/", "=",	Encryption Transmission Block (ETB)

5.104.2 <A110><B2> Expiration Date

Length	Format	Description
4 fixed	Numeric	This four-character numeric field contains the card/token expiration date in the form MMYT (month-month-year-year).

5.105 <A111> Chip Card Data

This field contains chip card data that is passed in a TLV format in the request and response messages.

TLV format Definition

This field contains a string of chip card datasets in a TLV format. Each dataset is composed of the following three data elements (represented in HEX format):

1. Chip Card Tag: This value contains the tag of the chip dataset.
2. Chip Card Data Length: This value indicates how many bytes of data constitute the value.
3. Value: This is the actual chip card data.

The TLV chip card datasets are preceded by a dataset length to indicate the length of the entire data string.

Chip card data may be included when <A10> Transaction Code is one of the following: Purchase, Cash advance, Quasi cash, Return, Balance Inquiry, and Authorization only.

Length	Format	Description
2-510 Characters	Hex (even number of characters)	Example of the chip card datasets in a TLV format: 1-byte dataset indicator (01 for Visa 00 for all other card brands) 2-byte dataset length (entire data string) TLV TLV

The following list of EMV tags are mandatory and must be sent on any transaction that contains Tag <A111>. If either or both of the tags are missing from tag <A111> the transaction will be rejected.

Tag	Tag name	Description
DF79	Kernel Version Number	This field contains the version number of the contact or contactless kernel used to process the chip data in the transaction. The kernel version can be found on the EMVco issued contact L2 kernel certification letter and the card brand issued contactless L2 kernel certification letters. Only A/N characters should be used. If a kernel version number is not listed on the card brand issued contactless L2 kernel certification letter, then the default value of "99999999" must be sent. (variable 1 - 32) Example: If the kernel version is "3.2.4B" then "324B" is all that should be sent.
DF78	Device Serial Number	The manufacturer's unique serial number of the device that interacts with the chip card. (variable 1 - 20)

Appendix B contains a list of EMV tags that are the primary tags used in transactions. Not all tags are used in every transaction. The tags being sent should be as close to the list in Appendix B as possible and in the order listed.

5.106 <A112> Card Acceptor Data

This data is required for all American Express transactions.

The Seller ID should be sent in Field <A50> Amex Seller/Vendor Code.

The DBA/common name of the Seller should be sent in Field <A48> Amex Merchant/DBA Name.

If the information is for an Oil Co. CAT, the merchant-assigned station identifier should be sent in Field <A49> Amex Station Location Code and Field <A50> should be left blank.

5.106.1 <A112><B1> Seller Telephone Number

Length	Format	Description
10	Numeric	This is the telephone number of the Seller. This field is required for Amex transactions that are submitted by an Aggregator / Payment Service Provider.

5.106.2 <A112><B2> Seller Email Address

Length	Format	Description
1-20	A/N and special characters	This is the email address of the Seller. This field is required for Amex transactions that are submitted by an Aggregator / Payment Service Provider.

5.106.3 <A112><B3> Seller Street Address

Length	Format	Description
1-20	A/N and special characters	This optional tag contains the Street Address of the Seller. Spaces should be included and the name should be truncated if necessary, rather than abbreviated. This field is required for Amex transactions that are submitted by an Aggregator / Payment Service Provider. Do not include this data for Oil Co. CATs

5.106.4 <A112><B4> Seller Postal Code

Length	Format	Description
5-10	A/N and special characters	This is the Postal Code of the Seller. The field should be left justified. This field is required for Amex transactions that are submitted by an Aggregator / Payment Service Provider or an Oil Co. CAT.

5.106.5 <A112><B5> Seller City

Length	Format	Description
1-13	A/N and special characters	This is the City of the Seller. This field is required for Amex transactions that are submitted by an Aggregator / Payment Service Provider. If operating in a CNP environment, this field should be populated with the location city associated with the Service Establishment number for the market. If the name is more than 13 characters, use proper and meaningful abbreviation(s) when possible. Do not truncate.

5.106.6 <A112><B6> Seller Region Code

Length	Format	Description
1-3	A/N and special characters	This is the Region, State, or Province Code of the Seller. It must correspond to the Country Code provided. This field is required for Amex transactions that are submitted by an Aggregator / Payment Service Provider.

5.106.7 <A112><B7> Seller Country Code

Length	Format	Description
1-3	A/N and special characters	This is the Country Code of the Seller. This field is required for Amex transactions that are submitted by an Aggregator / Payment Service Provider.

5.106.8 <A112><B8> PSP/Aggregator Name

Length	Format	Description
Up to 20	A/N and special characters	This is the name of the Payment Service Provider/Aggregator. It should contain NO spaces. This field is required for Amex transactions that are submitted by an Aggregator / Payment Service Provider.

5.107 <A113> Message Reason Code

This field has multiple usages. Refer to individual sections below. The values are applicable only for Visa transactions, but they can be sent for any card brand.

Reversals

This field is required in all reversal and reversal advice messages. It is not used in responses.

Code	Definition
2501	Transaction voided by customer
2502	Transaction has not completed (Request timed out or POS device malfunctioned)
2503	No confirmation from the point of sale
2504	POS partial reversal
2516	Premature chip card removal (after online request sent – before response received)
2517	Chip declined transaction after online issuer approved

*For TSYS internal use only, note that for Amex transactions, message reason codes 2502, 2516, and 2517 are system-generated reversals and not merchant-initiated reversals.

Adjustments

This field must be used for adjustments.

Code	Definition
2001	Transaction voided by customer
2002	Wrong amount
2003	Partial return
2007	Debit or Credit adjustment – no previous transaction NOTE: This message reason is valid only through Networks 2 (Visa) and 3 (Interlink)
2009	Debit or Credit adjustment – for previous transaction
2104	Acquirer Authorization Advice This value is required in AFD advice messages. It cannot be used with MCCs 6010 (Manual Cash) or 6011 (Automated Cash).
2140	Account Funding transaction – debit or credit adjustment

EBT

Code	Definition
5201	EBT Voucher

Deferred Authorizations

This field is required to identify deferred Visa credit and debit authorization requests. If the merchant's Point-of-Interaction (POI) authorization system is offline and cannot process transactions, the merchant often completes the transaction with the cardholder and will then defer the authorization until their POI system is back online.

Visa rules require the merchant submitting deferred authorizations to do so within the following times:

- Four days (96 hours) of the transaction date for the following merchant category codes (MCCs):
 - 4111 (Local and suburban commuter passenger transportation, including ferries)
 - 4112 (Passenger railways)
 - 4131 (Bus lines)
- One day (24 hours) of the transaction date for all other MCCs

Code	Definition
5206	Deferred Authorization

Visa Merchant Initiated Transactions

The Message Reason Code field will be used to identify Merchant Initiated Transactions [MIT] for Visa. A MIT is any transaction that relates to the previous consumer-initiated transaction but is conducted without the consumer being present, and without any cardholder validation performed.

MIT	Definition
Incremental	This is a continuation of the consumer purchase where the original approved amount can be modified to reflect the scope of the consumer and merchant agreement. Incremental authorization transactions are common in the hotel and car rental industries. Incremental authorization transactions are not supported on Amex.
Resubmission	This is an event that occurs when the original purchase occurred, but the merchant was not able to obtain authorization at the time the goods or services were provided. A resubmission is only valid when the original authorization was declined for insufficient funds. There are a limited number of merchant categories approved to utilize this type of MIT, and resubmission is only valid for a limited number of days after the consumer purchase.
Delayed Charges	A delayed charge is an account charge associated with an agreement between a cardholder and a merchant for services rendered. Delayed charges are typical in hotel and car rental industries.
Reauthorization	A reauthorization is a purchase made after the original purchase and can reflect a number of specific conditions. A split shipment is the most common type of reauthorization. A split shipment occurs when the goods are not available for shipment at the time of the consumer's purchase. A separate authorization is conducted to ensure that the consumer funds are available when the goods become available to ship. A reauthorization is also used when an estimated amount is presented, and the merchant wants to authorize the final amount.
No Show	A no show transaction is where the merchant is able to charge for service that the consumer entered into an agreement to purchase, but did not meet the terms of the agreement, typically used by hotels for a single night's stay.

Code	Definition
3900	Incremental Authorization
3901	Resubmission
3902	Delayed Charges
3903	Reauthorization
3904	No Show

5.108 <A114> Alternate Account ID 1

Length	Format	Description
1-28	Numeric	This field contains alternate account number information. For Mastercard transit and Mastercard mapping service transactions, this field will contain the Primary Account Number (PAN). It may be used in conjunction with the Mastercard mapping service in tag <A116>.

5.109 <A115> Mastercard Payment Indicators

Mastercard provides PayPass processing services to assist issuers processing PayPass transactions.

5.109.1 <A115><B1> Domain Server

This conditional one-character field indicates the Service Manager or program initiator of the Mastercard Mobile Remote Payments Program. It is applicable for Auth Request and Auth Advice transactions.

Domain value	Domain server
0	No domain
1	Issuer domain
2	Acquirer domain

5.109.2 <A115><B2> Device Type

This conditional two-character field provides information about the device type used to identify mobile-initiated (m-commerce) or other EMV Contact, Magnetic stripe Contactless, or M/Chip Contactless transactions. This is applicable in authorization request, advice, and reversal request transactions.

This field is also known as Form Factor. Where practicable, the new codes with values 20-33 must be used. The mobile device type is obtained from the card in contactless tag 9F6E, bytes five and six. This tag can be present on both Mastercard contactless EMV and Mastercard contactless MSR cards. The value from bytes five and six should be converted from hex to ASCII and passed in the mobile device type in Tag <115><B2>.

Device type value	Device description	Example
00	Card	
01	Mobile Network Operator (MNO) controlled removable secure element (SIM or UICC) personalized for use with a mobile phone or smartphone	
02	Key Fob	
03	Watch using a contactless chip or a fixed (non-removable) secure element not controlled by the MNO	
04	Mobile Tag	
05	Wristband	
06	Mobile Phone Case or Sleeve	
07	Mobile Phone or Smartphone with a fixed (non-removable) secure element controlled by the MNO (such as CDMA)	
08	Removable secure element not controlled by the MNO, for example, memory card personalized for use with a mobile phone or smartphone	
09	Mobile Phone or Smartphone with a fixed (non-removable) secure element not controlled by the MNO	
10	MNO controlled removable secure element (SIM or UICC) personalized for use with a tablet or e-book	
11	Tablet or E-Book with a fixed (non-removable) secure element controlled by the MNO	
12	Removable secure element not controlled by the MNO (such as a memory card personalized for use with a tablet or e-book)	
13	Tablet or E-Book with a fixed (non-removable) secure element not controlled by the MNO	

Device type value	Device description	Example
14	Mobile Phone or Smartphone with a payment application running in a host processor	
15	Tablet or E-Book with a payment application running in a host processor	
16	Mobile Phone or Smartphone with a payment application running in the TEE of a host processor	
17	Tablet or E-Book with a payment application running in the TEE of a host processor	
18	Watch with a payment application running in the TEE of a host processor	
19	Watch with a payment application running in a host processor	
20	Card	
21	Phone	Mobile phone
22	Tablet/e-reader	Tablet computer or e-reader
23	Watch/Wristband	Watch or wristband, including a fitness band, smart strap, disposable band, watch add-on, and security/ID band
24	Sticker	
25	PC	PC or laptop
26	Device Peripheral	Mobile phone case or sleeve
27	Tag	Key fob or mobile tag
28	Jewelry	Ring, bracelet, necklace, and cuff links
29	Fashion Accessory	Handbag, bag charm, and glasses
30	Garment	Dress
31	Domestic Appliance	Refrigerator, washing machine
32	Vehicle	Vehicle, including vehicle attached devices
33	Media/Gaming Device	Media or gaming device, including a set top box, media player, and television
34-99	These values reserved for future form factors. Any value in this range may occur within form factors and transaction data without prior notice.	

5.110 <A116> Mastercard Mapping Service

Mastercard provides a service to assist merchants and issuers that process transit transportation transactions. This optional transaction mapping service returns specific data to the POS device that is required to settle transit transactions. This tag may be included in some Mastercard authorization response messages, and may be used in conjunction with tag <A114> Alternate Account ID 1.

5.110.1 <A116><B1> Mapped PAN indicator

This conditional one-character field indicates the type of mapping account used and comes with the Mastercard mapping service.

5.110.2 <A116><B2> Mapped Product Code

This conditional three-character field represents the product code for the mapped account used and optionally comes with the Mastercard mapping service.

5.110.3 <A116><B3> Mapped Expiry Date

This conditional four-character numeric field contains the mapped card expiration date in the form MMYYY.

5.111 <A117> TSYS Acquiring Solutions Gateway ID

Length	Format	Description
10 fixed	A/N	The Gateway ID provides a unique value to identify a Gateway or Aggregator. This number is provided by TSYS Acquiring Solutions during the certification process. NOTE This field is optional.

5.112 <A118> Authorization Only Indicator

Length	Format	Description
9 fixed	A/N and special characters	Indicates the transaction is an authorization only. This must contain the value 'AUTH_ONLY'.

5.113 <A119> Visa Validation Code

Length	Format	Description
4 fixed	A/N	This field contains specific information generated by Visa. The POS device should not attempt to interpret the meaning of any data appearing in this field. Data returned in this field is recorded and submitted as part of the data capture settlement format.

5.114 <A120> Authorization Source Code

Length	Format	Description
1 fixed	A/N	This field contains a one-character code indicating the source of the authorization code. The received code must be stored and resubmitted in the data capture settlement record.
Code	Description	
6	Off line approval, POS device generated	
7	Acquirer approval	
9	No authorization source code; non-authorized transactions	
B	Response provided by Visa Transaction Advisor Service	
E	Off line approval: authorization code manually keyed	
F	CAFIS-generated response; post-authorization (Japan only)	
G	Issuer Approval, Post-Authorization (Japan only)	
H	Exceeds acquirer settlement risk exposure cap	
N	No authorization required: Below floor limit	
L	Late clearing – authorization previously obtained (clearing beyond 30 days)	
V	Authorization obtained via VisaNet (Issuer or STIP response)	
Space	Default value, if acquirer cannot define it	

5.115 <A121> Spend Qualified Indicator

Length	Format	Description
1	A/N	This is optionally sent by Visa on an authorization response message. The merchant's interchange rate may be determined by this value. Possible values: B - Base spend assessment has been met. N - Spend qualification threshold has not been met. Q - Spend qualification threshold has been met.

5.116 <A122> Goods Sold

Length	Format	Description
4	A/N	This is a code that allows the merchant to send an indicator for high-value merchandise such as gift cards, computers and other electronics that can be resold for cash. Possible values: 1000 - Gift Card

5.117 <A123> Gateway Administrative Message

Length	Format	Description
1-25	A/N	This is an optional variable length field. This field, when sent in the request, will be echoed in the response message unaltered. Other than inclusion in the response message, TSYS Acquiring Solutions does nothing with this data. NOTE : If this tag is received without data, it will not be included in the response message. In addition, if this field exceeds 25 characters, the transaction will be rejected.

5.118 <A124> Card Brand Tokenization

This tag contains sub tags that are used for Card Brand Tokenization. Please refer to the scenarios below to understand usage of the new fields. An overview of tokenization can be found in [Section 4.8 Using encryption and tokens](#).

Card Brand Tokenization may be used on all Visa, Mastercard, American Express and Discover token purchase transactions. This is recommended if the merchant uses or pays for card brand tokenization.

For card brand tokenization, the Token is a surrogate value for a PAN that is consistent with ISO 8583 message requirements and is a 13 to 19-digit numeric value that passes basic validation rules of an account number, including the LUHN check. Tokens are generated within a BIN range that has been designated as a token BIN range.

The following abbreviations are used in this description.

Abbreviation	Full Description
M	Mandatory - For a request, the field is required. For a response, the field will be present.
C	Conditional - For a request, the field will be required in some situations. For a response, the field may be present. Consult the field description for complete details.
O	Optional - The field may or may not be present. Consult the field description for complete details.
MC	Mastercard
V	Visa
D	Discover

Card Brand Token Purchase

This scenario is used to authorize a transaction using a card brand Token instead of a PAN. Mastercard, Visa, American Express and Discover offer this use case.

For this scenario, use an ordinary Transaction Code as used in any purchase transaction. The Token will be in tag <A25>, and the Token Expiration Date will be in tag <A26>.

Recurring and partial shipment purchase transactions that use a card brand token must be clearly identified. Please refer to tag <A21><B1> for more information.

NOTE Effective October 2017, the functions of Token Request and Token Maintenance for Card on File Tokens previously available for Mastercard were decommissioned. Mastercard Card on File Token users should note that this token type remains available for use but can be requested and maintained only through an API available to participating token requestors as part of the token services implementation. Contact Mastercard for more information.

5.118.1 DES Enhancement to support E-Commerce Transactions

The Dynamic Token Verification Value (DTVV) exists for merchants that process tokenized e-commerce transactions. The DTVV is a 3-digit value that is generated by Visa Checkout, and merchants will populate this value in existing Tag [≤A31>](#) (Verification Code CVC2). When DTVV is submitted, merchants should expect Visa to send the DTVV Verification Result Code in existing Tag [≤A97>](#) (Address Verification Result Code).

Merchants must be aware of the following rules for Token Authentication Verification Value (TAVV) and Dynamic Token Verification Value (DTVV) cryptograms for cardholder initiated token based card-on-file and e-commerce transactions, including in-app e-commerce. TAVV and DTVV cryptograms are generated by an approved entity, such as Visa Token Service, then passed to the merchant via the token requestor. The merchant must then submit an unchanged TAVV or DTVV in the authorization request.

- The TAVV and DTVV cryptograms must be new and unique for each authorization request.
- The TAVV and DTVV cryptograms are one-time use within the authorization window.
- The TAVV and DTVV must not be stored beyond the authorization request.
- The TAVV, DTVV, and electronic commerce indicator (ECI) values provided by the token requestor must be unchanged when submitted in the authorization request message.
- A resubmitted or stale TAVV or DTVV may result in a declined authorization request.
- Merchant-initiated transactions must follow the merchant-initiated transaction framework.

Mastercard enhanced MDES processing to help simplify the acceptance of Digital Secure Remote Payments (DSRP) Universal Cardholder Authentication Field (UCAF) transaction processing. This allows merchants that are not yet upgraded to support DSRP transactions with UCAF data or full EMV data to submit transactions containing MDES Secure Element tokens or cloud-based tokens when initiated from a Masterpass by Mastercard wallet.

The authentication option was introduced using two existing fields, a Dynamic Token Verification Value in Tag [≤A31>](#) (Verification Code CVC2) and Dynamic Expiration Date in Tag [≤A26>](#) (Manually Entered Expiry Date), enabling the transaction to process as a traditional e-commerce payment. Merchants should expect to receive Tag [≤A97>](#) (Address Verification Result Code) any time the Dynamic Token Verification Value is submitted.

Card Brand Token Purchase-Request

ID	Field Name	MC	V	D	AX
<A124><B1>	Card Brand Token Requestor ID	C	C	N/A	C
<A26>	Dynamic Expiration Date	C	N/A	N/A	N/A
<A31>	Dynamic Token Verification Value	C	C	N/A	N/A

Card Brand Token Purchase-Response

ID	Field Name	MC	V	D	AX
<A124><B2>	Card Brand Token Assurance Level	O	C	M	N/A
<A124><B3>	Card Brand Token PAN Last 4 Digits	O	C	M	M
<A124><B4>	Card Brand Token Account Range Status	N/A	C	N/A	N/A

NOTE Transit merchants will see additional information in tags <A114> and <A116><B3> for Mastercard transactions in this scenario.

5.118.2 <A124><B1> Card Brand Token Requestor ID

Length	Format	Description
11	AN	This field contains eleven digits that uniquely identify the pairing of token requestor with the token domain. It is assigned by the token service provider and is unique within the token vault. The Card Brand Token Requestor ID must be included in a reversal message if it was present in the original transaction.

5.118.3 <A124><B2> Card Brand Token Assurance Level

Length	Format	Description
2 fixed	AN	Defined by the token service provider, this optional Visa or Mastercard value indicates the assigned confidence level of the token-to-PAN/cardholder binding.

5.118.4 <A124><B3> Card Brand Token PAN Last Four Digits

Length	Format	Description
4	N	This field contains four characters that represent the last four digits of the actual cardholder PAN. This can be used to print on the paper receipt for cardholder reference.

5.118.5 <A124><B4> Card Brand Token Account Range Status

Length	Format	Description
1	AN	This value indicates the regulatory status of the actual card number for which the token represents. Valid values are shown below. Possible values: Space - Blank/no value R - Regulated N - Non-Regulated

5.119 <A125> Mastercard Masterpass

5.119.1 <A125><B1> Mastercard Wallet Identifier

Length	Format	Description														
3	A/N	This value provides information about transactions initiated through the Masterpass Online platform or through the Mastercard Digital Enablement Service (MDES) and indicates who the Wallet Provider was for the transaction. For MDES, Mastercard inserts the Wallet Identifier that is in the system, when available. For Masterpass, Mastercard passes the Wallet Identifier as sent in from the Point of Sale. NOTE The following values apply to both Masterpass and MDES transactions. <table><tr><th>Value</th><th>Description</th></tr><tr><td>101 -</td><td>Masterpass by Mastercard</td></tr></table> NOTE The following values only apply to MDES transactions. <table><tr><th>Value</th><th>Description</th></tr><tr><td>103 -</td><td>Apple Pay</td></tr><tr><td>216 -</td><td>Google Pay</td></tr><tr><td>217 -</td><td>Samsung Pay</td></tr><tr><td>327 -</td><td>Merchant tokenization program</td></tr></table>	Value	Description	101 -	Masterpass by Mastercard	Value	Description	103 -	Apple Pay	216 -	Google Pay	217 -	Samsung Pay	327 -	Merchant tokenization program
Value	Description															
101 -	Masterpass by Mastercard															
Value	Description															
103 -	Apple Pay															
216 -	Google Pay															
217 -	Samsung Pay															
327 -	Merchant tokenization program															

5.120 <A126> ISO Identifier

Length	Format	Description
11	NUM	This value, assigned by Mastercard, is assigned during registration via Mastercard Connect for a Service Provider designated as an “Independent Sales Organization”. This value must be provided on all Mastercard transactions where the merchant has a relationship with an ISO, and should be right justified with leading zeroes. For example, if the Identifier is ‘123456’, the value would be ‘00000123456’.

5.121 <A127> Merchant Name/Location Detail

Use this table to help with knowing which fields are required (R), optional (O), and illegal (X) for each usage described below.

Tag	Usage 1	Usage 2	Usage 3
B1- Payment Facilitator Identifier	R	R	X
B2 - Location Detail Name	R	R	O
B3 - Sub-Merchant ID	R	X	X
B4 - Location Detail City	R	X	O
B5 - Location Detail State/Province Code	R	X	O
B6 - Location Detail Country Code	R	X	O
B7 - Location Detail Postal Code	R	X	O
B8 - Foreign Retailer Indicator (Marketplace only)	O	O	O

Usage 1 – Payment Facilitators

Except as noted in each sub tag, this tag must be provided on all Visa, Mastercard, and Discover credit and/or debit transactions where the merchant has a relationship with a Payment Facilitator. If the tag is present in the message, all sub tags must be present and populated (cannot be all spaces) with the exception of sub tag B8, which is only required for Marketplaces.

Usage 2 – Visa Marketplaces

Visa classifies an entity as a marketplace if it handles payments on behalf of retailers through an online marketplace that brings together multiple buyers and retailers. This tag must be provided on all Visa credit and/or debit transactions for Marketplaces. Only the below fields apply for this usage, as noted in the descriptions:

- B1- Payment Facilitator Identifier
- B2 - Location Detail Name
- B8 - Foreign Retailer Indicator

Usage 3 – Dynamic Merchant Descriptor

This tag can be provided for any card brand for auth and capture. It is recommended that if any of these data fields are sent, all fields be sent. However, any field can be sent by itself. Only the below fields apply for this usage, as noted in the descriptions:

- B2 - Location Detail Name
- B4 - Location Detail City
- B5 - Location Detail State/Province Cod
- B6 - Location Detail Country Code
- B7 - Location Detail Postal Code

Table 5.13 Usage for Payment Facilitator Fields

Tag ID	Field Name	Visa	Mastercard	Discover
A126	ISO ID	N/A	Yes	N/A
A127.B1	Payment Facilitator Identifier	Yes	Yes	N/A
A127.B2	Location Detail Name	Yes	Yes	Yes
A127.B3	Sub-Merchant ID	Yes	Yes	N/A
A127.B4	Location Detail City	Yes	Yes	Yes
A127.B5	Location Detail State/Province Code	Yes	Yes	Yes
A127.B6	Location Detail Country Code	Yes	Yes	Yes
A127.B7	Location Detail Postal Code	Yes	Yes	Yes
A127.B8	Foreign Retailer Indicator (Marketplace only)	Yes	N/A	N/A

5.121.1 <A127><B1> Payment Facilitator Identifier

Length	Format	Description
11	NUM	Usage 1 - Payment Facilitator This 11-digit value is assigned to the Service Provider during registration with Mastercard or Visa. For Visa and Mastercard transactions, the value must be right justified and padded to 11 digits with leading zeros. For example, if the Identifier is '123456', the value would be '00000123456'. Discover: The value is not included; it is retrieved by the software from the merchant's profile. Usage 2 (Visa Marketplaces): This 11-digit value is assigned to the Service Provider during registration with Visa as the Marketplace Identifier. For Visa transactions, the value must be right justified and padded to 11 digits with leading zeros. For example, if the Identifier is '123456', the value would be '00000123456'.

5.121.2 <A127><B2>Location Detail Name

Length	Format	Description
5-25	A/N/S	This 25-character field has three usages. The Visa and Mastercard stipulations apply to any usage. Visa: For subscription merchants offering free trials or introductory promotions, inclusion of an enhanced descriptor (e.g., trial, trial period, free trial) is required effective April 2021 for the first transaction at the end of a trial period. It is not required for subsequent transactions. Mastercard: The name is required to be the same in authorization and clearing. Usage 1 - Payment Facilitator Name*Sub-Merchant Name For Payment Facilitators, the field contains two parts in a concatenation of two fields separated by an asterisk (*). The Payment Facilitator Name is three characters in length and is followed by the asterisk. The remainder of the field consists of the Sub-Merchant name. This usage is valid for Visa, Mastercard, and Discover. For example, if the Payment Facilitator is MONSTER FACILITATOR, and the Sub-Merchant is BOB'S CAR WASH AND AUTO WAX, then it would be MON*BOBS CAR WASH AND AUT". This field is required to be populated (not all spaces) for this usage. Usage 2 – Marketplace Name For Visa Marketplaces, the field contains the name of the Marketplace. Usage 3 – Location Detail Name This field is used to identify the merchant name that appears on the storefront and/or customer receipts and statements. It is commonly referred to as the DBA (Doing Business As) Name. If the name is more than 25 characters, use proper and meaningful abbreviation(s) when possible. Do not truncate.

5.121.3 <A127><B3> Sub-Merchant Identifier

Length	Format	Description
1-15	A/N	Usage 1 - Sub-Merchants Only: This 15-byte alphanumeric field, assigned by the Payment Facilitator or the Acquirer, must be provided on all Visa or Mastercard transactions where the merchant has a relationship with a Payment Facilitator. Discover does not provide a sub-merchant identifier.

5.121.4 <A127><B4> Location Detail City

Visa: If the name is more than 13 characters, use proper and meaningful abbreviation(s) when possible. Do not truncate. To qualify for the CPS/CNP PSIRF, this field must contain a merchant's customer service phone number. The telephone number must contain three numeric values followed by a hyphen and at least seven more nonblank characters. Or, if the MOTO/ECI value is 2, 5, 6, or 7, this field must contain a URL or an email address. If the MCC is 4816, this field must contain a URL or an email address and the MOTO/ECI can be any value except 8 or 9. For chargebacks and reversals, this field must be the same as in the original transaction.

Mastercard/Discover: If the name is more than 13 characters, use proper and meaningful abbreviation(s) when possible. Do not truncate. For CNP transactions, the card acceptor's customer service phone number must replace the Merchant City name.

Length	Format	Description
1-13	A/N/S	Usage 1 - Sub-Merchant: This 13-character field indicates the city of the Sub-Merchant location (not the Payment Facilitator location). For Discover transactions, enter the city where the transaction took place; this may or may not be the sub-merchant's city. For Direct Marketing merchants and preferred customer/passenger transport and card not present transactions, this field must contain a phone number for merchant customer service in format XXX-XXX-XXXX. If the phone number is not available, use the website URL. Usage 3: This field is used to identify the name of the city where the transaction took place. If the name is more than 13 characters, use proper and meaningful abbreviation(s) when possible. Do not truncate. For Mail Order, Phone Order, and Internet Industries, substitute the name of the city in which the merchant's order processing facility is located.

5.121.5 <A127><B5> Location Detail State/Province Code

Length	Format	Description
2	A/N	Usage 1 - Sub- Merchant: This 2-character field indicates the state or province code of the Sub-Merchant location (not the Payment Facilitator location). If the Sub-Merchant is non-U.S. and non-Canadian, this field should be '00'. This field is used for Visa, Mastercard, and Discover transactions. Usage 3: This field is used to identify the region code that corresponds to the state, province, or other country subdivision of the merchant location where the transaction took place. For example, the region code for California, USA is CA. If the Merchant is non-U.S. and non- Canadian, this field should be 00.

5.121.6 <A127><B6> Location Detail Country Code

Length	Format	Description
3	NUM	Usage 1 - Sub - Merchant: This three-digit field indicates the country of the Sub-Merchant location (not the Payment Facilitator location) using ISO-specified numeric codes. Usage 3: This three-digit field is used to identify the country code of the location where the transaction took place. For example, the country code for USA is 840.

5.121.7 <A127><B7> Location Detail Postal Code

Length	Format	Description
5-9	A/N	Usage 1 - Sub - Merchant: This nine-character field indicates the postal code of the Sub-Merchant location (not the Payment Facilitator location). Usage 3: This nine-character alphanumeric field is used to identify the postal/zip code of the location where the transaction took place.

5.121.8 <A127><B8> Foreign Retailer Indicator

Length	Format	Description
1-3	AN	This field is used by marketplaces to uniquely identify domestic transactions in which the retailer is in a different country than the marketplace. Visa classifies an entity as a marketplace if it handles payments on behalf of retailers through an online marketplace that brings together multiple buyers and retailers. This tag must be sent when the retailer is foreign. The tag is not required when the retailer is not foreign. If this tag is present, it must be left justified and space-filled to three bytes.

Code	Definition
<spaces>	All spaces indicate the retailer is not foreign
F	This value indicates the retailer is located in a different country than the marketplace

5.122 <A128> ACH Transaction Data

5.122.1 <A128><B1> ACH SEC Code

Length	Format	Description
3 fixed	A/N	The Standard Entry Class (SEC) Code is used to determine the “ACH Rules” that apply to the transaction. Please refer to the rules table below.

The transactions below are valid.

Table 5.14 ACH rules table

SEC Code	Purchase/Return	Check Reversal Indicator	ACH Telephone Number Field Text
PPD/CCD	Purchase	N	ACH Phone Number
	Purchase	Y	Insert text “REVERSAL”
	Return	N	ACH Phone Number
	Return	Y	Insert text “REVERSAL”
ARC/POP/TEL/WEB	Purchase	N	ACH Phone Number
	Return	Y	Insert text “REVERSAL”

Effective February 2020, the below disallowed transactions will be rejected.

Table 5.15 ACH disallowed rules table

SEC Code	Credit/Debit	Check Reversal Indicator	Comments
ARC/POP/TEL/WEB	Purchase	Y	These transactions are not allowed
	Return	N	

Code	Description
CCD	Cash Concentration or Disbursement - Can be either a credit or debit application where funds are either distributed or consolidated between corporate entities.
POP	Point of Sale Entry - Point of sale debit applications non-shared (POS) environment. These transactions are most often initiated by the consumer via a plastic access card.
PPD	Prearranged Payment and Deposits - Used to credit or debit a consumer account. Popularly used for payroll direct deposits and preauthorized bill payments.
TEL	Telephone-Initiated Entry - Used for the origination of a single entry debit transaction to a consumer's account pursuant to a verbal authorization obtained from the consumer via the telephone.
WEB	Internet (Web)-Initiated Entry - Used for the origination of debit entries (either Single or Recurring Entry) to a consumer's account pursuant to an authorization that is obtained from the Receiver via the Internet.

5.122.2 <A128><B2> ACH Routing Number

Length	Format	Description
9 fixed	Numeric	This field contains the receiving Depository Financial Institution (DFI) identification/routing number and check digit.

5.122.3 <A128><B3> ACH Check Number

Length	Format	Description
15 fixed	A/N	This field contains the individual check number. This field must be present for transactions with an SEC Code of POP. Otherwise, it must be space-filled.

5.122.4 <A128><B4> ACH Individual Name

Length	Format	Description
22 fixed	A/N	This field contains the individual or receiving company name.

5.122.5 <A128><B5> ACH Individual Reference ID

Length	Format	Description
10 fixed	A/N	This field may contain meaningful merchant and customer information.

5.122.6 <A128><B6> ACH Telephone Number

Length	Format	Description
10 fixed	A/N	This field contains one of three options: 1 The merchant / business contact phone number for customer inquiry <i>Table 5.16 REVERSAL indicates a reversal</i> <i>Table 5.17 REDEPCHECK indicates a returned check</i>

5.122.7 <A128><B7> ACH Terminal City

Length	Format	Description
4 fixed	A/N	This field represents the city, town, village, or township in which the merchant's terminal is located. It must be non-blank if the SEC Code is "POP".

5.122.8 <A128><B8> ACH Terminal State

Length	Format	Description
2 fixed	A/N	This field represents the state location for the merchant's terminal. It must be non-blank if the SEC Code is "POP".

5.122.9 <A128><B9> ACH Prenote Indicator

Length	Format	Description
1 fixed	Numeric	This field indicates whether the transaction is for pre-notification. It is used to aid in determination of the proper ACH Transaction Code.

Value	Description
0	Not Prenote
1	Prenote

5.122.10 <A128><B10> ACH Checking/Savings Indicator

Length	Format	Description
1 fixed	A/N	This field specifies the originating account type.

Value	Description
C	Checking
S	Savings
G	General Ledger (Valid only when SEC Code is "CCD")

5.122.11 <A128><B11> ACH Payment Type Indicator

Length	Format	Description
1 fixed	A/N	This field indicates the transaction payment type. A value of R or S is required when the ACH SEC Code is WEB.

Value	Description
Spaces	Used as a default for non-Web transactions
R	Recurring
S	Single

5.122.12

5.122.13 <A128><B12> ACH Returned Check Indicator

Length	Format	Description
1 fixed	A/N	This field indicates a returned check transaction.

Value	Description
Y	This transaction is a returned check
N	This transaction is not a returned check

5.122.14 <A128><B13> ACH Check Reversal Indicator

Length	Format	Description
1 fixed	A/N	This field indicates a check reversal transaction.

Value	Description
Y	This transaction is a check reversal
N	This transaction is not a check reversal

5.123 <A129> Tax Data

This tag can be used only for Mastercard Canada Intracountry Commercial Data. It can be included up to six times.

5.123.1 <A129><B1> Tax Amount Indicator

Length	Format	Description
1	A/N	This code indicates how additional tax information is captured and reported. Possible values: Y - Tax included in the total purchase amount N - Tax not included in total purchase amount

5.123.2 <A129><B2> Tax Amount

Length	Format	Description
12	NUM	This is the monetary amount of the additional tax. This value can be all zeroes.

5.123.3 <A129><B3> Tax Rate

Length	Format	Description
5	NUM	This is the detailed tax rate applied to the purchase. The Tax Rate and Tax Rate Exponent fields are used together to express the tax rate. For example, if the value in Tax Rate field is 12345 and the value in Tax Rate Exponent field is 3, then the amount in the Tax Rate field is read as 12.345.

5.123.4 <A129><B4> Tax Rate Exponent

Length	Format	Description
1	NUM	This is the decimal location of Tax Rate. Possible values: 0 to 5 The Tax Rate and Tax Rate Exponent fields are used together to express the tax rate. For example, if the value in Tax Rate field is 12345 and the value in Tax Rate Exponent field is 3, then the amount in the Tax Rate field is read as 12.345.

5.123.5 <A129><B5> Tax Type Applied

Length	Format	Description
4	A/N	This field uses Federal Government issued tax codes that may apply to specific domestic processing arrangements. This field is not required, unless directed otherwise by Mastercard.

5.123.6 <A129><B6> Tax Type Identifier

Length	Format	Description
2	A/N	This code identifies the type of tax that Tax Amount represents. Possible values: 00 - Unknown 01 - Federal or national sales tax 03 - State sales tax 04 - Local sales tax 05 - Municipal sales tax 06 - Other tax 10 - Value Added Tax (VAT) 11 - Goods and Services Tax (GST) 12 - Provincial Sales Tax (PST) 13 - Harmonized Sales Tax (HST) 14 - Quebec Sales Tax (QST) 20 - Room tax 21 - Occupancy tax 22 - Energy tax

5.124 <A130> Authorization Indicator

Length	Format	Description
1	A/N	This field defines the type of authorization request and must be included on all Mastercard authorization request transactions. It must not be included on zero-dollar card authentication (Transaction Code 20) requests. Purchase Return Authorization Requests: Value must be F

Code	Definition	Description
P	Pre-Authorization	This indicator is used when the transaction is a request for an estimated amount.
F	Final Authorization	This indicator is used when the transaction is not an estimated amount.
U	Undefined Authorization	This indicator may be used when the intent is unknown, and the transaction is neither a pre-authorization, nor a final authorization.

5.125 <A131> Multi-Clearing Indicator

Length	Format	Description
1	A/N	This field is used in situations where multiple clearing records will be sent with a single authorization Possible value: M - Previously approved authorization-partial amount, multi-clearing F - Previously approved authorization-partial amount, final clearing

5.126 <A132> Fraud Enhanced Data

5.126.1 <A132><B1> Registered User Indicator

Length	Format	Description
1 fixed	A/N	This field indicates whether the cardholder is a registered user on a merchant's website. Merchants must populate this value in Discover auth request/0100 messages for e-Commerce transactions. It may also be populated in some cases for merchants that cannot properly report e-commerce transactions. NOTE Discover usage only

5.126.2 <A132><B2> Registered User Last Profile Date Change

Length	Format	Description
8 fixed	Numeric	This field defines the date when the cardholder last voluntarily changed his or her registered profile. NOTE Discover usage only

5.127 <A133> Lane ID

This tag is effective April 2016.

Length	Format	Description
8 fixed	Numeric	This field uniquely identifies a terminal at the card acceptor location and must be 8 digits long. If the merchant's hierarchy inside the message does not uniquely identify a terminal/POS Application, the Lane ID is mandatory in order to not receive a data integrity error. The Lane ID may be printed on all POS transaction receipts where the terminal is capable of generating customer receipts. NOTE This field is currently used for Mastercard but can be sent for any card brand.

5.128 <A134> Payment Specific Data

5.128.1 <A134><B1> Payment Account Reference

Length	Format	Description
35 variable	A/N	The Payment Account Reference (PAR) is a value assigned by the BIN Controller, which is defined as either an issuer or card brand. This field is associated directly with the cardholder's account. The PAR is considered a non-financial value that cannot be used to initiate a financial transaction. This value enables merchants, acquirers, and payment processors to link a payment token to a cardholder's underlying payment account. It can be included on the authorization request if available. This field is valid for Visa, Mastercard, Amex, and Discover.

5.128.2 <A134><B2> PAN Reference Identifier

Length	Format	Description
35 variable	A/N	The PAN Reference Identifier (PRI) is a value assigned by Discover at the time of token provisioning and is associated with a specific mobile wallet.

5.129 <A135> POS Environment Indicator

Length	Format	Description
1 fixed	A/N	The Visa only field provides additional information about the transaction. The field must be used in the following scenarios.

This field provides additional information about Visa transactions.

Usage 1: Merchant stores payment credential for the first time

C - Credential on File

The merchant stores the payment credential for the first time with the intention of using it for subsequent Unscheduled Credentials on File (UCOF) transactions on the cardholder's behalf.

R - Recurring

The merchant stores the payment credential for the first time with the intention of using it for subsequent Recurring transactions on the cardholder's behalf.

I - Installment

The merchant stores the payment credential for the first time with the intention of using it for subsequent Installment transactions on the cardholder's behalf.

NOTE The value in <A21> -POS Entry Data - POS Data Code (SF3 - Card Data Input Mode) should be populated with a value that reflects the actual entry mode for the transaction (i.e. manual input, key entered).

Usage 2: Merchant initiates a transaction using a stored payment credential

C - Credential on File

The merchant submits a UCOF transaction on the cardholder's behalf.

R - Recurring

The merchant submits a Recurring transaction on the cardholder's behalf.

I - Installment

The merchant submits an Installment transaction on the cardholder's behalf

NOTE The value in <A21> - POS Entry Data (SF3 - Card Data Input Mode) should be populated with a value of 7 for Credential on File.

5.130 <A136> Business Application Identifier

The Business Application Identifier (BAI) identifies industry-specific business scenarios. This currently applies only to Visa transactions and is alphanumeric.

Consumer Bill Payment:

Consumer Bill Payment Service (CBPS) is an optional Visa service for third party consumer bill pay providers. Visa requires submitting the CB value in authorization and full-financial messages for purchase and credit voucher transactions from consumer bill-payment service providers. Refer to Section [3.10.5](#).

Funds Transfer:

The value of FT applies to Funds Transfer Transactions. For example, Samsung Pay Cash, Stored Value Digital Wallet (SVDW) is a feature that allows users to create a digital debit card that is stored in Samsung Pay. Users are able to easily add funds to this card either with an existing card within the app, or by transferring funds directly from a bank account. Please refer to section [Section 3.10.6 Account Funding Transactions](#).

Wallet Transfer:

The value of WT applies to Account Funding Transactions (AFT). For example, a Staged Digital Wallet Operator (SDWO) such as PayPal would generate an authorization transaction using the consumer's Visa payment credentials for loading the PayPal account (Account Funding), which resides in the consumer's digital wallet. This data element notifies Visa when a SDWO, such as PayPal, generates the transaction. This would be a BAI of Wallet Transfer (WT).

Table 5.18 Business Application Identifier Valid Values

Code	Description
CB	Consumer Bill Payment
FT	Funds Transfer
WT	Wallet Transfer

5.131 <A137> Alternate Hierarchy

This tag is reserved for contracted SMS Visa Cobrand transactions.

5.131.1 <A137><B1> Acquiring BIN

Length	Format	Description
6	NUM	This field is provided by the issuer of the Cobrand credit card product.

5.131.2 <A137><B2> Card Acceptor ID

Length	Format	Description
15	NUM	This field is provided by the issuer of the Cobrand credit card product.

5.132 <A138> American Express Travel Cruise Addendum

This Tag is used to report additional descriptive details for the American Express Travel/Cruise Industry. This information may become part of the descriptive bill on the Cardholder's statement or be used to resolve billing inquiries and disputes.

<A138><B1> Travel Package Indicator

Length	Format	Description
1	A/N	This required field contains the Travel Package Indicator code. If this transaction is for a travel package, this entry indicates if the package includes car rental, airline flight, both or neither. Possible value: C - Car rental reservation included A - Airline flight reservation included B - Both car rental and airline flight reservations included N - Unknown

<A138><B2> Lodging Check In Date

Length	Format	Description
8 fixed	A/N	This field contains the cruise departure date also known as the sail date. The format is: CCYYMMDD where CC = Century YY = Year MM = Month DD = Day For example, February 16, 2011, would be entered as: 20110216

<A138><B3>Lodging Check Out Date

Length	Format	Description
8 fixed	A/N	For cruise transactions, this tag contains the cruise return date also known as the sail end date. The format is CCYYMMDD, where CC = Century YY = Year MM = Month DD = Day For example, February 18, 2017 would be entered as 20170218

<A138><B4>Lodging Room Rate

Length	Format	Description
12 bytes	N	This field contains the total cost of the cruise. For onboard purchases or transactions occurring during cruise travel, this field must be zero filled.

<A138><B5>Lodging Stay Duration

Length	Format	Description
2 bytes	N	This field contains the length of the cruise in days. The number of days should align with the number of days indicated between the Check-In Date and the Check-Out Date. For onboard purchases or transactions occurring during cruise travel, this field must be zero filled.

<A138><B6>Lodging Name

Length	Format	Description
20	A/N	For cruise transactions, this field contains the ship name booked for the cruise.

<A138><B7> Lodging Region Code

Length	Format	Description
3	A/N	This field contains the region code that corresponds to the state, province, or other country subdivision of the lodging location. For example, the region code for California, USA is "CA"; Nova Scotia, Canada, is "NS"; and Colima, Mexico is "COL". Region code must correspond to the Country code provided. If the Lodging Country Code field is populated, the Lodging Region Code field will also contain data.

<A138><B8> Lodging Country Code

Length	Format	Description
3	AN	This field contains the country code of the lodging location. Country code must correspond to the Region code provided. If the Lodging Region Code field is populated, this Lodging Country Code field must also contain data.

<A138><B9> Lodging City Name

Length	Format	Description
18	A/N	This field contains the name of the city where the lodging property is located. This entry must be easily recognized by the Cardholder to avoid unnecessary inquiries. If the name is more than 18 characters, use proper and meaningful abbreviation, when possible. Do not truncate.

5.133 <A139> American Express Airline Addendum

This Tag is used to report additional descriptive details for the American Express Airline Industry. This information may become part of the descriptive bill on the Cardholder's statement or be used to resolve billing inquiries and disputes.

5.133.5 <A139><B5> Ticket Issue Date

Length	Format	Description
8	N	This required field contains the Ticket Issue Date. The format is: CCYYMMDD where CC = Century YY = Year MM = Month DD = Day For example, February 16, 2011, would be entered as: 20110216

5.133.6 <A139><B6> Number in Party

Length	Format	Description
3	N	This value corresponds to the number of people in the party and is required if <B1> Ticket Transaction Type is 'TKT' or 'EXC'.

5.133.7 <A139><B7> Conjunction Ticket indicator

Length	Format	Description
1	A/N	This code indicates whether the itinerary contains more than four segments of travel. This field is required if <B1> Ticket Transaction Type is 'TKT' or 'EXC'. Possible value: Y - Yes N - No

5.133.8 <A139><B8> Electronic Ticket indicator

Length	Format	Description
1	A/N	This field contains the Electronic Ticket Indicator, a code that indicates if an electronic ticket was issued. This field is required if <B1> Ticket Transaction Type is 'TKT' or 'EXC'. Possible value: Y - Yes N - No

5.133.9 <A139><B9> Original Transaction Currency

Length	Format	Description
3	A/N	This field contains the Original Currency Code and is required if the entry in <A15> Transaction Amount and the Currency Code were converted from another currency. If the Original Transaction Currency field is populated, the Original Transaction Amount field must also contain data.

5.133.10 <A139><B10> Original Transaction Amount

Length	Format	Description
12	A/N	This field contains the Original Transaction Amount and is required if the entry in <A15> Transaction Amount and the Currency Code were converted from another currency. If the Original Transaction Amount field is populated, the Original Transaction Currency field must also contain data.

5.133.11 <A139><B11> Original Ticket Number

Length	Format	Description
14	A/N	This field contains the original (exchanged) ticket number when this transaction is for a replacement ticket. It is required if Ticket Transaction Type is 'EXC'.

5.133.12 <A139><B12> Stopover Code 5

Length	Format	Description
1	A/N	This field contains Stopover Code 5, a value that indicates when a traveler spends more than four (4) hours at a destination before the next leg of travel begins. This field should only be used when Trip Legs 1 through 4 are used. Possible value: Space - Destination Point O - Stopover permitted (letter 'O', not zero) X - Connection The following example shows the usage of the field for travel between New York, NY and Sydney, Australia. The last segment of the trip should always be indicated with a space. • Stopover code 1: X Connection in NYC to DFW • Stopover code 2: O stopover in DFW on route to LAX • Stopover code 3: O stopover in LAX on route to HNL • Stopover code 4: O stopover in HNL on route to SYD • Stopover code 5: (space) terminate travel in SYD

5.134 <A140> Transaction Integrity Class

This Mastercard tag is used for purchase and purchase with cash back transaction only. It is not used PIN debit or reversal transactions.

Length	Format	Description																											
2	A/N	This value may be provided in the response of credit transactions by Mastercard. It encompasses the fundamental safety and security of credit transactions, including the assessment of both the validity of the card and the cardholder. Some transactions are inherently more secure than others. For example, EMV chip cards are more secure than magnetic stripe cards. There will be nuances across both the technology (card) and the Cardholder Verification Method (cardholder), but the combination will be assessed across the spectrum to determine the overall integrity of each transaction. NOTE This field is required when included by Mastercard. Effective in April 2019, Mastercard will incorporate the Transaction Integrity Class in the interchange process. Possible values: <table><tr><th>Classification</th><th>Description</th><th>Value</th></tr><tr><td rowspan="5">Card and Cardholder Present</td><td>EMV/Token in a Secure, Trusted Environment</td><td>A1</td></tr><tr><td>EMV/Chip Equivalent</td><td>B1</td></tr><tr><td>Mag Stripe</td><td>C1</td></tr><tr><td>Key Entered</td><td>E1</td></tr><tr><td>Unclassified</td><td>U0</td></tr><tr><td rowspan="6">Card and/or Cardholder Not Present</td><td>Digital Transactions</td><td>A2</td></tr><tr><td>Authenticated Checkout</td><td>B2</td></tr><tr><td>Transaction Validation</td><td>C2</td></tr><tr><td>Enhanced Data</td><td>D2</td></tr><tr><td>Generic Messaging</td><td>E2</td></tr><tr><td>Unclassified</td><td>U0</td></tr></table>	Classification	Description	Value	Card and Cardholder Present	EMV/Token in a Secure, Trusted Environment	A1	EMV/Chip Equivalent	B1	Mag Stripe	C1	Key Entered	E1	Unclassified	U0	Card and/or Cardholder Not Present	Digital Transactions	A2	Authenticated Checkout	B2	Transaction Validation	C2	Enhanced Data	D2	Generic Messaging	E2	Unclassified	U0
Classification	Description	Value																											
Card and Cardholder Present	EMV/Token in a Secure, Trusted Environment	A1																											
	EMV/Chip Equivalent	B1																											
	Mag Stripe	C1																											
	Key Entered	E1																											
	Unclassified	U0																											
Card and/or Cardholder Not Present	Digital Transactions	A2																											
	Authenticated Checkout	B2																											
	Transaction Validation	C2																											
	Enhanced Data	D2																											
	Generic Messaging	E2																											
	Unclassified	U0																											

5.135 <A141> Mastercard Service Parameters

5.135.1 <A141><B1> Canada Domestic Indicator

This tag must be used in every **credit** transaction for merchants that accept Canada region-issued Debit Mastercard cards.

Length	Format	Description
1	A/N	Canada Domestic Indicator Sending a value of 'Y' indicates the merchant accepts Canada domestic Debit Mastercard. This value must be sent in every Credit Authorization Request for merchants that accept Canada region-issued Debit Mastercard cards.

5.136 <A142> Authentication Data

Tag A142 is deprecated effective with the October 2019 release and must not be used for new development after that date. It will be retired October 2020; continued use may result in unexpected behavior and/or potential compliance fines.

This tag is required if the merchant uses Mastercard Identity Check. It is used to associate a Mastercard Identity Check transaction to its subsequent authorization. After data is received from the EMV 3DS Mastercard Directory Server during authentication, it is placed in the sub tags for authorization.

This group can be sent for any card brand but it will only be used for Mastercard transactions and will be ignored for other card brands.

5.136.1 <A142><B1> Program Protocol

Program Protocol is a one-byte alphanumeric field and must be provided by the merchant if they are participating in Mastercard Identity Check and have previously authenticated using the Mastercard Identity Check or SecureCode program. It identifies the type of 3D-Secure program.

Value	Description
1	3D Secure Version 1.0 [3DS 1.0]
2	EMV 3-D Secure [3DS 2.0]

5.136.2 <A142><B2> Directory Server Transaction ID

The Directory Server Transaction ID is a 36-byte alphanumeric value generated by the EMV 3DS Mastercard Directory Server during the authentication transaction and passed back to the merchant with the authentication results. This field allows the merchant to pass the Directory Server Transaction ID during authorization in order to link authentication and authorization data.

5.137 <A143> Merchant Assigned Reference

Length	Format	Description
12	A/N	The Merchant Assigned Reference is a value assigned by the merchant to reference the transaction. This is not a settlement field, but can be used for merchant or client reporting purposes. NOTE: This is not a settlement field; it will not appear on the cardholder billing statement. It can be used for merchant or client reporting purposes.

5.138 <A301> Batch Status

Length	Format	Description
1 fixed	A/N	This is an indicator of the status of the batch in the inquiry. A value of "O" indicates that the batch is still open. A value of "C" indicates that the batch has been closed and sent on to the settlement system.

5.139 <A302> Batch Response Code

Length	Format	Description
2 fixed	A/N	This is a coded response to a batch management request. It indicates whether or not the request itself was successful, whether there were any complications to the request / response, or any other abnormal indications. There will be a code value (Batch Response Code) and a text associated with that value (Batch Response Text).

<A302> Code	<A303> Response Text	Description
00	ACCEPTED or DATA RETRIEVED	If A10 is 201 (Batch Close), response text will be “ACCEPTED”. Otherwise, response text will be “DATA RETRIEVED”.
01	EXCEEDED LIMIT	The requested batch or transaction was not in the allowable limits (current, previous, or last 5 batches).
02	COULD NOT FIND	The host system could not find the batch or the transaction that was requested.
03	NOT ALLOWED	An invalid request was entered.
04	SYSTEM ERROR	The host system was not able to process the request due to some internal issue, maintenance, or other system error.
05	ALREADY CLOSED	During a batch close request, the batch was already found to have been closed. A batch summary will be provided with the pertinent information.
06	NOT IN BATCH	The requested Adjustment or Void was not in the current batch.

5.140 <A303> Batch Response Text

Length	Format	Description
1 to 32	A/N	The Batch Response text is associated with a specific <A302> Batch Response Code.

5.141 <A304> Local Batch Close Date

Length	Format	Description
6 fixed	Numeric	This is the date (in MMDDYY format) that the batch was closed. Note that this indicates when the batch close began, rather than when the batch close completed. The date is local based on the host-stored time zone differential. If this value is not available, the time will be in Pacific time.

5.142 <A305> Local Batch Close Time

Length	Format	Description
6 fixed	Numeric	This is the time (in HHMMSS format) that the batch was closed. The time is local based on the host-stored time zone differential. If this value is not available, the time will be in Pacific time.

5.143 <A307> Requested Transaction Count

Length	Format	Description
1	Numeric	This is the number of transactions that are requested in a batch detail request. If this value is not entered, five transactions will be the default number returned. The maximum number of Transactions that can be requested is five. The number of transactions returned in the response can not exceed the number of transactions in the requested Batch.

5.144 <A325> Batch Summary Group

This is the Batch Summary Group which contains 4 subfields of data. This data represents a type of summary detail, purchase counts for that type, refund counts for that type, and the net amount of transactions for that type. This group tag will be repeated for each summary type. Each supported summary type will be listed if it is present in a batch.

5.144.1 <A325><B1> Summary Type

Length	Format	Description
Variable	A/N	This is the type of summary being provided in the Batch Summary Inquiry. The following are current valid values, though this is to be considered a “free text” field. Possible value: VISA - The Visa summary totals Mastercard - The Mastercard summary totals AMEX - The Amex summary totals DISCOVER - The Discover summary totals PAYPAL - The PayPal summary totals JCB - The JCB summary totals ONLINE DEBIT - The Online Debit summary totals EBT - The EBT summary totals OTHER - All other card types summary totals, including private label BATCH TOTAL - The overall totals for the batch including all card types DELAY CAP - Total of all transactions being held for adjustment

5.144.2 <A325><B2> Purchase Count

Length	Format	Description
1 to 5	Numeric	This is the total count for the particular summary type of all purchases in the batch.

5.144.3 <A325><B3> Refunds Count

Length	Format	Description
1 to 5	Numeric	This is the total count for the particular summary type of all refunds in the batch.

5.144.4 <A325><B4> Net Amount

Length	Format	Description
1 to 16	Numeric	This is the net amount total for the particular summary type in the batch. This will be the total transaction amount for all purchases (with voids and adjustments taken into account) minus refunds.

5.145 <A326> Pending Transaction Count

Length	Format	Description
Variable	Numeric	This is the number of transactions that will not be captured in the current batch because they are waiting for adjustment. If they are adjusted before the current batch closes, they will be captured in the current batch. If they are not adjusted in the current batch, they will be captured in the next batch adjusted or not.

5.146 <A330> Batch Detail Group

This is the Batch Detail Group which contains 6 subfields of data. This data represents basic transaction data within the batch. This group tag will be repeated for each transaction returned in the response.

5.146.1 <A330><B1> Transaction Original Number (TON)

This is the TON from the transaction. See [<A17> Transaction Original Number \(TON\)](#) for TON description.

5.146.2 <A330><B2> Retrieval Reference Number (RRN)

This is the Retrieval Reference Number from the transaction. See [<A86> Retrieval Reference Number \(RRN\)](#) for RRN description.

5.146.3 <A330><B3> Card Type

This is the Card Type associated with the transaction. See [<A92> Card Product Codes](#) for Card Type description.

5.146.4 <A330><B4> Card Number Last 4 Digits

Length	Format	Description
4 fixed	Numeric	This is the last 4 digits of the customer card number.

5.146.5 <A330><B5> Transaction Code

This is the Transaction Code from the transaction. See [<A10> Transaction Code](#) for Transaction Code description.

5.146.6 <A330><B6> Transaction Amount

This is the Transaction Amount from the transaction. See [<A15> Transaction Amount](#) for Transaction Amount description. For partially authorized transaction, this field contains the original requested amount.

5.146.7 <A330><B7> Settlement Amount

This is the Settlement Amount from the transaction. See [<A68> Settlement Amount](#) for Settlement Amount description. For partially authorized transaction, this field contains the resulting authorized amount.

Chapter 6

Transaction Message Examples

This chapter provides detailed examples of each type of SGMF Transaction (Authorization, Batch, and Authentication) tagged data elements and conditions. The following table headers are used in the example descriptions:

Tag & name	Extensible tags identifier for identify message data elements and the English name of the Tag or Sub-Tag.
→	The Request Message M = Mandatory C = Conditional use O = Optional use
←	The Response Message (same notation as request).
Comment	Comments about the usage of the data element in a Transaction message.

6.1 Authenticate POS device

Tag name			Comment
<u><A1> Host Capture POS ID</u>	M	-	Must be the first tag in the request.
<u><A2> Authentication Code</u>	M	-	
<u><A3> GenKey</u>	-	C	On successful activation only
<u><A4> Authentication Factor</u>	M	-	
<u><A4><B1> Merchant Postal Code</u>	C	-	If <A4><B2> not present
<u><A4><B2>Merchant Phone Number</u>	C	-	If <A4><B1> not present
<u><A10> Transaction Code</u>	M	-	“1” Authentication
<u><A80> Batch Number</u>	-	M	Indicates the current open batch
<u><A83> Response Code</u>	-	M	
<u><A85> Authorization Response Text</u>	-	M	

6.2 Deactivate POS device

Tag name	→	←	Comment
<A1> Host Capture POS ID	M	-	
<A3> GenKey	M		
<A10> Transaction Code	M	-	“2” Deactivation
<A80> Batch Number	-	M	Indicates the current open batch
<A83> Response Code	-	M	“103”
<A85> Authorization Response Text	-	M	“DEACTIVATED

6.3 Retail credit card

Tag name	→	←	Comment
<A1> Host Capture POS ID	M	-	Must be the first tag in the request.
<A3> GenKey	M	-	
<A10> Transaction Code	M	-	“10” or “13”
<A12> Payment Code	M	-	“C” or “P”
<A14> Repeat Indicator	C	-	If a repeat message “RPT”
<A15> Transaction Amount	M	-	
<A16> Secondary Amount	O	-	
<A17> Transaction Original Number (TON)	M	M	
<A18> Merchant Category Code (MCC)	O	-	
<A19> Language Indicator	O	-	
<A21> POS Entry Data	M	-	
<A21><B1> Cardholder Present Data	M	-	
<A21><B2> Card Present Data	M	-	
<A21><B3> Card Data Input Mode	M	-	
<A21><B4> Cardholder Authentication Method	M	-	
<A21><B5> Authentication Entity	M	-	
<A21><B6> Card data input capability	M	-	

Tag name	→	←	Comment
<u><A43><B1> Account Type</u>	-	C	
<u><A43><B2> Amount Type</u>	-	C	
<u><A43><B3> Currency Code</u>	-	C	
<u><A43><B4> Sign</u>	-	C	
<u><A43><B5> Additional Amount</u>	-	C	
<u><A45> Free Form Text</u>	O	O	
<u><A46> Partial Authorization Indicator</u>	O	-	Use Discover/PayPal card values if needed
<u><A48> Amex Merchant DBA Name</u>	O	-	
<u><A49> Amex Station Location Code</u>	O	-	
<u><A50> Amex Seller/Vendor Code</u>	O	-	
<u><A51> Digital Entity Identifier</u>	C	-	If transaction comes though aggregator that is Visa Trusted Agent
<u><A51><B1> Digital Entity Identifier</u>	C	-	
<u><A51><B2> Reserved</u>	C	-	
<u><A53> Discover Promotional Code</u>	O	-	
<u><A56> Electromagnetic Signature</u>	C	-	If EM signature used
<u><A58> Visa Contactless Data Group</u>	C	-	If card was Visa Contactless
<u><A58><B1> Amount Authorized</u>	O	-	
<u><A58><B2> Application Cryptogram</u>	O	-	
<u><A58><B3> Application Transaction Counter</u>	O	-	
<u><A58><B4> Customer Exclusive Data</u>	O	-	
<u><A58><B5> Form Factor Indicator</u>	O	-	
<u><A58><B6> Issuer Application Data</u>	O	-	
<u><A58><B7> Unpredictable Number</u>	O	-	
<u><A58><B8> Card Sequence Number</u>	O	-	

Tag name	→	←	Comment
<A60> Tip Amount	O	-	
<A61> Tax Amount (Commercial Card)	C	-	A61, A62, A63 must be present for commercial card to settle in current batch without adjustment.
<A62> Tax Amount Identifier (Commercial Card)	C	-	A61, A62, A63 must be present for commercial card to settle in current batch without adjustment.
<A63> Purchase Order Number (Commercial Card)	C	-	A61, A62, A63 must be present for commercial card to settle in current batch without adjustment.
<A66> Invoice Number (Card Not Present)	O	-	
<A67> Soft Descriptor	O	-	
<A69> Transaction Fee Amount	O	-	
<A77><B1> Transit Transaction Type Indicator	O	-	
<A77><B2> Transportation Mode Indicator	O	-	
<A80> Batch Number	-	M	Indicates the current open batch
<A81> Local Transaction Date	-	M	
<A82> Local Transaction Time	-	M	
<A83> Response Code	-	M	
<A84> Approval Code	-	C	If <A83> indicates “approved”
<A85> Authorization Response Text	-	M	
<A86> Retrieval Reference Number (RRN)	-	M	
<A87> CCI Response (Commercial Card Indicator)	-	C	If card account is a commercial card
<A88> Card Verification Result (CID/CVC2/CVV2)	-	C	If <A31> is present
<A89> Transaction Data Condition Code	-	C	If Discover/PayPal card type
<A91> Merchant Advice Code	-	C	If Mastercard card type
<A92> Card Product Codes	-	C	If U.S. issued Visa card type
<A93> Card Type	-	M	
<A94> Requested ACI	O	-	
<A95> Return ACI	-	C	
<A97> Address Verification Result Code	-	C	If AVS was requested

Tag name	→	←	Comment
<A98> Delayed Capture Indicator	C	-	Send if this transaction should not settle at the next batch close. Send if this authorization will be followed by Level III data.
<A101> Amex Corporate Purchasing Card Group	C	-	If Amex CPC
<A101><B1> Requester Name	C	-	If Amex CPC
<A101><B2> Shipped to ZIP	C	-	If Amex CPC
<A101><B3> Charge Descriptor 1	C	-	If Amex CPC
<A101><B4> Charge Descriptor 2	C	-	If Amex CPC
<A101><B5> Charge Descriptor 3	C	-	If Amex CPC
<A101><B6> Charge Descriptor 4	C	-	If Amex CPC
<A102> Amex CAPN Retail Department Name	C	-	If Amex card type
<A103> Amex CAPN Retail Group	C	-	If Amex card type
<A104> Level III Enhanced Data Group	C	-	If Amex card type
<A103><B2> Retail Item Description	C	-	If Amex card type
<A103><B3> Retail Item Quantity	C	-	If Amex card type
<A111> Chip Card Data	C	C	If <A21><B3> is equal to 'C', 'F', or 'M'
<A112><B1> Seller Telephone Number	C	-	Required for Amex transactions that are submitted by an Aggregator.
<A112><B2> Seller Email Address	C	-	Required for Amex transactions that are submitted by an Aggregator.
<A114> Alternate Account ID 1	-	C	This will be returned in some Mastercard PayPass response messages.
<A116><B1> Mapped PAN indicator	-	C	This will be returned in some Mastercard PayPass response messages.
<A116><B2> Mapped Product Code	-	C	This may be returned in some Mastercard PayPass response messages.
<A116><B3> Mapped Expiry Date	-	C	This will be returned in some Mastercard PayPass response messages.
<A117> TSYS Acquiring Solutions Gateway ID	O	-	

6.4 Credit Return

Tag name	→	←	Comment
<A1> Host Capture POS ID	M	-	Must be the first tag in the request
<A3> GenKey	M	-	
<A10> Transaction Code	M	-	“13”
<A12> Payment Code	M	-	“C” or “P”
<A14> Repeat Indicator			If a repeat message “RPT”
<A15> Transaction Amount	M	-	
<A16> Secondary Amount	O	-	
<A17> Transaction Original Number (TON)	M	M	
<A18> Merchant Category Code (MCC)	O	-	
<A19> Language Indicator	O	-	
<A21> POS Entry Data	M	-	
<A21><B1> Cardholder Present Data	M	-	
<A21><B2> Card Present Data	M	-	
<A21><B3> Card Data Input Mode	M	-	
<A21><B4> Cardholder Authentication Method	M	-	
<A21><B5> Authentication Entity	M	-	
<A21><B6> Card data input capability	M	-	
<A21><B7> Cardholder authentication capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B8> Card capture capability	O	-	
<A21><B9> Operating environment	M	-	
<A21><B10> Card data output capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B11> Terminal data output capability	O	-	
<A21><B12> PIN capture capability	O	-	

Tag name	→	←	Comment
<u><A23> Track 1</u>	C		Either Track 1 or Track 2 should be sent
<u><A24> Track 2</u>	C	-	Either Track 1 or Track 2 should be sent
<u><A25> Manually Entered Card Number</u>	C		If Track 1 or Track 2 can not be read
<u><A26> Manually Entered Expiry Date</u>	C		if <A25> is present and expiration date available
<u><A41> TSYS Acquiring Solutions Developer ID</u>	M	-	
<u><A42> TSYS Acquiring Solutions Version ID</u>	M	-	
<u><A69> Transaction Fee Amount</u>	O	-	
<u><A80> Batch Number</u>	-	M	Indicates the current open batch
<u><A81> Local Transaction Date</u>		M	
<u><A82> Local Transaction Time</u>		M	
<u><A83> Response Code</u>	-	M	
<u><A84> Approval Code</u>	-	C	If <A83> indicated “approved”
<u><A85> Authorization Response Text</u>	-	M	
<u><A86> Retrieval Reference Number (RRN)</u>	-	M	
<u><A89> Transaction Data Condition Code</u>		C	If Discover/PayPal card type
<u><A91> Merchant Advice Code</u>		C	If Mastercard card type
<u><A92> Card Product Codes</u>		C	If U.S. issued Visa card type
<u><A93> Card Type</u>	-	M	
<u><A111> Chip Card Data</u>	C	C	If <A21><B3> is equal to 'C', 'F', or 'M'
<u><A117> TSYS Acquiring Solutions Gateway ID</u>	O	-	

6.5 Retail/Restaurant debit

Tag name	→	←	Comment
<A1> Host Capture POS ID	M	-	Must be the first tag in the request
<A3> GenKey	M	-	
<A10> Transaction Code	M	-	“10” or “13”
<A12> Payment Code	M	-	“D”
<A15> Transaction Amount	M	-	
<A16> Secondary Amount	O	-	Used for cashback
<A17> Transaction Original Number (TON)	M	M	
<A18> Merchant Category Code (MCC)	O	-	
<A19> Language Indicator	O	-	
<A20> Forwarding Institution ID (FIID)	C	-	
<A21> POS Entry Data	M	-	
<A21><B1> Cardholder Present Data	M	-	
<A21><B2> Card Present Data	M	-	
<A21><B3> Card Data Input Mode	M	-	
<A21><B4> Cardholder Authentication Method	M	-	
<A21><B5> Authentication Entity	M	-	
<A21><B6> Card data input capability	M	-	
<A21><B7> Cardholder authentication capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B8> Card capture capability	O	-	
<A21><B9> Operating environment	M	-	
<A21><B10> Card data output capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B11> Terminal data output capability	O	-	
<A21><B12> PIN capture capability	O	-	

Tag name	→	←	Comment
<A24> Track 2	M	-	
<A28> PIN Block	M	-	
<A29> Security Control Info	M	-	
<A38> Special Condition Indicator	O	-	
<A41> TSYS Acquiring Solutions Developer ID	M	-	
<A42> TSYS Acquiring Solutions Version ID	M	-	
<A45> Free Form Text	O	O	
<A60> Tip Amount	O	-	
<A67> Soft Descriptor	O	-	
<A69> Transaction Fee Amount	O	-	
<A80> Batch Number	-	M	Indicates the current open batch
<A81> Local Transaction Date	C	M	Transaction date of original request should be sent in return request; if original date not available then use current date.
<A82> Local Transaction Time	C	M	Transaction time of original request should be sent in return request; if original time not available then use current time.
<A83> Response Code	-	M	
<A84> Approval Code	-	C	If <A83> indicates “approved”
<A85> Authorization Response Text	-	M	
<A86> Retrieval Reference Number (RRN)	-	M	
<A93> Card Type	-	M	“Debit”
<A96> Network ID	-	M	
<A111> Chip Card Data	C	C	If <A21><B3> is equal to 'C', 'F', or 'M'
<A117> TSYS Acquiring Solutions Gateway ID	O	-	

6.6 Retail EBT

Tag name	→	←	Comment
<A1> Host Capture POS ID	M	-	Must be the first tag in the request
<A3> GenKey	M	-	
<A10> Transaction Code	M	-	“10, 11 and 13”
<A12> Payment Code	M	-	“E” or “F”
<A15> Transaction Amount	M	-	
<A16> Secondary Amount	O	-	
<A17> Transaction Original Number (TON)	M	M	
<A18> Merchant Category Code (MCC)	O	-	
<A19> Language Indicator	O	-	
<A20> Forwarding Institution ID (FIID)	C	-	
<A21> POS Entry Data	M	-	
<A21><B1> Cardholder Present Data	M	-	
<A21><B2> Card Present Data	M	-	
<A21><B3> Card Data Input Mode	M	-	
<A21><B4> Cardholder Authentication Method	M	-	
<A21><B5> Authentication Entity	M	-	
<A21><B6> Card data input capability	M	-	
<A21><B7> Cardholder authentication capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B8> Card capture capability	O	-	
<A21><B9> Operating environment	M	-	
<A21><B10> Card data output capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B11> Terminal data output capability	O	-	
<A21><B12> PIN capture capability	O	-	

Tag name	→	←	Comment
<u><A24> Track 2</u>	M	-	Track 2 should be sent
<u><A25> Manually Entered Card Number</u>	O	-	If Track 2 cannot be read
<u><A26> Manually Entered Expiry Date</u>	O	-	If Track 2 cannot be read and expiration date is available
<u><A28> PIN Block</u>	M	-	
<u><A29> Security Control Info</u>	M	-	
<u>3-D Secure CAVV</u>	C	-	
<u><A36> EBT Voucher Approval Code</u>	C	-	
<u><A41> TSYS Acquiring Solutions Developer ID</u>	M	-	
<u><A42> TSYS Acquiring Solutions Version ID</u>	M	-	
<u><A43> Additional Amounts Group</u>	-	C	Available balance
<u><A43><B1> Account Type</u>	-	C	
<u><A43><B2> Amount Type</u>	-	C	
<u><A43><B3> Currency Code</u>	-	C	
<u><A43><B4> Sign</u>	-	C	
<u><A43><B5> Additional Amount</u>	-	C	
<u><A60> Tip Amount</u>	O	-	
<u><A67> Soft Descriptor</u>	O	-	
<u><A69> Transaction Fee Amount</u>	O	-	If cash benefit; not food stamps
<u><A80> Batch Number</u>	-	M	Indicates the current open batch
<u><A81> Local Transaction Date</u>	C	M	Transaction date of original request should be sent in return request; if original date not available then use current date.
<u><A82> Local Transaction Time</u>	C	M	Transaction time of original request should be sent in return request; if original time not available then use current time.
<u><A83> Response Code</u>	-	M	
<u><A84> Approval Code</u>	-	C	If <A83> indicates “approved”
<u><A85> Authorization Response Text</u>	-	M	
<u><A86> Retrieval Reference Number (RRN)</u>	-	M	
<u><A93> Card Type</u>	-	M	“EBT”

Tag name	→	←	Comment
<A96> Network ID	-	M	
<A117> TSYS Acquiring Solutions Gateway ID	O	-	

6.7 Card not present credit card

Tag name	→	←	Comment
<A1> Host Capture POS ID	M	-	Must be the first tag in the request
<A3> GenKey	M	-	
<A10> Transaction Code	M	-	"10" or "13"
<A12> Payment Code	M	-	"C"
<A14> Repeat Indicator	C	-	If a repeat message "RPT"
<A15> Transaction Amount	M	-	
<A16> Secondary Amount	O	-	
<A17> Transaction Original Number (TON)	M	M	
<A18> Merchant Category Code (MCC)	O	-	
<A19> Language Indicator	O	-	
<A21> POS Entry Data	M	-	
<A21><B1> Cardholder Present Data	M	-	
<A21><B2> Card Present Data	M	-	
<A21><B3> Card Data Input Mode	M	-	
<A21><B4> Cardholder Authentication Method	M	-	
<A21><B5> Authentication Entity	M	-	
<A21><B6> Card data input capability	M	-	
<A21><B7> Cardholder authentication capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B8> Card capture capability	O	-	
<A21><B9> Operating environment	M	-	
<A21><B10> Card data output capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B11> Terminal data output capability	O	-	
<A21><B12> PIN capture capability	O	-	

Tag name	→	←	Comment
<A34> Cardholder Authentication for Secure e-Commerce Transactions	O	-	
<A37> Service Development Indicator	O	-	
<A38> Special Condition Indicator	O	-	
<A39> Electronic Commerce Security Level Indicator	O	-	If Mastercard
Response Value:	O	-	If Mastercard
<A41> TSYS Acquiring Solutions Developer ID	M	-	
<A42> TSYS Acquiring Solutions Version ID	M	-	
<A43> Additional Amounts Group	-	C	Partial approved amounts
<A43><B1> Account Type	-	C	
<A43><B2> Amount Type	-	C	
<A43><B3> Currency Code	-	C	
<A43><B4> Sign	-	C	
<A43><B5> Additional Amount	-	C	
<A45> Free Form Text	O	O	
<A46> Partial Authorization Indicator	O	-	Use Discover card values if needed
<A47> Amex ITD Data	C	-	Used for Amex only
<A47><B1> Secondary ID	O	-	
<A47><B2> Customer e-Mail	O	-	
<A47><B3> Customer Host Name	O	-	
<A47><B4> HTTP Browser Type	O	-	
<A47><B5> Ship to Country	O	-	
<A47><B6> Shipping Method	O	-	
<A47><B7> Merchant Product SKU	O	-	
<A47><B8> Customer IP Address	O	-	
<A48> Amex Merchant DBA Name	O	-	
<A50> Amex Seller/Vendor Code	O	-	

Tag name	→	←	Comment
<u><A51> Digital Entity Identifier</u>	O	-	
<u><A51><B1> Digital Entity Identifier</u>	O	-	
<u><A51><B2> Reserved</u>	O	-	
<u><A53> Discover Promotional Code</u>	O	-	
<u><A60> Tip Amount</u>	O	-	
<u><A61> Tax Amount (Commercial Card)</u>	C	-	A61, A62, A63 must be present for commercial card to settle in current batch without adjustment.
<u><A62> Tax Amount Identifier (Commercial Card)</u>	C	-	A61, A62, A63 must be present for commercial card to settle in current batch without adjustment.
<u><A63> Purchase Order Number (Commercial Card)</u>	C	-	A61, A62, A63 must be present for commercial card to settle in current batch without adjustment.
<u><A64> e-Commerce Goods Indicator</u>	O	-	If known at time of purchase
<u><A66> Invoice Number (Card Not Present)</u>	O	-	If known at time of purchase
<u><A67> Soft Descriptor</u>	O	-	
<u><A69> Transaction Fee Amount</u>	O	-	
<u><A73> Hold Capture Indicator</u>	O	-	"HOLD"; only for Hold Capture participants
<u><A74><B1>Billing ZIP Code</u>	-	C	
<u><A74><B2>Billing Street Match Code</u>	-	C	
<u><A74><B3>Billing Name Result Code</u>	-	C	
<u><A74><B4>Telephone Number Match Code</u>	-	C	
<u><A74><B5>E-mail Address Match Code</u>	-	C	
<u><A76><B1> EMS Service Request Indicator</u>	O	-	
<u><A76><B2> EMS Service Response Indicator</u>	-	C	If fraud scoring requested in <A76><B1>
<u><A76><B3> EMS Result Code</u>	-	C	If fraud scoring requested in <A76><B1>
<u><A76><B4> EMS Additional Information</u>	-	C	If fraud scoring requested in <A76><B1>

Tag name	→	←	Comment
<u><A76><B5> EMS Risk Score</u>	-	C	If fraud scoring requested in <A76><B1>
<u><A76><B6> EMS Risk Score Reason Code</u>	-	C	If fraud scoring requested in <A76><B1>
<u><A80> Batch Number</u>	-	M	Indicates the current open batch
<u><A81> Local Transaction Date</u>	-	M	
<u><A82> Local Transaction Time</u>	-	M	
<u><A83> Response Code</u>	-	M	
<u><A84> Approval Code</u>	-	C	If <A83> indicates “approved”
<u><A85> Authorization Response Text</u>	-	M	
<u><A86> Retrieval Reference Number (RRN)</u>	-	M	
<u><A87> CCI Response (Commercial Card Indicator)</u>	-	C	If card account is a commercial card
<u><A88> Card Verification Result (CID/CVC2/CVV2)</u>	-	C	If result sent from issuer
<u><A89> Transaction Data Condition Code</u>	-	C	If Discover card type
<u><A90> CAVV Result Code</u>	-	C	If CAVV response provided
<u><A91> Merchant Advice Code</u>	-	C	If Mastercard card type
<u><A92> Card Product Codes</u>	-	C	If U.S. issued Visa card type
<u><A93> Card Type</u>	-	M	“Credit”
<u><A94> Requested ACI</u>	O	-	
<u><A95> Return ACI</u>	-	C	
<u><A97> Address Verification Result Code</u>	-	C	If issuer provides
<u><A98> Delayed Capture Indicator</u>	O	-	
<u><A99> Multiple Clearing Sequence Number</u>	O	-	
<u><A100> Multiple Clearing Sequence Count</u>	O	-	
<u><A101> Amex Corporate Purchasing Card Group</u>	C	-	If Amex CPC
<u><A101><B1> Requester Name</u>	C	-	If Amex CPC
<u><A101><B2> Shipped to ZIP</u>	C	-	If Amex CPC
<u><A101><B3> Charge Descriptor 1</u>	C	-	If Amex CPC
<u><A101><B4> Charge Descriptor 2</u>	C	-	If Amex CPC

Tag name	→	←	Comment
<u><A101><B5> Charge Descriptor 3</u>	C	-	If Amex CPC
<u><A101><B6> Charge Descriptor 4</u>	C	-	If Amex CPC
<u><A102> Amex CAPN Retail Department Name</u>	C	-	If Amex card type
<u><A103> Amex CAPN Retail Group</u>	C	-	If Amex card type
<u><A104> Level III Enhanced Data Group</u>	C	-	If Amex card type
<u><A103><B2> Retail Item Description</u>	C	-	If Amex card type
<u><A103><B3> Retail Item Quantity</u>	C	-	If Amex card type
<u><A117> TSYS Acquiring Solutions Gateway ID</u>	O	-	

6.8 Credit authorization only or card authentication

Tag name	→	←	Comment
<A1> Host Capture POS ID	M	-	Must be the first tag in the request
<A3> GenKey	M	-	
<A10> Transaction Code	M	-	"20"
<A12> Payment Code	M	-	"C" or "P"
<A14> Repeat Indicator	C	-	If a repeat message "RPT"
<A15> Transaction Amount	C	-	May be absent, present and zero or present and non-zero
<A16> Secondary Amount	O	-	
<A17> Transaction Original Number (TON)	M	M	
<A18> Merchant Category Code (MCC)	O	-	
<A19> Language Indicator	O	-	
<A21> POS Entry Data	M	-	
<A21><B1> Cardholder Present Data	M	-	
<A21><B2> Card Present Data	M	-	
<A21><B3> Card Data Input Mode	M	-	
<A21><B4> Cardholder Authentication Method	M	-	
<A21><B5> Authentication Entity	M	-	
<A21><B6> Card data input capability	M	-	
<A21><B7> Cardholder authentication capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B8> Card capture capability	O	-	
<A21><B9> Operating environment	M	-	
<A21><B10> Card data output capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B11> Terminal data output capability	O	-	
<A21><B12> PIN capture capability	O	-	

Tag name	→	←	Comment
<A34> Cardholder Authentication for Secure e-Commerce Transactions	O	-	
<A37> Service Development Indicator	O	-	
<A38> Special Condition Indicator	O	-	
<A39> Electronic Commerce Security Level Indicator	O	-	
Response Value:	O	-	
<A41> TSYS Acquiring Solutions Developer ID	M	-	
<A42> TSYS Acquiring Solutions Version ID	M	-	
<A43> Additional Amounts Group	-	C	If additional amounts provided by issuer
<A43><B1> Account Type	-	C	
<A43><B2> Amount Type	-	C	
<A43><B3> Currency Code	-	C	
<A43><B4> Sign	-	C	
<A43><B5> Additional Amount	-	C	
<A45> Free Form Text	O	O	
<A46> Partial Authorization Indicator	O	-	Use Discover card values if needed
<A47> Amex ITD Data	O	-	
<A47><B1> Secondary ID	O	-	
<A47><B2> Customer e-Mail	O	-	
<A47><B3> Customer Host Name	O	-	
<A47><B4> HTTP Browser Type	O	-	
<A47><B5> Ship to Country	O	-	
<A47><B6> Shipping Method	O	-	
<A47><B7> Merchant Product SKU	O	-	
<A47><B8> Customer IP Address	O	-	
<A48> Amex Merchant DBA Name	O	-	
<A50> Amex Seller/Vendor Code	O	-	

Tag name	→	←	Comment
<u><A51> Digital Entity Identifier</u>	C	-	If transaction comes through aggregator that is Visa Trusted Agent
<u><A51><B1> Digital Entity Identifier</u>	C	-	
<u><A51><B2> Reserved</u>	C	-	
<u><A53> Discover Promotional Code</u>	O	-	
<u><A56> Electromagnetic Signature</u>	C	-	If EM signature used
<u><A57> NTIA UPC/SKU Data</u>	C	-	If <A12> is "P"
<u><A58> Visa Contactless Data Group</u>	C	-	If card was Visa Contactless
<u><A58><B1> Amount Authorized</u>	O	-	
<u><A58><B2> Application Cryptogram</u>	O	-	
<u><A58><B3> Application Transaction Counter</u>	O	-	
<u><A58><B4> Customer Exclusive Data</u>	O	-	
<u><A58><B5> Form Factor Indicator</u>	O	-	
<u><A58><B6> Issuer Application Data</u>	O	-	
<u><A58><B7> Unpredictable Number</u>	O	-	
<u><A58><B8> Card Sequence Number</u>	O	-	
<u><A60> Tip Amount</u>	O	-	
<u><A61> Tax Amount (Commercial Card)</u>	C	-	A61, A62, A63 must be present for commercial card to settle in current batch without adjustment.
<u><A62> Tax Amount Identifier (Commercial Card)</u>	C	-	A61, A62, A63 must be present for commercial card to settle in current batch without adjustment.
<u><A63> Purchase Order Number (Commercial Card)</u>	C	-	A61, A62, A63 must be present for commercial card to settle in current batch without adjustment.
<u><A64> e-Commerce Goods Indicator</u>	O	-	If known at time of purchase
<u><A66> Invoice Number (Card Not Present)</u>	O	-	If known at time of purchase
<u><A67> Soft Descriptor</u>	O	-	

Tag name	→	←	Comment
<u><A81> Local Transaction Date</u>	-	M	
<u><A82> Local Transaction Time</u>	-	M	
<u><A83> Response Code</u>	-	M	
<u><A84> Approval Code</u>	-	C	If <A83> indicates “approved”
<u><A85> Authorization Response Text</u>	-	M	
<u><A86> Retrieval Reference Number (RRN)</u>	-	M	
<u><A87> CCI Response (Commercial Card Indicator)</u>	-	C	If card account is a commercial card
<u><A88> Card Verification Result (CID/CVC2/CVV2)</u>	-	C	If <A31> present
<u><A89> Transaction Data Condition Code</u>	-	C	If Discover/PayPal card type
<u><A91> Merchant Advice Code</u>	-	C	If Mastercard card type
<u><A92> Card Product Codes</u>	-	C	If U.S. issued Visa card type
<u><A93> Card Type</u>	-	M	
<u><A94> Requested ACI</u>	O	-	
<u><A95> Return ACI</u>	-	C	
<u><A97> Address Verification Result Code</u>	-	C	If AVS was requested
<u><A101> Amex Corporate Purchasing Card Group</u>	C	-	If Amex CPC
<u><A101><B1> Requester Name</u>	C	-	If Amex CPC
<u><A101><B2> Shipped to ZIP</u>	C	-	If Amex CPC
<u><A101><B3> Charge Descriptor 1</u>	C	-	If Amex CPC
<u><A101><B4> Charge Descriptor 2</u>	C	-	If Amex CPC
<u><A101><B5> Charge Descriptor 3</u>	C	-	If Amex CPC
<u><A101><B6> Charge Descriptor 4</u>	C	-	If Amex CPC
<u><A102> Amex CAPN Retail Department Name</u>	C	-	If Amex card type
<u><A103> Amex CAPN Retail Group</u>	C	-	If Amex card type
<u><A104> Level III Enhanced Data Group</u>	C	-	If Amex card type
<u><A103><B2> Retail Item Description</u>	C	-	If Amex card type

Tag name	→	←	Comment
<u><A103><B3> Retail Item Quantity</u>	C	-	If Amex card type
<u><A117> TSYS Acquiring Solutions Gateway ID</u>	O	-	

6.9 Credit prepaid activation

Tag name	→	←	Comment
<A1> Host Capture POS ID	M	-	Must be the first tag in the request
<A3> GenKey	M	-	
<A10> Transaction Code	M	-	“15” Prepaid Activation
<A12> Payment Code	M	-	“C”
<A14> Repeat Indicator	C	-	If a repeat message “RPT”
<A15> Transaction Amount	C	-	May be absent, present and zero or present and non-zero
<A16> Secondary Amount	O	-	
<A17> Transaction Original Number (TON)	M	M	
<A18> Merchant Category Code (MCC)	O	-	
<A19> Language Indicator	O	-	
<A21> POS Entry Data	M	-	
<A21><B1> Cardholder Present Data	M	-	
<A21><B2> Card Present Data	M	-	
<A21><B3> Card Data Input Mode	M	-	
<A21><B4> Cardholder Authentication Method	M	-	
<A21><B5> Authentication Entity	M	-	
<A21><B6> Card data input capability	M	-	
<A21><B7> Cardholder authentication capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B8> Card capture capability	O	-	
<A21><B9> Operating environment	M	-	
<A21><B10> Card data output capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B11> Terminal data output capability	O	-	
<A21><B12> PIN capture capability	O	-	

Tag name	→	←	Comment
<A23> Track 1	C	-	Either Track 1 or Track 2 should be sent
<A24> Track 2	C	-	Either Track 1 or Track 2 should be sent
<A25> Manually Entered Card Number	C	-	If Track 1 or Track 2 cannot be read
<A26> Manually Entered Expiry Date	C	-	If Track 1 or Track 2 cannot be read
<A31> Verification Code (CID, CVC2, CVV2)	O	-	
<A41> TSYS Acquiring Solutions Developer ID	M	-	
<A42> TSYS Acquiring Solutions Version ID	M	-	
<A43> Additional Amounts Group	-	C	If additional amounts provided by issuer
<A43><B1> Account Type	-	C	
<A43><B2> Amount Type	-	C	
<A43><B3> Currency Code	-	C	
<A43><B4> Sign	-	C	
<A43><B5> Additional Amount	-	C	
<A48> Amex Merchant DBA Name	O	-	
<A51> Digital Entity Identifier	C	-	If transaction comes through aggregator that is Visa Trusted Agent
<A51><B1> Digital Entity Identifier	C	-	
<A51><B2> Reserved	C	-	
<A67> Soft Descriptor	O	-	
<A80> Batch Number	-	M	Indicates the current open batch
<A81> Local Transaction Date	-	M	
<A82> Local Transaction Time	-	M	
<A83> Response Code	-	M	
<A84> Approval Code	-	C	If <A83> indicates “approved”
<A85> Authorization Response Text	-	M	
<A86> Retrieval Reference Number (RRN)	-	M	
<A93> Card Type	-	M	
<A95> Return ACI	-	C	

Tag name	→	←	Comment
<A117> TSYS Acquiring Solutions Gateway ID	O	-	

6.10 Credit prepaid load

Tag name	→	←	Comment
<A1> Host Capture POS ID	M	-	Must be the first tag in the request
<A3> GenKey	M	-	
<A10> Transaction Code	M	-	“16” Prepaid Activation
<A12> Payment Code	M	-	“C”
<A14> Repeat Indicator	C	-	If a repeat message “RPT”
<A15> Transaction Amount	C	-	Must be >\$0.00
<A16> Secondary Amount	O	-	
<A17> Transaction Original Number (TON)	M	M	
<A18> Merchant Category Code (MCC)	O	-	
<A19> Language Indicator	O	-	
<A21> POS Entry Data	M	-	
<A21><B1> Cardholder Present Data	M	-	
<A21><B2> Card Present Data	M	-	
<A21><B3> Card Data Input Mode	M	-	
<A21><B4> Cardholder Authentication Method	M	-	
<A21><B5> Authentication Entity	M	-	
<A21><B6> Card data input capability	M	-	
<A21><B7> Cardholder authentication capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B8> Card capture capability	O	-	
<A21><B9> Operating environment	M	-	
<A21><B10> Card data output capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B11> Terminal data output capability	O	-	
<A21><B12> PIN capture capability	O	-	

Tag name	→	←	Comment
<A23> Track 1	C	-	Either Track 1 or Track 2 should be sent
<A24> Track 2	C	-	Either Track 1 or Track 2 should be sent
<A25> Manually Entered Card Number	C	-	If Track 1 or Track 2 cannot be read
<A26> Manually Entered Expiry Date	C	-	If Track 1 or Track 2 cannot be read
<A31> Verification Code (CID, CVC2, CVV2)	O	-	
<A41> TSYS Acquiring Solutions Developer ID	M	-	
<A42> TSYS Acquiring Solutions Version ID	M	-	
<A43> Additional Amounts Group	-	C	If additional amounts provided by issuer
<A43><B1> Account Type	-	C	
<A43><B2> Amount Type	-	C	
<A43><B3> Currency Code	-	C	
<A43><B4> Sign	-	C	
<A43><B5> Additional Amount	-	C	
<A48> Amex Merchant DBA Name	O	-	
<A51> Digital Entity Identifier	C	-	If transaction comes through aggregator that is Visa Trusted Agent
<A51><B1> Digital Entity Identifier	C	-	
<A51><B2> Reserved	C	-	
<A67> Soft Descriptor	O	-	
<A80> Batch Number	-	M	Indicates the current open batch
<A81> Local Transaction Date	-	M	
<A82> Local Transaction Time	-	M	
<A83> Response Code	-	M	
<A84> Approval Code	-	C	If <A83> indicates “approved”
<A85> Authorization Response Text	-	M	
<A86> Retrieval Reference Number (RRN)	-	M	
<A93> Card Type	-	M	
<A95> Return ACI	-	C	

Tag name	→	←	Comment
<A117> TSYS Acquiring Solutions Gateway ID	O	-	

6.11 Offline authorization

Tag name	→	←	Comment
<A1> Host Capture POS ID	M	-	Must be the first tag in the request
<A3> GenKey	M	-	
<A10> Transaction Code	M	-	“19” Offline Auth
<A11> Program Code	C	-	
<A12> Payment Code	M	-	
<A15> Transaction Amount	M	-	Must be >\$0.00
<A16> Secondary Amount	O	-	
<A17> Transaction Original Number (TON)	M	M	
<A18> Merchant Category Code (MCC)	O	-	
<A19> Language Indicator	O	-	
<A21> POS Entry Data	M	-	
<A21><B1> Cardholder Present Data	M	-	
<A21><B2> Card Present Data	M	-	
<A21><B3> Card Data Input Mode	M	-	
<A21><B4> Cardholder Authentication Method	M	-	
<A21><B5> Authentication Entity	M	-	
<A21><B6> Card data input capability	M	-	
<A21><B7> Cardholder authentication capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B8> Card capture capability	O	-	
<A21><B9> Operating environment	M	-	
<A21><B10> Card data output capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B11> Terminal data output capability	O	-	
<A21><B12> PIN capture capability	O	-	

Tag name	→	←	Comment
<u><A22> Market Specific Data Identifier</u>	O	-	
<u><A23> Track 1</u>	O	-	If track data can be gathered at time of offline authorization
<u><A24> Track 2</u>	O	-	If track data can be gathered at time of offline authorization
<u><A25> Manually Entered Card Number</u>	C	-	If track data was not available
<u><A26> Manually Entered Expiry Date</u>	C	-	If track data was not available
<u><A41> TSYS Acquiring Solutions Developer ID</u>	M	-	
<u><A42> TSYS Acquiring Solutions Version ID</u>	M	-	
<u><A60> Tip Amount</u>	O	-	
<u><A61> Tax Amount (Commercial Card)</u>	O	-	
<u><A62> Tax Amount Identifier (Commercial Card)</u>	O	-	
<u><A63> Purchase Order Number (Commercial Card)</u>	O	-	
<u><A64> e-Commerce Goods Indicator</u>	O	-	
<u><A65> Partial Shipment Indicator</u>	O	-	
<u><A65> Partial Shipment Indicator</u>	O	-	
<u><A67> Soft Descriptor</u>	O	-	
<u><A68> Settlement Amount</u>	O	-	
<u><A80> Batch Number</u>	-	M	Indicates the current open batch
<u><A81> Local Transaction Date</u>	M	M	
<u><A82> Local Transaction Time</u>	M	M	
<u><A83> Response Code</u>	-	M	
<u><A84> Approval Code</u>	M	M	
<u><A85> Authorization Response Text</u>	-	M	
<u><A86> Retrieval Reference Number (RRN)</u>	-	M	
<u><A91> Merchant Advice Code</u>	O	-	
<u><A92> Card Product Codes</u>	-	C	If U.S. issued Visa card type

Tag name	→	←	Comment
<u><A93> Card Type</u>	-	M	
<u><A95> Return ACI</u>	O	-	
<u><A97> Address Verification Result Code</u>	O	-	
<u><A99> Multiple Clearing Sequence Number</u>	O	-	
<u><A100> Multiple Clearing Sequence Count</u>	O	-	
<u><A101> Amex Corporate Purchasing Card Group</u>	C	-	If Amex CPC
<u><A101><B1> Requester Name</u>	C	-	If Amex CPC
<u><A101><B2> Shipped to ZIP</u>	C	-	If Amex CPC
<u><A101><B3> Charge Descriptor 1</u>	C	-	If Amex CPC
<u><A101><B4> Charge Descriptor 2</u>	C	-	If Amex CPC
<u><A101><B5> Charge Descriptor 3</u>	C	-	If Amex CPC
<u><A101><B6> Charge Descriptor 4</u>	C	-	If Amex CPC
<u><A102> Amex CAPN Retail Department Name</u>	C	-	If Amex card type
<u><A103> Amex CAPN Retail Group</u>	C	-	If Amex card type
<u><A104> Level III Enhanced Data Group</u>	C	-	If Amex card type
<u><A103><B2> Retail Item Description</u>	C	-	If Amex card type
<u><A103><B3> Retail Item Quantity</u>	C	-	If Amex card type
<u><A117> TSYS Acquiring Solutions Gateway ID</u>	O	-	

6.12 Account funding transaction

Tag name	→	←	Comment
<A1> Host Capture POS ID	M	-	Must be the first tag in the request
<A3> GenKey	M	-	
<A10> Transaction Code	M	-	“10” Purchase
<A11> Program Code	C	-	“W” Account Funding
<A12> Payment Code	M	-	“C” or “D”
<A14> Repeat Indicator	C	-	If a repeat message “RPT”
<A15> Transaction Amount	M	-	
<A16> Secondary Amount	O	-	
<A17> Transaction Original Number (TON)	M	M	
<A18> Merchant Category Code (MCC)	O	-	
<A19> Language Indicator	O	-	
<A20> Forwarding Institution ID (FIID)	C	-	If Debit
<A21> POS Entry Data	M	-	
<A21><B1> Cardholder Present Data	M	-	
<A21><B2> Card Present Data	M	-	
<A21><B3> Card Data Input Mode	M	-	
<A21><B4> Cardholder Authentication Method	M	-	
<A21><B5> Authentication Entity	M	-	
<A21><B6> Card data input capability	M	-	
<A21><B7> Cardholder authentication capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B8> Card capture capability	O	-	
<A21><B9> Operating environment	M	-	

Tag name	→	←	Comment
<u><A21><B10> Card data output capability</u>	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<u><A21><B11> Terminal data output capability</u>	O	-	
<u><A21><B12> PIN capture capability</u>	O	-	
<u><A22> Market Specific Data Identifier</u>	O	-	
<u><A23> Track 1</u>	C	-	Either Track 1 or Track 2 should be sent
<u><A24> Track 2</u>	C	-	Either Track 1 or Track 2 should be sent
<u><A25> Manually Entered Card Number</u>	C	-	If direct marketing, or track data cannot be read
<u><A26> Manually Entered Expiry Date</u>	C	-	If direct marketing, or track data cannot be read
<u><A28> PIN Block</u>	C	-	If Debit
<u><A30> Address Verification Data</u>	C	-	If Debit
<u><A30> Address Verification Data</u>	C	-	If cannot read track data then manual AVS
<u><A30><B1> Cardholder Billing Postal Code</u>	C	-	If cannot read track data then manual AVS
<u><A30><B2> Cardholder Billing Address</u>	C	-	If cannot read track data then manual AVS
<u><A30><B3> Cardholder Billing City</u>	O	-	
<u><A30><B4> Cardholder Billing State/Province</u>	O	-	
<u><A30><B5> Cardholder Billing Country</u>	O	-	
<u><A30><B6> Cardholder Billing First Name</u>	O	-	
<u><A30><B7> Cardholder Billing Last Name</u>	O	-	
<u><A30><B8> Cardholder Billing Phone Number</u>	O	-	
<u><A30><B9> Ship to Postal Code</u>	C	-	If Amex
<u><A30><B10> Ship to Address</u>	C	-	If Amex
<u><A30><B11> Ship to City</u>	O	-	
<u><A30><B12> Ship to State/Province</u>	O	-	
<u><A30><B13> Ship to Country</u>	O	-	

Tag name	→	←	Comment
<u><A30><B14> Ship to First Name</u>	O	-	
<u><A30><B15> Ship to Last Name</u>	O	-	
<u><A30><B16> Ship to Phone Number</u>	O	-	
<u><A31> Verification Code (CID, CVC2, CVV2)</u>	C	-	If direct marketing, or cannot read track data
<u><A32> MOTO/e-Commerce Indicator</u>	C	-	If direct marketing
<u><A33> Transaction ID (XID)</u>	O	-	
<u><A34> Cardholder Authentication for Secure e-Commerce Transactions</u>	O	-	
<u><A37> Service Development Indicator</u>	O	-	
<u><A39> Electronic Commerce Security Level Indicator</u>	O	-	
Response Value:	O	-	
<u><A41> TSYS Acquiring Solutions Developer ID</u>	M	-	
<u><A42> TSYS Acquiring Solutions Version ID</u>	M	-	
<u><A45> Free Form Text</u>	O	O	
<u><A47> Amex ITD Data</u>	O	-	
<u><A47><B1> Secondary ID</u>	O	-	
<u><A47><B2> Customer e-Mail</u>	O	-	
<u><A47><B3> Customer Host Name</u>	O	-	
<u><A47><B4> HTTP Browser Type</u>	O	-	
<u><A47><B5> Ship to Country</u>	O	-	
<u><A47><B6> Shipping Method</u>	O	-	
<u><A47><B7> Merchant Product SKU</u>	O	-	
<u><A47><B8> Customer IP Address</u>	O	-	
<u><A48> Amex Merchant DBA Name</u>	O	-	
<u><A50> Amex Seller/Vendor Code</u>	O	-	
<u><A51> Digital Entity Identifier</u>	O	-	

Tag name	→	←	Comment
<u><A51><B1> Digital Entity Identifier</u>	C	-	
<u><A51><B2> Reserved</u>	C	-	
<u><A53> Discover Promotional Code</u>	O	-	
<u><A54> Transaction Type Identifier</u>	C	-	If Mastercard
<u><A56> Electromagnetic Signature</u>	C	-	If EM signature used
<u><A58> Visa Contactless Data Group</u>	C	-	If card was Visa Contactless
<u><A58><B1> Amount Authorized</u>	O	-	
<u><A58><B2> Application Cryptogram</u>	O	-	
<u><A58><B3> Application Transaction Counter</u>	O	-	
<u><A58><B4> Customer Exclusive Data</u>	O	-	
<u><A58><B5> Form Factor Indicator</u>	O	-	
<u><A58><B6> Issuer Application Data</u>	O	-	
<u><A58><B7> Unpredictable Number</u>	O	-	
<u><A58><B8> Card Sequence Number</u>	O	-	
<u><A61> Tax Amount (Commercial Card)</u>	C	-	A61, A62, A63 must be present for commercial card to settle in current batch without adjustment.
<u><A62> Tax Amount Identifier (Commercial Card)</u>	C	-	A61, A62, A63 must be present for commercial card to settle in current batch without adjustment.
<u><A63> Purchase Order Number (Commercial Card)</u>	C	-	A61, A62, A63 must be present for commercial card to settle in current batch without adjustment.
<u><A67> Soft Descriptor</u>	O	-	
<u><A80> Batch Number</u>	-	M	Indicates the current open batch
<u><A81> Local Transaction Date</u>	-	M	
<u><A82> Local Transaction Time</u>	-	M	
<u><A83> Response Code</u>	-	M	
<u><A84> Approval Code</u>	-	C	If <A83> indicates “approved”

Tag name	→	←	Comment
<u><A85> Authorization Response Text</u>	-	M	
<u><A86> Retrieval Reference Number (RRN)</u>	-	M	
<u><A87> CCI Response (Commercial Card Indicator)</u>	-	C	If card account is a commercial card
<u><A88> Card Verification Result (CID/CVC2/CVV2)</u>	-	C	If <A31> present
<u><A89> Transaction Data Condition Code</u>	-	C	If Discover card type
<u><A90> CAVV Result Code</u>	-	C	If CAVV response present
<u><A91> Merchant Advice Code</u>	-	C	If Mastercard card type
<u><A92> Card Product Codes</u>	-	C	If U.S. issued Visa card type
<u><A93> Card Type</u>	-	M	
<u><A95> Return ACI</u>	-	C	
<u><A96> Network ID</u>	-	C	If Debit
<u><A97> Address Verification Result Code</u>	-	C	If AVS was requested
<u><A117> TSYS Acquiring Solutions Gateway ID</u>	O	-	

6.13 Cardholder funds transfer

Tag name	→	←	Comment
<u><A1> Host Capture POS ID</u>	M	-	Must be the first tag in the request
<u><A3> GenKey</u>	M	-	
<u><A10> Transaction Code</u>	M	-	“10” Purchase
<u><A11> Program Code</u>	C	-	“X” Funds Transfer
<u><A12> Payment Code</u>	M	-	“C” or “D”
<u><A14> Repeat Indicator</u>	C	-	If a repeat message “RPT”
<u><A15> Transaction Amount</u>	M	-	
<u><A16> Secondary Amount</u>	O	-	
<u><A17> Transaction Original Number (TON)</u>	M	M	
<u><A18> Merchant Category Code (MCC)</u>	O	-	
<u><A19> Language Indicator</u>	O	-	
<u><A20> Forwarding Institution ID (FIID)</u>	C	-	If Debit
<u><A21> POS Entry Data</u>	M	-	
<u><A21><B1> Cardholder Present Data</u>	M	-	
<u><A21><B2> Card Present Data</u>	M	-	
<u><A21><B3> Card Data Input Mode</u>	M	-	
<u><A21><B4> Cardholder Authentication Method</u>	M	-	
<u><A21><B5> Authentication Entity</u>	M	-	
<u><A21><B6> Card data input capability</u>	M	-	
<u><A21><B7> Cardholder authentication capability</u>	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<u><A21><B8> Card capture capability</u>	O	-	
<u><A21><B9> Operating environment</u>	M	-	

Tag name	→	←	Comment
<u><A21><B10> Card data output capability</u>	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<u><A21><B11> Terminal data output capability</u>	O	-	
<u><A21><B12> PIN capture capability</u>	O	-	
<u><A22> Market Specific Data Identifier</u>	O	-	
<u><A23> Track 1</u>	C	-	Either Track 1 or Track 2 should be sent
<u><A24> Track 2</u>	C	-	Either Track 1 or Track 2 should be sent
<u><A25> Manually Entered Card Number</u>	C	-	If direct marketing, or track data cannot be read
<u><A26> Manually Entered Expiry Date</u>	C	-	If direct marketing, or track data cannot be read
<u><A28> PIN Block</u>	C	-	If Debit
<u><A30> Address Verification Data</u>	C	-	If Debit
<u><A30> Address Verification Data</u>	C	-	If cannot read track data then manual AVS
<u><A30><B1> Cardholder Billing Postal Code</u>	C	-	If cannot read track data then manual AVS
<u><A30><B2> Cardholder Billing Address</u>	C	-	If cannot read track data then manual AVS
<u><A30><B3> Cardholder Billing City</u>	O	-	
<u><A30><B4> Cardholder Billing State/Province</u>	O	-	
<u><A30><B5> Cardholder Billing Country</u>	O	-	
<u><A30><B6> Cardholder Billing First Name</u>	O	-	
<u><A30><B7> Cardholder Billing Last Name</u>	O	-	
<u><A30><B8> Cardholder Billing Phone Number</u>	O	-	
<u><A30><B9> Ship to Postal Code</u>	C	-	If Amex
<u><A30><B10> Ship to Address</u>	C	-	If Amex
<u><A30><B11> Ship to City</u>	O	-	
<u><A30><B12> Ship to State/Province</u>	O	-	
<u><A30><B13> Ship to Country</u>	O	-	

Tag name	→	←	Comment
<u><A30><B14> Ship to First Name</u>	O	-	
<u><A30><B15> Ship to Last Name</u>	O	-	
<u><A30><B16> Ship to Phone Number</u>	O	-	
<u><A31> Verification Code (CID, CVC2, CVV2)</u>	C	-	If cannot read track data
<u><A37> Service Development Indicator</u>	O	-	
<u><A38> Special Condition Indicator</u>	O	-	
<u><A39> Electronic Commerce Security Level Indicator</u>	O	-	
Response Value:	O	-	
<u><A41> TSYS Acquiring Solutions Developer ID</u>	M	-	
<u><A42> TSYS Acquiring Solutions Version ID</u>	M	-	
<u><A45> Free Form Text</u>	O	O	
<u><A48> Amex Merchant DBA Name</u>	O	-	
<u><A53> Discover Promotional Code</u>	O	-	
<u><A56> Electromagnetic Signature</u>	C	-	If EM signature used
<u><A58> Visa Contactless Data Group</u>	C	-	If card was Visa Contactless
<u><A58><B1> Amount Authorized</u>	O	-	
<u><A58><B2> Application Cryptogram</u>	O	-	
<u><A58><B3> Application Transaction Counter</u>	O	-	
<u><A58><B4> Customer Exclusive Data</u>	O	-	
<u><A58><B5> Form Factor Indicator</u>	O	-	
<u><A58><B6> Issuer Application Data</u>	O	-	
<u><A58><B7> Unpredictable Number</u>	O	-	

6.14 Bill pay credit/debit card

Tag name	→	←	Comment
<A1> Host Capture POS ID	M	-	Must be the first tag in the request
<A3> GenKey	M	-	
<A10> Transaction Code	M	-	“10” Purchase
<A11> Program Code	C	-	“Y” Bill Pay
<A12> Payment Code	M	-	“C”
<A14> Repeat Indicator	C	-	If a repeat message “RPT”
<A15> Transaction Amount	M	-	
<A16> Secondary Amount	O	-	
<A17> Transaction Original Number (TON)	M	M	
<A18> Merchant Category Code (MCC)	O	-	
<A19> Language Indicator	O	-	
<A20> Forwarding Institution ID (FIID)	C	-	If Debit
<A21> POS Entry Data	M	-	
<A21><B1> Cardholder Present Data	M	-	
<A21><B2> Card Present Data	M	-	
<A21><B3> Card Data Input Mode	M	-	
<A21><B4> Cardholder Authentication Method	M	-	
<A21><B5> Authentication Entity	M	-	
<A21><B6> Card data input capability	M	-	
<A21><B7> Cardholder authentication capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B8> Card capture capability	O	-	
<A21><B9> Operating environment	M	-	

Tag name	→	←	Comment
<u><A51> Digital Entity Identifier</u>	O	-	
<u><A51><B1> Digital Entity Identifier</u>	O	-	
<u><A51><B2> Reserved</u>	O	-	
<u><A56> Electromagnetic Signature</u>	C	-	If EM signature used
<u><A58> Visa Contactless Data Group</u>	C	-	If card was Visa Contactless
<u><A58><B1> Amount Authorized</u>	O	-	
<u><A58><B2> Application Cryptogram</u>	O	-	
<u><A58><B3> Application Transaction Counter</u>	O	-	
<u><A58><B4> Customer Exclusive Data</u>	O	-	
<u><A58><B5> Form Factor Indicator</u>	O	-	
<u><A58><B6> Issuer Application Data</u>	O	-	
<u><A58><B7> Unpredictable Number</u>	O	-	
<u><A58><B8> Card Sequence Number</u>	O	-	
<u><A60> Tip Amount</u>	O	-	
<u><A61> Tax Amount (Commercial Card)</u>	C	-	A61, A62, A63 must be present for commercial card to settle in current batch without adjustment.
<u><A62> Tax Amount Identifier (Commercial Card)</u>	C	-	A61, A62, A63 must be present for commercial card to settle in current batch without adjustment.
<u><A63> Purchase Order Number (Commercial Card)</u>	C	-	A61, A62, A63 must be present for commercial card to settle in current batch without adjustment.
<u><A67> Soft Descriptor</u>	O	-	
<u><A69> Transaction Fee Amount</u>	O	-	If debit
<u><A80> Batch Number</u>	-	M	Indicates the current open batch
<u><A81> Local Transaction Date</u>	-	M	
<u><A82> Local Transaction Time</u>	-	M	
<u><A83> Response Code</u>	-	M	
<u><A84> Approval Code</u>	-	C	If <A83> indicates “approved”

Tag name	→	←	Comment
<u><A85> Authorization Response Text</u>	-	M	
<u><A86> Retrieval Reference Number (RRN)</u>	-	M	
<u><A87> CCI Response (Commercial Card Indicator)</u>	-	C	If card account is a commercial card
<u><A88> Card Verification Result (CID/CVC2/CVV2)</u>	-	C	If <A31> present
<u><A90> CAVV Result Code</u>	-	C	If CAVV response present
<u><A92> Card Product Codes</u>	-	C	If U.S. issued Visa card type
<u><A93> Card Type</u>	-	M	
<u><A94> Requested ACI</u>	O	-	
<u><A95> Return ACI</u>	-	C	
<u><A96> Network ID</u>	-	C	If Debit
<u><A97> Address Verification Result Code</u>	-	C	If AVS was requested
<u><A99> Multiple Clearing Sequence Number</u>	O	-	
<u><A100> Multiple Clearing Sequence Count</u>	O	-	
<u><A117> TSYS Acquiring Solutions Gateway ID</u>	O	-	

6.15 Quasi cash credit card

Tag name	→	←	Comment
<A1> Host Capture POS ID	M	-	Must be the first tag in the request
<A3> GenKey	M	-	
<A10> Transaction Code	M	-	"12" Quasi-Cash
<A12> Payment Code	M	-	"C"
<A14> Repeat Indicator	C	-	If a repeat message "RPT"
<A15> Transaction Amount	M	-	
<A16> Secondary Amount	O	-	
<A17> Transaction Original Number (TON)	M	M	
<A18> Merchant Category Code (MCC)	O	-	
<A19> Language Indicator	O	-	
<A21> POS Entry Data	M	-	
<A21><B1> Cardholder Present Data	M	-	
<A21><B2> Card Present Data	M	-	
<A21><B3> Card Data Input Mode	M	-	
<A21><B4> Cardholder Authentication Method	M	-	
<A21><B5> Authentication Entity	M	-	
<A21><B6> Card data input capability	M	-	
<A21><B7> Cardholder authentication capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B8> Card capture capability	O	-	
<A21><B9> Operating environment	M	-	
<A21><B10> Card data output capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B11> Terminal data output capability	O	-	
<A21><B12> PIN capture capability	O	-	

Tag name	→	←	Comment
<u><A81> Local Transaction Date</u>	-	M	
<u><A82> Local Transaction Time</u>	-	M	
<u><A83> Response Code</u>	-	M	
<u><A84> Approval Code</u>	-	C	If <A83> indicates “approved”
<u><A85> Authorization Response Text</u>	-	M	
<u><A86> Retrieval Reference Number (RRN)</u>	-	M	
<u><A88> Card Verification Result (CID/CVC2/CVV2)</u>	-	C	If <A31> present
<u><A89> Transaction Data Condition Code</u>	-	C	If Discover card type
<u><A91> Merchant Advice Code</u>	-	C	If Mastercard card type
<u><A92> Card Product Codes</u>	-	C	If U.S. issued Visa card type
<u><A93> Card Type</u>	-	M	
<u><A97> Address Verification Result Code</u>	-	C	If AVS was requested
<u><A101> Amex Corporate Purchasing Card Group</u>	C	-	If Amex CPC
<u><A101><B1> Requester Name</u>	C	-	If Amex CPC
<u><A101><B2> Shipped to ZIP</u>	C	-	If Amex CPC
<u><A101><B3> Charge Descriptor 1</u>	C	-	If Amex CPC
<u><A101><B4> Charge Descriptor 2</u>	C	-	If Amex CPC
<u><A101><B5> Charge Descriptor 3</u>	C	-	If Amex CPC
<u><A101><B6> Charge Descriptor 4</u>	C	-	If Amex CPC
<u><A102> Amex CAPN Retail Department Name</u>	C	-	If Amex card type
<u><A103> Amex CAPN Retail Group</u>	C	-	If Amex card type
<u><A104> Level III Enhanced Data Group</u>	C	-	If Amex card type
<u><A103><B2> Retail Item Description</u>	C	-	If Amex card type
<u><A103><B3> Retail Item Quantity</u>	C	-	If Amex card type
<u><A111> Chip Card Data</u>	C	C	If <A21><B3> is equal to 'C', 'F', or 'M'

Tag name	→	←	Comment
<A117> TSYS Acquiring Solutions Gateway ID	O	-	

6.16 Level III line item detail transaction

Tag name	→	←	Comment
<A1> Host Capture POS ID	M	-	Must be the first tag in the request
<A3> GenKey	M	-	
<A10> Transaction Code	M	-	“105” Level III Line Item Detail
<A15> Transaction Amount	M	-	
<A19> Language Indicator	O	-	
<A80> Batch Number	-	M	Indicates the current open batch
<A81> Local Transaction Date	-	M	
<A82> Local Transaction Time	-	M	
<A83> Response Code	-	M	
<A85> Authorization Response Text	-	M	
<A86> Retrieval Reference Number (RRN)	M	M	
<A105> Level III Line Item Details Group	M	-	
<A105><B1> Item Commodity Code	O	-	
<A105><B2> Item Description	O	-	
<A105><B3> Item Product Code	O	-	
<A105><B4> Item Quantity	M	-	
<A105><B5> Item Unit of Measure	O	-	
<A105><B6> Item Unit Cost	O	-	
<A105><B7> Item Discount Amount	C	-	If item Discount Indicator is present.
<A105><B8> Item Total Amount	M	-	
<A105><B9> Item Alternate Tax Identifier	O	-	
<A105><B10> Item VAT/TAX Rate	O	-	
<A105><B11> Item Tax Type	O	-	
<A105><B12> Item VAT/Tax Amount	O	-	
<A105><B13> Net/Gross Indicator	M	-	Value must be either “Y” or “N”
<A105><B14> Extended Item Amount	M	-	

Tag name	→	←	Comment
<u><A105><B15> Debit/Credit Indicator</u>	M	-	Value must be either “C” or “D”
<u><A105><B16>Item Discount Indicator</u>	C	-	Value must be either “Y” or “N” If present then Item Discount Amount must be present.
<u><A105><B17>Item Discount Rate</u>	O	-	
<u><A105><B18>Quantity Exponent Indicator</u>	O	-	
<u><A105><B19> Discount Rate Exponent</u>	O	-	

6.17 Level III enhanced data transaction

If the Level III Enhanced Data requires a change, resubmit the entire A104 Level III Enhanced Data group transaction.

Tag name	→	←	Comment
<u><A1> Host Capture POS ID</u>	M	-	Must be the first tag in the request
<u><A3> GenKey</u>	M	-	
<u><A10> Transaction Code</u>	M	-	“104” Level III Enhanced Data
<u><A15> Transaction Amount</u>	M	-	
<u><A19> Language Indicator</u>	O	-	
<u><A80> Batch Number</u>	-	M	Indicates the current open batch
<u><A81> Local Transaction Date</u>	-	M	
<u><A82> Local Transaction Time</u>	-	M	
<u><A83> Response Code</u>	-	M	
<u><A85> Authorization Response Text</u>	-	M	
<u><A86> Retrieval Reference Number (RRN)</u>	M	M	
<u><A104> Level III Enhanced Data Group</u>	M	-	
<u><A104><B1> National Tax Amount</u>	O	-	
<u><A104><B2> Freight Amount</u>	O	-	
<u><A104><B3> Duty Amount</u>	O	-	
<u><A104><B4> Ship from Postal/ZIP Code</u>	O	-	

Tag name	→	←	Comment
<u><A104><B5> Destination Postal/ZIP Code</u>	O	-	
<u><A104><B6>Destination Country Code</u>	O	-	
<u><A104><B7>Line Item Count</u>	C	-	If line items have been sent
<u><A104><B8>Merchant VAT Registration Number</u>	O	-	
<u><A104><B9> Customer VAT Registration Number</u>	O	-	
<u><A104><B10>Summary Commodity Code</u>	O	-	
<u><A104><B11>Discount Amount</u>	O	-	
<u><A104><B12>Unique VAT Invoice Reference Number</u>	O	-	
<u><A104><B13>Order Date</u>	O	-	
<u><A104><B14>VAT/TAX Amount (Frt/Ship)</u>	O	-	
<u><A104><B15>VAT/TAX Rate (Frt/Ship)</u>	O	-	
<u><A104><B16>Alternate Tax Amount</u>	O	-	

6.18 Amex Airline Passenger Transport (APD) Transaction

Tag name	→	←	Comment
<A1> Host Capture POS ID	M	-	Must be the first tag in the request
<A3> GenKey	M	-	
<A10> Transaction Code	M	-	"10" or "13"
<A12> Payment Code	M	-	"C" or "P"
<A14> Repeat Indicator	C	-	If a repeat message "RPT"
<A15> Transaction Amount	M	-	
<A16> Secondary Amount	O	-	
<A17> Transaction Original Number (TON)	M	M	
<A18> Merchant Category Code (MCC)	O	-	
<A19> Language Indicator	O	-	
<A21> POS Entry Data	M	-	
<A21><B1> Cardholder Present Data	M	-	
<A21><B2> Card Present Data	M	-	
<A21><B3> Card Data Input Mode	M	-	
<A21><B4> Cardholder Authentication Method	M	-	
<A21><B5> Authentication Entity	M	-	
<A21><B6> Card data input capability	M	-	
<A21><B7> Cardholder authentication capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B8> Card capture capability	O	-	
<A21><B9> Operating environment	M	-	
<A21><B10> Card data output capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B11> Terminal data output capability	O	-	
<A21><B12> PIN capture capability	O	-	

Tag name	→	←	Comment
<u><A23> Track 1</u>	C	-	Either Track 1 or Track 2 should be sent
<u><A24> Track 2</u>	C	-	Either Track 1 or Track 2 should be sent
<u><A25> Manually Entered Card Number</u>	C	-	If Track1 or Track2 cannot be read
<u><A26> Manually Entered Expiry Date</u>	C	-	If<A25> is present and expiration date available
<u><A30> Address Verification Data</u>	C	-	If cannot read track data then manual AVS
<u><A30><B1> Cardholder Billing Postal Code</u>	C	-	If cannot read track data then manual AVS
<u><A30><B2> Cardholder Billing Address</u>	C	-	If cannot read track data then manual AVS
<u><A30><B9> Ship to Postal Code</u>	O	-	Amex only if cannot read track data
<u><A30><B10> Ship to Address</u>	O	-	Amex only if cannot read track data
<u><A31> Verification Code (CID, CVC2, CVV2)</u>	C	-	If cannot read track data
<u><A37> Service Development Indicator</u>	O	-	
<u><A38> Special Condition Indicator</u>	O	-	
<u><A41> TSYS Acquiring Solutions Developer ID</u>	M	-	
<u><A42> TSYS Acquiring Solutions Version ID</u>	M	-	
<u><A43> Additional Amounts Group</u>	-	C	If additional amounts provided by Issuer
<u><A43><B1> Account Type</u>	-	C	
<u><A43><B2> Amount Type</u>	-	C	
<u><A43><B3> Currency Code</u>	-	C	
<u><A43><B4> Sign</u>	-	C	
<u><A43><B5> Additional Amount</u>	-	C	
<u><A45> Free Form Text</u>	O	O	
<u><A46> Partial Authorization Indicator</u>	O	-	
<u><A48> Amex Merchant DBA Name</u>	O	-	
<u><A49> Amex Station Location Code</u>	O	-	
<u><A50> Amex Seller/Vendor Code</u>	O	-	

Tag name	→	←	Comment
<u><A56> Electromagnetic Signature</u>	O	-	If EM signature used
<u><A60> Tip Amount</u>	O	-	
<u><A66> Invoice Number (Card Not Present)</u>	O	-	
<u><A67> Soft Descriptor</u>	O	-	
<u><A71> Amex Airline Data</u>	M	-	
<u><A71><B1> Secondary ID</u>	M	-	
<u><A71><B2> Departure Date</u>	M	-	
<u><A71><B3> Passenger Name</u>	M	-	
<u><A71><B4> Cardmember Name</u>	M	-	
<u><A71><B5> Origin (Origin Airport)</u>	M	-	
<u><A71><B6> Dest (First Segment Travel Destination Airport)</u>	M	-	
<u><A71><B7> Routing Cities</u>	M	-	At least two sub-tags.
<u><A71><B8> Airline Carriers</u>	M	-	At least one sub-tag.
<u><A71><B9> Fare Basis</u>	M	-	
<u><A71><B10> Number of Passengers</u>	M	-	
<u><A71><B11> E-Ticket Indicator</u>	O	-	
<u><A71><B12> Reservation Code</u>	M	-	
<u><A80> Batch Number</u>	-	M	Indicates the current open batch
<u><A81> Local Transaction Date</u>	-	M	
<u><A82> Local Transaction Time</u>	-	M	
<u><A83> Response Code</u>	-	M	
<u><A84> Approval Code</u>	-	C	If <A83> indicates “approved”
<u><A85> Authorization Response Text</u>	-	M	
<u><A86> Retrieval Reference Number (RRN)</u>	-	M	
<u><A88> Card Verification Result (CID/CVC2/CVV2)</u>	-	C	If <A31> present
<u><A93> Card Type</u>	-	M	
<u><A94> Requested ACI</u>	O	-	
<u><A95> Return ACI</u>	-	C	

Tag name	→	←	Comment
<A97> Address Verification Result Code	-	C	If AVS was requested
<A98> Delayed Capture Indicator	O	-	
<A102> Amex CAPN Retail Department Name	C	-	If Amex card type
<A103> Amex CAPN Retail Group	C	-	If Amex card type
<A104> Level III Enhanced Data Group	C	-	If Amex card type
<A103><B2> Retail Item Description	C	-	If Amex card type
<A103><B3> Retail Item Quantity	C	-	If Amex card type
<A117> TSYS Acquiring Solutions Gateway ID	O	-	

6.19 Passenger Transport Data Transaction

If the Passenger Transport Data transaction requires a change, resubmit the entire A72 Passenger Transport Data group in the adjustment transaction.

Tag name	→	←	Comment
<A1> Host Capture POS ID	M	-	Must be the first tag in the request
<A3> GenKey	M	-	
<A10> Transaction Code	M	-	"10" or "13"
<A12> Payment Code	M	-	"C" or "P"
<A14> Repeat Indicator	C	-	If a repeat message "RPT"
<A15> Transaction Amount	M	-	
<A16> Secondary Amount	O	-	
<A17> Transaction Original Number (TON)	M	M	
<A18> Merchant Category Code (MCC)	O	-	
<A19> Language Indicator	O	-	
<A21> POS Entry Data	M	-	
<A21><B1> Cardholder Present Data	M	-	
<A21><B2> Card Present Data	M	-	

Tag name	→	←	Comment
<A21><B3> Card Data Input Mode	M	-	
<A21><B4> Cardholder Authentication Method	M	-	
<A21><B5> Authentication Entity	M	-	
<A21><B6> Card data input capability	M	-	
<A21><B7> Cardholder authentication capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B8> Card capture capability	O	-	
<A21><B9> Operating environment	M	-	
<A21><B10> Card data output capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B11> Terminal data output capability	O	-	
<A21><B12> PIN capture capability	O	-	
<A23> Track 1	C	-	Either Track 1 or Track 2 should be sent
<A24> Track 2	C	-	Either Track 1 or Track 2 should be sent
<A25> Manually Entered Card Number	C	-	If Track1 or Track2 cannot be read
<A26> Manually Entered Expiry Date	C	-	If<A25> is present and expiration date available
<A30> Address Verification Data	C	-	If cannot read track data then manual AVS
<A30><B1> Cardholder Billing Postal Code	C	-	If cannot read track data then manual AVS
<A30><B2> Cardholder Billing Address	C	-	If cannot read track data then manual AVS
<A31> Verification Code (CID, CVC2, CVV2)	C	-	If cannot read track data
<A37> Service Development Indicator	O	-	
<A38> Special Condition Indicator	O	-	
<A41> TSYS Acquiring Solutions Developer ID	M	-	

Tag name	→	←	Comment
<A42> TSYS Acquiring Solutions Version ID	M	-	
<A43> Additional Amounts Group	-	C	If additional amounts provided by Issuer
<A43><B1> Account Type	-	C	
<A43><B2> Amount Type	-	C	
<A43><B3> Currency Code	-	C	
<A43><B4> Sign	-	C	
<A43><B5> Additional Amount	-	C	
<A45> Free Form Text	O	O	
<A46> Partial Authorization Indicator	O	-	Use Discover/PayPal card values if needed
<A48> Amex Merchant DBA Name	O	-	
<A56> Electromagnetic Signature	C	-	If EM signature used
<A60> Tip Amount	O	-	
<A61> Tax Amount (Commercial Card)	C	-	A61, A62, A63 must be present for commercial card to settle in current batch without adjustment.
<A62> Tax Amount Identifier (Commercial Card)	C	-	A61, A62, A63 must be present for commercial card to settle in current batch without adjustment.
<A63> Purchase Order Number (Commercial Card)	C	-	A61, A62, A63 must be present for commercial card to settle in current batch without adjustment.
<A66> Invoice Number (Card Not Present)	O	-	
<A67> Soft Descriptor	O	-	
<A72> Passenger Transportation Data Group	C	-	If available
<A72><B1>Ticket Number	C	-	If available
<A72><B2>Passenger Name	C	-	If available
<A72><B3>Restricted Ticket Indicator	C	-	If available
<A72><B4>Departure Date	C	-	If available
<A72><B5>Origination City/ Airport Code	C	-	If available
<A72><B6>Trip Leg Information	C	-	If available At least 1, can repeat up to 4 times
<A80> Batch Number	-	M	Indicates the current open batch

Tag name	→	←	Comment
<u><A81> Local Transaction Date</u>	-	M	
<u><A82> Local Transaction Time</u>	-	M	
<u><A83> Response Code</u>	-	M	
<u><A84> Approval Code</u>	-	C	If <A83> indicates “approved”
<u><A85> Authorization Response Text</u>	-	M	
<u><A86> Retrieval Reference Number (RRN)</u>	-	M	
<u><A87> CCI Response (Commercial Card Indicator)</u>	-	C	If card account is a commercial card
<u><A88> Card Verification Result (CID/CVC2/CVV2)</u>	-	C	If <A31> present
<u><A89> Transaction Data Condition Code</u>	-	C	If Discover/PayPal card type
<u><A91> Merchant Advice Code</u>	-	C	If Mastercard type
<u><A92> Card Product Codes</u>	-	C	If U.S. issued Visa card type
<u><A93> Card Type</u>	-	M	
<u><A94> Requested ACI</u>	O	-	
<u><A95> Return ACI</u>	-	C	
<u><A97> Address Verification Result Code</u>	-	C	If AVS was requested
<u><A98> Delayed Capture Indicator</u>	O	-	
<u><A117> TSYS Acquiring Solutions Gateway ID</u>	O	-	

6.20 Transaction reversal

Tag name	→	←	Comment
<u><A1> Host Capture POS ID</u>	M	-	Must be the first tag in the request
<u><A3> GenKey</u>	M	-	
<u><A10> Transaction Code</u>	M	-	Original transaction code
<u><A11> Program Code</u>	C	-	If program code used in original
<u><A12> Payment Code</u>	M	-	
<u><A13> Reversal Indicator</u>	M	-	
<u><A15> Transaction Amount</u>	M	-	
<u><A16> Secondary Amount</u>	O	-	
<u><A17> Transaction Original Number (TON)</u>	C	M	Only use the TON if the RRN is not known.

Tag name	→	←	Comment
<u><A18> Merchant Category Code (MCC)</u>	O	-	
<u><A19> Language Indicator</u>	O	-	
<u><A41> TSYS Acquiring Solutions Developer ID</u>	M	-	
<u><A42> TSYS Acquiring Solutions Version ID</u>	M	-	
<u><A48> Amex Merchant DBA Name</u>	O	-	
<u><A67> Soft Descriptor</u>	O	-	
<u><A69> Transaction Fee Amount</u>	C	-	If present in original debit transaction
<u><A80> Batch Number</u>	-	M	Indicates the current open batch
<u><A81> Local Transaction Date</u>	-	M	
<u><A82> Local Transaction Time</u>	-	M	
<u><A83> Response Code</u>	-	M	
<u><A85> Authorization Response Text</u>	-	M	
<u><A86> Retrieval Reference Number (RRN)</u>	C	M	Always use the RRN if it is known.
<u><A93> Card Type</u>	-	C	If the transaction is found in the host.
<u><A117> TSYS Acquiring Solutions Gateway ID</u>	O	-	

6.21 Adjust transaction

Tag name	→	←	Comment
<u><A1> Host Capture POS ID</u>	M	-	Must be the first tag in the request
<u><A3> GenKey</u>	M	-	
<u><A10> Transaction Code</u>	M	-	"101" Adjustment
<u><A15> Transaction Amount</u>	M	-	Original transaction amount
<u><A17> Transaction Original Number (TON)</u>	C	M	Use TON only if RRN is not known, or if application designed for TON uniqueness.
<u><A19> Language Indicator</u>	O	-	
<u><A61> Tax Amount (Commercial Card)</u>	C	-	Commercial card
<u><A62> Tax Amount Identifier (Commercial Card)</u>	C	-	Commercial card
<u><A63> Purchase Order Number (Commercial Card)</u>	C	-	Commercial card
<u><A64> e-Commerce Goods Indicator</u>	O	-	
<u><A66> Invoice Number (Card Not Present)</u>	O	-	Direct marketing
<u><A67> Soft Descriptor</u>	O	-	
<u><A68> Settlement Amount</u>	O	-	Amount to be settled. Used for financial adjustments only.
<u><A72> Passenger Transportation Data Group</u>	O	-	Resend entire group when adjusting Passenger Transport fields
<u><A72><B1>Ticket Number</u>	O	-	
<u><A72><B2>Passenger Name</u>	O	-	
<u><A72><B3>Restricted Ticket Indicator</u>	O	-	
<u><A72><B4>Departure Date</u>	O	-	
<u><A72><B5>Origination City/ Airport Code</u>	O	-	
<u><A72><B6>Trip Leg Information</u>	O	-	At least 1, can repeat up to 4 times
<u><A73> Hold Capture Indicator</u>	O	-	"CAPTURE"; only for Hold Capture participants
<u><A75><B1> Custom Identifier Type</u>	O	-	Mastercard
<u><A75><B2> Custom Identifier Detail</u>	O	-	
<u><A80> Batch Number</u>	-	M	Indicates the current open batch
<u><A81> Local Transaction Date</u>	-	M	

Tag name	→	←	Comment
<u><A82> Local Transaction Time</u>	-	M	
<u><A83> Response Code</u>	-	M	
<u><A85> Authorization Response Text</u>	-	M	
<u><A86> Retrieval Reference Number (RRN)</u>	C	M	Always use the RRN if it is known
<u><A99> Multiple Clearing Sequence Number</u>	O	-	
<u><A100> Multiple Clearing Sequence Count</u>	O	-	
<u><A101> Amex Corporate Purchasing Card Group</u>	C	-	If Amex CPC
<u><A101><B1> Requester Name</u>	C	-	If Amex CPC
<u><A101><B2> Shipped to ZIP</u>	C	-	If Amex CPC
<u><A101><B3> Charge Descriptor 1</u>	C	-	If Amex CPC
<u><A101><B4> Charge Descriptor 2</u>	C	-	If Amex CPC
<u><A101><B5> Charge Descriptor 3</u>	C	-	If Amex CPC
<u><A101><B6> Charge Descriptor 4</u>	C	-	If Amex CPC

6.22 Tip adjustment

Tag name	→	←	Comment
<u><A1> Host Capture POS ID</u>	M	-	Must be the first tag in the request
<u><A3> GenKey</u>	M	-	
<u><A10> Transaction Code</u>	M	-	“102” Tip Adjustment
<u><A15> Transaction Amount</u>	M	-	Original transaction amount
<u><A17> Transaction Original Number (TON)</u>	C	M	Use TON only if RRN is not known, or if application designed for TON uniqueness.
<u><A19> Language Indicator</u>	O	-	
<u><A60> Tip Amount</u>	M	-	
<u><A67> Soft Descriptor</u>	O	-	
<u><A71> Amex Airline Data</u>	-	M	
<u><A81> Local Transaction Date</u>	-	M	
<u><A82> Local Transaction Time</u>	-	M	
<u><A83> Response Code</u>	-	M	
<u><A85> Authorization Response Text</u>	-	M	
<u><A86> Retrieval Reference Number (RRN)</u>	C	M	Always use the RRN if it is known

6.23 Void transaction

Tag name	→	←	Comment
<u><A1> Host Capture POS ID</u>	M	-	Must be the first tag in the request
<u><A3> GenKey</u>	M	-	
<u><A10> Transaction Code</u>	M	-	“103” Void
<u><A15> Transaction Amount</u>	M	-	Original transaction amount
<u><A17> Transaction Original Number (TON)</u>	C	M	Only use the TON if the RRN is not known.
<u><A18> Merchant Category Code (MCC)</u>	O	-	
<u><A19> Language Indicator</u>	O	-	
<u><A67> Soft Descriptor</u>	O	-	
<u><A71> Amex Airline Data</u>	-	M	
<u><A81> Local Transaction Date</u>	-	M	
<u><A82> Local Transaction Time</u>	-	M	
<u><A83> Response Code</u>	-	M	

Tag name	→	←	Comment
<u><A85> Authorization Response Text</u>	-	M	
<u><A86> Retrieval Reference Number (RRN)</u>	C	M	Always use the RRN if it is known.

6.24 Batch summary inquiry

Tag name	→	←	Comment
<u><A1> Host Capture POS ID</u>	M	-	Must be the first tag in the request
<u><A3> GenKey</u>	M	-	
<u><A10> Transaction Code</u>	M	-	“202” Batch Summary
<u><A19> Language Indicator</u>	O	-	
<u><A76> Mastercard EMS Fraud Service Group</u>	O	M	If not present in request, current open batch will be used
<u><A114> Alternate Account ID 1</u>	-	M	
<u><A302> Batch Response Code</u>	-	M	
<u><A303> Batch Response Text</u>	-	M	
<u><A304> Local Batch Close Date</u>	-	C	If batch closed
<u><A305> Local Batch Close Time</u>	-	C	If batch closed
<u><A325> Batch Summary Group</u>	-	M	
<u><A325><B1> Summary Type</u>	-	M	
<u><A325><B2> Purchase Count</u>	-	M	
<u><A325><B3> Refunds Count</u>	-	M	
<u><A325><B4> Net Amount</u>	-	M	
<u><A326> Pending Transaction Count</u>	-	C	If any pending transactions

6.25 Batch detail inquiry

Tag name	→	←	Comment
<u><A1> Host Capture POS ID</u>	M	-	Must be the first tag in the request
<u><A3> GenKey</u>	M	-	
<u><A10> Transaction Code</u>	M	-	“203” Batch Detail
<u><A19> Language Indicator</u>	O	-	
<u><A80> Batch Number</u>	-	M	Indicates the current open batch
<u><A86> Retrieval Reference Number (RRN)</u>	O	-	If not present, the most recent transactions will be sent. If present, the transactions prior to RRN will be sent
<u><A114> Alternate Account ID 1</u>	-	M	
<u><A302> Batch Response Code</u>	-	M	

Tag name	→	←	Comment
<u><A303> Batch Response Text</u>	-	M	
<u><A304> Local Batch Close Date</u>	-	C	If batch closed
<u><A305> Local Batch Close Time</u>	-	C	If batch closed
<u><A307> Requested Transaction Count</u>	O	-	1-5, default is 5
<u><A330> Batch Detail Group</u>	-	M	As many times as requested
<u><A330><B1> Transaction Original Number (TON)</u>	-	M	
<u><A330><B2> Retrieval Reference Number (RRN)</u>	-	M	
<u><A330><B3> Card Type</u>	-	M	
<u><A330><B4> Card Number Last 4 Digits</u>	-	M	
<u><A330><B5> Transaction Code</u>	-	M	
<u><A330><B6> Transaction Amount</u>	-	M	Original authorized amount
<u><A330><B7> Settlement Amount</u>	-	M	

6.26 Transaction detail inquiry

Tag name	→	←	Comment
<u><A1> Host Capture POS ID</u>	M	-	Must be the first tag in the request
<u><A3> GenKey</u>	M	-	
<u><A10> Transaction Code</u>	M	-	“204” Transaction Detail
<u><A17> Transaction Original Number (TON)</u>	C	-	If RRN not known
<u><A19> Language Indicator</u>	O	-	
<u><A32> MOTO/e-Commerce Indicator</u>	-	C	If data present
<u><A60> Tip Amount</u>	-	C	If data present
<u><A61> Tax Amount (Commercial Card)</u>	-	C	If data present
<u><A62> Tax Amount Identifier (Commercial Card)</u>	-	C	If data present
<u><A63> Purchase Order Number (Commercial Card)</u>	-	C	If data present
<u><A64> e-Commerce Goods Indicator</u>	-	C	If data present
<u><A65> Partial Shipment Indicator</u>	-	C	If data present
<u><A66> Invoice Number (Card Not Present)</u>	-	C	If data present
<u><A68> Settlement Amount</u>	-	M	
<u><A69> Transaction Fee Amount</u>	-	C	If data present
<u><A80> Batch Number</u>	-	M	Indicates the current open batch
<u><A81> Local Transaction Date</u>	-	M	
<u><A82> Local Transaction Time</u>	-	M	
<u><A83> Response Code</u>	-	M	
<u><A84> Approval Code</u>	-	M	
<u><A86> Retrieval Reference Number (RRN)</u>	C	-	If RRN known
<u><A88> Card Verification Result (CID/CVC2/CVV2)</u>	-	C	If data present
<u><A89> Transaction Data Condition Code</u>	-	C	If data present
<u><A91> Merchant Advice Code</u>	-	C	If data present
<u><A92> Card Product Codes</u>	-	C	If data present
<u><A93> Card Type</u>	-	M	

Tag name	→	←	Comment
<u><A97> Address Verification Result Code</u>	-	C	If AVS was requested
<u><A94> Requested ACI</u>	-	C	If data present
<u><A95> Return ACI</u>	-	C	If data present
<u><A96> Network ID</u>	-	C	If data present
<u><A114> Alternate Account ID 1</u>	-	M	
<u><A302> Batch Response Code</u>		M	
<u><A303> Batch Response Text</u>		M	
<u><A330> Batch Detail Group</u>	-	M	
<u><A330><B1> Transaction Original Number (TON)</u>	-	M	
<u><A330><B2> Retrieval Reference Number (RRN)</u>	-	M	
<u><A330><B3> Card Type</u>	-	M	
<u><A330><B4> Card Number Last 4 Digits</u>	-	M	
<u><A330><B5> Transaction Code</u>	-	M	Original auth transaction code
<u><A330><B6> Transaction Amount</u>	-	M	
<u><A330><B7> Settlement Amount</u>	-	M	

6.27 Manual close batch

Tag name	→	←	Comment
<A1> Host Capture POS ID	M	-	Must be the first tag in the request
<A3> GenKey	M	-	
<A10> Transaction Code	M	-	“201” Batch Close
<A19> Language Indicator	O	-	
<A80> Batch Number	M	M	Indicates the current open batch
<A302> Batch Response Code	-	M	
<A303> Batch Response Text	-	M	
<A304> Local Batch Close Date	-	M	
<A305> Local Batch Close Time	-	M	

6.28 Encryption and tokenization

6.28.1 Credit Request with Token Request - Card Not Present - Manually Entered

This is a sample retail credit request that is also requesting a token.

Tag name	→	←	Comment
<A1> Host Capture POS ID	M	-	Must be the first tag in the request.
<A3> GenKey	M	-	
<A10> Transaction Code	M	-	“10” or “13”(“5T” is simple token request)
<A12> Payment Code	M	-	“C”
<A14> Repeat Indicator	C	-	If a repeat message “RPT”
<A15> Transaction Amount	M	-	
<A16> Secondary Amount	O	-	
<A17> Transaction Original Number (TON)	M	M	
<A18> Merchant Category Code (MCC)	O	-	
<A19> Language Indicator	O	-	
<A21> POS Entry Data	M	-	
<A21><B1> Cardholder Present Data	M	-	
<A21><B2> Card Present Data	M	-	
<A21><B3> Card Data Input Mode	M	-	

Tag name	→	←	Comment
<A21><B4> Cardholder Authentication Method	M	-	
<A21><B5> Authentication Entity	M	-	
<A21><B6> Card data input capability	M	-	
<A21><B7> Cardholder authentication capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B8> Card capture capability	O	-	
<A21><B9> Operating environment	M	-	
<A21><B10> Card data output capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B11> Terminal data output capability	O	-	
<A21><B12> PIN capture capability	O	-	
<A22> Market Specific Data Identifier	O	-	
<A30> Address Verification Data	C	-	If AVS requested
<A30><B1> Cardholder Billing Postal Code	C	-	If AVS requested
<A30><B2> Cardholder Billing Address	C	-	If AVS requested
<A30><B9> Ship to Postal Code	O	-	If Amex
<A30><B10> Ship to Address	O	-	If Amex
<A31> Verification Code (CID, CVC2, CVV2)	O	-	
<A37> Service Development Indicator	O	-	
<A38> Special Condition Indicator	O	-	
<A41> TSYS Acquiring Solutions Developer ID	M	-	
<A42> TSYS Acquiring Solutions Version ID	M	-	

Tag name	→	←	Comment
<A85> Authorization Response Text	-	M	
<A86> Retrieval Reference Number (RRN)	-	M	
<A87> CCI Response (Commercial Card Indicator)	-	C	If card account is a commercial card
<A88> Card Verification Result (CID/CVC2/CVV2)	-	C	If result sent from issuer
<A89> Transaction Data Condition Code	-	C	If Discover card type
<A91> Merchant Advice Code	-	C	If Mastercard card type
<A92> Card Product Codes	-	C	If U.S. issued Visa card type
<A93> Card Type	-	M	"Credit"
<A94> Requested ACI	O	-	
<A95> Return ACI	-	C	
<A97> Address Verification Result Code	-	C	If issuer provides
<A98> Delayed Capture Indicator	O	-	
<A101> Amex Corporate Purchasing Card Group	C	-	If Amex CPC
<A101><B1> Requester Name	C	-	If Amex CPC
<A101><B2> Shipped to ZIP	C	-	If Amex CPC
<A101><B3> Charge Descriptor 1	C	-	If Amex CPC
<A101><B4> Charge Descriptor 2	C	-	If Amex CPC
<A101><B5> Charge Descriptor 3	C	-	If Amex CPC
<A101><B6> Charge Descriptor 4	C	-	If Amex CPC
<A102> Amex CAPN Retail Department Name	C	-	If Amex card type
<A103> Amex CAPN Retail Group	C	-	If Amex card type
<A104> Level III Enhanced Data Group	C	-	If Amex card type
<A103><B2> Retail Item Description	C	-	If Amex card type
<A103><B3> Retail Item Quantity	C	-	If Amex card type

Tag name	→	←	Comment
<A117> TSYS Acquiring Solutions Gateway ID	O	-	

6.28.2 Encrypted retail credit

Tag name	→	←	Comment
<A1> Host Capture POS ID	M	-	Must be the first tag in the request.
<A3> GenKey	M	-	
<A10> Transaction Code	M	-	“10” or “13”
<A12> Payment Code	M	-	“C” or “P”
<A14> Repeat Indicator			If a repeat message “RPT”
<A15> Transaction Amount	M	-	
<A16> Secondary Amount	O	-	
<A17> Transaction Original Number (TON)	M	M	
<A18> Merchant Category Code (MCC)	O	-	
<A19> Language Indicator	O	-	
<A21> POS Entry Data	M	-	
<A21><B1> Cardholder Present Data	M	-	
<A21><B2> Card Present Data	M	-	
<A21><B3> Card Data Input Mode	M	-	
<A21><B4> Cardholder Authentication Method	M	-	
<A21><B5> Authentication Entity	M	-	
<A21><B6> Card data input capability	M	-	
<A21><B7> Cardholder authentication capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B8> Card capture capability	O	-	
<A21><B9> Operating environment	M	-	

Tag name	→	←	Comment
<A21><B10> Card data output capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B11> Terminal data output capability	O	-	
<A21><B12> PIN capture capability	O	-	
<A30> Address Verification Data	C	-	If AVS requested
<A30><B1> Cardholder Billing Postal Code	C	-	If AVS requested
<A30><B9> Ship to Postal Code	C	-	Amex only if cannot read track data
<A30><B10> Ship to Address	C	-	Amex only if cannot read track data
<A37> Service Development Indicator	O	-	
<A38> Special Condition Indicator	O	-	
<A41> TSYS Acquiring Solutions Developer ID	M	-	
<A42> TSYS Acquiring Solutions Version ID	M	-	
<A43> Additional Amounts Group	-	C	If additional amounts provided by Issuer
<A43><B1> Account Type	-	C	
<A43><B2> Amount Type	-	C	
<A43><B3> Currency Code	-	C	
<A43><B4> Sign	-	C	
<A43><B5> Additional Amount	-	C	
<A45> Free Form Text	O	O	
<A46> Partial Authorization Indicator	O	-	Use Discover card values if needed
<A48> Amex Merchant DBA Name	O	-	
<A49> Amex Station Location Code	O	-	
<A50> Amex Seller/Vendor Code	O	-	
<A51> Digital Entity Identifier	O	-	If transaction comes through aggregator that is Visa Trusted Agent
<A51><B1> Digital Entity Identifier	C	-	

Tag name	→	←	Comment
<u><A51><B2> Reserved</u>	C	-	
<u><A53> Discover Promotional Code</u>	O	-	
<u><A56> Electromagnetic Signature</u>	C	-	If EM signature used
<u><A58> Visa Contactless Data Group</u>	C	-	If card was Visa Contactless
<u><A58><B1> Amount Authorized</u>	O	-	
<u><A58><B2> Application Cryptogram</u>	O	-	
<u><A58><B3> Application Transaction Counter</u>	O	-	
<u><A58><B4> Customer Exclusive Data</u>	O	-	
<u><A58><B5> Form Factor Indicator</u>	O	-	
<u><A58><B6> Issuer Application Data</u>	O	-	
<u><A58><B7> Unpredictable Number</u>	O	-	
<u><A58><B8> Card Sequence Number</u>	O	-	
<u><A60> Tip Amount</u>	O	-	
<u><A61> Tax Amount (Commercial Card)</u>	C	-	A61, A62, A63 must be present for commercial card to settle in current batch without adjustment.
<u><A62> Tax Amount Identifier (Commercial Card)</u>	C	-	A61, A62, A63 must be present for commercial card to settle in current batch without adjustment.
<u><A63> Purchase Order Number (Commercial Card)</u>	C	-	A61, A62, A63 must be present for commercial card to settle in current batch without adjustment.
<u><A66> Invoice Number (Card Not Present)</u>	O	-	
<u><A67> Soft Descriptor</u>	O	-	
<u><A77><B1> Transit Transaction Type Indicator</u>	O	-	
<u><A77><B2> Transportation Mode Indicator</u>	O	-	
<u><A80> Batch Number</u>	-	M	Indicates the current open batch
<u><A81> Local Transaction Date</u>	-	M	
<u><A82> Local Transaction Time</u>	-	M	

Tag name	→	←	Comment
<u><A83> Response Code</u>	-	M	
<u><A84> Approval Code</u>	-	C	If <A83> indicates “approved”
<u><A85> Authorization Response Text</u>	-	M	
<u><A86> Retrieval Reference Number (RRN)</u>	-	M	
<u><A87> CCI Response (Commercial Card Indicator)</u>	-	C	If card account is a commercial card
<u><A88> Card Verification Result (CID/CVC2/CVV2)</u>	-	C	If <A31> is present
<u><A89> Transaction Data Condition Code</u>	-	C	If Discover/PayPal card type
<u><A91> Merchant Advice Code</u>	-	C	If Mastercard card type
<u><A92> Card Product Codes</u>	-	C	If U.S. issued Visa card type
<u><A93> Card Type</u>	-	M	
<u><A94> Requested ACI</u>	O	-	
<u><A95> Return ACI</u>	-	C	
<u><A97> Address Verification Result Code</u>	-	C	If AVS was requested
<u><A98> Delayed Capture Indicator</u>	C	-	Send if this transaction should not settle at the next batch close. Send if this authorization will be followed by Level III data.
<u><A101> Amex Corporate Purchasing Card Group</u>	C	-	If Amex CPC
<u><A101><B1> Requester Name</u>	C	-	If Amex CPC
<u><A101><B2> Shipped to ZIP</u>	C	-	If Amex CPC
<u><A101><B3> Charge Descriptor 1</u>	C	-	If Amex CPC
<u><A101><B4> Charge Descriptor 2</u>	C	-	If Amex CPC
<u><A101><B5> Charge Descriptor 3</u>	C	-	If Amex CPC
<u><A101><B6> Charge Descriptor 4</u>	C	-	If Amex CPC
<u><A102> Amex CAPN Retail Department Name</u>	C	-	If Amex card type
<u><A103> Amex CAPN Retail Group</u>	C	-	If Amex card type
<u><A104> Level III Enhanced Data Group</u>	C	-	If Amex card type

Tag name	→	←	Comment
<u><A103><B2> Retail Item Description</u>	C	-	If Amex card type
<u><A103><B3> Retail Item Quantity</u>	C	-	If Amex card type
<u><A106> Encrypted Transaction</u>	C	-	
<u><A106><B1>Encryption Type</u>	C	-	
<u><A106><B2>Encryption Transmission Block (ETB)</u>	C	-	
<u><A107> Encrypted Track 1</u>	C	-	
<u><A108> Encrypted Track 2</u>	C	-	
<u><A109> Encrypted Manually Entered Data</u>	C	-	
<u><A109><B1> Encrypted Manually Entered Card Number</u>	C	-	
<u>< A109><B2>Manually Entered Expiration Date</u>	C	-	
<u>< A109><B3> Encrypted Manually Entered Verification Code (CID, CVC2, CVV2)</u>	C	-	
<u><A117> TSYS Acquiring Solutions Gateway ID</u>	O	-	

6.28.3 Encrypted card not present credit card

Tag name	→	←	Comment
<A1> Host Capture POS ID	M	-	Must be the first tag in the request.
<A3> GenKey	M	-	
<A10> Transaction Code	M	-	“10” or “13”
<A12> Payment Code	M	-	“C” or “P”
<A14> Repeat Indicator	C	-	If a repeat message “RPT”
<A15> Transaction Amount	M	-	
<A16> Secondary Amount	O	-	
<A17> Transaction Original Number (TON)	M	M	
<A18> Merchant Category Code (MCC)	O	-	
<A19> Language Indicator	O	-	
<A21> POS Entry Data	M	-	
<A21><B1> Cardholder Present Data	M	-	
<A21><B2> Card Present Data	M	-	
<A21><B3> Card Data Input Mode	M	-	
<A21><B4> Cardholder Authentication Method	M	-	
<A21><B5> Authentication Entity	M	-	
<A21><B6> Card data input capability	M	-	
<A21><B7> Cardholder authentication capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B8> Card capture capability	O	-	
<A21><B9> Operating environment	M	-	
<A21><B10> Card data output capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B11> Terminal data output capability	O	-	
<A21><B12> PIN capture capability	O	-	

Tag name	→	←	Comment
<A30> Address Verification Data	C	-	If AVS requested
<A30><B1> Cardholder Billing Postal Code	C	-	If AVS requested
<A30><B2> Cardholder Billing Address	C	-	If AVS requested
<A30><B9> Ship to Postal Code	C	-	Amex only if cannot read track data
<A30><B10> Ship to Address	C	-	Amex only if cannot read track data
<A37> Service Development Indicator	O	-	
<A38> Special Condition Indicator	O	-	
<A41> TSYS Acquiring Solutions Developer ID	M	-	
<A42> TSYS Acquiring Solutions Version ID	M	-	
<A43> Additional Amounts Group	-	C	If additional amounts provided by Issuer
<A43><B1> Account Type	-	C	
<A43><B2> Amount Type	-	C	
<A43><B3> Currency Code	-	C	
<A43><B4> Sign	-	C	
<A43><B5> Additional Amount	-	C	
<A45> Free Form Text	O	O	
<A46> Partial Authorization Indicator	O	-	Use Discover card values if needed
<A48> Amex Merchant DBA Name	O	-	
<A49> Amex Station Location Code	O	-	
<A50> Amex Seller/Vendor Code	O	-	
<A51> Digital Entity Identifier	C	-	If transaction comes through aggregator that is Visa Trusted Agent
<A51><B1> Digital Entity Identifier	C	-	
<A51><B2> Reserved	C	-	
<A53> Discover Promotional Code	O	-	

Tag name	→	←	Comment
<A56> Electromagnetic Signature	C	-	If EM signature used
<A58> Visa Contactless Data Group	C	-	If card was Visa Contactless
<A58><B1> Amount Authorized	O	-	
<A58><B2> Application Cryptogram	O	-	
<A58><B3> Application Transaction Counter	O	-	
<A58><B4> Customer Exclusive Data	O	-	
<A58><B5> Form Factor Indicator	O	-	
<A58><B6> Issuer Application Data	O	-	
<A58><B7> Unpredictable Number	O	-	
<A58><B8> Card Sequence Number	O	-	
<A60> Tip Amount	O	-	
<A61> Tax Amount (Commercial Card)	C	-	A61, A62, A63 must be present for commercial card to settle in current batch without adjustment.
<A62> Tax Amount Identifier (Commercial Card)	C	-	A61, A62, A63 must be present for commercial card to settle in current batch without adjustment.
<A63> Purchase Order Number (Commercial Card)	C	-	A61, A62, A63 must be present for commercial card to settle in current batch without adjustment.
<A66> Invoice Number (Card Not Present)	O	-	
<A67> Soft Descriptor	O	-	
<A77><B1> Transit Transaction Type Indicator	O	-	
<A77><B2> Transportation Mode Indicator	O	-	
<A80> Batch Number	-	M	Indicates the current open batch
<A81> Local Transaction Date	-	M	
<A82> Local Transaction Time	-	M	
<A83> Response Code	-	M	
<A84> Approval Code	-	C	If <A83> indicates “approved”

Tag name	→	←	Comment
<A85> Authorization Response Text	-	M	
<A86> Retrieval Reference Number (RRN)	-	M	
<A87> CCI Response (Commercial Card Indicator)	-	C	If card account is a commercial card
<A88> Card Verification Result (CID/CVC2/CVV2)	-	C	If <A31> is present
<A89> Transaction Data Condition Code	-	C	If Discover card type
<A91> Merchant Advice Code	-	C	If Mastercard card type
<A92> Card Product Codes	-	C	If U.S. issued Visa card type
<A93> Card Type	-	M	
<A94> Requested ACI	O	-	
<A95> Return ACI	-	C	
<A97> Address Verification Result Code	-	C	If AVS was requested
<A98> Delayed Capture Indicator	C	-	Send if this transaction should not settle at the next batch close. Send if this authorization will be followed by Level III data.
<A101> Amex Corporate Purchasing Card Group	C	-	If Amex CPC
<A101><B1> Requester Name	C	-	If Amex CPC
<A101><B2> Shipped to ZIP	C	-	If Amex CPC
<A101><B3> Charge Descriptor 1	C	-	If Amex CPC
<A101><B4> Charge Descriptor 2	C	-	If Amex CPC
<A101><B5> Charge Descriptor 3	C	-	If Amex CPC
<A101><B6> Charge Descriptor 4	C	-	If Amex CPC
<A102> Amex CAPN Retail Department Name	C	-	If Amex card type
<A103> Amex CAPN Retail Group	C	-	If Amex card type
<A104> Level III Enhanced Data Group	C	-	If Amex card type
<A103><B2> Retail Item Description	C	-	If Amex card type
<A103><B3> Retail Item Quantity	C	-	If Amex card type

Tag name	→	←	Comment
<u><A106> Encrypted Transaction</u>	C	-	
<u><A106><B1>Encryption Type</u>	C	-	
<u><A106><B2>Encryption Transmission Block (ETB)</u>	C	-	
<u><A109> Encrypted Manually Entered Data</u>	C	-	
<u><A109><B1> Encrypted Manually Entered Card Number</u>	C	-	
<u>< A109><B2>Manually Entered Expiration Date</u>	C	-	
<u>< A109><B3> Encrypted Manually Entered Verification Code (CID, CVC2, CVV2)</u>	C	-	
<u><A117> TSYS Acquiring Solutions Gateway ID</u>	O	-	

6.29 ACH Transaction

Tag name	→	←	Comment
<A1> Host Capture POS ID	M	-	Must be the first tag in the request.
<A3> GenKey	M	-	
<A10> Transaction Code	M	-	Must be “10” for purchase Must be “13” for return
<A12> Payment Code	M	-	Must be “A”
<A14> Repeat Indicator	C	-	If a repeat message “RPT”
<A15> Transaction Amount	M	-	
<A17> Transaction Original Number (TON)	M	M	
<A18> Merchant Category Code (MCC)	O	-	
<A19> Language Indicator	O	-	
<A21> POS Entry Data	M	-	
<A21><B1> Cardholder Present Data	M	-	
<A21><B2> Card Present Data	M	-	
<A21><B3> Card Data Input Mode	M	-	
<A21><B4> Cardholder Authentication Method	M	-	
<A21><B5> Authentication Entity	M	-	
<A21><B6> Card data input capability	M	-	
<A21><B7> Cardholder authentication capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B8> Card capture capability	O	-	
<A21><B9> Operating environment	M	-	
<A21><B10> Card data output capability	O	-	These Tags are optional because they can be augmented by the Host System if not sent by the POS Software. This data will be accurate on the Host only if the Acquirer keeps the data up to date in the merchant boarding profile. In most cases, this information is best transmitted by the POS Software.
<A21><B11> Terminal data output capability	O	-	
<A21><B12> PIN capture capability	O	-	

Tag name	→	←	Comment
<u><A25> Manually Entered Card Number</u>	M	-	
<u><A32> MOTO/e-Commerce Indicator</u>	O	-	
<u><A37> Service Development Indicator</u>	O	-	
<u><A38> Special Condition Indicator</u>	O	-	
<u><A41> TSYS Acquiring Solutions Developer ID</u>	M	-	
<u><A42> TSYS Acquiring Solutions Version ID</u>	M	-	
<u><A45> Free Form Text</u>	O	O	
<u><A66> Invoice Number (Card Not Present)</u>	O	-	
<u><A67> Soft Descriptor</u>	O	-	
<u><A80> Batch Number</u>	-	M	Indicates the current batch open
<u><A81> Local Transaction Date</u>	O	M	
<u><A82> Local Transaction Time</u>	O	M	
<u><A83> Response Code</u>	-	M	
<u><A84> Approval Code</u>	-	C	If <A83> indicates “approved”
<u><A85> Authorization Response Text</u>	-	M	
<u><A86> Retrieval Reference Number (RRN)</u>	-	M	
<u><A93> Card Type</u>	-	M	
<u><A117> TSYS Acquiring Solutions Gateway ID</u>	O	-	
<u><A128><B1> ACH SEC Code</u>	M	-	
<u><A128><B2> ACH Routing Number</u>	M	-	
<u><A128><B3> ACH Check Number</u>	O	-	
<u><A128><B4> ACH Individual Name</u>	O	-	
<u><A128><B5> ACH Individual Reference ID</u>	O	-	
<u><A128><B6> ACH Telephone Number</u>	O	-	
<u><A128><B7> ACH Terminal City</u>	O	-	

Tag name	→	←	Comment
<u><A128><B8> ACH Terminal State</u>	O	-	
<u><A128><B9> ACH Prenote Indicator</u>	M	-	
<u><A128><B10> ACH Checking/Savings Indicator</u>	O	-	
<u><A128><B11> ACH Payment Type Indicator</u>	O	-	
<u><A128><B12> ACH Returned Check Indicator</u>	O	-	
<u><A128><B13> ACH Check Reversal Indicator</u>	O	-	

6.30 3D Secure Transaction

Tag name	→	←	Comment
<u><A1> Host Capture POS ID</u>	M	-	Must be the first tag in the request.
<u><A3> GenKey</u>	M	-	
<u><A10> Transaction Code</u>	M	-	Must be “10” for purchase Must be “13” for return
<u><A12> Payment Code</u>	M	-	Must be “C”
<u><A14> Repeat Indicator</u>	C	-	If a repeat message “RPT”
<u><A15> Transaction Amount</u>	M	-	
<u><A17> Transaction Original Number (TON)</u>	M	M	
<u><A18> Merchant Category Code (MCC)</u>	O	-	If not provided, it will be augmented from the user profile.
<u><A21> POS Entry Data</u>	M	-	
<u><A21><B1> Cardholder Present Data</u>	M	-	5 - Cardholder not present, electronic
<u><A21><B2> Card Present Data</u>	M	-	0 - Card not present (For Amex, if from a Digital Wallet, should use value 'Z'.)
<u><A21><B3> Card Data Input Mode</u>	M	-	S - Electronic commerce, no security, channel encrypted, or SET without cardholder certificate
<u><A21><B4> Cardholder Authentication Method</u>	M	-	0 - Not authenticated
<u><A21><B5> Authentication Entity</u>	M	-	0 - Not authenticated
<u><A21><B6> Card data input capability</u>	M	-	6 - Key entry only capability
<u><A21><B7> Cardholder authentication capability</u>	O	-	If not provided, will be augmented from the user profile.
<u><A21><B8> Card capture capability</u>	O	-	If not provided, will be augmented from the user profile.
<u><A21><B9> Operating environment</u>	M	-	Q - Off card acceptor premises; cardholder mobile environment, including home PC, mobile phone, PDA
<u><A21><B10> Card data output capability</u>	O	-	If not provided, will be augmented from the user profile.
<u><A21><B11> Terminal data output capability</u>	O	-	If not provided, will be augmented from the user profile.
<u><A21><B12> PIN capture capability</u>	O	-	If not provided, will be augmented from the user profile.

Tag name	→	←	Comment
<A25> Manually Entered Card Number	M	-	
<A26> Manually Entered Expiry Date	M	-	
<A30> Address Verification Data	C	-	If AVS requested
<A30><B1> Cardholder Billing Postal Code	C	-	If AVS requested
<A30><B2> Cardholder Billing Address	C	-	If AVS requested
<A30><B3> Cardholder Billing City through <A30><B16> Ship to Phone Number	O	-	If Amex
<A31> Verification Code (CID, CVC2, CVV2)	O	-	If CID, CVC2, or CVV2 is available.
<A32> MOTO/e-Commerce Indicator	O	-	5 - Fully authenticated CAVV 6 - Non-authenticated, but attempted 3-D secure authentication 7 - Non-authenticated security transaction 8 - Non-secure e-commerce transaction A - In-App authentication (usually Apple Pay) for Discover, Mastercard, and Amex
<A33> Transaction ID (XID)	O	-	This is the XID for Visa, Discover, and Amex, or the Token Block B for Amex Tokens.
<A34> Cardholder Authentication for Secure e-Commerce Transactions	O	-	This is the CAVV for Visa, Discover, and Amex, or the Token Block A for Amex Tokens.
<A39> Electronic Commerce Security Level Indicator	O	-	If Mastercard - this is sent instead of A33/A34
<A40> Authentication Data	O	-	If Mastercard
<A41> TSYS Acquiring Solutions Developer ID	M	-	
<A42> TSYS Acquiring Solutions Version ID	M	-	
<A51> Digital Entity Identifier	C	-	If Visa Checkout is used
<A78> Transaction ID	C	-	If provided
<A80> Batch Number	-	M	Indicates the current batch open
<A81> Local Transaction Date	O	M	

Tag name	→	←	Comment
<u><A82> Local Transaction Time</u>	O	M	
<u><A83> Response Code</u>	-	M	
<u><A84> Approval Code</u>	-	C	If <A83> indicates “approved”
<u><A85> Authorization Response Text</u>	-	M	
<u><A86> Retrieval Reference Number (RRN)</u>	-	M	
<u><A87> CCI Response (Commercial Card Indicator)</u>	-	C	If card account is commercial card
<u><A88> Card Verification Result (CID/CVC2/CVV2)</u>	-	C	If result sent from issuer
<u><A90> CAVV Result Code</u>	-	C	If CAVV response provided
<u><A92> Card Product Codes</u>	-	M	
<u><A93> Card Type</u>	-	M	
<u><A94> Requested ACI</u>	O	-	Y
<u><A95> Return ACI</u>	-	C	
<u><A97> Address Verification Result Code</u>	-	C	If issuer provides
<u><A118> Authorization Only Indicator</u>	C	-	AUTH_ONLY If the boarding profile at TSYS is set-up as Authorization Only, then this is not necessary.
<u><A119> Visa Validation Code</u>	-	C	If provided
<u><A120> Authorization Source Code</u>	-	C	If provided
<u><A130> Authorization Indicator</u>	C	-	If Mastercard

6.31 Hold Capture Original Request Message

Tag name	→	←	Comment
<u><A1> Host Capture POS ID</u>	M	-	Must be the first tag in the request.
<u><A3> GenKey</u>	M	-	Value obtained from an Authentication request
<u><A10> Transaction Code</u>	M	-	Must be "10" for purchase
<u><A12> Payment Code</u>	M	-	Must be "C"
<u><A15> Transaction Amount</u>	M	-	
<u><A17> Transaction Original Number (TON)</u>	M	M	The TON must be a unique value to identify the transaction. The Request TON is echoed back in the response.
<u><A18> Merchant Category Code (MCC)</u>	O	-	If not provided, it will be augmented from the user profile.
<u><A21> POS Entry Data</u>	M	-	
<u><A21><B1> Cardholder Present Data</u>	M	-	5 - Cardholder not present, electronic
<u><A21><B2> Card Present Data</u>	M	-	0 - Card not present (For Amex, if from a Digital Wallet, should use value 'Z'.)
<u><A21><B3> Card Data Input Mode</u>	M	-	S - Electronic commerce, no security, channel encrypted, or SET without cardholder certificate
<u><A21><B4> Cardholder Authentication Method</u>	M	-	0 - Not authenticated
<u><A21><B5> Authentication Entity</u>	M	-	0 - Not authenticated
<u><A21><B6> Card data input capability</u>	M	-	6 - Key entry only capability
<u><A21><B9> Operating environment</u>	M	-	Q - Off card acceptor premises; cardholder mobile environment, including home PC, mobile phone, PDA
<u><A25> Manually Entered Card Number</u>	M	-	
<u><A26> Manually Entered Expiry Date</u>	M	-	
<u><A30> Address Verification Data</u>	C	-	If AVS requested
<u><A30><B1> Cardholder Billing Postal Code</u>	C	-	If AVS requested
<u><A30><B2> Cardholder Billing Address</u>	C	-	If AVS requested

Tag name	→	←	Comment
<A30><B3> Cardholder Billing City through <A30><B16> Ship to Phone Number	O	-	For Amex, see SGMF specification for further details on these sub tags.
<A31> Verification Code (CID, CVC2, CVV2)	O	-	If CID, CVC2, or CVV2 is available.
<A32> MOTO/e-Commerce Indicator	O	-	5 - Fully authenticated CAVV 6 - Non-authenticated, but attempted 3-D secure authentication 7 - Non-authenticated security transaction 8 - Non-secure e-commerce transaction A - In-App authentication (usually Apple Pay) for Discover, Mastercard, and Amex
<A33> Transaction ID (XID)	C	-	This is the XID for Visa, Discover, and Amex, or the Token Block B for Amex Tokens.
<A34> Cardholder Authentication for Secure e-Commerce Transactions	C	-	This is the CAVV for Visa, Discover, and Amex, or the Token Block A for Amex Tokens.
<A39> Electronic Commerce Security Level Indicator	C	-	This is the MasterCard UCAF Collection Indicator. If UCAF Data is not present with this value, MasterCard may downgrade the transaction on the response, and this will include the three downgraded UCAF indicator values in the response.
<A40> Authentication Data	C	-	Include if MasterCard UCAF (3-D Secure) data is available.
<A41> TSYS Acquiring Solutions Developer ID	M	-	This is the Developer ID assigned by TSYS during certification.
<A42> TSYS Acquiring Solutions Version ID	M	-	This is the software Version ID assigned by TSYS during certification.
<A51> Digital Entity Identifier	C	-	If Visa Checkout is used
<A73> Hold Capture Indicator	O	-	HOLD Use this if the transaction should be held (not settled) until the user is ready to capture and settle the transaction.
<A78> Transaction ID	-	C	If provided
<A80> Batch Number	-	M	
<A81> Local Transaction Date	-	M	
<A82> Local Transaction Time	-	M	

Tag name	→	←	Comment
<A83> Response Code	-	M	
<A84> Approval Code	-	C	If approved
<A85> Authorization Response Text	-	M	
<A86> Retrieval Reference Number (RRN)	-	M	
<A87> CCI Response (Commercial Card Indicator)	-	C	If card account is commercial card
<A88> Card Verification Result (CID/CVC2/CVV2)	-	C	If result sent from issuer
<A90> CAVV Result Code	-	C	If CAVV response provided
<A92> Card Product Codes	-	M	
<A93> Card Type	-	M	Contains the text representation of the card type
<A94> Requested ACI	O	-	Y, if applicable
<A95> Return ACI	-	C	If provided
<A119> Visa Validation Code	-	C	If provided
<A120> Authorization Source Code	-	C	If provided
<A124> Card Brand Tokenization	O	-	
<A124><B1> Card Brand Token Requestor ID	O	-	Required for Amex transactions in order to send the appropriate Tokenized format
<A130> Authorization Indicator	C	-	Mandatory for Mastercard transactions; See the SGMF Specification for more details. Values here will depend upon the nature of your transactions.

6.31.1 Hold Capture Original Request Sample

```
H2.999999<SGREQ><A1>123456789012345</A1><A3>1234567890ABCDEF12345678</A3><A10>10</A10><A12>C</A12><A15>10000</A15><A17>00001</A17><A18>5960</A18><A21><B1>5</B1><B2>0</B2><B3>S</B3><B4>0</B4><B5>0</B5><B6>6</B6><B9>Q</B9></A21><A25>123456*****7890</A25><A26>1231</A26><A32>5</A32><A34>FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF</A34><A41>SAMPLE</A41><A42>1000</A42><A73>HOLD</A73><A94>Y</A94><A124><B1>FFFFFFFFFFFF</B1></A124></SGREQ>
```

6.31.2 Hold Capture Original Response Sample

I2.<SGRSP><A80>61</A80><A17>00001</A17><A81>020617</A81><A82>100526</A82><A83>00</A83><A84>TAS001</A84><A88>M</A88><A95>N</A95><A78>0000000000000001</A78><A86>123456789000</A86><A93>VISA</A93><A92>A</A92><A85>APPROVAL</A85><A120> </A120><A119>1234</A119></SGRSP>

6.32 Split Shipment Request Message

Tag name	→	←	Comment
<u><A1> Host Capture POS ID</u>	M	-	Must be the first tag in the request.
<u><A3> GenKey</u>	M	-	Value obtained from an Authentication request
<u><A10> Transaction Code</u>	M	-	Must be 110 for split shipment
<u><A15> Transaction Amount</u>	M	-	This amount must match the original transaction
<u><A17> Transaction Original Number (TON)</u>	M	M	The TON must be a unique value to identify the transaction. The Request TON is echoed back in the response.
<u><A68> Settlement Amount</u>	M	-	Must reflect the amount to be settled
<u><A80> Batch Number</u>	-	M	
<u><A81> Local Transaction Date</u>	-	M	
<u><A82> Local Transaction Time</u>	-	M	
<u><A83> Response Code</u>	-	M	200 - If good 201 - If not good (original could not be found, or validation failed)
<u><A85> Authorization Response Text</u>	-	M	COMPLETED - if successfully matched and processed. NOT COMPLETED - if not successfully matched or processed.
<u><A86> Retrieval Reference Number (RRN)</u>	M	M	This must match the value returned on the original Authorization response. This value will be used to lookup the original transaction. In the response, it will be a unique number generated to match the Split Shipment Request.
<u><A99> Multiple Clearing Sequence Number</u>	M	-	This value reflects which payment or split shipment this is in a series.
<u><A100> Multiple Clearing Sequence Count</u>	C	-	This value reflects the total number of shipments or payments. If the number of shipments or payments is not known, this can be the value from A99 + 1 to reflect that this is not the final shipment. If this is the final shipment, then the A100 value should equal the A99 value.

6.32.1 Split Shipment Request Sample

H2.999999<SGREQ><A1>123456789012345</A1><A3>1234567890ABCDEF12345678</A3><A10>110</A10><A15>10000</A15><A17>00002</A17><A68>2500</

A68><A86>123456789000</A86><A99>1</A99><A100>2</A100></SGREQ>

6.32.2 Split Shipment Response Sample

I2.<SGRSP><A80>61</A80><A17>00002</A17><A81>020617</A81><A82>100526</A82><A83>200</A83><A86>123456789000</A86><A85>COMPLETED</A85></SGRSP>

6.33 Recurring / Installment

Tag name	→	←	Comment
<A1> Host Capture POS ID	M	-	Must be the first tag in the request.
<A3> GenKey	M	-	
<A10> Transaction Code	M	-	“10”
<A12> Payment Code	M	-	“C”
<A14> Repeat Indicator	C	-	If a repeat message “RPT”
<A15> Transaction Amount	M	-	
<A16> Secondary Amount	O	-	
<A17> Transaction Original Number (TON)	M	M	
<A18> Merchant Category Code (MCC)	O	-	
<A19> Language Indicator	O	-	
<A21> POS Entry Data	M	-	
<A21><B1> Cardholder Present Data	M	-	4 – Recurring 8 – MC Installment
<A21><B2> Card Present Data	M	-	0 – Card not present
<A21><B3> Card Data Input Mode	M	-	7 – Payment Credential stored on file
<A21><B4> Cardholder Authentication Method	M	-	0 – Not authenticated
<A21><B5> Authentication Entity	M	-	0 – Not authenticated
<A21><B6> Card data input capability	M	-	
<A21><B7> Cardholder authentication capability	M	-	0 – No electronic authentication capability
<A21><B8> Card capture capability	M	-	0 – No capture capability
<A21><B9> Operating environment	M	-	0 - No terminal used; Voice auth/ARU only; Mastercard recurring/installment transactions 4 – Off card acceptor premises; unattended
<A21><B10> Card data output capability	M	-	1 – None
<A21><B11> Terminal data output capability	M	-	3 – Display capability only
<A21><B12> PIN capture capability	M	-	0 – No PIN capture capability

Tag name	→	←	Comment
<u><A21><B13> Chip condition code</u>	-	-	
<u><A21><B14> Transaction status</u>	C	-	
<u><A25> Manually Entered Card Number</u>	M	-	
<u><A26> Manually Entered Expiry Date</u>	M	-	
<u><A30> Address Verification Data</u>	C	-	If AVS requested
<u><A30><B1> Cardholder Billing Postal Code</u>	C	-	If AVS requested
<u><A30><B2> Cardholder Billing Address</u>	C	-	If AVS requested
<u><A30><B3> Cardholder Billing City</u>	O	-	If Amex
<u><A30><B4> Cardholder Billing State/Province</u>	O	-	If Amex
<u><A30><B5> Cardholder Billing Country</u>	O	-	If Amex
<u><A30><B6> Cardholder Billing First Name</u>	O	-	If Amex
<u><A30><B7> Cardholder Billing Last Name</u>	O	-	If Amex
<u><A30><B8> Cardholder Billing Phone Number</u>	O	-	If Amex
<u><A30><B9> Ship to Postal Code</u>	O	-	If Amex
<u><A30><B10> Ship to Address</u>	O	-	If Amex
<u><A30><B11> Ship to City</u>	O	-	If Amex
<u><A30><B12> Ship to State/Province</u>	O	-	If Amex
<u><A30><B13> Ship to Country</u>	O	-	If Amex
<u><A30><B14> Ship to First Name</u>	O	-	If Amex
<u><A30><B15> Ship to Last Name</u>	O	-	If Amex
<u><A30><B16> Ship to Phone Number</u>	O	-	If Amex

Tag name	→	←	Comment
<u><A32> MOTO/e-Commerce Indicator</u>	O	-	2 - Recurring mail/telephone order transaction 3 - Installment payment of a mail / telephone order transaction R - First recurring SecureCode phone order transaction (Mastercard Only)
<u><A38> Special Condition Indicator</u>	O	-	
<u><A41> TSYS Acquiring Solutions Developer ID</u>	M	-	
<u><A42> TSYS Acquiring Solutions Version ID</u>	M	-	
<u><A43> Additional Amounts Group</u>	-	C	Partial approved amounts
<u><A43><B1> Account Type</u>	-	C	
<u><A43><B2> Amount Type</u>	-	C	
<u><A43><B3> Currency Code</u>	-	C	
<u><A43><B4> Sign</u>	-	C	
<u><A43><B5> Additional Amount</u>	-	C	
<u><A45> Free Form Text</u>	C	C	
<u><A46> Partial Authorization Indicator</u>	O	-	Use Discover card values if needed
<u><A48> Amex Merchant DBA Name</u>	C	-	Use for Amex transactions
<u><A73> Hold Capture Indicator</u>	O	-	"HOLD"; only for Hold Capture participants
<u><A74><B1>Billing ZIP Code</u>	-	C	
<u><A74><B2>Billing Street Match Code</u>	-	C	
<u><A74><B3>Billing Name Result Code</u>	-	C	
<u><A74><B4>Telephone Number Match Code</u>	-	C	
<u><A74><B5>E-mail Address Match Code</u>	-	C	
<u><A80> Batch Number</u>	-	M	Indicates the current open batch
<u><A81> Local Transaction Date</u>	-	M	
<u><A82> Local Transaction Time</u>	-	M	
<u><A83> Response Code</u>	-	M	
<u><A84> Approval Code</u>	-	C	If <A83> indicates “approved”

Tag name	→	←	Comment
<A85> Authorization Response Text	-	M	
<A86> Retrieval Reference Number (RRN)	-	M	
<A87> CCI Response (Commercial Card Indicator)	-	C	If card account is a commercial card
<A88> Card Verification Result (CID/CVC2/CVV2)	-	C	If result sent from issuer
<A89> Transaction Data Condition Code	-	C	If Discover card type
<A90> CAVV Result Code	-	C	If CAVV response provided
<A91> Merchant Advice Code	-	C	If Mastercard card type
<A92> Card Product Codes	-	C	If U.S. issued Visa card type
<A94> Requested ACI	O	-	
<A95> Return ACI	-	C	
<A97> Address Verification Result Code	-	C	If issuer provides
<A98> Delayed Capture Indicator	O	-	
<A135> POS Environment Indicator	M	-	C - Credential on File R – Recurring I - Installment

Appendix A

Table A.1 Summary of transaction types

Description	<A10> Transaction code	<A11> Program code	<A12> Payment code	<A13> Reversal indicator	<A21><B1> Cardholder present data	<A14> Repeat indicator
Purchase	10		C			RPT
Cash advance	11		C			RPT
Card not present	10		C		1-5, 8	RPT
Quasi cash	12		C			RPT
Card Authentication	20		C			RPT
Credit Return	13		C			RPT
Online Authorization Reversal	10		C	RVR		RPT
Credit bill pay	10	Y	C			RPT
Account funding credit purchase	10	W	C			RPT
Account funding card not present credit purchase	10	W	C		1-5, 8	RPT
Funds transfer credit purchase	10	X	C		1-5, 8	RPT
Funds transfer card not present credit purchase	10	X	C		1-5, 8	RPT
Credit/Stored value balance inquiry	14		C			RPT
Eligibility inquiry	14	Z	C			RPT
Balance inquiry reversal	14		C	RVR		RPT
Check guarantee	20	V				
Private label purchase	10		P			
PL cash advance	11		P			
PL card not present	10		P		1-5, 8	
PL quasi cash	12		P			
PL card authentication	20		P			

Description	<A10> Transaction code	<A11> Program code	<A12> Payment code	<A13> Reversal indicator	<A21><B1> Cardholder present data	<A14> Repeat indicator
Food stamp return	13		F			
Debit purchase	10		D			
Debit return	13		D			
Cash benefits cash withdrawal	11		E			
FS purchase	10		F			
Debit balance inquiry	14		D			
Debit bill pay	10	y	D			
FS voucher purchase	10		F			
CB purchase with cash back	10		E			
Account funding debit purchase	10	W	D			
Account funding debit return	13	W	D			
Funds transfer debit purchase	10	X	D			
Funds transfer debit return	13	X	D			
FS balance inquiry	14		F			
CB balance inquiry	14		E			
Debit purchase reversal	10		D	RVR		
Debit return reversal	13		D	RVR		
GC close card	18		G			
GC balance inquiry	14		G			
GC purchase	10		G			
GC return	13		G			
GC add value/load card	16		G			
GC decrease value/unload card	17		G			

Description	<A10> Transaction code	<A11> Program code	<A12> Payment code	<A13> Reversal indicator	<A21><B1> Cardholder present data	<A14> Repeat indicator
GC stand alone tip	10		G			
GC issue gift card	15		G			
GC purchase reversal	10		G	RVR		
GC cash back	11		G			
Credit prepaid card activation	15		C			RPT
Credit prepaid card activation reversal	15		C	RVR		
Credit prepaid card load	16		C			RPT
Debit prepaid card activation	15		D			
Debit prepaid card activation reversal	15		D	RVR		
Debit prepaid card load	16		D			
Debit prepaid card load reversal	16		D	RVR		

Appendix B

This section contains information related to EMV. It contains the following subsections:

- Mandatory Tags for Tag <A111>
- Summary Chip Card Transaction Data Elements
- Tags NOT to be sent in Tag <A111>
- Tags used in Contactless Transactions
- Expected Response Tags

8.0.1 Mandatory Tags for Tag <A111>

Table B.1 below contains a list of mandatory tags to be sent in Tag <A111>. If either or both of the tags are missing from Tag <A111>, the transaction will be rejected.

Table B.1 Mandatory Tags for Tag `<A111>`

Tag	Tag name	Description
DF79	Kernel Version Number	This tag specifies the version number of the kernel used to process the chip data in the transaction. (variable 1-32)
DF78	Device Serial Number	This tag specifies the manufacturer's unique serial number of the device that interacts with the chip card. (variable 1 - 20)

8.0.2 Summary Chip Card Transaction Data Elements

The following list of EMV tags are the primary tags used in a chip card transaction. Not all tags are used in every transaction. The tags being sent should be as close to the following list as possible and in the order listed where possible.

Table B.2 Summary Chip Card Transaction Data Elements

Tag	Tag name	Description
9F40	Additional Terminal Capabilities	This tag indicates the data input and output capabilities of the terminal.
9F02	Amount, Authorized	This tag specifies the authorized amount of the transaction (excluding adjustments).
9F03	Amount, Other	This tag specifies the secondary amount associated with the transaction representing a cashback amount
9F26	Application Cryptogram	This tag contains the cryptogram returned by the ICC in response of the GENERATE AC command.

Tag	Tag name	Description
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4F	Application Identifier (AID) - ICC	This tag identifies the application as described in ISO/IEC 7816-5.
9F06	Application Identifier (AID) - Terminal	This tag identifies the application as described in ISO/IEC 7816-5.
82	Application Interchange Profile	This tag contains the mnemonic associated with the AID according to ISO/IEC 7816-5.
9F36	Application Transaction Counter	This tag contains the counter maintained by the application in the ICC (incrementing the ATC is managed by the ICC).
9F34	Cardholder Verification Method (CVM) Results	This tag identifies a method of verification of the cardholder supported by the application.
9F27	Cryptogram Information Data	This tag indicates the type of cryptogram and the actions to be performed by the terminal.
9F39	POS Entry Mode	This tag indicates the method by which the PAN was entered, according to the first two digits of the ISO 8583:1987 POS Entry Mode.
9F33	Terminal Capabilities	This tag indicates the card data input, CVM, and security capabilities of the terminal.
9F1A	Terminal country Code	This tag indicates the country of the terminal, represented according to ISO 3166.
9F35	Terminal Type	This tag indicates the environment of the terminal, its communications capability, and its operational control.
95	Terminal Verification Results	This tag contains the status of the different functions as seen from the terminal.
5F2A	Transaction Currency Code	This tag indicates the currency code of the transaction according to ISO 4217.
9A	Transaction Date	This tag contains the local date that the transaction was authorized.
9B	Transaction Status Information	This tag indicates the functions performed in a transaction.
9F21	Transaction Time	This tag contains the local time that the transaction was authorized.
Tag	Tag name	Description
9C	Transaction Type	This tag indicates the type of financial transaction, represented by the first two digits of ISO 8583:1987 Processing Code. The actual values to be used for the Transaction Type data element are defined by the relevant payment system. Refer to the table below for a list of valid values.

Value	Description	Payment System
19	Fee Collection	V, MC, D
20	Return of Goods (Credit)	V, MC, D, AX
22	Adjustment - Credit	V, MC, D
29	Funds Disbursement	V, MC, D
30	Available Funds Inquiry	V, MC, D
40	Cardholder Account Transfer	V, MC, D

8.0.3 Tags not sent in Tag <A111>

The table below lists EMV tags that must not be sent in Tag <A111>.

Table B.4 Tags not sent in Tag <A111>

Tag	Tag name
5A	Primary Account Number (PAN)
57	Track 2 Equivalent Data
5F24	Application Expiration date
5F30	Service Code
9F1F	Track 1 Discretionary Data
9F20	Track 2 Discretionary Data

8.0.4 Tags used in Contactless Transactions

The following list of EMV tags are only used in contactless transactions. If the tag is transaction specific, this is stated in the description.

Table B.5 Tags used in Contactless Transactions

Tag	Tag name	Description
9F4C	ICC Dynamic Number	ICC PIN Encipherment Public Key certified by the issuer (Contactless EMV only)
9F6B	Track 2 Data (Mastercard)	The Track 2 Data is present in the file read using the READ RECORD command during a PayPass - Mag Stripe transaction. The PayPass reader copies the required digits of the UN (Unsigned Numeric), CVC3TRACK2, ATC and UN into the discretionary data field of the Track 2 Data and stores the modified Track 2 Data in the Data Record to be sent to the terminal. (Mastercard Contactless mag stripe only)
9F6E	Form Factor Indicator (FFI)	The PayPass Third Party Data contains proprietary information from a third party. (contactless mag stripe only)

9F7C	Customer Exclusive Data (CED) (Visa)	This tag contains data for transmission to the Issuer in Magnetic Stripe Data (MSD) transactions with a cryptogram. The MSD chip may contain customer data to be sent to the issuer in a contactless MSD transaction containing a cryptogram. For Visa only, the issuer may change this customer data using the Issuer Script in the response message.
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8.0.5 Expected Response Tags

The following is a list of expected response tags in a typical response message. Tags 71 and 72 are optional and if one is received, the other will not be received. Other tags may be included in the response message.

Table B.6 Expected Response Tags

Tag	Tag name	Description
8A	Authorization Response Code	This tag contains a code that defines the disposition of a message. (response)
91	Issuer Authentication Data	This tag contains data sent to the ICC for online issuer authentication.
71	Issuer Script Template 1	This tag contains proprietary issuer data for transmission to the ICC before the second GENERATE AC command. (response)
72	Issuer Script Template 2	This tag contains proprietary issuer data for transmission to the ICC after the second GENERATE AC command. (response)

Appendix C

Table C.1 Visa card product codes

Length	Format	Description
2	A/N	This field contains a code created by Visa during the authorization process. This field is conditional in the response. It will only be returned if the values are non-zero. Possible values: A^- Visa Traditional AX- American Express Card B^- Visa Traditional Rewards C^- Visa Signature D^- Visa Signature Preferred DI- Discover Card DN- Diners Card E^- Proprietary ATM F^- Visa Classic G^- Visa Business G1- Visa Signature Business G2- Reserved G3- Visa Business Enhanced Visa Platinum Business G4- Visa Infinite Business Visa Infinite Privilege Business (Canada) G5- Visa Business Rewards H^- Reserved I^- Visa Infinite [New Consumer Credit Product] I1- Visa Infinite Privilege I2- [Ultra High Net Worth]

Length	Format	Description
		J^- Reserved
		J1- Reserved
		J2- Reserved
		J3- Visa Healthcare
		J4- Reserved
		JC- JCB Card
		K^- Visa Corporate T & E
		K1- Visa GSC Corporate T & E
		L^- Electron
		M^- Mastercard
		N^- Visa Platinum
		N1- Visa Rewards
		N2- Visa Select
		P^- Visa Gold
		Q^- Private Label
		Q1- Reserved
		Q2- Private Label Basic
		Q3- Private Label Standard
		Q4- Private Label Enhanced
		Q5- Private Label Specialized
		Q6- Private Label Premium
		R^- Proprietary
		S^- Visa Purchasing
		S1- Visa Purchasing with Fleet (outside of Canada) Visa Fleet (cards issued in Canada)
		S2- Visa GSA Purchasing
		S3- Visa GSA Purchasing with Fleet
		S4- Commercial Loan
		S5- Commercial Transport EBT

Length	Format	Description
		S6- Business Loan
		S7- Reserved
		T^- Reserved
		U^- Visa Travel Money
		V^- V Pay
		V1- Reserved
		W^- Reserved
		X^- Visa B2B Virtual Payments
		Y^- Reserved
		Z^- Reserved

NOTE: The new names in brackets ([-]) are tentative; and may be changed with very short notice.

Appendix D

The chart below lists the suggested values for each subfield when authorizing a Mastercard transaction.

MasterCard												
Transaction Type	POS Entry Data Subfields											
	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12
e-Commerce	5	0	S	0	0	6	0	1	4-6,M-R	0	0	0,4
Mail Order/Telephone Order	1,2,3	0	1,6	0	0	6	0	1	0	0	0	0,4
Retail magnetic stripe read	0	1	2,B	5	0	2,B	0	1	1	0	0-4	0,4
Retail keyed	0	0,1	6	5	0	6,B	0	1	1	0	0-4	0,4
Retail Contactless	0	1	A,M	0,5	0	A	0	1	1	0	0-4	0,4
Recurring Payments	4,R	0	1	0	0	6	0	1	0	0	0	0,4
Installment Payments	4,R	0	1	0	0	6	0	1	0	0	0	0,4

Appendix E

11.0.1 Introduction - Control & Character Sets

The following subsections define the authorization request record character set and the character sets used for Track one and Track two data encoded on the magnetic stripes.

The authorization request records are generated with characters defined by ANSI X3.4-1986. The data stored on the cardholder's card in magnetic or optical form must be converted to the ANSI X3.4 character set before transmission to TSYS Acquiring Solutions. The following is a description of each subparagraph:

- Track One character set definition
- Track Two character set definition
- ANSI X3.4-1986 and ISO 646 character set definitions
- Cross-reference between the Track one, Track two, and ANSI X3.4 character sets
- Description of the method for generating and checking the Mod-10 Luhn check digit for credit card account numbers
- Description of the method for generating the LRC byte for the authorization request message and for testing the magnetic stripe LRC byte

The POS device must perform the following operations on Track read data before it can be used in an authorization request message.

1. The LRC must be calculated for the data read from the Track and compared to the LRC read from the Track. The Track data is assumed to be read without errors when no character parity errors are detected and the calculated and read LRCs match.
2. The starting sentinel, ending sentinel, and LRC are discarded.
3. The character codes read from the magnetic stripe must be converted from the encoded character set to the set used for the authorization request message. The characters encoded on Track One are six bit plus parity codes and the characters encoded on Track Two are four bit plus parity codes, with the character set used for the request message defined as seven bit plus parity code.

All characters read from a Track must be converted to the request message character set and transmitted as part of the request. The converted Track data cannot be modified by adding or deleting non-framing characters and must be a one for one representation of the characters read from the Track.

11.0.2 Track one character definition

Each character in Track One is defined by the six bit codes listed in [Table E. 1](#). Track One can be encoded with up to 79 characters as shown in [Table E. 2](#).

[Table E. 2](#) provides the ISO 7811-2 Track One character encoding definitions. This standards format is a sample guideline for expected credit card Track encoding. Private issue cards can differ in the way they are encoded.

Table E.1 Track One Character Definition

Field	Description	Length	Format
SS	Start Sentinel "%"	1	A/N
FC	Format Code ("B" for credit cards)	1	A/N
PAN	Primary Account Number	19 max	NUM
FS	Field Separator "^"	1	A/N
Name	Cardholder Name	2-26 max	A/N
FS	Field Separator "^"	1	A/N
Date	Expiration Date (YYMM)	4	NUM
SVC CD	Service Code	3	A/N
Discretionary Data	Optional Issuer Data	Variable	A/N
ES	End Sentinel "?"	1	A/N
LRC	Longitudinal Redundancy Check	1	
	Total cannot exceed 79 bytes	79 max	

Table E.2 Track One Character Encoding Set

BIT NUMBER				b6	0	0	1	1
				b5	0	1	0	1
b4	b3	b2	b1	ROW/COL	0	1	2	3
0	0	0	0	0	SP	0	(a)	P
0	0	0	1	1	(a)	1	A	Q
0	0	1	0	2	(a)	2	B	R
0	0	1	1	3	(c)	3	C	S
0	1	0	0	4	\$	4	D	T
0	1	0	1	5	(%)	5	E	U
0	1	1	0	6	(a)	6	F	V
0	1	1	1	7	(a)	7	G	W
1	0	0	0	8	(	8	H	X
1	0	0	1	9	)	9	I	Y
1	0	1	0	A	(a)	(a)	J	Z
1	0	1	1	B	(a)	(a)	K	(b)
1	1	0	0	C	(a)	(a)	L	(b)
1	1	0	1	D	-	(a)	M	(b)
1	1	1	0	E	.	(a)	N	(^)
1	1	1	1	F	/	(?)	O	(a)

(a) These character positions are for hardware use only

(b) These characters are for country use only, not for international use

(c) These characters are reserved for added graphic use

(%) Start sentinel
(?) End sentinel
(^) Field Separator
/ Surname separator
. Title separator
SP Space

PAR	MSB	B5	B4
	Most Significant Bit		
	Parity Bit (ODD)		
	Read LSB first		

11.0.3 Track two character definition

Each character in Track Two is defined by the four bit codes listed in [Table E. 3](#). Track Two can be encoded with up to 40 characters as shown in [Table E. 4](#).

[Table E. 4](#) provides the ISO 7811-2 Track Two character encoding definitions. This standards format is a sample guideline for expected credit card Track encoding. Private issue cards can differ in the way they are encoded.

Table E.3 Track Two Character Definition

Field	Description	Length	Format
SS	Start Sentinel "%"	1	A/N
PAN	Primary Account Number	19 max	NUM
FS	Field Separator "^"	1	A/N
Date	Expiration Date (YYMM)	4	NUM
SVC CD	Service Code	3	A/N
Discretionary Data	Optional Issuer Data	Variable	A/N
ES	End Sentinel "?"	1	A/N
LRC	Longitudinal Redundancy Check	1	A/N
	Total cannot exceed 40 bytes	40 max	

Table E.4 Track Two Character Encoding Set

b4	b3	b2	b1	COL		
0	0	0	0	0	0	(a) These characters are for hardware use only
0	0	0	1	1	1	
0	0	1	0	2	2	(B) Starting Sentinel
0	0	1	1	3	3	
0	1	0	0	4	4	(D) Field Separator
0	1	0	1	5	5	
0	1	1	0	6	6	(F) Ending Sentinel
0	1	1	1	7	7	
1	0	0	0	8	8	
1	0	0	1	9	9	
1	0	1	0	A	(a)	
1	0	1	1	B	(B)	
1	1	0	0	C	(a)	
1	1	0	1	D	(D)	
1	1	1	0	E	(a)	
1	1	1	1	F	(F)	

PAR	MSB	b3	b2	LSB
-----	-----	----	----	-----

(a) These characters are for hardware use only
 (B) Starting Sentinel
 (D) Field Separator
 (F) Ending Sentinel
 Most Significant Bit
 Parity Bit (ODD)
 Read LSB first

Appendix F

This table is used with <A139><B2> Document Type and <A139><B1> Ticket Transaction Type. An X in the column indicates the Document Type code (column 1) may be used with the Ticket Transaction Type code (other column headings) for the Amex Airline Industry.

Table 12.1 Document Type Code

Code	Description	TKT	REF	EXC	MSC
01	Passenger Ticket	X	X	X	
02	Additional Collection		X		X
03	Excess Baggage		X		X
04	Misc. Charge Order (MCO) / Prepaid Ticket Auth.	X	X	X	X
05	Special Service Ticket	X	X	X	X
06	Supported Refund		X		X
07	Unsupported Refund		X		X
08	Lost Ticket Application		X		X
09	Tour Order Voucher	X	X	X	X
10	Ticket by Mail	X	X	X	
11	Undercharge Adjustment		X		X
12	Group Ticket	X	X	X	
13	Exchange Adjustment		X	X	
14	SPD/Air Freight		X		X
15	In-flight Adjustment		X		X
16	Agency Passenger Ticket		X		X
17	Agency Tour Order/Voucher	X	X	X	
18	Agency Misc. Charge Order (MCO)	X	X	X	X
19	Agency Exchange Order		X		X
20	Agency Group Ticket	X	X	X	
21	Debit Adjustment Duplicate Refund/Use		X		X
22	In-flight Merchandise Ordered		X		X
23	Catalogue Merchandise Ordered		X		X
24	In-flight Phone Charges		X		X
25	Frequent Flyer Fee/Purchase		X		X
26	Kennel Charge		X		X
27	Animal Transportation Charge		X		X
28	Firearms Case		X		X
29	Upgrade Charge		X		X
30	Credit Unused Transportation		X		X
31	Credit Class of Service Adjustment		X		X
32	Credit Denied Boarding		X		X
33	Credit Misc. Refund		X		X
34	Credit Lost Ticket Refund		X		X
35	Credit Exchange Refund		X		X
36	Credit Overcharge Adjustment		X		X
37	Credit Multiple Unused Tickets		X		X
38	Exchange Order		X	X	X
39	Self-Service Ticket(s)	X	X	X	

Code	Description	TKT	REF	EXC	MSC
41	In-flight Duty Free Purchase		X		X
42	Senior Citizen Discount Booklets	X	X	X	X
43	Club Membership Fee		X		X
44	Coupon Book	X	X	X	X
45	In-flight Charges		X		X
46	Tour Deposit		X		X
47	Frequent Flyer Overnight Delivery Charge		X		X
48	Frequent Flyer Fulfillment		X		X
49	Small Package Delivery		X		X
50	Vendor Sale		X		X
51	Miscellaneous Tax(es) Fee(s)		X		X
52	Travel Agency Fee		X		X
60	Vendor Refund Credit		X		X
64	Duty Free Sale		X		X
65	Preferred Seat Upgrade	X		X	X
66	Cabin Upgrade	X		X	X
67	Lounge/Club Access or Day Pass	X		X	X
68	Agent Assisted Reservation. Ticketing Fee	X		X	X
69	Ticket Change or Cancel Fee	X		X	X
70	Trip Insurance	X		X	X
71	Unaccompanied Minor	X	X	X	X
72	Standby Fee	X		X	X
73	Curbside Baggage	X		X	X
74	Inflight Medical Equipment	X		X	X
75	Ticket or Pass Print Fee	X		X	X
76	Checked Sporting/Special Equipment	X		X	X
77	Dry Ice Fee	X		X	X
78	Mail/Postage Fee	X		X	X
79	Club Membership Fee – Temporary/Trial	X		X	X
80	Frequent Flyer Activation/Reinstatement	X		X	X
81	Gift Certificate	X		X	X
82	Onboard/Inflight Prepaid Voucher	X		X	X
83	Optional Services Fee	X		X	X
84	Advanced Purchase – Excess Baggage	X		X	X
85	Advanced Purchase – Preferred Seat Upgrade	X		X	X
86	Advanced Purchase – Cabin Upgrade	X		X	X
87	Advanced Purchase – Optional Services	X		X	X
88	Wi-Fi	X		X	X
89	Packages	X		X	X
90	Inflight Entertainment/Internet Access	X		X	X
91	Overweight Bag Fee	X		X	X
92	Sleep Sets	X		X	X
93	Special Purchase Fee	X		X	X